



NATIONAL TECHNICAL UNIVERSITY OF ATHENS

School of Naval Architecture & Marine Engineering

DIPLOMA THESIS

Dynamic Modeling & Predictive Control of a Hydrofoil Vessel

Ioannis Skounakis – Kounavis

Supervisor: Associate Prof. G. Papalambrou

Committee Members:

Prof. G. Grigoropoulos
Associate Prof. C. Papadopoulos

October 2025

Abstract

Recent advances in high-speed marine vehicles have renewed interest in hydrofoil craft as energy-efficient alternatives to displacement hulls, particularly under tightening emission and fuel consumption regulations. Fully submerged hydrofoils, however, are inherently unstable and require precise automatic control to maintain flight stability and ride comfort.

This thesis investigates model-based control strategies for a small-scale fully submerged hydrofoil platform, while also establishing a foundation for scaling to larger systems. A parametric hydrodynamic model is developed and linearized over a range of forward speeds to capture the system's nonlinear dependence on velocity. The resulting family of linear state-space models forms the basis for controller design: a Linear Quadratic Regulator (LQR) with integral action and actuator saturation serves as a baseline, while linear and linear parameter-varying Model Predictive Control (MPC and LPV-MPC) schemes are developed to accommodate varying dynamics.

The proposed approaches are validated in simulation and benchmarked against the baseline LQR controller, demonstrating improved robustness and trajectory tracking across operating conditions. The work contributes a systematic framework for LPV-MPC design tailored to marine hydrofoils, bridging established optimal control theory with emerging applications in high-speed, energy-efficient waterborne transport.

Acknowledgements

This work marks the completion of my studies at the School of Naval Architecture and Marine Engineering. It proved especially challenging, as it required extensive research and a multidisciplinary approach, bringing together diverse fields that ultimately became areas of great personal interest.

I would like to express my sincere gratitude to my supervisor, Associate Professor George Papalambrou, for introducing me to the field of Control Systems, instructing me throughout this work and for his continuous effort to demonstrate the value of computational methods in engineering applications, a perspective that has accompanied me throughout my studies.

I am deeply thankful to my friends, who patiently endured my fixation on this topic and offered invaluable feedback whenever asked.

I am grateful to my family for their unwavering support and encouragement throughout my studies.

Last but certainly not least, I wish to acknowledge the many academics and researchers worldwide who generously share their work in the public domain, contributing selflessly to the advancement of knowledge without artificial barriers.

Copyright Statement

The author gratefully acknowledges permission to reproduce copyrighted material as follows:

- Figs. 2.5 and 2.6 Cambridge University Press, license ID 110477 (PLSclear).
- Figs. 2.8 and 5.19, Cambridge University Press, license ID 110716 (PLSclear).
- Fig. 4.6, Cambridge University Press, license ID 110478 (PLSclear).
- Fig. 4.2 John Wiley & Sons, license ID 1661025-1 (CCC).

Contents

Abstract	iii
Acknowledgements	iv
Preface	xiii
Nomenclature	xiv
1 Introduction	1
1.1 Motivation	1
1.2 Literature Review	6
1.2.1 Theoretical Background	6
1.2.2 State of the Art	6
1.3 Objectives & Outline	7
2 Dynamic Model	9
2.1 Prerequisites	9
2.1.1 Parametric Model	10
2.1.2 Physical Parameters	11
2.2 Kinematics	11
2.2.1 Coordinate systems & reference frames	13
2.2.2 Coordinate Transforms	15
2.3 Rigid Body Kinetics	16
2.4 Environmental Forces	18
2.4.1 Gravity	18
2.4.2 Waves	18
2.5 Actuation Forces	23
2.5.1 Hydrofoils & Struts	23
2.5.2 Propulsion	40
2.6 Parametric Simulink Model	43
3 Linearization & System Analysis	45
3.1 Model Reduction	45
3.1.1 State Selection	46

3.2	Linearization	46
3.2.1	Static Stability - Trimming	47
3.2.2	Automatic Linearization	48
3.3	Linear Model	49
3.4	System Analysis	52
3.4.1	Characteristics	52
4	Control System Design	59
4.1	Problem Formulation	59
4.1.1	Objectives	59
4.1.2	Frequency Domain	60
4.1.3	Time Domain	61
4.2	Optimal Control	62
4.2.1	Continuous-Time LQR	62
4.2.2	Discrete-Time LQR	67
4.2.3	Designs & Tuning Parameters	68
4.3	Constrained Optimal Control	69
4.3.1	Constrained Finite Time Optimal Control	69
4.3.2	Quadratic Program Solution	70
4.3.3	Receding Horizon Control	71
4.4	Model Predictive Control	74
4.4.1	Controller Structure	74
4.4.2	Extensions to the MPC Scheme	77
4.4.3	Designs & Tuning Parameters	79
4.5	State Estimation	80
4.5.1	Sensor Measurements	80
4.5.2	Discrete Kalman Filter	85
4.5.3	Tuning Parameters	87
5	Simulation Results	89
5.1	Setup & Objectives	89
5.1.1	Performance Metrics	91
5.2	Reference Tracking	91
5.2.1	Step Response	92
5.2.2	Multiple Output Reference	95
5.3	Disturbance Rejection	98
5.3.1	Frequency Response	98
5.3.2	Waves	98
5.3.3	Additional Scenarios	100
5.3.4	Sensor Feedback	103
5.4	Model Uncertainty	105
5.5	Discussion	110
6	Conclusions & Future Work	112
6.1	Dynamic Modelling & Simulation Environment	112
6.2	Proposed Control Algorithms	112
6.3	Next Steps	113

Appendices	115
A Models, Parameters, Concepts	116
A.1 Test Vessel Parameters	116
A.2 Selection of Wave Parameters	118
A.3 Nominal Model Matrices	120
A.4 Mathematical Concepts	121
A.5 Vessel Design & Dynamic Stability	121
B Developed Simulation Environment	124
B.1 Parametric Model	124
B.2 Control & Estimation	124
B.3 Main Functionalities	125

List of Figures

1.1	Boeing Jetfoil 929.	1
1.3	Modern types of hydrofoil ferries ¹	2
1.4	Typical resistance curve of a hydrofoil vessel.	3
1.2	USS Plainview in 1969	3
1.5	Control surface arrangement on a passenger jet aircraft (Reproduced from [6]).	4
1.6	Control surface arrangement for the Jetfoil (Reproduced from [5]).	4
1.7	Typical diagram of control systems architecture (Based on works presented in [7]).	5
2.1	Modeling process: physical system, coordinate definition, and governing equations (image of the Clafis Victron Solar Boat Team taken from the web).	10
2.2	Earth-Centered Earth-Fixed (ECEF) frame shown to rotate with an angular rate ω_e with respect to Earth-Centered Inertial (ECI). Also shown are the North-East-Down (NED) and BODY frames (Reproduced from [10]).	12
2.3	Representation of coordinate systems, vessel design parameters and acting forces as a visual guide to the dynamic model.	12
2.4	Regular wave	19
2.5	Orbital motion of water particles induced by a harmonic wave in deep water (Adapted from [46]).	19
2.6	Frequencies and periods of the vertical motions of the ocean surface (Reproduced from [46]).	21
2.7	Short-crested wind waves simulated by 10 regular components, International Towing Tank Committee (ITTC) spectrum, $H_s = 0.3\text{ m}$, $\omega_{\text{peak}} = 2\text{ rad} \cdot \text{s}^{-1}$, 5 directions in grid.	22
2.8	Foil and wing geometry definitions (Reproduced from [9]).	24
2.9	The two-dimensional lifting problem characteristics.	25
2.13	First order delay response to step input. Also visible are the τ and 3τ time instants achieving 63% and 95% of reference respectively.	26
2.10	Hydrodynamic angle definitions and coordinate frames involved in control surface modelling. Also visible are strut quantities used in the simulation.	27
2.11	Inflow velocity and angle of attack change due to vessel's steady speed, incident waves, and velocity perturbations.	28

2.12	Perspective view of the axes system used to model hydrofoils (Reproduced from [19]).	28
2.14	Reynolds sweep results for NACA 63412, $n_{\text{crit}} = 5$, Mach = 0.	30
2.15	Wing tip vortices and their effect on effective angle of attack due to downwash (Reproduced from [49]).	31
2.16	Lift vector tilt due to downwash on a finite wing (Reproduced from [8]). . .	31
2.17	Foil & strut assembly, VLM (linear) results for $\alpha = 10^\circ$	33
2.18	Comparison of calculated polars by XFRL5 for an elliptical wing versus direct application of Prandtl's Lifting Line Theory ($AR = 12$).	33
2.19	Polar plots of Theodorsen and Sears functions. Exact values shown for $k = 0.1, 0.2, 0.5, 1, 2, 5$	36
2.20	Block diagram showing the convolution of input angle signal with the derivative of Wagner's step response in transfer function form.	37
2.21	Angle of attack changes due to roll rate, inducing lateral damping.	40
2.22	Propellers developed by Solar Boat Twente (From the team's website [59]) .	41
2.23	3D model of the propeller geometry selected for the nominal simulation model.	41
2.24	Thrust, torque and efficiency coefficients used in the simulation model. . . .	42
2.25	Block diagram description of the non-linear model as it is implemented in Simulink. Also visible are the points for automatic linearization.	44
3.1	Forces & moments affecting longitudinal stability.	48
3.2	Result of successive linearization over a speed range $U_0 = 5 \text{ m} \cdot \text{s}^{-1}$ to $10 \text{ m} \cdot \text{s}^{-1}$ for nominal model parameters.	49
3.3	Comparison of linear vs non-linear model response to sinusoidal control input on all control surfaces.	52
3.4	Response to impulse input from each input channel.	54
3.5	Response to step input from each input channel.	55
3.6	Maximum and minimum singular values of the model described by Eqs. (A.3.1) and (A.3.2).	57
4.1	Typical controller-plant block diagram	60
4.2	Qualitative frequency response of multivariable loop transfer function $\mathbf{L} = \mathbf{G}\mathbf{K}$ by its minimum and maximum singular values (Reproduced from [11]).	61
4.3	Transfer functions L, S, T for a Single Input Single Output (SISO) system showing results of feedback and objective regions. (Adapted from [65]). . . .	62
4.4	Block diagram form of tracking LQR controller.	65
4.5	Block diagram form of LQI controller with saturation and anti-windup. . .	66
4.6	The working principle of Receding Horizon Control (Reproduced from [12]).	72
4.7	Internal model used for MPC state estimation and prediction. All sub-models are included in the state space description of Eqs. (4.4.3) and (4.4.4).	75
4.8	State estimation within one computation interval. Notice the distinct actions labeled <i>prediction</i> : One is used within the KF and one within the QP context.	76
4.9	Differences between different motion tracking products and naming conventions.	81

4.10	Spectral densities of white and band limited noise, and the time-domain signals they produce.	82
4.11	Sonar sensor placement and beam measurements. Shown magnified, is the approximation error due to the simplifications made.	84
5.1	Response to step change of +0.2m in heave tracking for LQI controllers. Visible saturation and integrator windup effects.	92
5.2	Response to step change of +0.2m in heave tracking for MPC controllers with no reference previewing. Generated input signals are omitted as these are not final designs.	93
5.3	Response to step change of +0.2m in heave tracking for MPC controllers with reference previewing active. Significantly smaller settling time.	94
5.4	Effect of actuator delay modelling in MPC response. Controller output and optimal input predictions are shown at $f = 5$ Hz.	96
5.5	Effect of different reference trajectories.	97
5.6	Response for multiple output reference tracking, LQI-SW. The dotted lines are trajectories generated for zero pitch reference.	97
5.7	Comparison of singular values for different designs. Vertical line signifies the Nyquist frequency $\omega_N \approx 31.4 \text{ rad} \cdot \text{s}^{-1}$ for $f_s = 10$ Hz.	99
5.8	Comparison for regular wave disturbance on a frequency sweep, head and following sea, x -axis referring to encounter frequency for $V_s = 7 \text{ m} \cdot \text{s}^{-1}$. . .	100
5.9	Response amplitudes for various wave frequencies with respect to encountered wavelength.	101
5.10	Controller performance at a take-off/touchdown maneuver with regular wave disturbance of $\omega_e = 4 \text{ rad} \cdot \text{s}^{-1}$ and wave direction 0°	102
5.11	Controller performance at a disturbance in K moment, applied for $t \in [4, 5.5]$ s and regular wave disturbance of $\omega_e = 4 \text{ rad} \cdot \text{s}^{-1}$ for wave direction 0°	102
5.12	Comparison of ground truth, estimated and measured quantities, occurring at different sample rates.	103
5.13	Effect of initial guess on eventual DKF convergence.	104
5.14	Boxplot analysis of different control designs at $N = 400$ trials. Visible outliers and vertical scale offset for LQI-SW.	106
5.15	Response for reference tracking and disturbance rejection under model uncertainty, LPV-MPC-IP.	107
5.16	Response for reference tracking and disturbance rejection under model uncertainty, LQI-SW.	108
5.17	Correlation of mass and horizontal Center of Gravity (CG) position to performance: LPV-MPC-IP.	109
5.18	Correlation of mass and horizontal CG position to performance: LQI-SW. .	109
5.19	Heave and pitch amplitudes at increasing encountered wavelength. Uncontrolled case (Reproduced from [9]).	110
A.1	Geographical window used for attaining sea state parameters.	118
A.2	Sea state over the summer of 2024 averaged over a specified geographical domain in the Mediterranean.	119
A.3	Pitch stiffness (Reproduced from [3]).	122

A.4	Vessel C_{m_a} for varying CG position, vertical lines corresponding to design choices simulated in Fig. A.5.	123
A.5	Heave response for elevator impulse input at different CG placements (distance d from fore foils).	123
B.1	Simulink model	126

List of Tables

2.1	Six degrees of freedom for marine vessel dynamics.	13
2.2	Definitions of Society of Naval Architects & Marine Engineers (SNAME) notation within Fossen’s coordinate vector framework.	14
4.1	Controller naming legend for LQR family.	68
4.2	Tuning parameters for LQR design.	69
4.3	Controller naming legend for MPC family.	79
4.4	Tuning parameters for MPC design.	80
4.5	Noise characteristics for different measured quantities.	83
5.1	Final controller parameters for LQI.	90
5.2	Final controller parameters for MPC.	90
5.3	Step response performance metrics for LQI controllers.	93
5.4	Step response performance metrics for MPC controllers without reference previewing.	94
5.5	Step response performance metrics for MPC controllers with reference pre- viewing active.	95
5.6	Parameter bounds for probabilistic stability assessment.	105
A.1	Parameter definitions for nominal simulation model.	117

Preface

The idea for the present work arose from my involvement in the Monaco Energy Boat Challenge (MEBC), as a member of the Energy Class team Oceanos NTUA. It was my first encounter with the practical application of engineering concepts, where principles from mechanical and marine engineering intersected with electronics, feedback control, and performance analytics within a complex system. The Solar Class boats competing in MEBC, constrained by the low power density of solar panels, relied on hydrofoils to reduce total power consumption efficiently. However, this introduced the need for active control, increasing the system's complexity, but at the same time creating opportunities to implement advanced strategies such as optimized energy preservation, constraint handling, and even granting a degree of autonomy to the vessel.

Through this project, I was allowed to combine my interests in hydrodynamics, programming, and control, by developing an implementable solution to a real-world problem. It also strengthened my appreciation for interdisciplinary collaboration and the balance between rigorous theoretical derivations and their translation into practical applications.

Nomenclature

Symbols

g	Gravitational acceleration	9.81	m/s^2
μ	Viscosity	1.077×10^{-3}	$\text{Pa} \cdot \text{s}$
ν	Kinematic viscosity	1.0508×10^{-6}	m^2/s
ρ	Density of seawater	1024.8103	$\text{kg} \cdot \text{m}^{-3}$

Acronyms

AR	Aspect Ratio
CFTOC	Constrained Finite Time Optimal Control
CG	Center of Gravity
CO	Center of Origin
DCM	Direction Cosine Matrix
DKF	Discrete Kalman Filter
DOF	Degrees of Freedom
ECEF	Earth-Centered Earth-Fixed
ECI	Earth-Centered Inertial
EOM	Equations of Motion
GPS	Global Positioning System
IAE	Integral of the Absolute Error
IMU	Inertial Measurement Unit
ISE	Integral of the Square of the Error
ITAE	Integral of the Time weighted Absolute Error
ITTC	International Towing Tank Committee
LPV	Linear Parameter Varying
LQI	Linear Quadratic Integral
LQR	Linear Quadratic Regulator
MEBC	Monaco Energy Boat Challenge
MIMO	Multiple Input Multiple Output
MPC	Model Predictive Control

MSS	Marine Systems Simulator
NACA	National Advisory Committee for Aeronautics
NED	North-East-Down
SISO	Single Input Single Output
SNAME	Society of Naval Architects & Marine Engineers

Chapter 1

Introduction

1.1 Motivation

In recent years, maritime transport has faced unprecedented pressure to reduce fuel consumption and greenhouse gas emissions. The International Maritime Organization (IMO) has mandated steep cuts in CO₂ intensity through its 2023 strategy, targeting a 40% reduction by 2030 and net-zero by 2050. Similarly, the European Union’s “Fit for 55” and the FuelEU Maritime Regulation extend emissions trading and efficiency requirements to coastal and inland vessels. At the same time, the transition to electric propulsion, limited by current battery energy density, has made drag reduction and hydrodynamic efficiency paramount. Every percent of resistance saved translates directly to extended range or smaller, lighter batteries.

Against this backdrop, hydrofoils have re-emerged as one of the most promising technologies for high-speed, zero-emission vessels. By lifting the hull out of the water, they can drastically reduce wetted area and wave making drag, as evidenced in Fig. 1.4, allowing small electric boats and ferries to achieve long ranges and smooth rides that would otherwise be impossible. Hydrofoils are hardly a new concept, as they emerged before the 20th century by the Italian inventor Enrico Forlanini. In subsequent years the concept took off, with such seagoing vessel design seeming very promising both for military and passenger use. During the mid 1960s, the US Navy placed significant effort into constructing hydrofoil military ships cooperating with Lockheed Shipbuilding, resulting in the USS Plainview, shown in Fig. 1.2. Just 4 years before Plainview’s decommission, Boeing launched the Jetfoil 929 passenger hydrofoil ferry (Fig. 1.1), used to this day in Hong-Kong and Japan.



Figure 1.1: Boeing Jetfoil 929.

It was not by chance, that two of the most iconic hydrofoil vessels were built by companies specializing in aircrafts. The operational principles when foilborne are similar to any flying machine, and for the particular designs mentioned before, stability is an impor-

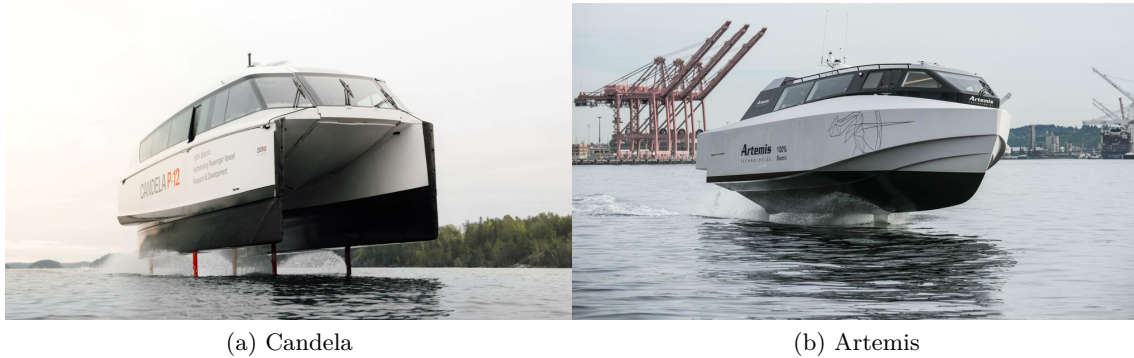


Figure 1.3: Modern types of hydrofoil ferries ¹.

tant concern. Many hydrofoil designs relied on a passive control system, by incorporating surface-piercing hydrofoils at various arrangements to ensure dynamic stability. Such designs are characteristic of the vessels developed in the Soviet Union such as the *Cometa* type, which still operates in Greece. Relying on such a system however, albeit simpler, negates a part of the benefits in resistance, raising the lowest part of the curve in Fig. 1.4. Such caveats, combined with their inherently complicated design and elevated operational costs caused these ships to lose popularity. To be precise, that does not negate their tremendous efficiency at high speeds but for the shipowners, deciding to operate a slower fleet has proven more cost effective than high-maintenance alternatives. As this technology re-emerges, providing a solution for the limited energy density of battery-electric comes with the added complexity of designing the flight control system. Shown in Fig. 1.3 for example, are two modern designs of hydrofoil ferries developed by leading companies in the field. Both have opted for fully submerged inverted-T shaped foils, requiring active control. In this context and as was mentioned in the Preface, student competitions such as MEBC offer a relatively low-cost way to implement and test such technologies. For this reason, it was decided that simulations were to be based on a nominal model that conforms to the competition's design parameters. The nominal model, which will be analyzed in the following chapters, is based on an older participation [1].

When designing aircraft control systems, the engineer is typically working on stability augmentators, aiming to affect unstable modes such as the phugoid or dutch roll [2], [3], [4]. In order to achieve this, airplanes are usually designed in way that optimizes for decentralized control. This allows for classic methods like root-locus or simple pole placement, since unstable modes are seen as SISO systems with their respective control surface inputs. Taking a look at Fig. 1.5 provides a good view of how this specification is met: Flaps, which engage for take-off and landing are separated from stabilizing surfaces like ailerons (roll) and elevators (pitch). While generally a desired arrangement, it isn't always implementable, seeing as how the Jetfoil [5] control surfaces in Fig. 1.6 appear to be largely simplified. Even in that case however, labels 28, 30 and 32, 34 show that pitch regulation can be separated from roll motions (and vice versa). This is not the case for the studied system.

¹From thenextweb.com/news/swedish-flying-boat-startup-candela-funding and seattletimes.com/opinion/editorials/a-peek-into-the-future-as-foil-ferries-take-flight/.

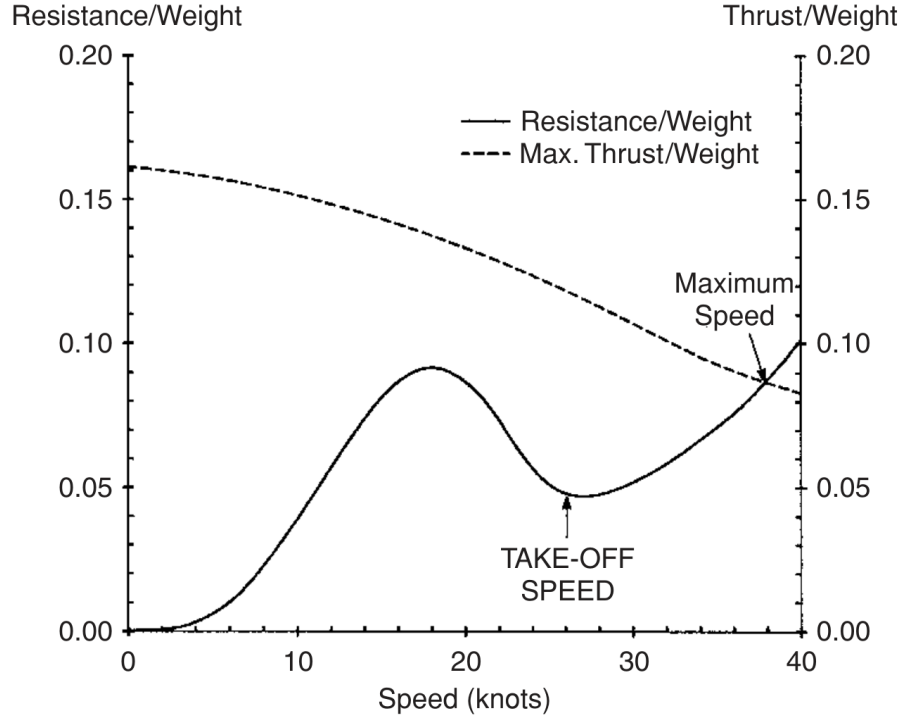


Figure 1.4: Typical resistance curve of a hydrofoil vessel.

To mitigate this issue, controller synthesis can be automated to *optimally* consider all of the system's Degrees of Freedom (DOF), and work on a high-level hierarchy generating signals for the controlled input actuators. Those signals are then interpreted by simpler, SISO low-level controllers designed to track the given command. Figure 1.7 shows a diagram view of this idea, and pinpoints the exact scope of the present work: Design a dynamic optimizer that will stabilize a given system, track a pre-generated trajectory and enforce constraints.



Figure 1.2: USS Plainview in 1969

Problem Description The above discussion sets the problem which the present work aims to tackle: Firstly, creating a tool that will allow for modelling the dynamics of such a system, reducing them to a linear representation usable in the control design procedure. Secondly select and implement an appropriate control method for the given system. Through the process of reviewing the available literature, the thesis objectives became clearer and are presented in Section 1.3.

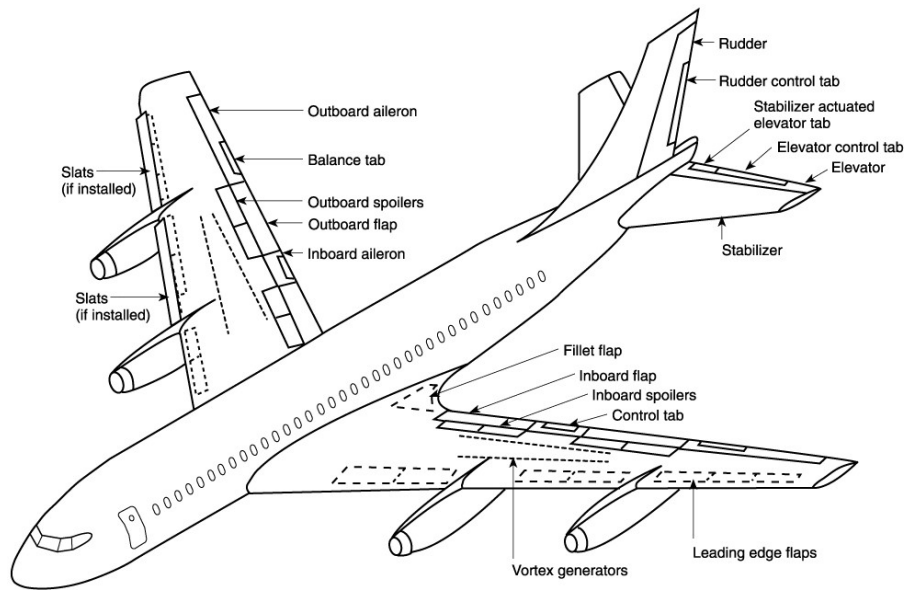


Figure 1.5: Control surface arrangement on a passenger jet aircraft (Reproduced from [6]).

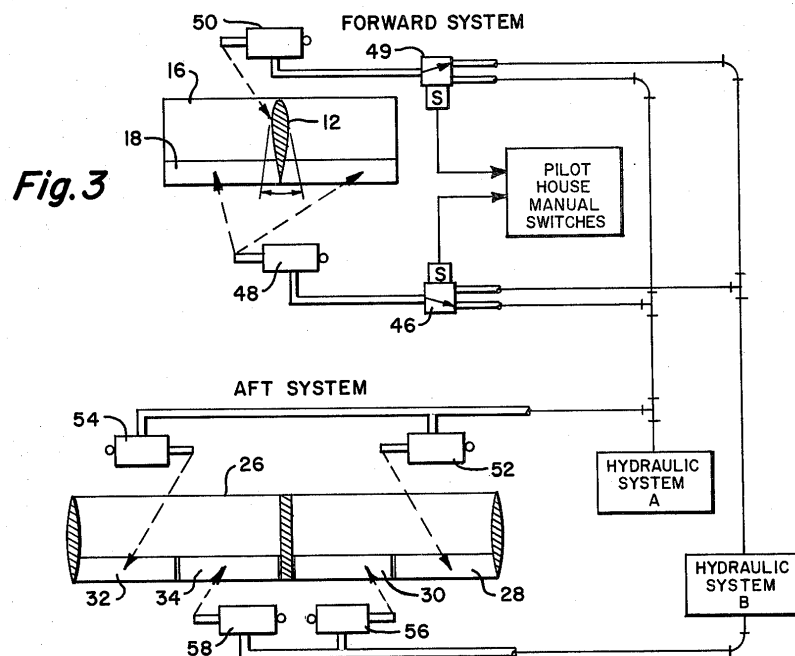


Figure 1.6: Control surface arrangement for the Jetfoil (Reproduced from [5]).

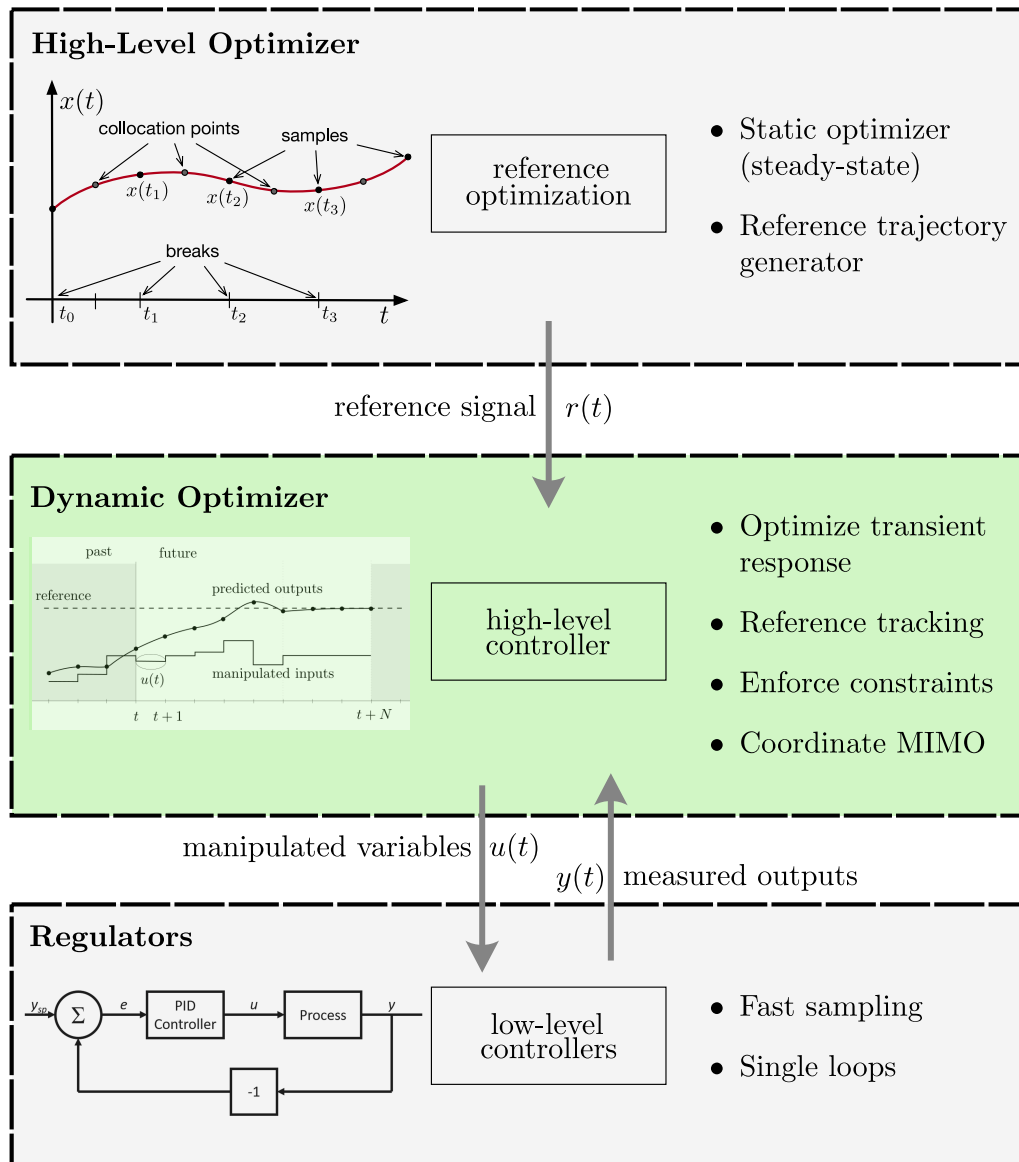


Figure 1.7: Typical diagram of control systems architecture (Based on works presented in [7]).

1.2 Literature Review

Covering the available literature on the subject, may be viewed through different lenses. From a historical standpoint, specific technologies may be viewed as more favorable and possibly feasible, but from a scientific point of view, some approaches will seem obsolete, or simply a product of their time. At the same time, since the subject is highly applied, it is important to track progress made both in academia and industry. While this section ought to address the state of the art relating to the subject of interest, due to its multidisciplinary nature, it was decided to also include specific textbooks and material used to set up the theoretical part.

1.2.1 Theoretical Background

Hydrodynamic Modelling To create mathematical models of motion and forces arising from interactions between lifting surfaces and a fluid, a combination of sources on aerodynamics and hydrodynamics was used. Theory of flight stability and control is presented in detail by Etkin [3] and Roskam [2] with Beard & McLain [4] implementing it in code, while Anderson [8] provides the important background in aerodynamics. Extended consideration of high-speed marine crafts and hydrodynamic properties are analyzed by Faltinsen [9]. Results from all these separate fields have been accumulated and re-derived specifically for marine crafts with considerations for control, by Fossen [10]. For that reason, his framework is the one adopted throughout this text.

Control & Estimation Applying control theory on a real system required synthesizing material from several authors. Classical multivariable control theory is presented primarily using the framework of Skogestad and Postlethwaite [11], while model predictive control (MPC) concepts are introduced following Borrelli et al. [12] and further developed using Maciejowski [13]. Concepts of optimal control and digital applications follow Franklin et al. [14] with advanced elements adopted from Kirk [15] and Hespanha [16]. Finally, the formulation of state estimation via Kalman filtering follows Simon [17].

1.2.2 State of the Art

We now switch our focus towards the combined problem of modelling and control. The first systematic studies on hydrofoil control date back to the USS Plainview program, where M. Martin and colleagues documented modeling and stability analyses of a fully submerged, large-scale hydrofoil craft [18], [19]. Similar empirical results were obtained by Keuning [20], where unsteady effects were taken into consideration based on results by Theodorsen and Garrick [21]. These reports established a framework for representing foil dynamics and the principles of automatic ride-control systems, laying the foundation for later experimental vessels.

During the following decades, industrial and academic attention centered on the Jetfoil, developed in the early 70s by the Boeing Company. Many of the works found in journals, involve the Jetfoil linearized dynamics initially presented by Y. Saito and T. Ikebuchi. The works of Kim & Yamato ranging from mid-90s to 2005 provided a comprehensive overview of the control problem, especially that of switching modes between platforming

and contouring using optimal servo theory [22]. In 2004 and 2005 they worked on optimal preview control and extended results to random seas, including experimental set-ups to support simulation findings [23], [24]. Their publications address hydrodynamic modeling, ride-control algorithms, and experimental validation on passenger craft, marking important integration of control theory with commercial hydrofoil operation.

Academic developments at MIT led by P. Slavounos and M. Triantafyllou connected hydrofoil dynamics with marine vehicle stability and sailing technology often forming non-linear control methods [25], influencing high-performance designs such as their involvement on the America's Cup. Advised by Slavounos, Hatzakis [26] completed preliminary modelling and control simulations for pitch-heave stabilization of military hydrofoil crafts using LQR. Similar work for the Luna Rossa team of America's Cup was carried out by Bencatel et. al [27], combining LQR with integral action and state estimation for hydrofoiling catamaran sailboats. These findings represent a transition from classical to optimal control formulations for hydrofoils.

Renewed academic interest emerged through university projects in the 2010s. At TU Delft, the Hydro Motion team and the University spin-off Flying Fish contributed detailed modeling and control design studies for small-scale autonomous hydrofoils, including the works of Marrewijk & Schonebaum [28] and later experimentations on the *Hydrofoil Education and Research Platform* by Kapnisis [29]. Their approaches range from self-stabilizing to robust control implementations such as $\mathcal{H}_\infty$. Research groups at EPFL and ETH Zürich have contributed in sailing hydrofoils, approaching the stabilization problem by LQR or MPC [30]. Notably with Hydroptère which in 2009 broke the record of 50 knot foil assisted flight [31] there was use of computer vision to assess foil immersion.

Another similar initiative, emerged at KTH, namely the FoilCart, allowing students to implement their work. Many variations of control techniques were studied, such as robust [32], use of Kalman Filter for estimation [33] and LQ augmentations [34]. Parallel efforts at Texas A&M University, including studies by Yuchen Shang and Juan Horrillo, explored autonomous foiling surface vehicles and sensor-based stabilization bridging CFD with stability assessments [35], [36]. At the same time, world leading companies such as Candela (Sweden), Artemis (UK) and Navier (USA) are currently pushing the limits on hydrofoil shipping, but unfortunately little is known about their product's inner workings.

Recent doctoral work, notably J. M. Kleven Godo's PhD at NTNU [37], consolidates these developments through comprehensive modeling, simulation, and control validation on modern experimental platforms. Selected publications included in the dissertation, experimented with application of Reinforcement Learning algorithms, compared to PID as a baseline. Together, these contributions trace a clear evolution from early analog stabilization toward rigorous, model-based control of hydrofoil craft across both industrial and academic contexts.

1.3 Objectives & Outline

The extensive literature study made above, solidified the research questions that the present work aims to explore.

- Can we approach the modelling problem in a general, abstract sense, allowing for the description of different vessels as well as the parametric uncertainty of a specific one?

- Can we automate the trimming and linearization process to create linear models?
- Can we utilize optimal control approaches that can serve as high level optimizers with a prescribed level of robustness?

The following are clearly divided into two separate procedures by industry standards. The process of mathematically describing the system and implementing it in a computer for simulation concerns the modelling phase. Subsequent linearization, identification of model structure and control system design to improve, or enforce, stability relate to the part covered by control theory. Outside of the current work's scope, fall the initial and final steps that tie the model to it's real world counterpart: system identification and hardware implementation of the control algorithm. Although steps are taken to address the latter, this is the reason why there will be no attempt to assert a specific design as superior to others. Such certainty can only come through experimenting on the physical plant. All things considered, subsequent chapters will present the theoretical foundations for the proposed solutions, adhering to the following structure.

Chapter 2: Theoretical background of hydrodynamic modelling and development of a parametric, non-linear simulation environment including realistic sea generation.

Chapter 3: General linearization framework and automatic derivation of linear state-space models from the previously defined non-linear ones. Connection to flight dynamics.

Chapter 4: Theoretical background of optimal control methods, development of various controller architectures based on the Linear Quadratic Regulator and Model Predictive Control. Injection of noise in the measured states, and recovery of estimated signals by a Kalman Filter.

Chapter 5: Simulation results relating to typical control requirements, namely reference tracking, disturbance rejection and robustness.

Chapter 6: Conclusions reached by the conducted research, next steps for future work and possible implementation.

Chapter 2

Dynamic Model

This chapter aims to present the theoretical background needed to develop a general purpose non-linear hydrodynamic model for vessels equipped with hydrofoils, mathematically modeling its Equations of Motion (EOM). Largely based on Fossen's kinetics and kinematics framework, the general rigid body dynamics equations are developed in Sections 2.2 and 2.3. Having defined the LHS of the motion equations, we move to modelling of forces divided between external (or environmental) in Section 2.4 and internal, specifically actuator forces in Section 2.5. All theoretical results are combined to implement the parametric Simulink model presented in Section 2.6.

2.1 Prerequisites

Before delving into mathematical facts and derivations, it is important to set a scope of what the proposed model will simulate, which will also expose certain limitations that will be formally presented in Chapter 6. Figure 2.1 provides a qualitative view of what constitutes dynamic modelling within the scope of the present work. A complex mechanical system existing as a continuous medium and interacting with its environment, is compactly described by forces and moments acting on its CG. By this assumption, all fluid-structure interactions that induce pressure distributions, are replaced with pre-calculated forces acting at single points. The equations of Fig. 2.1 (Eq. (2.3.1)) are none other than Newton's second law of motion applied on translational and rotational movements within an inertial frame. The objectives which the final design ought to satisfy were for the model to be

- Configurable: Ability to describe a range of vessels through parameters and modular code.
- Computationally simple: Motion simulation without need of full CFD evaluation to calculate forces.
- Automatically linearizeable: An important property of the simulation environment, that requires it being capable of differentiating the force-producing equations.

To further simplify the analysis and acquire general results, EOM are derived with respect to the stationary Center of Origin (CO), while a non-dimensional approach is used

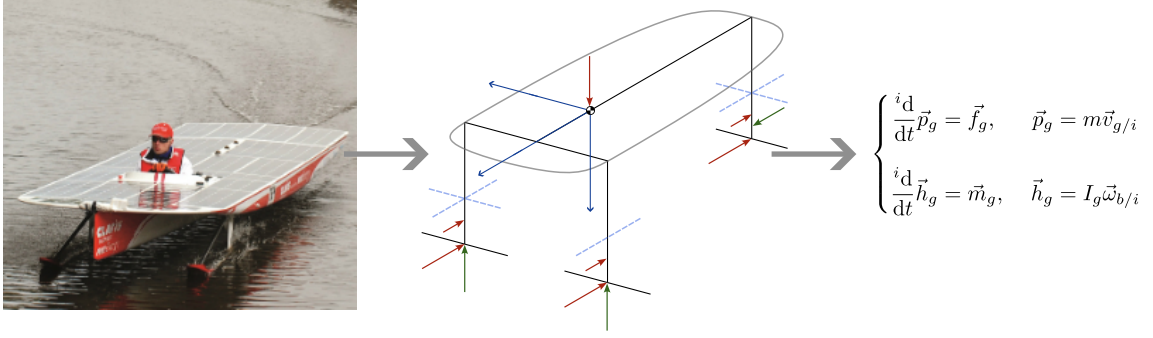


Figure 2.1: Modeling process: physical system, coordinate definition, and governing equations (image of the Clafis Victron Solar Boat Team taken from the [web](#)).

for lifting surface definitions. Namely, Reynolds number (the ratio of inflow velocity times characteristic length to kinematic viscosity) is used to calculate a distribution of pressure applicable to different velocities and fluids. The results are then non-dimensionalized by the product of the freestream *dynamic pressure* q_∞ and planform area S . Also important is the Froude number, used to describe gravitational phenomena such as wave formation. The characteristic length used is the chord c , or the foil's submergence h (also called submergence Froude number).

$$\text{Re} = \frac{V_\infty c}{\nu}, \quad \text{Fr} = \frac{V_\infty}{\sqrt{gc}}, \quad \text{Fr}_h = \frac{V_\infty}{\sqrt{gh}} \quad (2.1.1)$$

$$q_\infty = \frac{1}{2} \rho V_\infty^2 \quad (2.1.2)$$

2.1.1 Parametric Model

Setting up an outline of how physical phenomena are modelled acts as a guide to subsequent sections. An extensive review of the developed environment is given in Appendix B. The basic blocks that make up the simulation are:

- Sea state generation on a coordinate frame.
- Lifting surfaces, including hydrofoils (producing actuation forces) and supporting columns (struts).
- Thrust producing devices, in this case a propeller.
- Main vessel body defined by CG and an inertia matrix.

Position and attitude is tracked by integration of translational and rotational rates, following movements of the CG. Hydrodynamic data are pre-calculated and loaded as look-up tables or direct coefficients. This means that we can estimate hydrodynamic loads separately and then construct appropriate vectors to describe how they act with respect to the CG.

2.1.2 Physical Parameters

Lastly, model parameters and working conditions of the vessel are presented. Nominal parameters are collected in Appendix A at Table A.1. Those are combined to obtain dimensional data and run time domain simulations of the model.

Geometry Most of the works cited in the literature review provide only linearized coefficients for their lifting surface arrangements, usually not including exact dimensions. Even if rough estimates are included, it was decided to perform an extensive study on hydrodynamic behavior and appropriate sizing. This was conducted using airfoil data provided by the National Advisory Committee for Aeronautics (NACA), specifically NACA-63412 and NACA-0012, as well as standard software packages. This is the focus of Section 2.5.1, while the exact foil and strut geometry is specified at Table A.1.

Environmental properties Simulations are based on a rough descriptive parameters of the modelled environment, which are presented here and gathered in the Nomenclature. Fluid properties are based on data by the International Towing Tank Conference [38] for saltwater and a temperature of 20 °C, as a mean sea surface temperature for the Mediterranean region, an assumption supported by data from the Poseidon System of the Hellenic Center for Marine Research.

- Temperature: 20 °C
- Density ρ : 1024.8103 kg/m³
- Gravitational acceleration g : 9.81 m/s²
- Viscosity μ : 1.077×10^{-3} Pa · s
- Kinematic Viscosity ν : 1.0508×10^{-6} m²/s

The geometric and environmental properties are used to evaluate ranges for the non-dimensional numbers. The studied vessel will be in foilborne mode starting from around 9 kn with speeds reaching up to 27 kn. An average speed lies around the 14 kn mark due to power availability requirements.

$$\text{Re} = 3.5 \times 10^5 \text{ to } 1 \times 10^6, \quad \text{Fr} = 5.23 \text{ to } 15.68, \quad \text{Fr}_h = 1.77 \text{ to } 5.3$$

2.2 Kinematics

The following sections will revolve around Figs. 2.2 and 2.3, attempting to define motion both geometrically and dynamically, while also laying the foundations to include sensor models. This study begins with kinematics, the geometric characterization of motion. A freely moving rigid body defined by a point in space (yet still having volume, for rotations to apply) experiences motion in 6 degrees of freedom, characterized by coordinates, forces and moments as shown in Table 2.1. To apply vector mathematics, there is a need to introduce appropriate axes on which the vectors defining previous quantities, can be decomposed.

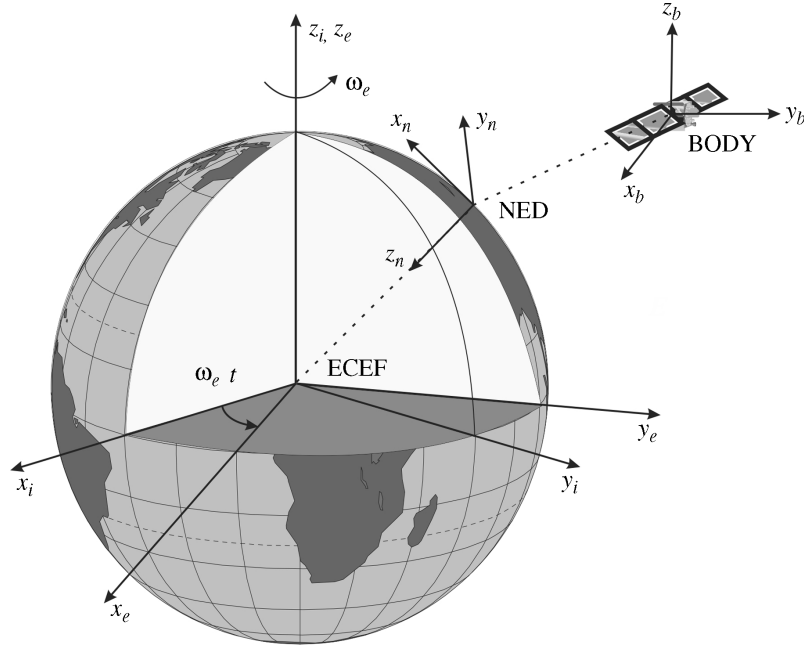


Figure 2.2: ECEF frame shown to rotate with an angular rate ω_e with respect to ECI. Also shown are the NED and BODY frames (Reproduced from [10]).

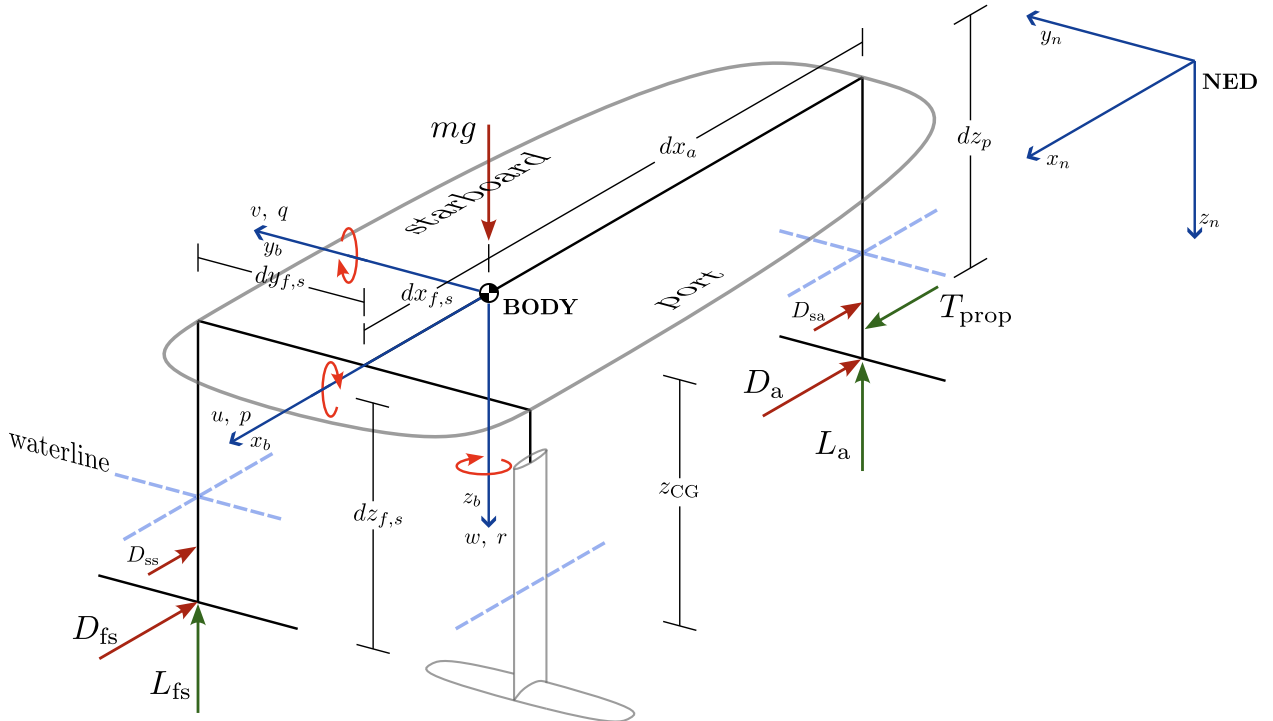


Figure 2.3: Representation of coordinate systems, vessel design parameters and acting forces as a visual guide to the dynamic model.

DOF	Description	Forces and moments	Linear and angular velocities	Positions and Euler angles
1	motion in the x_b direction (surge)	X	u	x
2	motion in the y_b direction (sway)	Y	v	y
3	motion in the z_b direction (heave)	Z	w	z
4	rotation about the x_b axis (roll)	K	p	ϕ
5	rotation about the y_b axis (pitch)	M	q	θ
6	rotation about the z_b axis (yaw)	N	r	ψ

Table 2.1: Six degrees of freedom for marine vessel dynamics.

2.2.1 Coordinate systems & reference frames

Even though they present fundamental concepts in vector mechanics, they are not uniformly defined among disciplines. The intuition provided herein was developed by standard Flight Dynamics courses and presentations given for software developed by NASA [39], but this does not mean that the terminology is consistent among other sources. A reference frame is defined by its origin, and the mutually orthogonal unit-length direction vectors. A coordinate system is a means of absolutely measuring space, thus specifying a mechanism for locating points within the frame.

A usual caveat arising from incomplete understanding, is how a body’s velocity when viewed from the stationary earth, is expressed with respect to the body-fixed frame. An observer moving with the body perceives zero velocity, but the free vector describing the motion can be expressed in the body-fixed frame. This is how *with respect to*, which entails relative motion, differs from *expressed in* which implies vector decomposition.

Below, and according to [40, Ch. 2], we define two earth-centered and two geographic reference frames (reference and coordinate frames are mostly used interchangeably within the specific literature), that are also pictured in Fig. 2.2. The frames of importance for this work are $\{n\}$ and $\{b\}$, as shown in Fig. 2.3.

- The Earth-Centered Inertial (ECI) frame $\{i\} = (x_i, y_i, z_i)$ with origin o_i at the center of the earth. Viewed as “fixed to the stars” in the sense that it is inertial, thus Newton’s laws of motion apply unaltered.
- The Earth-Centered Earth-Fixed (ECEF) frame $\{e\} = (x_e, y_e, z_e)$ with origin o_e fixed to the center of the earth, but with axes rotating relative to $\{i\}$ at $\omega_e = 7.2921 \times 10^{-5}$ rad/s. For low speed terrestrial navigation, it is also considered inertial.
- The North-East-Down (NED) frame $\{n\} = (x_n, y_n, z_n)$ with origin o_n relative to earth’s reference ellipsoid. It is a tangent plane on the surface of the earth with positive axes pointing North, East and Down towards earth’s center such that $\frac{\vec{g}}{|\vec{g}|} = z_n$. From this point on, $\{n\}$ will be considered inertial (following a flat earth navigation system) and vessel position is calculated relative to o_n as (x, y, z) within $\{n\}$. According to the previous categorization, it can also be treated as a coordinate system attached to $\{e\}$.
- The Body-Fixed frame $\{b\} = (x_b, y_b, z_b)$ with origin o_b translates and rotates with the vessel it is fixed on. The position and orientation of this frame (and by extension,

the vessel's) are described relative to some inertial frame, in this case $\{n\}$. The origin o_b is picked as the general stationary point CO instead of the possibly time dependent CG.

Note how both NED and BODY adopt *down* as the positive z axis, which once again may appear to conflict other references, but is compliant with [2], [3], [10], [40].

Having defined the appropriate coordinate systems, free vectors can be expressed in any frame of interest, and the quantities introduced in Table 2.1 are formally presented below. Regarding notation, since it was decided to adhere to Fossen's model, notation is explained by Eq. (2.2.1). Keep in mind that notation differs among authors, with most flight dynamics literature following Etkin's [3] standards, but the underlying equations and described quantities remain unchanged.

NED position	$\mathbf{p}_{b/n}^n = \begin{bmatrix} N \\ E \\ D \end{bmatrix}$	Attitude	${}_{nb} = \begin{bmatrix} \phi \\ \theta \\ \psi \end{bmatrix}$
Body-fixed linear velocity	$\mathbf{v}_{b/n}^b = \begin{bmatrix} u \\ v \\ w \end{bmatrix}$	Body-fixed angular velocity	$\boldsymbol{\omega}_{b/n}^b = \begin{bmatrix} p \\ q \\ r \end{bmatrix}$
Body-fixed force	$\mathbf{f}_b^b = \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$	Body-fixed moment	$\mathbf{m}_b^b = \begin{bmatrix} K \\ M \\ N \end{bmatrix}$

Table 2.2: Definitions of SNAME notation within Fossen's coordinate vector framework.

$$\mathbf{v}_{\{\text{refers to}\}/\{\text{with respect to}\}}^{\{\text{expressed in}\}} \quad (2.2.1)$$

For example:

- Body-fixed linear velocity $\mathbf{v}_{b/n}^b$ is a vector that *refers to* $\{b\}$ velocity, *with respect to* $\{n\}$ and *expressed in* $\{b\}$.
- NED position $\mathbf{p}_{b/n}^n$ is a vector tracking the crafts position, *referring to* CO and thus $\{b\}$ motion, *with respect to* $\{n\}$ (as it tracks translations relative to it) and *expressed in* $\{n\}$.
- Attitude gives the euler angles between $\{n\}$ and $\{b\}$.

Building up to a vectorial representation, elements of Table 2.2 are concatenated row-wise to form general coordinate vectors expressed in $\{b\}$ as shown below.

- $\boldsymbol{\eta} = [N, E, D, \phi, \theta, \psi]$, the generalized position-orientation vector.
- $\boldsymbol{\nu} = [u, v, w, p, q, r]^\top$, the generalized velocity vector.
- $\boldsymbol{\tau}_{RB} = [X, Y, Z, K, M, N]^\top$, the generalized external force-moment vector.

2.2.2 Coordinate Transforms

To complete this section, we need a formal way to perform decomposition of vectors between frames. This is important not only for modelling, but for practical reasons as various sensor devices produce readings bound to specific frames. An example is how an Inertial Measurement Unit (IMU) tracks acceleration and velocity components in the $\{b\}$ frame. For this reason, the *rotation matrix* $\mathbf{R}$ is defined as an element in $SO(3)$, the special orthogonal group of order 3 shown in Eq. (2.2.2). Serial multiplication of rotation matrices describing sequential rotations around each axis defines the Direction Cosine Matrix (DCM), an important construct for vector mechanics.

$$SO(3) = \{\mathbf{R} | \mathbf{R} \in \mathbb{R}^{3 \times 3}, \mathbf{R}\mathbf{R}^\top = \mathbf{R}^\top\mathbf{R} = \mathbf{I} \text{ and } \det \mathbf{R} = 1\} \quad (2.2.2)$$

A direct corollary of Eq. (2.2.2) is that $\mathbf{R}^{-1} = \mathbf{R}^\top$, an important property for switching between frames. The most useful DCM we can produce, is the one between $\{n\}$ and $\{b\}$. First, principal rotation matrices are obtained as in Eq. (2.2.3) corresponding to x, y, z axes and the respective rotations around them. Trigonometric functions are abbreviated as $s \cdot = \sin(\cdot)$ and $c \cdot = \cos(\cdot)$. Then, a rotational convention is used, specifically zyx for $\{n\} \rightarrow \{b\}$, which results in $\mathbf{R}_b^n$, adhering to the convention in Eq. (2.2.4).

$$\mathbf{R}_{x,\phi} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\phi & -s\phi \\ 0 & s\phi & c\phi \end{bmatrix}, \quad \mathbf{R}_{y,\theta} = \begin{bmatrix} c\theta & 0 & s\theta \\ 0 & 1 & 0 \\ -s\theta & 0 & c\theta \end{bmatrix}, \quad \mathbf{R}_{z,\psi} = \begin{bmatrix} c\psi & -s\psi & 0 \\ s\psi & c\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (2.2.3)$$

$$\mathbf{v}^{\text{to}} = \mathbf{R}_{\text{from}}^{\text{to}} \mathbf{v}^{\text{from}} \quad (2.2.4)$$

The transformation matrix defined by Eq. (2.2.5) has the full form of Eq. (2.2.6), and in practice defines a way to rotate the NED frame axes sequentially to reach the BODY frame. That is

- A yaw rotation about the z_n axis: $(x_n, y_n, z_n) \rightarrow (x'_n, y'_n, z_n)$
- A pitch rotation about the y'_n axis: $(x'_n, y'_n, z_n) \rightarrow (x''_n, y'_n, z'_n)$
- A roll rotation about the x''_n axis: $(x''_n, y'_n, z'_n) \rightarrow (x''_n, y''_n, z''_n)$

With the new coordinate system now corresponding to $\{b\}$.

$$\mathbf{R}_b^n(\mathbf{\Theta}_{nb}) := \mathbf{R}_{z,\psi} \mathbf{R}_{y,\theta} \mathbf{R}_{x,\phi} \quad (2.2.5)$$

$$\mathbf{R}_b^n(\mathbf{\Theta}_{nb}) = \begin{bmatrix} c\psi c\theta & -s\psi c\phi + c\psi s\theta s\phi & s\psi s\phi + c\psi c\phi s\theta \\ s\psi c\theta & c\psi c\phi + s\phi s\theta s\psi & -c\psi s\phi + s\theta s\psi c\phi \\ -s\theta & c\theta s\phi & c\theta c\phi \end{bmatrix} \quad (2.2.6)$$

Equation (2.2.6) is especially useful, because it produces the differential equation connecting body-fixed velocity components and flight path of the vehicle according to Eq. (2.2.7). This equation is how the Simulink model creates vessel trajectories.

$$\dot{\mathbf{p}}_{b/n}^n = \mathbf{R}_b^n(\mathbf{\Theta}_{nb}) \mathbf{v}_{b/n}^b \quad (2.2.7)$$

A second differential equation will accompany Eq. (2.2.7), in order to acquire attitude information. Once again, one cannot simply integrate body-fixed angular rates to compute Euler angle information. Vessel orientation is calculated by Eq. (2.2.9) by use of the transformation matrix shown in Eq. (2.2.8). The matrix $\mathbf{T}_\Theta$ does not satisfy the properties of Eq. (2.2.2), meaning $\mathbf{T}_\Theta^{-1} \neq \mathbf{T}_\Theta^\top$. Instead refer to Eq. (2.2.8) for the inverse relationship.

$$\mathbf{T}_\Theta(\textit{nb}) = \begin{bmatrix} 1 & s\phi t\theta & c\phi t\theta \\ 0 & c\phi & -s\phi \\ 0 & \frac{s\phi}{c\theta} & \frac{c\phi}{c\theta} \end{bmatrix}, \quad \mathbf{T}_\Theta(\textit{nb})^{-1} = \begin{bmatrix} 1 & 0 & -s\theta \\ 0 & c\phi & c\theta s\phi \\ 0 & -s\phi & c\theta c\phi \end{bmatrix} \quad (2.2.8)$$

$$\dot{\textit{nb}} = \mathbf{T}_\Theta(\textit{nb})\boldsymbol{\omega}_{b/n}^b \quad (2.2.9)$$

Also notice that $\mathbf{T}_\Theta$ is undefined for $\theta = \pm 90^\circ$, but this is of little importance in this scope. Equations (2.2.7) and (2.2.9) are readily combined in Eq. (2.2.10) expressed in vector form.

$$\dot{\boldsymbol{\eta}} = \begin{bmatrix} \dot{\mathbf{p}}_{b/n}^n \\ \dot{\textit{nb}} \end{bmatrix} = \begin{bmatrix} \mathbf{R}_b^n(\textit{nb}) & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & \mathbf{T}_\Theta(\textit{nb}) \end{bmatrix} \begin{bmatrix} \mathbf{v}_{b/n}^b \\ \boldsymbol{\omega}_{b/n}^b \end{bmatrix} = \mathbf{J}_\Theta(\boldsymbol{\eta})\boldsymbol{\nu} \quad (2.2.10)$$

Before moving on to the next section, a cross-product matrix operator $\mathbf{S}$ is introduced as in Eq. (2.2.11). Following [10], $\mathbf{S} \in SS(n)$, the set of skew-symmetric matrices of order n such that $\mathbf{S} = -\mathbf{S}^\top$.

$$\boldsymbol{\lambda} \times \boldsymbol{\alpha} := \mathbf{S}(\boldsymbol{\lambda})\boldsymbol{\alpha} = \begin{bmatrix} 0 & -\lambda_3 & \lambda_2 \\ \lambda_3 & 0 & -\lambda_1 \\ -\lambda_2 & \lambda_1 & 0 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} \quad (2.2.11)$$

2.3 Rigid Body Kinetics

At this point, we seek to form the *general equations of unsteady motion* [3, Ch. 4] for any rigid, moving vessel. What distinguishes an ocean-going vehicle from an aircraft or a car is simply how the right hand side of the EOM describes forces and moments entering the system dynamics. Equation (2.3.1) is also referred to as *Newton-Euler equations*, and describes translational and rotational motion around the CG in an inertial frame (denoted by the left superscript i on the differential operator).

$$\frac{^i d}{dt} \vec{p}_g = \vec{f}_g, \quad \vec{p}_g = m\vec{v}_{g/i} \quad (2.3.1a)$$

$$\frac{^i d}{dt} \vec{h}_g = \vec{m}_g, \quad \vec{h}_g = I_g \vec{\omega}_{b/i} \quad (2.3.1b)$$

These equations can be rewritten with respect to CO and the (non-inertial) $\{b\}$ frame and compactly presented in vectorial form. To avoid lengthy re-derivations, results from [10], [41] are directly presented. We have already defined the needed vectors for force $\mathbf{f}_b^b$, moment $\mathbf{m}_b^b$ and velocities $\mathbf{v}_{b/n}^b$, $\boldsymbol{\omega}_{b/n}^b$ according to SNAME in Eq. (2.2.1). The vector from o_b to CG expressed in $\{b\}$ is $\mathbf{r}_g^b = [x_g, y_g, z_g]^\top$.

Translation Differentiation for a moving frame shall include angular velocity terms (coupling) as shown in Eq. (2.3.2).

$$m[\dot{\mathbf{v}}_{b/n}^b + \mathbf{S}(\dot{\boldsymbol{\omega}}_{b/n}^b)\mathbf{r}_g^b + \mathbf{S}(\boldsymbol{\omega}_{b/n}^b)\mathbf{v}_{b/n}^b + \mathbf{S}^2(\boldsymbol{\omega}_{b/n}^b)\mathbf{r}_g^b] = \mathbf{f}_b^b \quad (2.3.2)$$

Rotation Also needed, is an expression for the inertia matrix about CO, by the Huygens–Steiner Theorem [10, Ch. 3.2.2], giving $\mathbf{I}_b$ in Eq. (2.3.3). This is used for rotational motion around CO described by Eq. (2.3.4), which is once again coupled to translational if there's an offset $\mathbf{r}_g^b$.

$$\mathbf{I}_b = \mathbf{I}_g - m\mathbf{S}^2(\mathbf{r}_g^b) = \mathbf{I}_g - m\left(\mathbf{r}_g^b(\mathbf{r}_g^b)^\top - (\mathbf{r}_g^b)^\top \mathbf{r}_g^b \mathbf{I}_{3 \times 3}\right) \quad (2.3.3)$$

$$\mathbf{I}_b \dot{\boldsymbol{\omega}}_{b/n}^b + \mathbf{S}(\boldsymbol{\omega}_{b/n}^b) \mathbf{I}_b \boldsymbol{\omega}_{b/n}^b + m\mathbf{S}(\mathbf{r}_g^b) \dot{\mathbf{v}}_{b/n}^b + m\mathbf{S}(\mathbf{r}_g^b) \mathbf{S}(\boldsymbol{\omega}_{b/n}^b) \mathbf{v}_{b/n}^b = \mathbf{m}_b^b \quad (2.3.4)$$

Equations (2.3.2) and (2.3.4) can be combined in a vectorial setting as shown below, which is the main equation describing the vessel's dynamics within the developed tool. For the sake of completeness, involved matrices are expanded in Appendix A.4.

$$\mathbf{M}_{RB} \dot{\boldsymbol{\nu}} + \mathbf{C}_{RB}(\boldsymbol{\nu}) \boldsymbol{\nu} = \boldsymbol{\tau}_{RB} \quad (2.3.5)$$

Upon closer inspection of Eq. (A.4.1), $\mathbf{M}_{RB}$ is a square matrix with linearly independent columns and rows, making it invertible [42, Ch. 11.2]. This shall be the base of simulating vessel motion in Simulink.

$$\dot{\boldsymbol{\nu}} = \mathbf{M}_{RB}^{-1}(-\mathbf{C}_{RB}(\boldsymbol{\nu}) \boldsymbol{\nu} + \boldsymbol{\tau}_{RB}) \quad (2.3.6)$$

The focus is now turned to the RHS of Eq. (2.3.5). The following sections will introduce forces and resulting moments from the encountered environment, the lifting surface arrangement and the attached propulsive devices. Any force $\mathbf{f}^b = [F_x, F_y, F_z]^\top$ applied at a distance $\mathbf{r}^b = [l_x, l_y, l_z]^\top$ from the center of origin can be used to construct a generalized force-moment vector $\boldsymbol{\tau}$ as in Eq. (2.3.7).

$$\boldsymbol{\tau} = \begin{bmatrix} \mathbf{f}^b \\ \mathbf{r}^b \times \mathbf{f}^b \end{bmatrix} = \begin{bmatrix} F_x \\ F_y \\ F_z \\ l_y F_z - l_z F_y \\ l_z F_x - l_x F_z \\ l_x F_y - l_y F_x \end{bmatrix} \quad (2.3.7)$$

The generalized force vector $\boldsymbol{\tau}_{RB}$ is used to describe “externally” applied forces, with “internal” meaning ones resulting from hydrodynamic phenomena proportional to velocities and accelerations which should be grouped at the LHS of Eq. (2.3.5). In this context however, it was decided to treat all forces as *external*, since they are added equally in the simulation step. Damping and added mass components of hydrodynamic interaction are some examples that will be handled as external, making for a slight violation in notation.

2.4 Environmental Forces

Forces arising from environment interactions generally include gravity, waves and wind. For small scale marine vessels, wind forces are usually neglected as they tend to be minimal compared to submerged body resistance and wave loads. But these assumptions are sometimes misleading. For a steadily moving vessel, when thrust counteracts resistance fully in the x_b direction, wind gusts can significantly affect stability and inflict path deviations in the remaining DOF. We shall focus on gravity and waves in the modelling section and wind gusts will be discussed in Chapter 5 again.

2.4.1 Gravity

By the axis convention introduced earlier, gravity vector is always pointing at z_n (positive downwards in $\{n\}$). The weight of the vessel is a force acting on its CG as $W = mg$.

$$\mathbf{f}_g^n = \begin{bmatrix} 0 \\ 0 \\ W \end{bmatrix} \quad (2.4.1)$$

By use of the rotational matrix introduced with Eq. (2.2.6), the force vector is expressed in $\{b\}$ as $\mathbf{f}_g^b = \mathbf{R}_n^b \mathbf{f}_g^n$. To account for moments introduced by $\mathbf{r}_g^b$, the general force vector will also include an $\mathbf{m}_g^b$ term, equal to the cross product operation $\mathbf{m}_g^b = \mathbf{r}_g^b \times \mathbf{f}_g^b$. Expanding this, gives Eq. (2.4.2).

$$\boldsymbol{\tau}_g = \begin{bmatrix} \mathbf{R}_n^b \mathbf{f}_g^n \\ \mathbf{r}_g^b \times \mathbf{R}_n^b \mathbf{f}_g^n \end{bmatrix} = \begin{bmatrix} -W \sin \theta \\ W \cos \theta \sin \phi \\ W \cos \theta \cos \phi \\ y_g W \cos \theta \cos \phi - z_g W \cos \theta \sin \phi \\ -z_g W \sin \theta - x_g W \cos \theta \cos \phi \\ x_g W \cos \theta \sin \phi + y_g W \sin \theta \end{bmatrix} \quad (2.4.2)$$

Note that in Fossen's work, $\boldsymbol{\tau}_g$ is referred to as $\mathbf{g}(\boldsymbol{\eta})$, moved to the LHS of Eq. (2.3.5).

2.4.2 Waves

The free surface deformation encountered by the vessel is an important factor affecting seakeeping behavior. From a systems perspective, waves affect the model's stability by introducing a periodic disturbance. Realistic sea state can be studied as the superposition of regular wave components that are described by sinusoidal terms [43, Ch. 10]. This allows for simple expressions regarding response to regular waves, to be generalized for irregular seas [9]. For small scale craft operating in relatively calm seas, waves might be completely overlooked for models concerned with control [1], [28]. Response to regular waves can be used to acquire an idea of disturbance rejection [29], [44]. More involved studies include [45]. Results and derivations are largely based on Holthuijsen [46].

Regular Waves

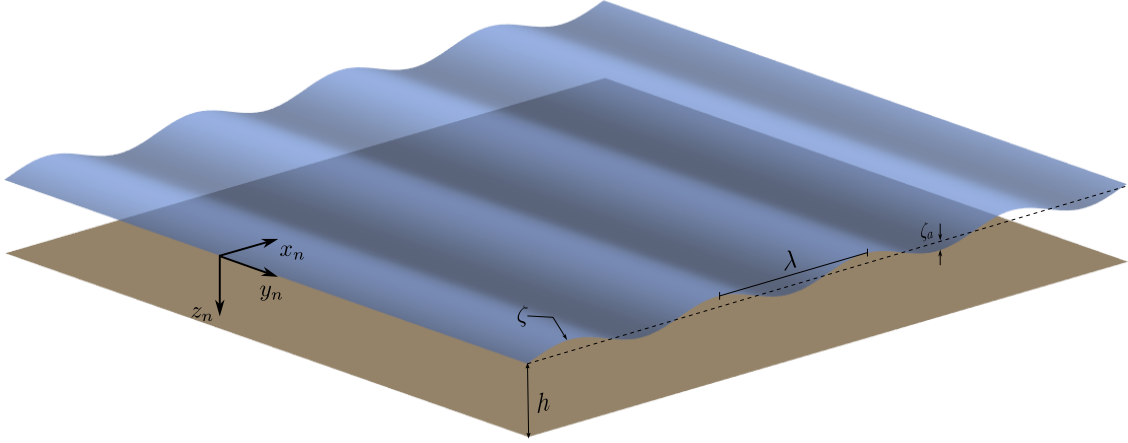


Figure 2.4: Regular wave

A propagating harmonic wave in two dimensions deforms the free surface η in a cylindrical long crested form visualized in Fig. 2.4. This behavior is mathematically defined by Eq. (2.4.3). Note that contrary to various resources, including Holthuijsen, we select the positive z axis pointing downwards for easier derivations in $\{n\}$. All equations are presented exactly as in the available literature, but the simulation inverts z sign on runtime.

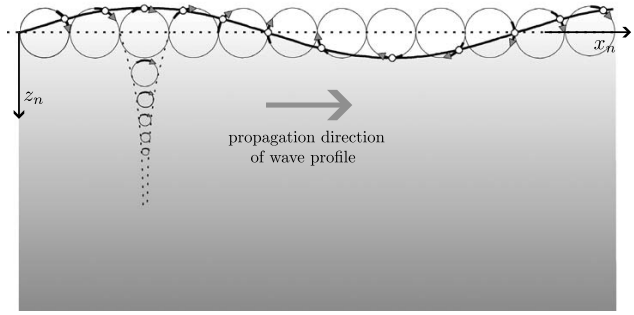


Figure 2.5: Orbital motion of water particles induced by a harmonic wave in deep water (Adapted from [46]).

$$\zeta(x, t) = \zeta_a \sin(\omega t - kx) \quad (2.4.3)$$

A wave amplitude ζ_a describes the free surface elevation (half of the wave height from crest to trough), the wave number $k = 2\pi/\lambda$ corresponds to the spatial frequency over a unit distance, and a wave frequency $\omega = 2\pi/T$ dictates the time dependent oscillation. Figure 2.5 provides a visualization of the characteristic water particle motion created by a regular wave, showing both the sinusoidal free surface disturbance and the depth-dependent exponentially decaying motion. By assuming linear theory (small-amplitude approximation), a velocity potential is introduced, defined as a function with spatial derivatives being equal to water particle velocities Eq. (2.4.4)[46, Ch. 5.3.4]. This means that $(u_w, v_w, w_w) = (\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}) \cdot \Phi$. Also shown are the infinite depth approximations, by calculating the limit as $h \rightarrow \infty$.

$$\begin{aligned} \Phi(x, y, z, t) &= \frac{\omega \zeta_a}{k} \frac{\cosh[k(h+z)]}{\sinh(kh)} \sin(\omega t - kx) \\ &\xrightarrow{h \rightarrow \infty} \frac{\omega \zeta_a}{k} e^{kz} \sin(\omega t - kx) \end{aligned} \quad (2.4.4)$$

Another important relationship is that between ω and k , based on the condition that the fluid surface is impervious.

$$\omega^2 = gk \tanh(kh) \xrightarrow{h \rightarrow \infty} gk \quad (2.4.5)$$

The wave frequency and wave length are dependent quantities, since the waveform moves one wavelength λ per period T , giving the phase velocity of Eq. (2.4.6). Combining this with previous definitions yields Eq. (2.4.7) connecting wavelength with frequency (assuming deep water).

$$c = \frac{\lambda}{T} = \frac{\omega}{k} \quad (2.4.6)$$

$$\lambda = \frac{2\pi g}{\omega^2} \quad (2.4.7)$$

Finally, waves do not necessarily travel parallel to x_n , but might propagate at angle ψ and include a phase term β . By these assumptions, we arrive at the full description of regular waves in Eqs. (2.4.8) to (2.4.12) exactly as they are implemented by the MSS Toolbox in the *wave velocitycalc* block. The equations below assume NED coordinates, which explains the negative sign on z and w_w .

$$X = y \sin(\psi) + x \cos(\psi) \quad (2.4.8)$$

$$\zeta = \zeta_a \sin(\omega t - kX + \beta) \quad (2.4.9)$$

$$u_w = \omega \zeta_a e^{-kz} \cos(\psi) \sin(\omega t - kX + \beta) \quad (2.4.10)$$

$$v_w = \omega \zeta_a e^{-kz} \sin(\psi) \sin(\omega t - kX + \beta) \quad (2.4.11)$$

$$w_w = -\omega \zeta_a e^{-kz} \cos(\psi) \cos(\omega t - kX + \beta) \quad (2.4.12)$$

As wave orbitals correspond to the later defined sinusoidal disturbances in angle of attack perceived by the lifting surface, an important property is reminded. Similarly to most wave phenomena, the apparent frequency for a moving observer is different to that of the source due to the Doppler effect. All wave-induced ship motions that are dependent on frequency, should use the transformation of Eq. (2.4.13).

$$\omega_e = \left| \omega_0 - \omega_0^2 \frac{V_s}{g} \cos(\psi) \right| \quad (2.4.13)$$

This correction is not applied within the simulation, as the perceived frequency implicitly changes as the vessel advances through $\{n\}$ at steady speed V_s , exactly as it would physically happen. When calculating amplitude of responses for given sea states however, it is important to include the correct frequency.

Irregular Waves

As it turns out, Eqs. (2.4.8) to (2.4.12) are especially important, because they are able to describe irregular waves by the principle of *superposition*. A realistic sea state emerges as the result of a Fourier series based on harmonic (regular) terms. Such a description relies on statistical analysis of sea surface elevation, and is the result of many previous theoretical steps [46, Ch. 3.5], that lead to an elegant representation of elevation $\zeta(x, y, t)$

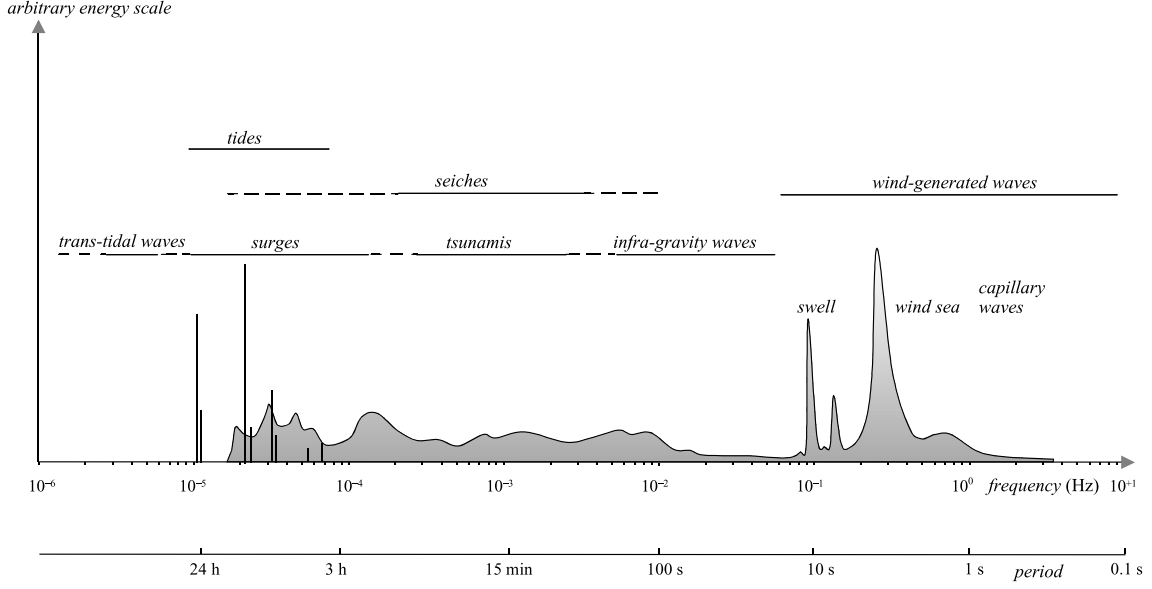


Figure 2.6: Frequencies and periods of the vertical motions of the ocean surface (Reproduced from [46]).

in Eq. (2.4.14), applicable to the remaining terms of interest such as wave orbital velocities. Figure 2.6 provides an arbitrary view of vertical motion energy content at a period sweep between seconds and days.

$$\zeta(x, y, t) = \sum_{i=1}^N \sum_{j=1}^M \zeta_{i,j} \cos(\omega_i t - k_i(x \cos(\psi_j) + y \sin(\psi_j)) + \beta_{i,j}) \quad (2.4.14)$$

For the one-dimensional case the random-phase/amplitude model for $\zeta(t)$ combines a Rayleigh distribution for amplitudes ζ_i with a uniform one for phase β_i . Acquiring the expected value $E[\frac{1}{2}\zeta_i^2]$ over the frequency range yields the *variance density spectrum* $S(\omega)$ [46] such that the total variance is calculated by:

$$\text{Var}(\zeta) = \int_0^\infty S(\omega) d\omega \quad (2.4.15)$$

Now that we have a spectrum capable of defining amplitudes, introducing direction ψ generates a 2D wave spectrum $S(\omega, \psi)$ which finally explains Eq. (2.4.14), summing N frequency components of M directions. The shape of $S(\omega, \psi)$ can be decoupled and used as $S(\omega)D(\psi; \omega)$, with $D(\psi; \omega)$ being the directional distribution. The most widely used form is that of Eq. (2.4.16) [46, Ch. 6.3.4], with ψ defined relative to the mean wave direction, usually 0° .

$$D(\psi) = \begin{cases} \frac{2}{\pi} \cos^2(\psi), & -\pi/2 \leq \psi \leq \pi/2 \\ 0, & \text{elsewhere} \end{cases} \quad (2.4.16)$$

Regarding irregular sea state generation, this is achieved by:

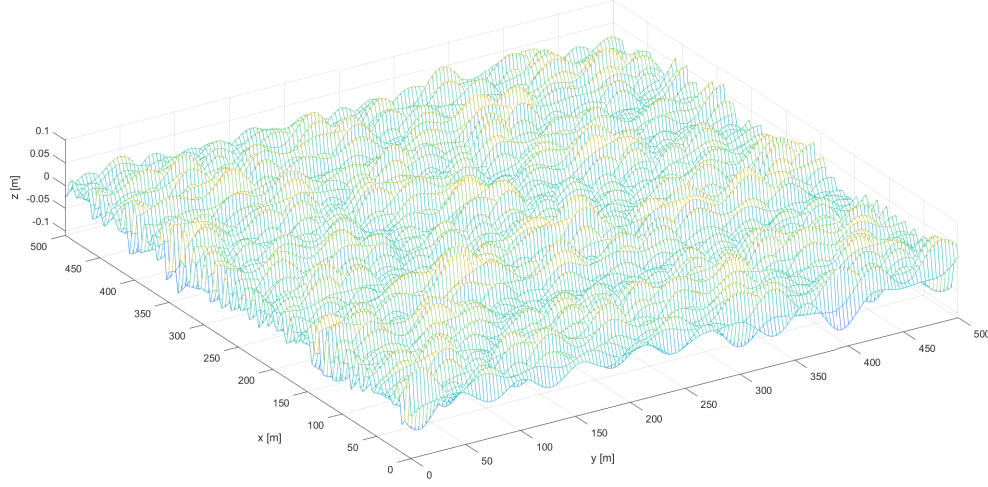


Figure 2.7: Short-crested wind waves simulated by 10 regular components, ITTC spectrum, $H_s = 0.3$ m, $\omega_{\text{peak}} = 2 \text{ rad} \cdot \text{s}^{-1}$, 5 directions in grid.

- Discretization of the spectrum into frequency bins ω_i , $i = 1, \dots, N$, spacing $\Delta\omega$.
- Selection of direction components (usually uniform) ψ_j , $j = 1, \dots, M$.
- Phase generation as $\beta_{i,j} \sim U(0, 2\pi)$.
- Assignment of amplitudes $A_{i,j} = \sqrt{2S(\omega_j)D(\psi)\Delta\omega\Delta\psi}$.
- Application to Eq. (2.4.14).

All this is automatically generated by Marine Systems Simulator (MSS) Toolbox, with an example visualized in Fig. 2.7. There are however some limitation into applying a fully irregular short-crested sea, arising from actuator dynamics. We shall revisit this in the next section.

Parameter Selection

The problem of generating a sea state, reduces to the selection of parameters corresponding not only to possible sea scenarios, but also the characteristics of the studied vessel. A first insight is given in Fig. 2.6, showing that wind generated wave frequencies lie between the range of 1 to 10 Hz. A more involved analysis was conducted for the vessel's operational profile, which is included in Appendix A.2.

- Amplitude ζ_0 : 0.1 m to 0.3 m
- Frequency ω : 1.8 rad/s to 7 rad/s
- Wavelength λ : ≈ 1.25 m to 19 m

2.5 Actuation Forces

Accurately modelling actuator dynamics is possibly the most important aspect of a control-oriented model as it directly affects input-output relationships. Same as for an aircraft, the available commands concern control surface deflections and thrust generation.

A study on foil lift and drag characteristics both for the 2D and the 3D case will yield results covering hydrofoils and struts. Only the former pose as actuators for the system, with struts governing internal dynamics. This observation will be revisited in Chapters 3 and 4. The hydrofoil and strut system will provide the input from which wave excitation enters the vessel dynamics, closely linking it with the previous results.

Means of propulsion are modelled simply as a force attached to where the system is installed, shown as T_{prop} in Fig. 2.3. We are interested in how this thrust is produced relative to the pilot's control input and the electro-mechanical setup of the propulsive device. In the current context a propeller drive is used, contrary to larger vessels where there is a need for mechanically complex waterjet installations.

2.5.1 Hydrofoils & Struts

Definitions

The geometry of foils and wings is clearly defined in literature, and Fig. 2.8 describes the most important characteristics. Adopting standard notation, the symbols for wing span and *projected*, or planform, area (not to be confused with *wetted* area) will be b and S respectively. We also define Aspect Ratio (AR), useful for the analysis of the 3-dimensional case.

$$\text{AR} = \frac{b^2}{S}$$

The physical characteristics of the problem are showcased in Fig. 2.9. The sources of hydrodynamic forces and moments on any submerged body are the pressure and shear stress distributions, which can be integrated over the body's surface. This results to a reaction (or resultant) force R and a moment M . The angle between the foil chord line and the inflow direction, is called *angle of attack* and is usually denoted by α .

The resultant force can be described by a component vertical to the inflow velocity and one parallel to it, namely Lift and Drag. The point at which those forces are applied is chosen such that the effects of the forces are identical to the ones due to the distributed loads. One point would be the *center of pressure*, where $M = 0$, but it is more convenient to use a stationary point of reference along the chord line, usually the quarter-chord point. For symmetric foils, it turns out that these two definitions are equivalent, but that does not hold for the non symmetric case. Combining the produced forces and moments with factors defined at the beginning of this chapter, the resulting we acquire non-dimensional coefficients:

- Lift Coefficient, C_L : $\frac{L}{q_\infty S}$
- Drag Coefficient, C_D : $\frac{D}{q_\infty S}$
- Moment Coefficient, C_M : $\frac{M}{q_\infty c S}$

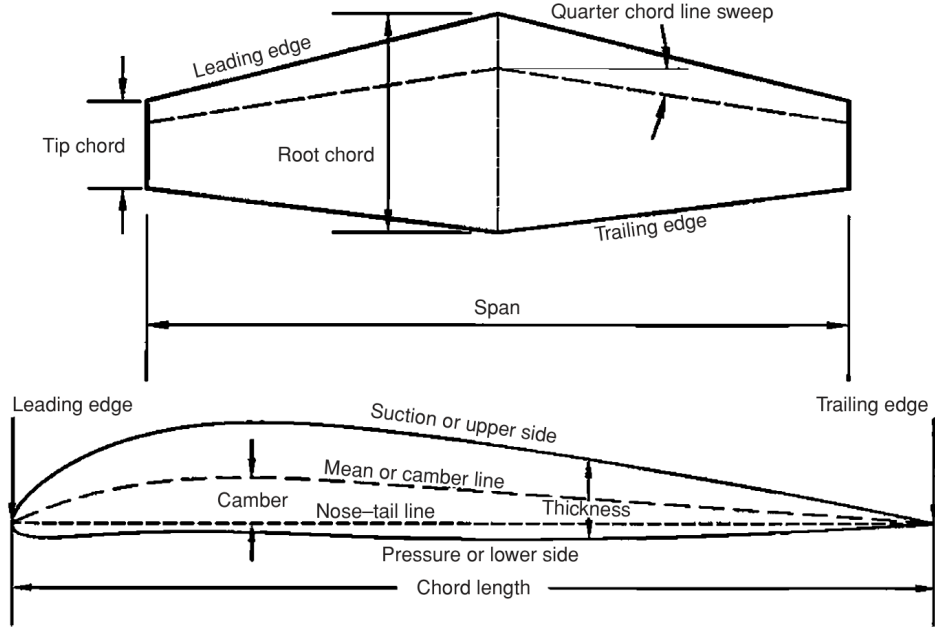


Figure 2.8: Foil and wing geometry definitions (Reproduced from [9]).

Lift The force perpendicular to the inflow velocity, the component most accurately predicted by the methods analyzed in the following. It is also the most important factor affecting vessel stability through free-surface effects and unsteadiness due to motion and waves, which will be touched upon later.

Drag This component is not as important for the control case, but it is also largely under predicted by the mostly inviscid methods. Specifically drag can be attributed to six main components:

$$D_{OA} = D_{sf} + D_p + D_{ind} + D_{sp} + D_w + D_{int} \quad (2.5.1)$$

- Skin friction, D_{sf} : Calculated by integration of fluid shear forces and dependent on the type of flow (laminar or turbulent, the latter creates larger drag).
- Pressure, D_p : Calculated by integration of pressure differences on the foil surface.
- Induced, D_{ind} : Attributed to 3D flow effects on wings.
- Spray, D_{sp} : Apparent only on surface piercing components, such as struts.
- Wave, D_w : Due to wave orbital velocities, or changes in strut submergence.
- Interference, D_{int} : Created due to flow inconsistencies around connections of lifting surfaces (e.g struts and foils).

The first 3 components are directly calculated by computational and semi-empirical methods while the last 3 are only approximated.

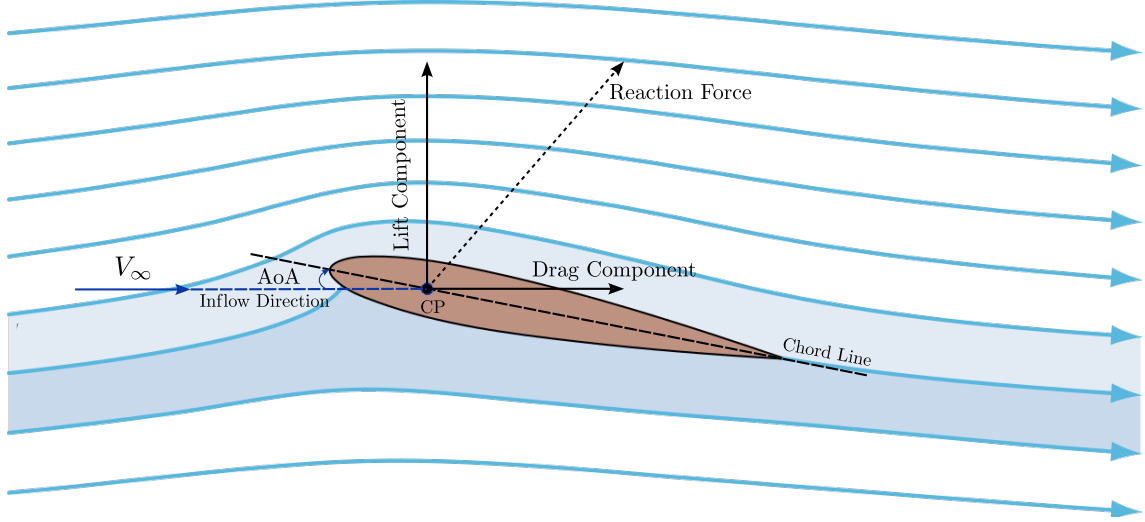


Figure 2.9: The two-dimensional lifting problem characteristics.

Control Surface Kinematics

I decided to present this particular part before deriving any force-related equations, mirroring the way EOM are analyzed. Throughout this brief explanation, Fig. 2.10 along with Fig. 2.3 function as visualizations for what eventually goes into the simulation. Figure 2.12 generalizes the problem in 3D space, and introduces sideslip angle β as a factor affecting hydrodynamic performance. Throughout this work, the mechanism considered for actuation is the one rotating the wing as a whole and not by the use of flaps. The hinge point is assumed to be arbitrarily close to the center of pressure, such that no need for correction of moment arm due to deflection is needed.

Hydrodynamic Angle Because lift and drag forces are calculated with respect to the hydrodynamic angles α and β , we need a way to analytically provide this angle on runtime. These results are readily available by Beard et. al [4, Ch. 4], or for this specific case, by Martin [19]. The full derivations are case specific and require a geometric approach.

Taking a look at Fig. 2.10, there are three frames defined, namely $\{n\}$, $\{b\}$ and $\{f\}$. The foil frame, $\{f\}$ results as a translation of the body frame to the hinge position of the foil. The wing is free to rotate about this axis denoting a control deflection α_c . The hydrodynamic angle is the sum of the control deflection, and the inflow angle α_∞ , which will be extensively studied later. The velocity vector V_∞ is a relative speed, different for each foil and can be decomposed in the foil frame. When the vessel moves through a calm sea at zero pitch and roll, this vector is the opposite of the vessel's speed: it's as if the vessel is stationary facing the moving fluid.

The physical characteristics of a foil's movement through realistic seas can be seen in Fig. 2.11. The velocities that characterize the motion are divided into three parts, based on their origin. Each part is responsible for a change in the hydrodynamic angles. Specifically,

- The craft's CO velocity components $[u, v, w]$, with u sometimes denoted as U since it is the largest component for level flight. Expressed in the body frame, therefore

it is readily used in the foil frame as well.

- Velocities due to the position of the foil relative to CO, calculated by the cross product $\boldsymbol{\omega}^b \times \mathbf{r}_f^b = [p, q, r] \times [x_f, y_f, z_f]$. Marked as perturbations $[\delta u, \delta v, \delta w]$.
- Velocities due to wave orbital motion, shown as a vector field with components $[u_w, v_w, w_w]$ in $\{n\}$. Note the importance of sign and vector addition.

Hydrodynamic angle derivations

Building upon the intuition given by Figs. 2.10 and 2.11, the angle of attack α is calculated based on foil relative velocity V_∞^f . For a positive vessel w , angle α is also positive, while a positive (downward) w_w orbital velocity reduces α . This leads to Eq. (2.5.2), with orbital velocity components in $\{b\}$.

$$\begin{aligned} V_\infty^f &= \mathbf{v}_{b/n}^b + \boldsymbol{\omega}_{b/n}^b \times \mathbf{r}_f^b - \mathbf{R}_n^b \mathbf{v}_w^n \\ &= \begin{bmatrix} u + qz_f - ry_f - u_w^b \\ v + rx_f - pz_f - v_w^b \\ w + py_f - qx_f - w_w^b \end{bmatrix} = \begin{bmatrix} u_f \\ v_f \\ w_f \end{bmatrix} \end{aligned} \quad (2.5.2)$$

Finally, utilizing results from Eq. (2.5.2), angle of attack α and sideslip angle β are calculated by Eq. (2.5.3).

$$\alpha = \tan^{-1}\left(\frac{w_f}{u_f}\right), \quad \beta = \sin^{-1}\left(\frac{v_f}{\sqrt{u_f^2 + v_f^2 + w_f^2}}\right) \quad (2.5.3)$$

Pitch angle θ affects the hydrodynamic angle by the decomposition of the vessel velocity vector in the $\{b\}$ frame. This is shown in Fig. 2.10, where if vessel pitch were to be zero at the same velocity vector, the w component (shown in gray) will appear reduced, lowering the effective α .

Dynamic Response The mechanical assembly driving the foils cannot possibly react instantaneously to commands, and is thus modelled as a first order lag.

$$\frac{d}{dt}u_a = -\frac{1}{\tau}u_a + \frac{1}{\tau}u_c \quad (2.5.4)$$

$$g_f(s) = \frac{U_a(s)}{U_c(s)} = \frac{1}{\tau s + 1} \quad (2.5.5)$$

As any control command will be a digital signal, and could be drastically different from the current actuator position, the transfer function $g_f(s)$ of Eq. (2.5.5) can be used to filter the controller output [29], [45]. The time constant τ is chosen as 0.1s, which

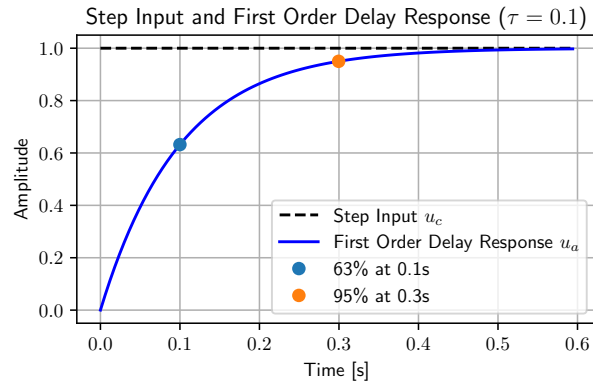


Figure 2.13: First order delay response to step input. Also visible are the τ and 3τ time instants achieving 63% and 95% of reference respectively.

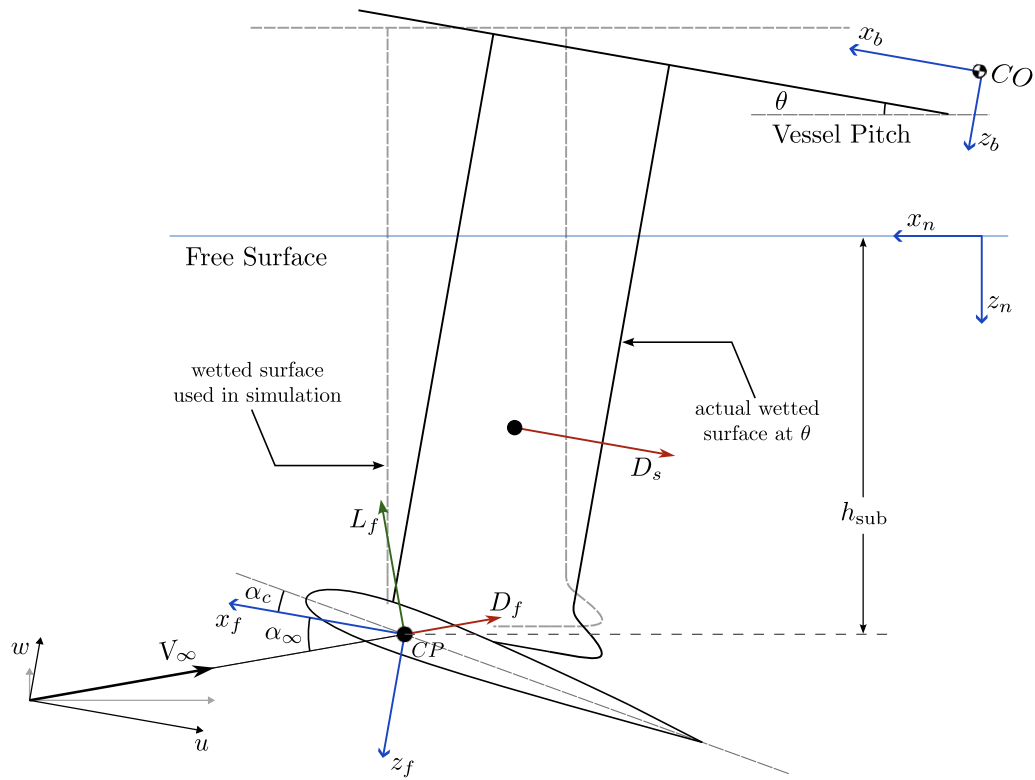


Figure 2.10: Hydrodynamic angle definitions and coordinate frames involved in control surface modelling. Also visible are strut quantities used in the simulation.

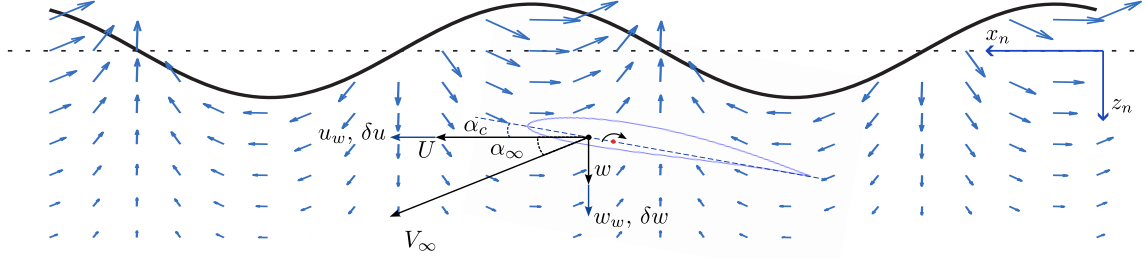


Figure 2.11: Inflow velocity and angle of attack change due to vessel's steady speed, incident waves, and velocity perturbations.

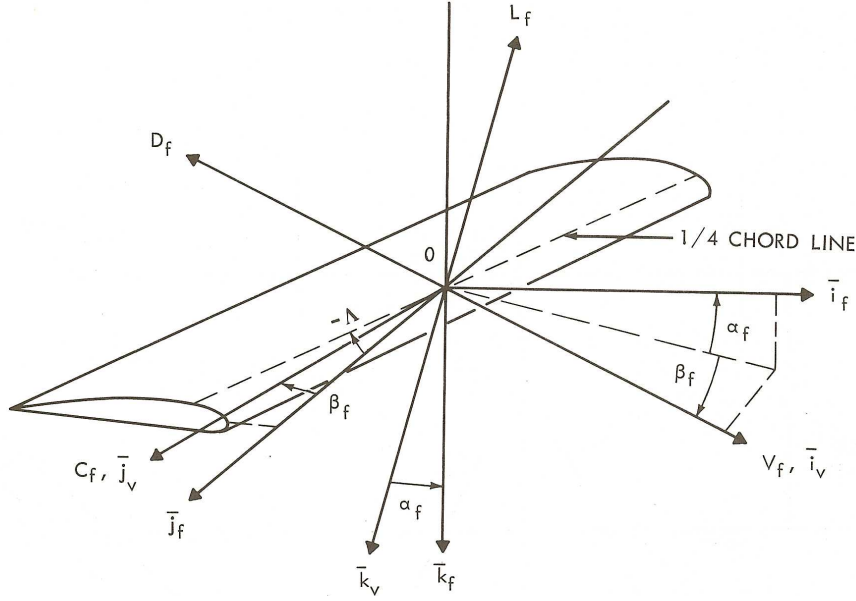


Figure 2.12: Perspective view of the axes system used to model hydrofoils (Reproduced from [19]).

translates to control surfaces reaching $u_a = 63\%$ of the commanded u_c in τ seconds as illustrated in Fig. 2.13.

Submergence & Strut Force Submergence, h_{sub} refers to the vertical distance in $\{n\}$ of the hydrofoil's reference point, CP from the free surface, including elevation from waves. The exact position of CP in $\{n\}$ is given by Eq. (2.5.6), and by isolating the z component and adding wave elevation, Eq. (2.5.7) yields total submergence.

$$\mathbf{p}_{f/n}^n = \mathbf{p}_{b/n}^n + \mathbf{R}_b^n \mathbf{r}_f^b \quad (2.5.6)$$

$$h_{\text{sub}} = (\mathbf{p}_{f/n}^n)_3 + \zeta(\mathbf{p}_{f/n}^n) \quad (2.5.7)$$

Forces arising from struts will be revisited, but in general they are modelled as symmetrical foil prisms with u and v velocity components governing drag and side-force phe-

nomena. These are respectively D_s and L_s with D_s being the most important part for this application. The actual wetted surface versus the one used for simulation purposes are both visible in Fig. 2.10. To reduce computation time, projected surface is calculated as if the vessel operated at zero pitch and roll, making S_{strut} equal to chord times submergence, or $c_s h_{\text{sub}}$. Adhering to the above, strut drag acts at the submergence midpoint, acquiring, a moment arm equal to $(\mathbf{r}_f^b)_3 - \frac{1}{2}h_{\text{sub}}$.

Hydrodynamic Force Vector The total force arising from hydrofoils is calculated as the vectors L_f and D_f of Fig. 2.10 and the cross force C_f due to sideslip. To express this in the body frame, we can observe from Fig. 2.12 that there needs to be a rotation about the hydrodynamic angles [19].

$$\mathbf{f}_f^b = \begin{bmatrix} X_f \\ Y_f \\ Z_f \end{bmatrix} = \begin{bmatrix} \cos \alpha_\infty & 0 & -\sin \alpha_\infty \\ 0 & 1 & 0 \\ \sin \alpha_\infty & 0 & \cos \alpha_\infty \end{bmatrix} \begin{bmatrix} \cos \beta_\infty & -\sin \beta_\infty & 0 \\ \sin \beta_\infty & \cos \beta_\infty & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -D_f \\ -C_f \\ -L_f \end{bmatrix} \quad (2.5.8)$$

Utilizing Eq. (2.3.7) and the foil position with respect to CO vector, $\mathbf{r}_f^b$, we construct the full hydrodynamic force and moment vector for foils, $\boldsymbol{\tau}_{\text{hyd},f}$.

Two Dimensional Hydrofoil

Two dimensional analysis (applicable to an infinite span) is considered firstly. While the results are directly obtained from widely available solvers, a quick overview will be presented. Regarding model accuracy, these methods tend to overestimate lift and underestimate drag (quite the opposite of what we'd hope for) but are relatively accurate, considering the simplifications that lead to their development. The methods are described in-depth in references [8] and [47]. To avoid confusion, 2D lift and drag coefficients will be c_l and c_d respectively.

Thin Airfoil Theory Thin airfoil theory treats the infinite wing as a *vortex sheet*, with a strength $\gamma(s)$ per unit length, aligned with the camber line (thickness is neglected). The total circulation is $\Gamma = \gamma ds$. The results are based on experimental observations formed as mathematical theorems, particularly:

- The Kutta-Joukowski Theorem: $L' = \rho_\infty V_\infty \Gamma$
- The Kutta Condition, that the value of Γ around the airfoil allows the flow to smoothly leave the trailing edge.
- Kelvin's Circulation Theorem: $\frac{D\Gamma}{Dt} = 0$, a conservation of circulation (which will be revisited in the unsteady case) that connects circulation around the airfoil, to the opposite one created by the starting vortex.

This leads to a solution for a cambered foil:

$$c_l = 2\pi(a - a_{L=0}) \quad (2.5.9)$$

Where $a_{L=0}$ is the angle of attack that generates zero lift. While this produces a reasonable approximation, there's no consideration for viscous phenomena and by extend, drag forces.

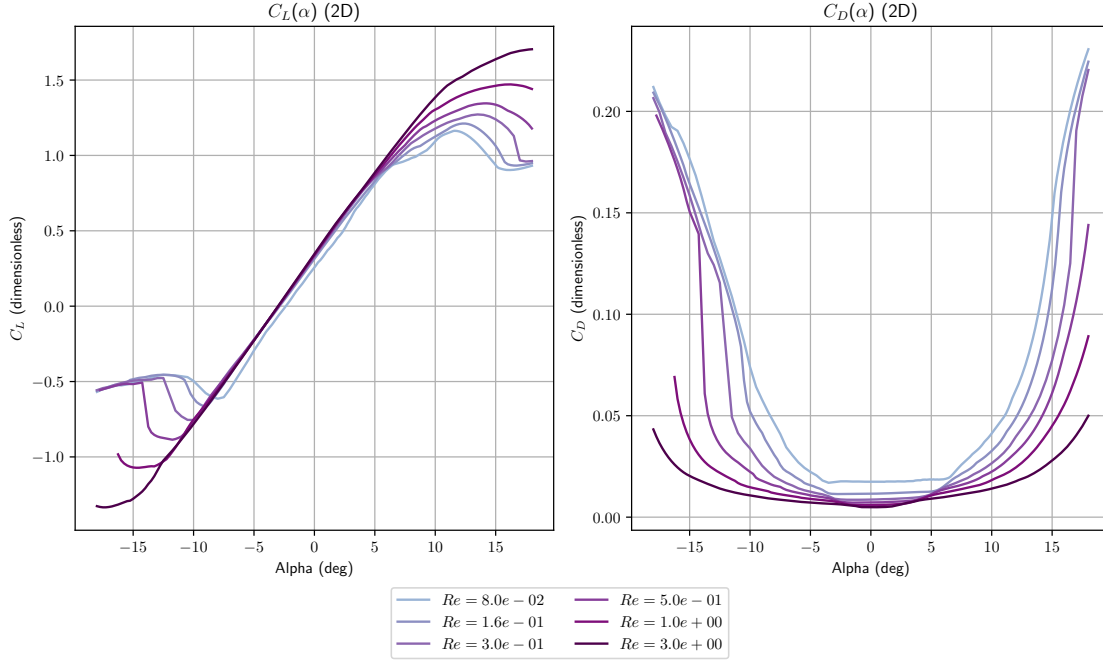


Figure 2.14: Reynolds sweep results for NACA 63412, $n_{crit} = 5$, Mach = 0.

Panel Methods The above described drawbacks led to the development of numerical solutions able to take into account viscous effects and complex geometries. These methods follow the philosophy of wrapping the vortex sheet over the foil surface (approximated by an arrangement of straight *panels*), asserting the physical observation that the body boundary is a streamline of the flow. One such solver for the 2D case is XFOIL [48], which accounts for viscous effects, such as boundary layer and flow separation that leads to wing stall and sudden loss of lift. In a few words, it solves the inviscid, irrotational, incompressible problem while adding a mix of empirical and theoretical turbulence and transition models.

Finite Wings

A lifting surface found in real-life applications will not behave the same way as a theoretical wing of infinite span. An examination of Fig. 2.15 provides a first understanding on how limited span allows interaction between the flow of the upper and lower surface. Around the wingtips, high-pressure lower region flow is forced towards the low-pressure upper surface, forming wing-tip vortices which in turn induce a downward air velocity (appearing as a spanwise distribution). This qualitative observation, appears in a more quantified way in Fig. 2.16. While the flow velocity V is horizontal and the expected angle of attack is α , there is a downward vertical component w at an angle α_i . This leads to the representation of lift with respect to an *effective* angle of attack.

$$\alpha_{eff} = \alpha - \alpha_i \quad (2.5.10)$$

The implications of this mechanism is a tilt of the lift vector towards the drag direction, which is named *induced drag* and symbolized by $C_{D,i}$.

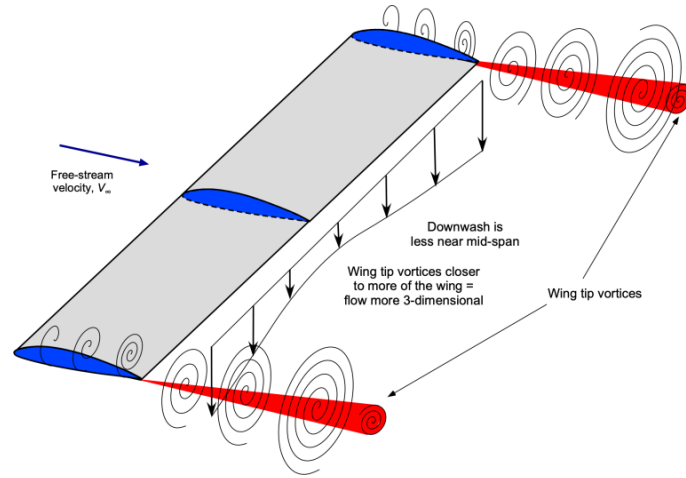


Figure 2.15: Wing tip vortices and their effect on effective angle of attack due to downwash (Reproduced from [49]).

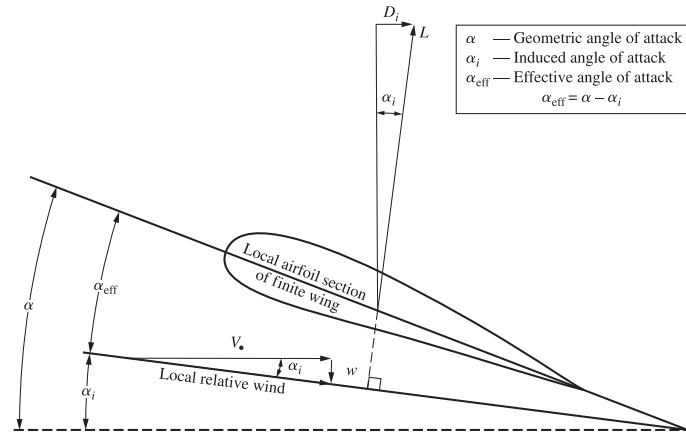


Figure 2.16: Lift vector tilt due to downwash on a finite wing (Reproduced from [8]).

Lifting Line Theory

Prandtl's Analytic Solution To approach this new problem, an important tool is introduced, the *lift distribution* along the span of a lifting surface. By using the lift per unit span L' at a spanwise location y with chord c the total lift generated can be calculated as:

$$L = \int_{-b/2}^{+b/2} L'(y) dy = \rho_\infty V_\infty \int_{-b/2}^{+b/2} \Gamma(y) dy \quad (2.5.11)$$

And in terms of lift coefficients:

$$L = \frac{1}{2} \rho_\infty V_\infty^2 \int_{-b/2}^{+b/2} c(y) c_l(\alpha_{eff}) dy \quad (2.5.12)$$

With α_{eff} varying with y due to the dependence of α_i on the spanwise downwash distribution w .

$$\alpha_i(y) = \tan^{-1} \left(\frac{w(y)}{V_\infty} \right) \quad (2.5.13)$$

The solution to this problem once again requires the value of $\Gamma(y)$, to acquire the distribution of w . An important simplification is the assumption of elliptical lift distribution which gives an exact solution for $\Gamma(y)$ and interestingly results in constant w , directly giving an analytic result for induced angle α_i and drag $C_{D,i}$.

$$\alpha_i = \frac{C_L}{\pi e AR}, \quad \text{and} \quad C_{D,i} = \frac{C_L^2}{\pi e AR} \quad (2.5.14)$$

Where e is the *Oswald efficiency factor*, ranging between 0.8 and 1 ([4], [8]) accounting for the general case (not only elliptic). Regarding the lift coefficient, LLT does not account for phenomena close to stall angles, providing info only for the linear case around small α . An approximation for the lift slope using LLT is found using the results for the foil profile $a_0 = dc_l/d\alpha$ [8, 9].

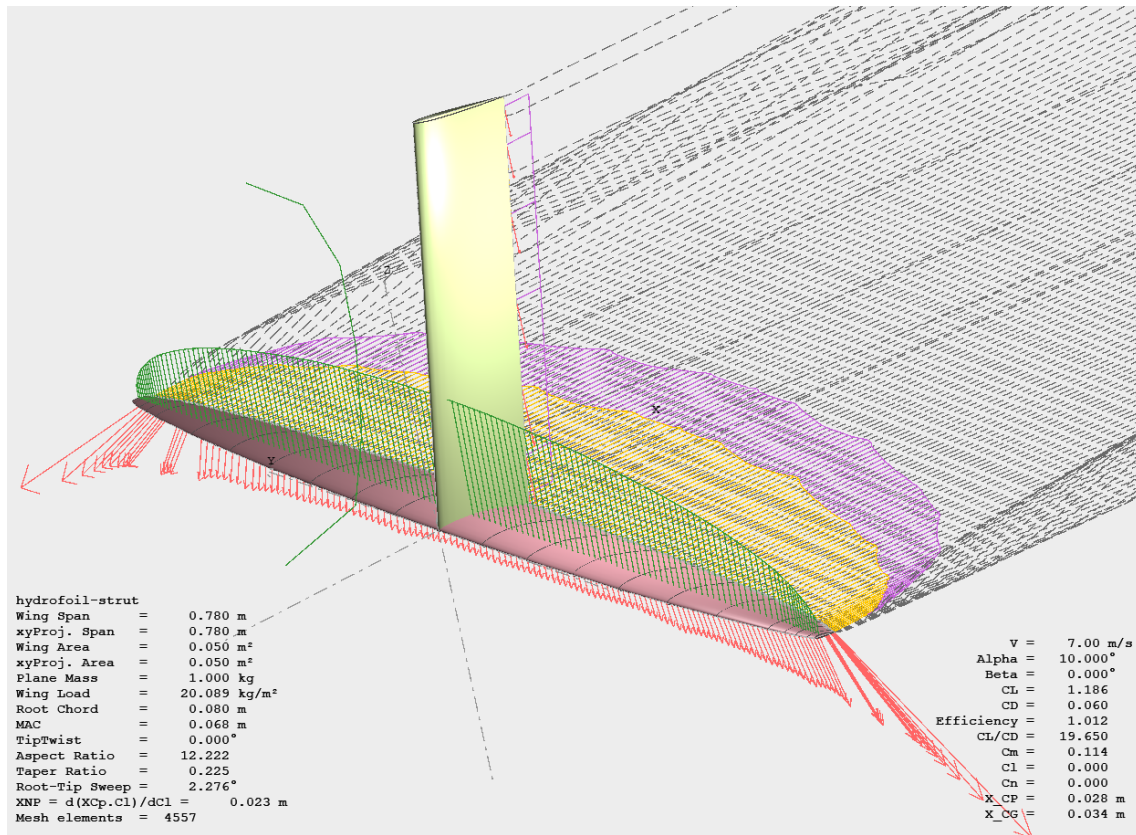
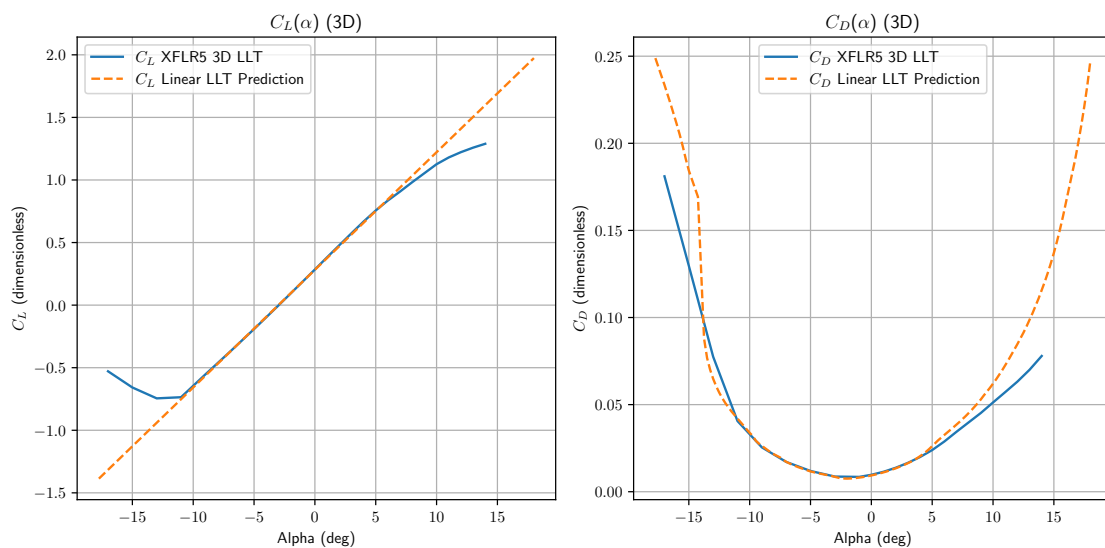
$$\frac{dC_L}{d\alpha} = \frac{a_0}{1 + \frac{a_0}{\pi AR}} \quad (2.5.15)$$

And a slightly more conservative equation is given by [4]:

$$\frac{dC_L}{d\alpha} = \frac{AR/2}{1 + \sqrt{1 + (AR/2)^2}} \quad (2.5.16)$$

Computational Approach The 2D results from XFOIL can be applied to a finite wing by methods of superposition (especially for the linear part of C_L) and interpolation (specifically to account for viscous effects). One such software, XFLR5, is used to construct and analyze model airplanes and in our case can be generalized for a preliminary hydrofoil model. The elliptical planform shall be used both because it is generic enough to give results close to similar designs, but also to provide a comparison to analytic results.

Simulations were run with and without including the strut, utilizing LLT (non linear) and VLM (linear) methods. In the linear region, both methods provide identical results, but VLM overpredicts lift when separation starts to occur. Due to software limitations, interaction effects are not accurately modeled, and the inclusion of strut only adds to the drag component. In Fig. 2.17 a visualization of VLM result is shown. Also visible are the downwash vectors (red), lift (green), viscous and induced drag components (purple and yellow respectively) and the stream. In Fig. 2.18 a comparison is made between a direct application of Prandtl's findings, and LLT results from XFLR5 which will be used for the simulation.

Figure 2.17: Foil & strut assembly, VLM (linear) results for $\alpha = 10$ deg.Figure 2.18: Comparison of calculated polars by XFLR5 for an elliptical wing versus direct application of Prandtl's Lifting Line Theory ($AR = 12$).

Free Surface Effects

Contrary to usual aerodynamic analysis, a hydrofoil craft will perform close to a free surface, altering the foil's hydrodynamic characteristics as a function of its submergence. This is a complex phenomenon that is best predicted by computational methods [50], but can be approximately described by a reduction in lift and increase in drag inversely proportional to a submergence ratio h/c . Based on findings by [20], a formula to determine lift reduction based on submergence ratio is:

$$C_{L,h} = C_{L,\infty} \cdot (1 - 0.422e^{-1.454h/c}) \quad (2.5.17)$$

The presence of free surface will also alter the induced drag coefficient, specifically by increasing is as submergence decreases [20].

$$C_{D,i} = \left(2 - \frac{33(h/s)^{3/2}}{33(h/s)^{3/2} + 1} \right) \cdot \frac{C_L^2}{\pi e AR} \quad (2.5.18)$$

The coefficient described above reaches 2 for $h/s \rightarrow 0$ and 1 for $h/s \rightarrow \infty$.

Unsteady Effects

The study of wings in transient or generally unsteady states is the domain of flight dynamics. In general, analytic solutions to such problems are complex or outright impossible, and are sometimes reduced to dynamic phenomena in a steady state sense, for example oscillatory disturbances and responses. The sources of non-constant motion can be divided into two categories based on the source of motion: emerging from the movement of the vehicle, or from the movement of the fluid surrounding it. For the first category, the problems that led to the development of the field are mainly aerodynamic flutter (and flapping wing flight by extension) relating to the field of aeroelasticity. The second category usually describes the sinusoidal movement of wind gusts, or wave orbital motion.

Incorporating the above definitions, we can form a general equation for the quasi-steady effective angle of attack [29] [19].

$$\alpha_{\text{eff}} = \alpha_c + \alpha_\infty = \alpha_c + \tan^{-1} \left(\frac{w + \delta_w + w_w}{U + \delta_u + u_w} \right) \quad (2.5.19)$$

Utilizing the results from the cross product, namely that $\delta_u = qz_f - ry_f$ and $\delta_w = py_f - qx_f$ and $r = 0$ while also making sure that the components of wave disturbances are transformed to the body axes:

$$\alpha_{\text{eff}} = \alpha_c + \tan^{-1} \left(\frac{w + \delta_w + py_f - qx_f + w_w^b}{U + qz_f + u_w^b} \right) \quad (2.5.20)$$

While the changes in angle of attack describe the circulatory part of lift, there is also the effect of added (or apparent) mass. This results in a somewhat arbitrary [51] separation of dynamics into circulatory (quasi-steady) and added mass parts, roughly explained by Eq. (2.5.21).

$$L = q_\infty S C_L^{qs} a_{\text{eff}}^{qs} + \pi \rho (\ddot{z} - U \dot{\alpha}) \quad (2.5.21)$$

It is important to remember that this equation is more of a representation of the dynamics behind unsteady lift generation, and not an exact mathematical result (or even

approximation). The quasi-steady marked terms depend on oscillatory movements (and thus dynamic change of α due to velocity), while the time and acceleration dependent last term, refers to added mass effects [47].

Unsteady Lift

To derive a model useful for simulation we shall use a quasi-steady approach as a response to dynamic changes of angle of attack, and some frequency domain corrections. This analysis is based on derivations from Newman [47]. Let us first consider sinusoidal body and wave motion. Each angle of attack component shall be represented using the complex amplitude α_0 such that $\alpha(t) = \text{Re} \{ \alpha_0 e^{i\omega t} \}$.

$$\alpha_{\text{eff}}^{qs}(t) = \text{Re} \left\{ \left(\alpha_{c,0} + \alpha_{b,0}^{qs} + \alpha_{w,0}^{qs} \right) e^{i\omega t} \right\} \quad (2.5.22)$$

Also introducing the reduced frequency k_r , a dimensionless number that characterizes the movement (resembling the classic Strouhal number $\text{St} = fL/U$).

$$k_r = \frac{\omega c}{2U} \quad (2.5.23)$$

This break down will allow us to use well known frequency domain techniques to model lift response.

Sectional Lift Following the methods for hydrodynamic force modelling by Belibassakis & Politis [52] and DeLaurier [53], one can approach the problem as a span-wise integration of sectional properties. Each wing strip of chord length $c(y)$

$$C_L^{2d} = C_{L,0} + C_{L_a} [\mathbb{C}(k_r)(\alpha_c + \alpha_b^{qs}) + \mathbb{S}(k_r)(\alpha_w^{qs})] \quad (2.5.24)$$

Where $\mathbb{C}(k_r)$ and $\mathbb{S}(k_r)$ are the Theodorsen and Sears functions, respectively. These are complex-valued transfer functions that pose as solutions to the problem lift amplitude attenuation and phase difference due to wake vorticity. They "scale" the appropriate component of quasi-steady angle as they are solutions for vessel movements and sinusoidal gusts respectively. In Fig. 2.19, a polar plot for these functions is presented. Notice that they tend to be equal for sufficiently small k_r .

$$\mathbb{C}(k_r) = \frac{H_1^{(2)}(k_r)}{H_1^{(2)}(k_r) + iH_0^{(2)}(k_r)} \quad (2.5.25) \quad \mathbb{S}(k_r) = \frac{2i/\pi k_r}{H_1^{(2)}(k_r) + iH_0^{(2)}(k_r)} \quad (2.5.26)$$

Where $H_n^{(2)}$ is the Hankel function of the second kind of n-th order.

Wing Lift It follows that for an elliptic wing, the lift coefficient will be a superposition of the steady state Prandtl's solution, and unsteady theory [52].

$$C_L^{3d} = \frac{AR}{AR + 2} C_L^{2d} \quad (2.5.27)$$

While this is a very concise and generally accepted result, it hides a very bold assumption: that the three different movement components are **sinusoidal** in nature, and of the **same**

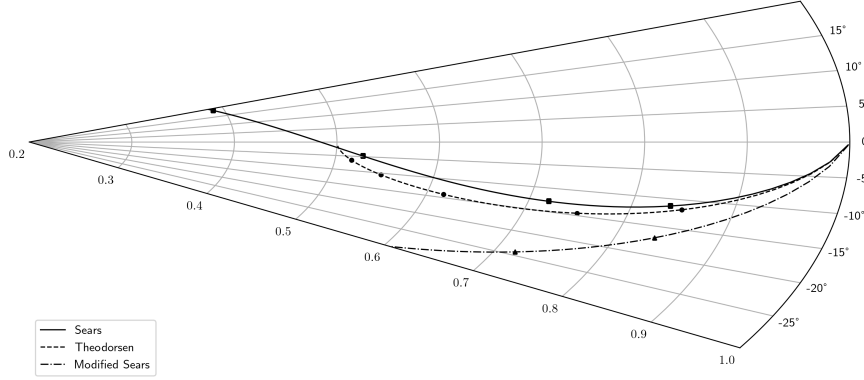


Figure 2.19: Polar plots of Theodorsen and Sears functions. Exact values shown for $k = 0.1, 0.2, 0.5, 1, 2, 5$.

frequency. To describe arbitrary time-dependent motion, techniques of dynamic systems response by convolution must be used.

Time Domain We shall introduce the concept of non-dimensional time $\tau = 2Ut/c$ (traveled distance in semi-chord lengths) and Wagner's function $\phi(\tau)$, which describes the response of lift due to a step change in the angle of attack. This function ties to Theodorsen's findings since they form a Fourier Transform pair [51] as shown in Eq. (2.5.28).

$$\phi(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{\mathbb{C}(k_r)}{ik_r} e^{ik_r\tau} dk_r \quad \text{and} \quad \mathbb{C}(k_r) = ik_r \int_{-\infty}^{\infty} \phi(\tau) e^{-ik_r\tau} d\tau \quad (2.5.28)$$

Which satisfies the important property that $\phi(\tau = 0) = \lim_{k_r \rightarrow \infty} |\mathbb{C}(k_r)| = 1/2$, meaning that half of the lift is produced **instantaneously**. Equation (2.5.28) is not easily applied, which is why approximations have been made. One such approximation is the exponential decay model by R. T. Jones [54] shown in Eq. (2.5.29). The coefficients A_1 and A_2 must always add up to $1/2$, and specifically in this context $A_1 = 0.165$, $A_2 = 0.335$. The exponential coefficients describe response time as $b_1 = 0.0455$ and $b_2 = 0.3$. Note that these are to be scaled with the model dimensions.

$$\phi(\tau) \approx 1 - A_1 e^{-b_1\tau} - A_2 e^{-b_2\tau} \quad (2.5.29)$$

Treating the known function as a step response for the circulatory lift dynamics and by applying the Duhamel principle [21], we can obtain a complete unsteady lift representation.

$$C_L(t) = 2\pi \left[\alpha^{qs}(0)\phi(t) + \int_0^t \frac{d\alpha(\sigma)}{d\sigma} \phi(t - \sigma) d\sigma \right] \quad (2.5.30)$$

Constantly derivating the input angle is not computationally preferable, which is why we shall rely on signals and systems theory [55] to use the impulse response instead. The convolution of the derivative of the input signal with the step response, is equivalent to

the convolution of the original input signal with the system's impulse response. This can also be shown through integration by parts.

$$\begin{aligned}
 \alpha^{qs}(0)\phi(t) + \int_0^t \frac{d\alpha^{qs}(\sigma)}{d\sigma} \phi(t-\sigma) d\sigma &= \\
 \alpha^{qs}(0)\phi(t) + [\alpha^{qs}(\sigma)\phi(t-\sigma)]_0^t - \int_0^t \alpha^{qs} \frac{d\phi(t-\sigma)}{d\sigma} d\sigma &= \\
 \alpha^{qs}(t)\phi(0) - \int_0^t \alpha^{qs} \frac{d\phi(t-\sigma)}{d\sigma} d\sigma
 \end{aligned}$$

To perform the convolution, a Laplace transform of the impulse response shall be taken, to use for multiplication in frequency domain. This result is shown in block diagram format in Fig. 2.20, where it is shown precisely how the above analytic result can be used in a simulation environment.

$$\mathcal{L}\{\dot{\phi}(\tau)\} = \frac{A_1 b_1}{s + b_1} + \frac{A_2 b_2}{s + b_2} \quad (2.5.31)$$

The part of α_w^{qs} that corresponds to wave disturbances can remain as is, scaled by $\mathbb{S}(k_r)$ as long as the disturbance is a simple harmonic wave.

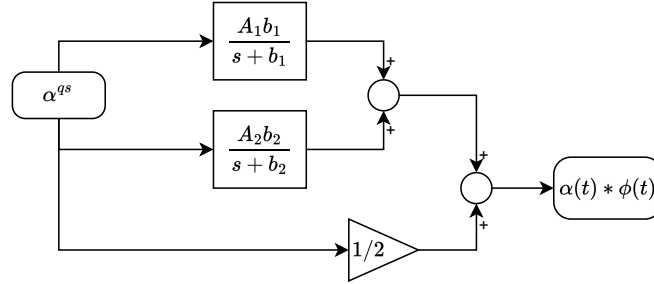


Figure 2.20: Block diagram showing the convolution of input angle signal with the derivative of Wagner's step response in transfer function form.

Added Mass

Regarding the effects of acceleration, linear added mass coefficients will be approximated. Following the same logic as Touw [44] and Scherer [56], the added mass of any wing can be approximated by that of an ellipse with the same thickness and chord. The coefficients shall be used to populate the added mass matrix $\mathbf{M}_A$.

$$\mathbf{M}_A = \begin{bmatrix} X_{\dot{u}} & X_{\dot{w}} & X_{\dot{p}} & X_{\dot{q}} \\ Z_{\dot{u}} & Z_{\dot{w}} & Z_{\dot{p}} & Z_{\dot{q}} \\ K_{\dot{u}} & K_{\dot{w}} & K_{\dot{p}} & K_{\dot{q}} \\ M_{\dot{u}} & M_{\dot{w}} & M_{\dot{p}} & M_{\dot{q}} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{13} & A_{14} & A_{15} \\ A_{31} & A_{33} & A_{34} & A_{35} \\ A_{41} & A_{43} & A_{44} & A_{45} \\ A_{51} & A_{53} & A_{54} & A_{55} \end{bmatrix} \quad (2.5.32)$$

For an ellipse of width c and height t , coinciding with local axis x and z respectively, the added mass coefficients for movement along each axis shall be:

$$a_{11} = a_{xx} = \frac{\pi \rho t^2}{4}, \quad a_{22} = a_{zz} = \frac{\pi \rho c^2}{4} \quad (2.5.33)$$

Expressing each foil section as an ellipse of width equal to chord and height equal to thickness and denoting wing semi-span $s = b/2$ for simplicity, we have the following:

$$c(y) = c_0 \sqrt{1 - \left(\frac{y}{s}\right)^2} \quad (2.5.34)$$

$$a_{11} = \int_{-s}^s \frac{\pi \rho t(y)^2}{4} dy = \int_{-s}^s \frac{\pi \rho}{4} \left(p c_0 \sqrt{1 - \left(\frac{y}{s}\right)^2} \right)^3 dy = \frac{\pi \rho}{6} b p^2 c_0^2 \quad (2.5.35)$$

$$a_{33} = \int_{-s}^s \frac{\pi \rho c(y)^2}{4} dy = \int_{-s}^s \frac{\pi \rho}{4} \left(c_0 \sqrt{1 - \left(\frac{y}{s}\right)^2} \right)^2 dy = \frac{\pi \rho}{6} b c_0^2 \quad (2.5.36)$$

According to Scherer [56], a correction factor must be added to account for AR effects. Specifically, each coefficient is scaled by the following quantity:

$$a_{ij}^{3d} = \frac{\text{AR}}{\sqrt{1 + \text{AR}^2}} a_{ij} \quad (2.5.37)$$

Care must be taken when applying the linear coefficients in order to calculate vessel response: while the lowercase a_{ij} coefficients are expressed in axes orthogonal to the foil chord, the partial derivatives shown in Eq. (2.5.32) are expressed in body fixed axes.

Additional Drag Components

Consideration of different drag sources has already been explained in Section 2.5.1. Based on analysis by [57] and [9] the additional coefficients are described by semi-empirical methods.

Wave

$$C_{D,w} = C_L^2 \left(\frac{1}{2 \text{Fr}_h^2} \left(\frac{c}{h} \right) e^{-\frac{2}{\text{Fr}_h^2}} \right) \quad (2.5.38)$$

Interaction Interaction effects between strut and foil shall be approximated via the curve-fitting of experimental data by [57].

$$C_{D,\text{int}} = \frac{D_{\text{int}}}{q_\infty t^2} = 0.75 \left(\frac{t}{c} \right) - 0.0003 \left(\frac{t}{c} \right)^{-2} \quad (2.5.39)$$

Spray

$$C_{D,\text{sp}} = \frac{D_{\text{sp}}}{q_\infty t^2} = 0.24 \quad (2.5.40)$$

Lateral Damping

In this section, an attempt shall be made to approximate damping forces due to roll movement. This ties to the general problem of lateral equations of motion, and the coupling that exists between them. Lateral dynamics are strongly coupled, meaning for example that changes in roll will affect side force and yawing moment. While this effect is important, it is neglected for the sake of simplicity. This approach slightly affects the longitudinal equations, since the p rotational velocity component in the cross products is implicitly accounted for in the lateral derivatives.

A rough outline of how roll affects the angle of attack α experienced by foils and struts is shown in Fig. 2.21. While the distribution of α is linear, the forces depend on lifting surface geometry. A general approach is once again span wise integration of chord segments. Equation (2.5.41) shows how this is analytically calculated for each lifting surface.

$$L_{\text{damp},f} = - \int y \delta \ell, \quad L_{\text{damp},s} = - \int z \delta \ell \quad (2.5.41)$$

Following an approach similar to Etkin [3], we start with the horizontal wings.

$$\begin{aligned} L_{\text{damp},f} &= - \int y (q_{\infty} \delta S(y) C_{L_{\alpha}} \Delta \alpha(y)) \\ &\stackrel{S(y)=c(y)dy}{=} -q_{\infty} C_{L_{\alpha}} \int y \Delta \alpha(y) c(y) dy \\ &\stackrel{\Delta \alpha=p y/V}{=} -q_{\infty} C_{L_{\alpha}} V p \int y^2 c(y) dy \end{aligned}$$

The last integral can be calculated for each lifting surface separately to give the final sum of moments. Specifically for the case of the rear foil that is centered with respect to vessel axes, the integral limits shall be $[-s, s]$ and the result is the same as in [28]).

$$L_{\text{damp},fa} = -\frac{1}{2} \rho V C_{L_{\alpha}} p c_0 \frac{\pi b^3}{64} \quad (2.5.42)$$

Continuing with the derivations for the struts, but using a height reduction h_r as is shown in Fig. 2.21, the integration shall be made in the range $[z_{WL}, z_s]$ where $z_{WL} = z_{CG} + h_r$ and z_s the strut's lowest point.

$$\begin{aligned} L_{\text{damp},s} &= - \int z (q_{\infty} \delta S(z) C_{L_{\alpha}} \Delta \alpha(z)) \\ &\stackrel{S(z)=c_s dz}{=} -q_{\infty} C_{L_{\alpha}} \int z \Delta \alpha(z) c_s dz \\ &\stackrel{\Delta \alpha=p z/V}{=} -q_{\infty} C_{L_{\alpha}} V p \int z^2 c_s dz \end{aligned}$$

Using the previously defined limits:

$$L_{\text{damp},s} = -q_{\infty} C_{L_{\alpha}} V p \int_{z_{WL}}^{z_s} z^2 c_s dz = -\frac{1}{2} \rho V C_{L_{\alpha}} p c_s \frac{(z_s^3 - z_{WL}^3)}{3} \quad (2.5.43)$$



Figure 2.22: Propellers developed by Solar Boat Twente (From the team's website [59])

To acquire open-water coefficients, we use the polynomial regression data by Bernitsas et. al [60], that have been ported to Matlab by Fossen within MSS [40]. To select a diagram we require a pitch ratio P/D , expanded area ratio A_E/A_O , number of blades z , and the diameter D to dimensionalize results. To select an appropriately sized propeller, a brief research on propeller designs in similar boats, such as the ones in Fig. 2.22 or reports in publicly available repositories [61] was conducted. The general idea in such vessels is to achieve maximum motor efficiency at high rpm and low torque. It was found that diameter D generally ranges from 0.2 to 0.3 meters, expanded area ratio is kept low and 2 or 3 bladed configurations were preferred. Nominal model data are again collected in Table A.1. Using an online tool for wageningen-b geometry generation [62], a 3D visualization of the selected propeller is shown in Fig. 2.23, which appears similar to ones used in MEBC.

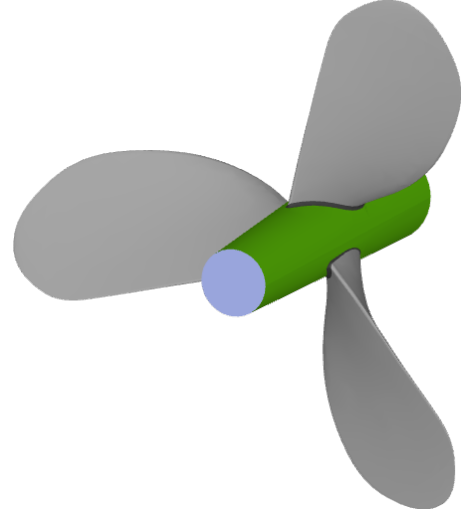


Figure 2.23: 3D model of the propeller geometry selected for the nominal simulation model.

Figure 2.24 presents the calculated coefficient curves for the selected parameters. These are used to acquire a quadratic polynomial fit of 3 coefficients for each curve ($[p_1, p_2, p_3]$ for K_T and K_Q) that are then used as input to the *open-water propeller characteristics* block from the MSS Toolbox. The block calculates torque required and thrust produced, according to Eqs. (2.5.47) and (2.5.48).

$$T = \rho n |n| D^4 (p_{kT,1} J^2 + p_{kT,2} J + p_{kT,3}) \quad (2.5.47)$$

$$Q = \rho n |n| D^5 (p_{kQ,1} J^2 + p_{kQ,2} J + p_{kQ,3}) \quad (2.5.48)$$

Since thrust is directed on the x_b axis, there's only an X force component, and an M

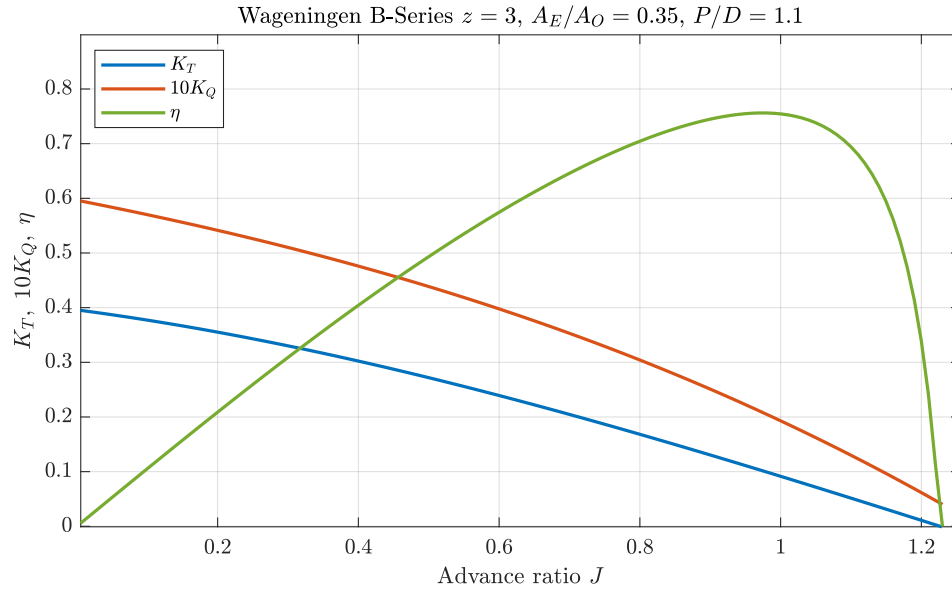


Figure 2.24: Thrust, torque and efficiency coefficients used in the simulation model.

moment equal to $dz_p T$ that will also affect trimming conditions. The generalized force vector is constructed accordingly.

2.6 Parametric Simulink Model

To avoid presenting the developed simulation environment in a rather cumbersome fashion by dividing it between chapters of modelling and control, an in-depth presentation is made within Appendix B. In this section a rough overview of the environment's structure is given with possible shortcomings discussed in Chapter 6.

Figure 2.25 combines the contents of this chapter in a single diagram, providing a parametrically defined model of both vessel and environment. Dashed outlines indicate parts where model accuracy is deemed insufficient, or simply ill-defined unless real-world tests are conducted. The available parameters are roughly the following:

- Distance of CG to CO, $\mathbf{r}_g^b$.
- Moments of inertia around CG, with automated change of origin around CO.
- Vessel mass m , gravitational acceleration g and fluid density ρ .
- Foil placement as $\mathbf{r}_f^b$ for each lifting surface, which indirectly defines struts.
- Foil and strut chord and thickness, as well as hydrodynamic performance characteristics in look-up table form.
- Propeller placement $\mathbf{r}_p^b$ and data as given in Table A.1.
- Wagner unsteady lift model coefficients.
- Hydrodynamic derivatives where applicable, such as roll damping L_p .

All simulations and linear model derivations are based on a *reduced* model of the original 6DOF formulation. In this reduced state-space, the longitudinal dynamics remain realistic, lateral degrees of freedom however do not. There is strong coupling between roll rate/angle and changes in side force and heading. For the purpose of control system benchmarks however, the approach of isolating roll dynamics and using them in conjunction with longitudinal is utilized in literature [29], [45]. This allows us to evaluate performance in tasks where control surfaces need to move independently, to counteract disturbances in different planes.

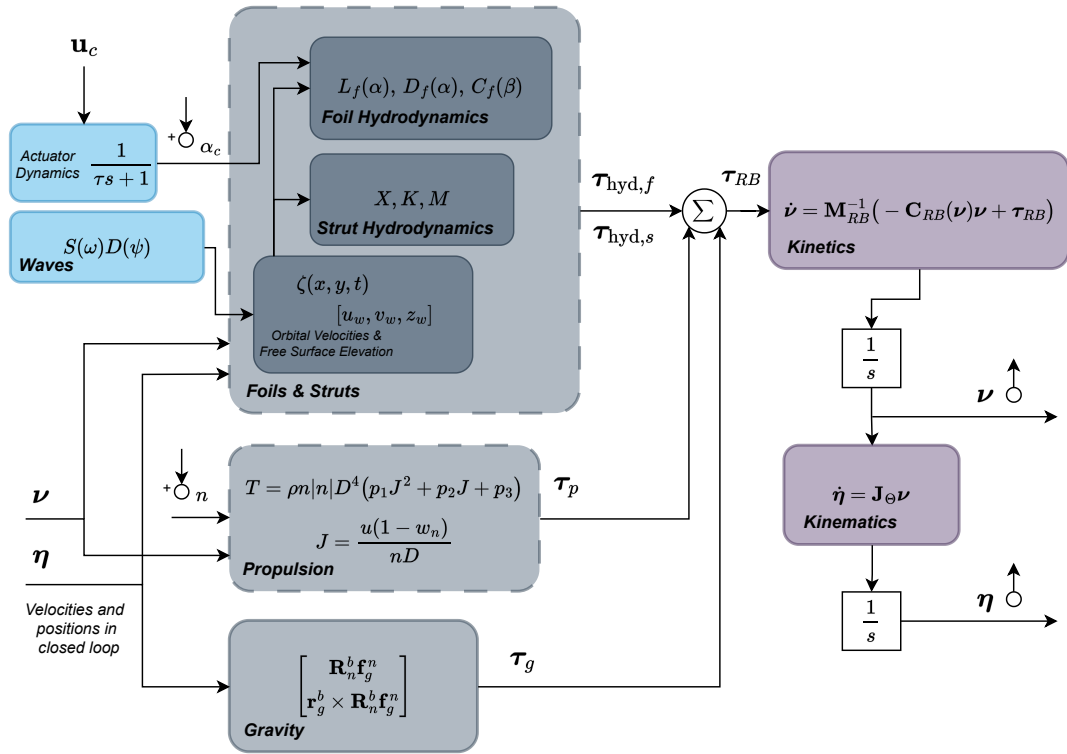


Figure 2.25: Block diagram description of the non-linear model as it is implemented in Simulink. Also visible are the points for automatic linearization.

Chapter 3

Linearization & System Analysis

While the nonlinear Simulink model captures the system's dynamics with high fidelity, it is computationally too heavy to serve as a prediction model and too complex to permit rigorous nonlinear control design with guaranteed stability margins. To enable analysis and controller synthesis, a reduced-order linear representation is therefore extracted around selected trim conditions. This linear model provides mathematical tractability and direct comparability with results found in literature, such as Etkin's linearized EOM. In practice, a single linearization is insufficient to cover the full flight envelope, so an array of linearized models is generated at different operating velocities, yielding a piecewise-linear approximation that balances fidelity with analytic convenience.

3.1 Model Reduction

For the purposes of developing a control algorithm, a model of the system was created in Simulink, with its general structure shown in Fig. 2.25. As this system operates in more DOF than needed for a seakeeping controller, we will proceed with presenting a reduced order non-linear representation. Software-wise, this is performed by eliminating signal paths that pertain to the undesirable states.

Assumptions Specific simplifying assumptions are applied to the model of Fig. 2.25, in order to construct the reduced, non-linear model. These are,

- Heading, yaw rate and side force are considered to be constant and zero, thus eliminating sway and yaw DOF (realistically not possible, but in actual conditions a pilot handles the rudder control inputs to retain such conditions).
- Air resistance is neglected as it is minimal compared to fluid loads.
- Body-fixed axes are centered on vessel *CG*, simplifying the equations of motion.
- All products of inertia are set to zero, considering a symmetrical body (this is not always accurate for I_{xz}).

3.1.1 State Selection

In the particular case, control objectives drive state choice. Since we develop a stabilizing controller in heave, pitch and roll, z, ϕ, θ (a subset of $\boldsymbol{\eta}$) seem like the appropriate choice of states. The dynamics are governed by second order differential equations with respect to the position coordinates, which means that the state representation also requires the velocities of the selected states. One choice would be the appropriate subset of $\dot{\boldsymbol{\eta}}$, but body-fixed velocities $\boldsymbol{\nu}$ are readily measurable by existing equipment (this will be revisited in Section 4.5) presenting a more favorable choice also supported by literature [1], [29], [45]. The linearized system will involve the following quantities:

$$[z, \phi, \theta, w, p, q]$$

For the time being, all states shall be considered *measurable*. The reduced model will be a function f , mapping $\mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n$, consisting of n states and m inputs as described by Eq. (3.1.1).

$$\dot{\mathbf{x}}(t) = f(\mathbf{x}(t), \mathbf{u}(t)) \quad (3.1.1)$$

Adhering to the assumptions made and solving for rates, the resulting equations of motion appearing in Eq. (3.1.2) and Eq. (3.1.3) can be viewed as a less compact form of Eq. (3.1.1). Quantities expressed by Eq. (3.1.3) are in practice augmentations to the existing state-space using rotation-translation matrices.

$$\dot{u} = -\frac{qw}{m} + \frac{X_{\text{hyd}}}{m} - g \sin \theta \quad (3.1.2a)$$

$$\dot{w} = \frac{qu}{m} + \frac{Z_{\text{hyd}}}{m} + g \cos \theta \cos \phi \quad (3.1.2b)$$

$$\dot{p} = \frac{K}{I_x} \quad (3.1.2c)$$

$$\dot{q} = \frac{M}{I_y} \quad (3.1.2d)$$

$$\dot{z} = -u \sin \theta + w \cos \phi \cos \theta \quad (3.1.3a)$$

$$\dot{\phi} = p + q \sin \phi \tan \theta \quad (3.1.3b)$$

$$\dot{\theta} = q \cos \phi \quad (3.1.3c)$$

As is the case with many control applications, an important tool both for analysis and design is a linearized model of the original system's equations.

3.2 Linearization

To develop the linear model, we shall use a Jacobian linearization about a specific operating point (also called, the equilibrium or trim point). Consider the general model described by a set of non-linear differential equations that was introduced by Eq. (3.1.1). We seek the equilibrium points $\mathbf{x}_o \in \mathbb{R}^n$ for which there are specific equilibrium inputs $\mathbf{u}_o \in \mathbb{R}^m$ such that

$$f(\mathbf{x}_o, \mathbf{u}_o) = \mathbf{0}_n, \quad \forall t \quad (3.2.1)$$

We now define the state and input perturbations from an equilibrium point as

$$\begin{aligned}\delta \mathbf{x}(t) &= \mathbf{x}(t) - \mathbf{x}_o \\ \delta \mathbf{u}(t) &= \mathbf{u}(t) - \mathbf{u}_o\end{aligned}$$

and by substitution, Eq. (3.1.1) becomes

$$\delta \dot{\mathbf{x}}(t) = f(\mathbf{x}_o + \delta \mathbf{x}(t), \mathbf{u}_o + \delta \mathbf{u}(t)) \quad (3.2.2)$$

The concept of the Jacobian enters by taking the Taylor approximation of the right hand side and ignoring all terms higher than first order resulting in

$$\delta \dot{\mathbf{x}}(t) \approx f(\mathbf{x}_o, \mathbf{u}_o) + \left. \frac{\partial f}{\partial \mathbf{x}} \right|_{\mathbf{x}_o, \mathbf{u}_o} \delta \mathbf{x}(t) + \left. \frac{\partial f}{\partial \mathbf{u}} \right|_{\mathbf{x}_o, \mathbf{u}_o} \delta \mathbf{u}(t) \quad (3.2.3)$$

Considering that, as stated in Eq. (3.2.1), $f(\mathbf{x}_o, \mathbf{u}_o) = 0$, dropping the deltas and by a change of notation, such that

$$\mathbf{A} := \left. \frac{\partial f}{\partial \mathbf{x}} \right|_{\mathbf{x}_o, \mathbf{u}_o} \in \mathbb{R}^{n \times n}, \quad \mathbf{B} := \left. \frac{\partial f}{\partial \mathbf{u}} \right|_{\mathbf{x}_o, \mathbf{u}_o} \in \mathbb{R}^{n \times m} \quad (3.2.4)$$

We reach the well known state space form (3.2.5). Since the Jacobian matrices $\mathbf{A}$ and $\mathbf{B}$ are not dependent on time, this form provides a Linear, Time Invariant system commonly abbreviated as LTI [55]. Output matrices are usually picked as the identity matrix for $\mathbf{C}$ and 0 (no input feedthrough) for $\mathbf{D}$.

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \quad (3.2.5a)$$

$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u} \quad (3.2.5b)$$

An important remark is that this form does not allow derivative terms on the right hand side, which has a significant impact on hydrodynamic modelling.

3.2.1 Static Stability - Trimming

Moving away from the abstract notion of equilibrium in the mathematical sense, the system studied here is more straightforward. An equilibrium point is picked as **steady, level** flight, meaning that the linearization will be performed around a steady forward speed $u = U_0$, flight height $z = z_h$ and all other states will be 0. To find the equilibrium inputs, a glance at Fig. 3.1 will provide us with a clear understanding of the problem needed to be solved¹. Due to vessel symmetry, the fore foils will provide the same amount of lift. Using the subscripts w for wing and t for tail (making it easier to follow derivations also made in [3]) we need the input vector $\mathbf{u}_o = [\delta_w, \delta_w, \delta_t, T_p]$ such that the system remains stable at $\mathbf{x}_o = [z_h, 0, 0, U_0, 0, 0, 0]$.

Assuming that the selected z_h is enough to diminish any free surface effects, the problem stated above reduces to a force-moment equilibrium. Starting from forces

$$L_w + L_t = mg \quad (3.2.6)$$

¹While the figure might seem unnecessarily convoluted since $\alpha_\infty = 0$ by default in the equilibrium case, it plays an important role in stability analysis later on, especially in understanding the physical properties of $\frac{\partial C_m}{\partial \alpha}$.

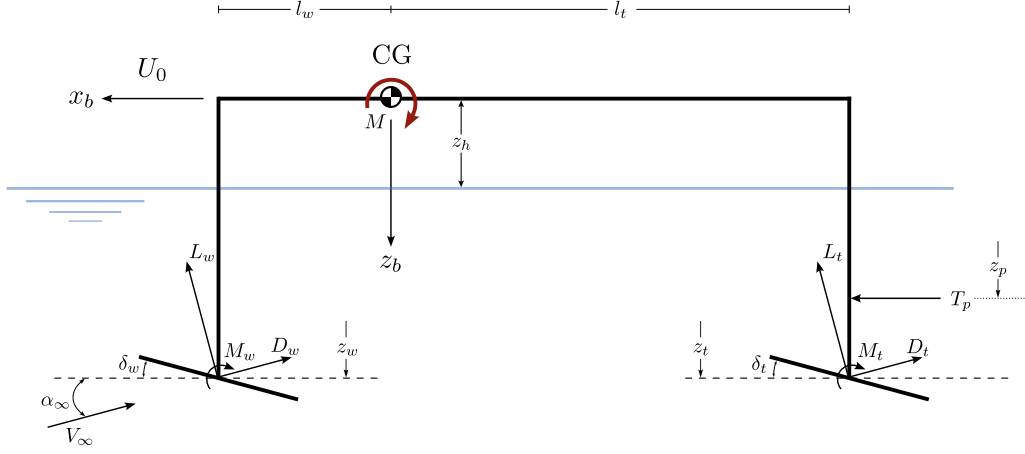


Figure 3.1: Forces & moments affecting longitudinal stability.

And by substitution of the known function between angle and force produced by the lifting surface

$$q_\infty (2S_w C_L(\alpha_\infty + \delta_w) + S_t C_L(\alpha_\infty + \delta_t)) = mg \quad (3.2.7)$$

Following the same logic for the moments, the general case is slightly more complex.

$$M_w + M_t + M_p = 0 \quad (3.2.8)$$

Where,

$$M_w = M_{ac,w} + (L_w \cos \alpha_\infty + D_w \sin \alpha_\infty) l_w + (L_w \sin \alpha_\infty - D_w \cos \alpha_\infty) z_w \quad (3.2.9)$$

$$M_t = M_{ac,t} - (L_t \cos \alpha_\infty + D_t \sin \alpha_\infty) l_t - (D_t \cos \alpha_\infty - L_t \sin \alpha_\infty) z_t \quad (3.2.10)$$

Which, by assuming that $a_\infty = 0$, reduces to

$$M_{ac,w} + L_w l_w - D_w z_w + M_{ac,t} - L_t l_t - D_t z_t + T_p z_p = 0 \quad (3.2.11)$$

To avoid lengthy computations by hand, this *operating point* is automatically calculated in MATLAB, by setting appropriate starting values and ranges for the states and inputs. Different speeds as well as different vessel states (e.g, a moved center of gravity) will yield different $\mathbf{u}_o$.

3.2.2 Automatic Linearization

The points selected for linearization analysis that govern input to output relationships are also highlighted in Fig. 2.25 with their appropriate symbols (Simulink notation). By pushing test signals through states and inputs and recording the system's response on the output, we are able to numerically determine the behavior of Eq. (3.1.1). Two steps are performed to acquire the linear model matrices:

1. A trimming of the model around a set operating point as it was defined above, yielding the trim inputs $\mathbf{u}_o$.
2. A linearization around that point, using the active states and the input/output signals marked in Fig. 2.25.

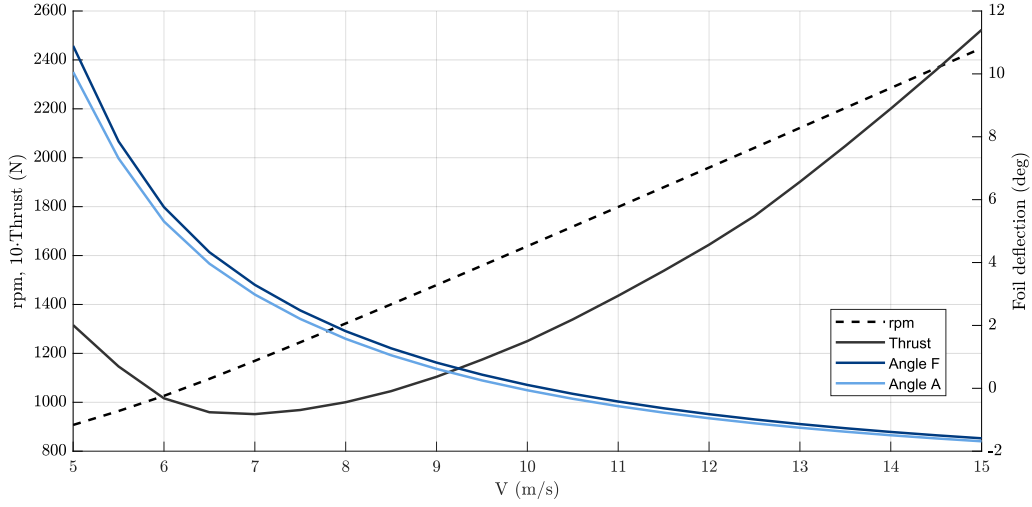


Figure 3.2: Result of successive linearization over a speed range $U_0 = 5 \text{ m} \cdot \text{s}^{-1}$ to $10 \text{ m} \cdot \text{s}^{-1}$ for nominal model parameters.

Two functions were written to carry out these tasks automatically, in order to come up with an array of trim inputs and linear models covering vessel speeds U_0 between $5 \text{ m} \cdot \text{s}^{-1}$ and $15 \text{ m} \cdot \text{s}^{-1}$. The results of the successive linearizations are pictured in Fig. 3.2, showing control deflections as well as propeller rpm. The behavior of required thrust is especially important, as it verifies the characteristics of hydrofoil vessel resistance, where shortly after take-off there is a local minima in total resistance, followed by a polynomial increase [9, Ch. 6.3].

3.3 Linear Model

The Jacobian matrices described by Eq. (3.2.4) have been automatically calculated. The result is a 7×7 state matrix $\mathbf{A}$ and a 7×4 input matrix $\mathbf{B}$. Notice that even if the final LTI system does not include forward velocity as a state or propeller rpm as an input, they are still included in the linearization step. Linearizing around a steady forward speed without taking into account the pitching moment created by the propulsive forces will lead to an invalid trim point. You can imagine this as a trim point that would exist only if the thrust force was acting through the center of gravity, which is mostly valid for aircrafts but does not hold for hydrofoil vessels.

To acquire an intuitive sense of how the states interact, we can look at the state matrices of the linear case, found in Etkin [3, p. 112, p. 229], tweaked to contain the already defined operating point $[\mathbf{x}_o, \mathbf{u}_o]$ as well as the various assumptions. Partial derivatives with respect to $\dot{w}$ are retained in this analysis with the most important reason being the modified m due to vertical added mass effects contained in the $m - Z_{\dot{w}}$ terms.

Looking at Eqs. (3.3.1) and (3.3.2), we get a clear look on what the numerically calculated model should look like. For example, Z_{δ_p} is expected to be 0 due to propeller orientation, $K_{\delta_{fs}}$ should be the opposite of $K_{\delta_{fp}}$ again due to foil placement. By thinking in terms of the vessel's physics, one can also guess relative magnitude and sign of most quantities. At this point, we can also simply derive these partial derivatives based on

hydrofoil dynamics described in previous sections (as was the case in [28], [29], [45]) but that would beat the purpose of using Simulink to create a general model of a large class of vessels.

The numerically computed nominal model matrices are collected in Appendix A.3 (Eqs. (A.3.1) and (A.3.2)). Through the previously mentioned series of linearizations, 21 of these state space representations were created to investigate both the trim point behavior and the effects that the prediction model has on the designed controller.

Equation (3.3.1): Longitudinal equations of motion, linearized around operating point, in partial derivative form augmented with CG height z . To avoid repetition, subscript i in $\mathbf{B}$ matrix corresponds to identical columns for each control input $i = [\delta_{fs}, \delta_{fa}, \delta_e, \delta_p]$

$$\begin{bmatrix} \dot{z} \\ \dot{\theta} \\ \dot{u} \\ \dot{w} \\ \dot{q} \end{bmatrix} = \begin{bmatrix} 0 & -U_0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & -g & \frac{X_w}{m} & 0 \\ 0 & 0 & 0 & \frac{Z_w}{m - Z_{\dot{w}}} & \frac{Z_q + mU_0}{m - Z_{\dot{w}}} \\ 0 & \frac{1}{I_y} \left[M_u + \frac{M_{\dot{w}}Z_u}{m - Z_{\dot{w}}} \right] & \frac{1}{I_y} \left[M_w + \frac{M_{\dot{w}}Z_w}{m - Z_{\dot{w}}} \right] & \frac{1}{I_y} \left[M_q + \frac{M_{\dot{w}}(Z_q + mU_0)}{m - Z_{\dot{w}}} \right] & 0 \end{bmatrix} \begin{bmatrix} z \\ \theta \\ u \\ w \\ q \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{X_i}{m} \\ \frac{Z_i}{m - Z_{\dot{w}}} \\ \frac{M_i}{I_y} + \frac{M_{\dot{w}}Z_i}{I_y(m - Z_{\dot{w}})} \end{bmatrix} \begin{bmatrix} \delta_{fs} \\ \delta_{fa} \\ \delta_e \\ \delta_p \end{bmatrix} \quad (3.3.1)$$

Equation (3.3.2): Lateral equations of motion, linearized around operating point, in partial derivative form. Notice the appearance of roll damping K_p .

$$\begin{bmatrix} \dot{\phi} \\ \dot{p} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \frac{K_p}{I_x} & 0 \end{bmatrix} \begin{bmatrix} \phi \\ p \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 \\ \frac{K_{\delta_{fs}}}{I_x} & \frac{K_{\delta_{fp}}}{I_x} & 0 & 0 \end{bmatrix} \begin{bmatrix} \delta_{fs} \\ \delta_{fa} \\ \delta_e \\ \delta_p \end{bmatrix} \quad (3.3.2)$$

3.4 System Analysis

In order to design well performing control systems, one must first analyze the properties of the system to be stabilized. This is most easily performed in the narrow region described by the linear system. A first analysis is done by comparing how closely the non-linear case is modeled under certain actuation input. Since it is important to stay within the linear region, and no feedback control has been introduced yet, we can excite the system inputs by a sinusoidal control signal described in Eq. (3.4.1).

$$u_i = u_{o,i} + A \sin \omega t, i = [\delta_{fs}, \delta_{fa}, \delta_e] \quad (3.4.1)$$

The amplitude A is selected as 3° and a frequency of 2 Hz, and the resulting time evolution for both models (for CG height and pitch) is shown in Fig. 3.3. For this simulation, no actuator delay is modeled, Eq. (3.4.1) describes the exact control deflection. The linear case shown makes use of the *reduced* model, which does not take into account the u state, since it is usually harder to measure.

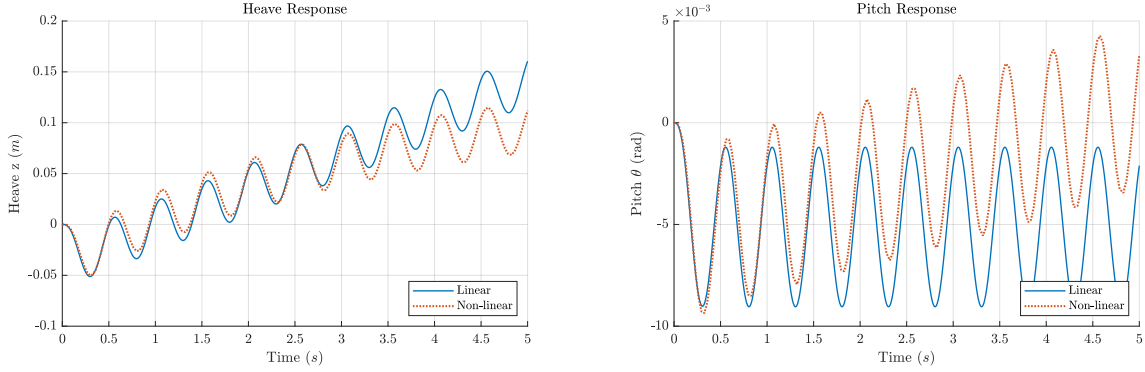


Figure 3.3: Comparison of linear vs non-linear model response to sinusoidal control input on all control surfaces.

The full model noticeably deviates from linear case, showing a steady drift². An explanation for this behavior can be attributed to inertia effects along with the incorporation of Wagner’s equation in lift forces resulting to a delay in the total energy entering the system. We shall now focus on the linearized dynamics, and most importantly, on the unactuated system’s characteristics.

3.4.1 Characteristics

For linear dynamic models, there are various mathematical tools used in order to acquire formal results regarding system characteristics. We shall revisit some fundamentals results here. The general solution for the linear differential equation described by Eq. (3.2.5) is found as a superposition of the homogeneous solution and a convolution with the forcing term, as shown in Eq. (3.4.2). For zero initial conditions, this leaves of course only the second term.

²Still, the model provides accurate approximation for a short period of time, which is important when used in conjunction with repeated initialization at measured states for optimization purposes. This will be revisited in controller design.

$$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0) + \int_0^t e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau) d\tau \quad (3.4.2)$$

This solution form starts to expose the underlying system dynamics, and how the state matrix $\mathbf{A}$ dictates time evolution both in the free response, and forced case. The term $e^{\mathbf{A}t}$ is also defined as the *state transition matrix* (t) . It has been proven [63] that for a linear system of order n described by a matrix $\mathbf{A}$ with n distinct eigenvalues, λ_j , the homogeneous response from an initial state $\mathbf{x}(0)$ is formed as a linear combination of n modal components $e^{\lambda_j t}$, such that

$$x_i(t) = \sum_{j=1}^n m_{ij} e^{\lambda_j t} \quad (3.4.3)$$

Manipulation of the above equation through linear algebra yields the following result, in matrix form.

$$\mathbf{x}_h = [\mathbf{M}e^{\mathbf{M}^{-1}}]\mathbf{x}(0) \quad (3.4.4)$$

Where $\mathbf{M} = [\mathbf{m}_1, \mathbf{m}_2, \dots, \mathbf{m}_n]$ is an $n \times n$ matrix of eigenvectors (coefficients) and $\mathbf{M}^{-1} = \text{diag}[\lambda_1, \lambda_2, \dots, \lambda_n]$ is the diagonal matrix of eigenvalues. An interesting case is the one of *repeated eigenvalues*, where the modal matrix is non-singular (the eigenvectors are not linearly independent), and $\mathbf{A}$ is not diagonalizable but can acquire Jordan block form (thoroughly explained in Ch. 3.5 of [63]). For a pair of repeated eigenvalues, the matrix exponential will introduce pairs of components $e^{\lambda_i t}$ and $te^{\lambda_i t}$, a result which will be revisited shortly.

Input Response Considering now the actuated system, a first result is an impulse response from each input channel. In a mathematical sense, the impulse response is calculated using Eq. (3.4.2) with

$$\mathbf{u}(t) = \mathbf{K}\delta(t) = \begin{bmatrix} k_1 \\ k_2 \\ \vdots \\ k_m \end{bmatrix} \delta(t) \quad (3.4.5)$$

Where all components of $\mathbf{K}$ are 1, $\delta(t)$ is the *dirac delta* function, and by its sifting property the problem reduces to a simple response to initial conditions, governed by the same dynamics as Eq. (3.4.4).

$$\mathbf{x}(t) = (t)(\mathbf{x}(0) + \mathbf{B}\mathbf{K}) \quad (3.4.6)$$

The simulated state evolution presents in Fig. 3.4 as expected, with the fore foils affecting all states, while the aft (elevator) foil only affecting heave and pitch. Two very important remarks can be made about these results. Firstly, heave responses are *unbounded*, revealing system instability and the need for control. Secondly elevator to heave response graph presents with an important property, the *non-minimum phase* behavior of the longitudinal subsystem. This is a fundamental result because it characterizes systems

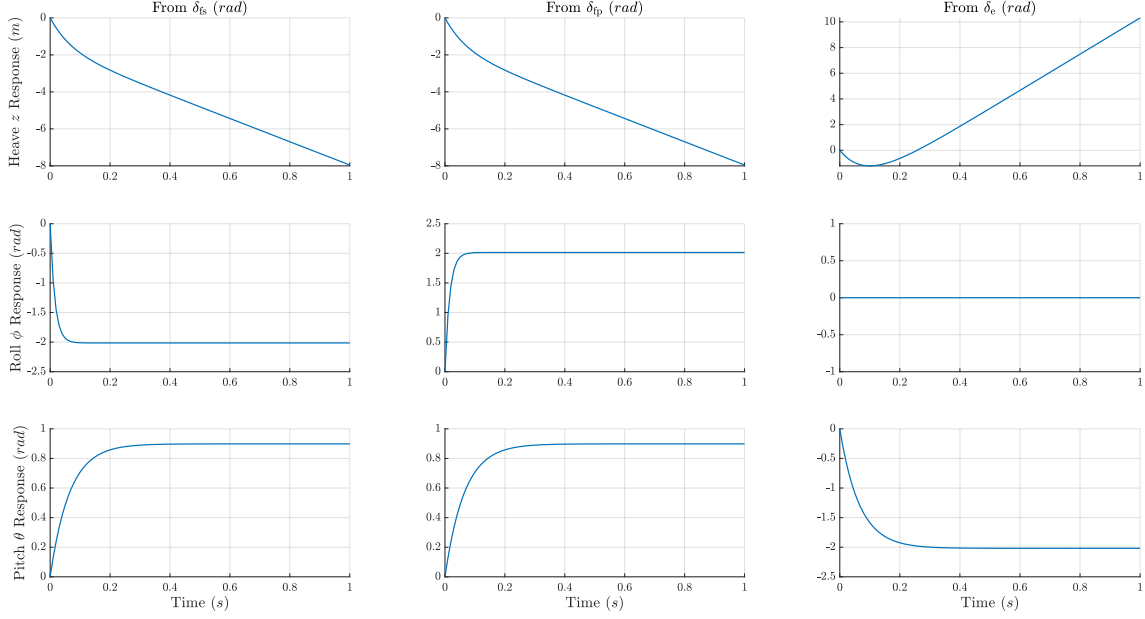


Figure 3.4: Response to impulse input from each input channel.

that tend to initiate by “moving the opposite way” in their responses, such as aircraft elevator to height dynamics.

At the same time, a *step response* has been calculated and plotted in Fig. 3.5 for a scaled system of 1° , meaning the unit step is transformed to represent a response to 0.01745 rad actuator deflection. Once again, non-minimum phase behavior from elevator to heave is apparent, while all *affected* outputs remain unbounded. What makes Figs. 3.4 and 3.5 extremely important, is that they represent the building blocks of the prediction dynamics that form the basis for a predictive controller. Specifically, at each operation point, current values act as initial conditions, and subsequent control commands utilize the step response.

Poles & Zeros As it has already been stated, results in Fig. 3.4 are linear combinations of modal components. Using the frequency domain transform of Eq. (3.2.5) (assuming zero initial conditions), we can define for MIMO systems, $\mathbf{G}(s)$, an $l \times m$ transfer function matrix (n states, but l outputs) where each entry $g_{ij}(s)$ is a polynomial fraction of s describing the sinusoidal response from input j to output i [11]. The numerically evaluated transfer function matrix for the previously derived model is found in Eq. (A.3.3).

$$\mathbf{G}(s) = \frac{\mathbf{Y}(s)}{\mathbf{U}(s)} = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D} \quad (3.4.7)$$

Poles are defined as the frequencies p_i where any of the $\mathbf{G}(p_i)$ entries have a singularity (in less technical terms, the system’s response contains pole frequencies, even if they are not contained in the inputs). Equation (3.4.7) explains how the eigenvalues of $\mathbf{A}$ are the poles of the transfer function matrix. In general, the poles of $\mathbf{G}$ are equal in both *location* and *multiplicity* to the reachable and observable eigenvalues of $\mathbf{A}$ [64].

Zeros for a SISO system are fairly straight-forward, defined as the generalized input frequencies z_i where the output of the system is zero, what we call the “blocking effect”.

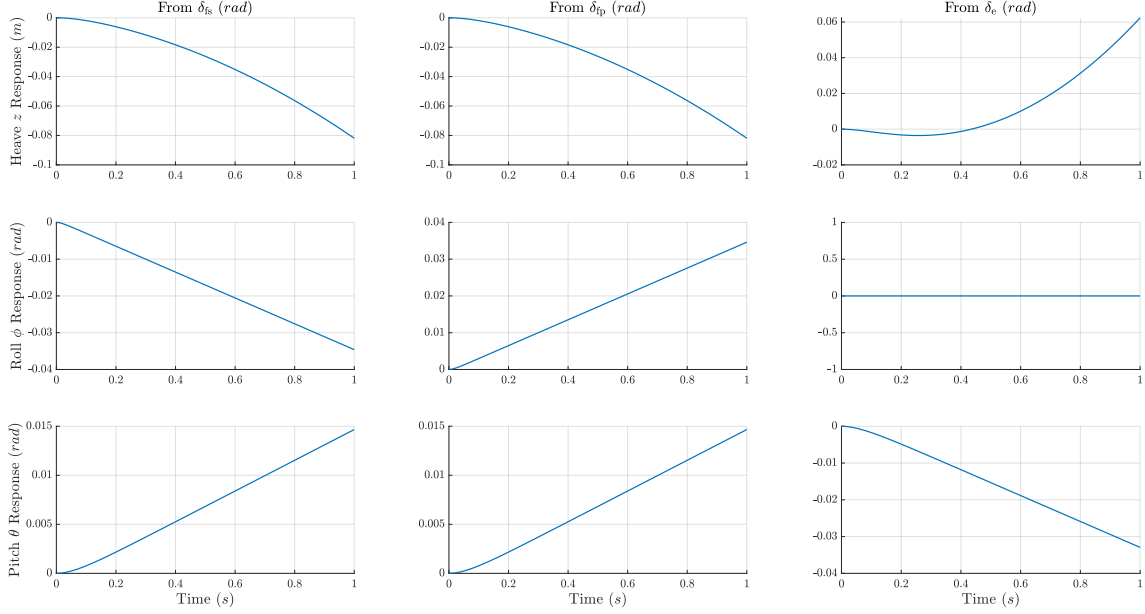


Figure 3.5: Response to step input from each input channel.

For MIMO systems, *transmission* zeros (characterizing the whole system) are defined as frequencies blocked from any output. Thus, z_i is a zero of the transfer matrix if $\text{rank}(\mathbf{G}(z_i)) < \text{rank}(\mathbf{G}(s))$ [11]. Finally, RHP zeros are responsible for non-minimum phase behavior.

For the system in Eqs. (A.3.1) and (A.3.2), the poles are

$$[-64.358, -15.489, -12.572, -1.615, 0, 0.196]$$

with 0 having multiplicity of 2. The slightly positive eigenvalue of the first model is due to the instability produced by speed perturbations, introducing a drift from the operating point. For the reduced model, not taking into account thrust and horizontal speed, Matlab yields

$$[-64.358, -15.489, -12.572, 0]$$

with 0 having a multiplicity of 3. While individual components of $\mathbf{G}(s)$ described by Eq. (A.3.3) present with zeros, no transmission zeros exist ($\mathbf{G}(s)$ does not lose rank for any frequency). These results are in strong agreement with existing literature such as [29], [45], peer reviewed studies by Kim & Yamato [23] and real-world tested models for America's Cup [27]. In these works, there are no transmission zeros, most poles appear to be over-damped on the LHP (no complex conjugate pairs, does not hold for [27] but will be discussed shortly) and height dynamics present with unstable RHP poles (or marginally stable zeros).

Stability Regarding internal stability, we shall again refer to the state-space LTI system of Eq. (3.2.5). Such a model is stable if and only if all of its poles lie in the open left-hand plane (LHP), since positive real parts will give rise to unbounded response. A more rigorous approach given by [63], studies two discrete cases:

- Marginal (internal) stability (or stable in the sense of Lyapunov) iff $\text{Re}\{\lambda\} \leq 0$, $\forall \lambda$, and $\lambda = 0$ appear as simple roots of the characteristic polynomial of $\mathbf{A}$.
- Asymptotic (internal) stability iff $\text{Re}\{\lambda\} < 0$, $\forall \lambda$, denoting $\mathbf{A}$ as “stable” or Hurwitz.

The first case will produce bounded outputs, while the second will always decay to zero as $t \rightarrow \infty$.

The previous theoretical remarks can now be used to effectively understand Fig. 3.4. While $\mathbf{G}(s)$ does not contain transmission zeros, $\mathbf{g}_{ij}$ components present with blocked frequencies. Specifically, it is evident from Eq. (A.3.3) that the elevator to heave I/O relationship contains RHP zeros, explaining the non-minimum phase plot. As it was previously discussed, most states present with eigenvalues being real negative or zero with multiplicity 1 (an integrator). Heave however, due to the s^2 term in the denominator, is always unstable. This raises a concern, regarding the validity of the model.

Most models found in literature are at least *marginally stable*, adhering to the definition given above. This deviation can either mean that the provided non-linear block diagram in Fig. 2.25 is invalid, or (which is a bolder assumption), that there is a problematic design variable in the physical system chosen. To avoid lengthy computations and digressing from the topics of interest, an attempt to address any misgivings is made in Appendix A.5. A brief summary is that attempting to move the selected center of gravity closer to the fore foils, introduces complex conjugate poles and produces the impulse response shown in Fig. A.5, showing a stable system.

Frequency Response How the system behaves over a range of frequencies is especially important when the disturbances “live” in the frequency domain as well (e.g, waves). This is also tied to Multiple Input Multiple Output (MIMO) directionality, since we can’t describe behavior by a single Bode plot as with SISO systems. These important findings are covered in depth by Skogestad [11], and will find use in controller analysis. Some important math facts are included in Appendix A.4, to keep the present text concise.

The main limitation lies in the fact that for a MIMO system $\mathbf{G}(s)$ given an input $\mathbf{d}(\omega)$ as a phasor, then the gain of the system given by Eq. (3.4.8) is dependent on the direction of $\mathbf{d}$ but not the actual magnitude.

$$\frac{\|\mathbf{y}(\omega)\|_2}{\|\mathbf{d}(\omega)\|_2} = \frac{\|\mathbf{G}(j\omega)\mathbf{d}(\omega)\|_2}{\|\mathbf{d}(\omega)\|_2} \quad (3.4.8)$$

The maximum gain for *any* input direction is called the *maximum singular value* of $\mathbf{G}$, while the minimum attainable gain is the *minimum singular value* with symbols $\bar{\sigma}$ and $\underline{\sigma}$ respectively.

$$\max_{\mathbf{d} \neq 0} \frac{\|\mathbf{G}\mathbf{d}\|_2}{\|\mathbf{d}\|_2} = \bar{\sigma}(\mathbf{G}) \quad (3.4.9)$$

$$\min_{\mathbf{d} \neq 0} \frac{\|\mathbf{G}\mathbf{d}\|_2}{\|\mathbf{d}\|_2} = \underline{\sigma}(\mathbf{G}) \quad (3.4.10)$$

These are calculated by the Singular Value Decomposition of $\mathbf{G}(s)$, at all frequencies $j\omega$ of interest such that $\mathbf{G}(j\omega) = \mathbf{U} \mathbf{V}^*$. To allow for better understanding, a plot of

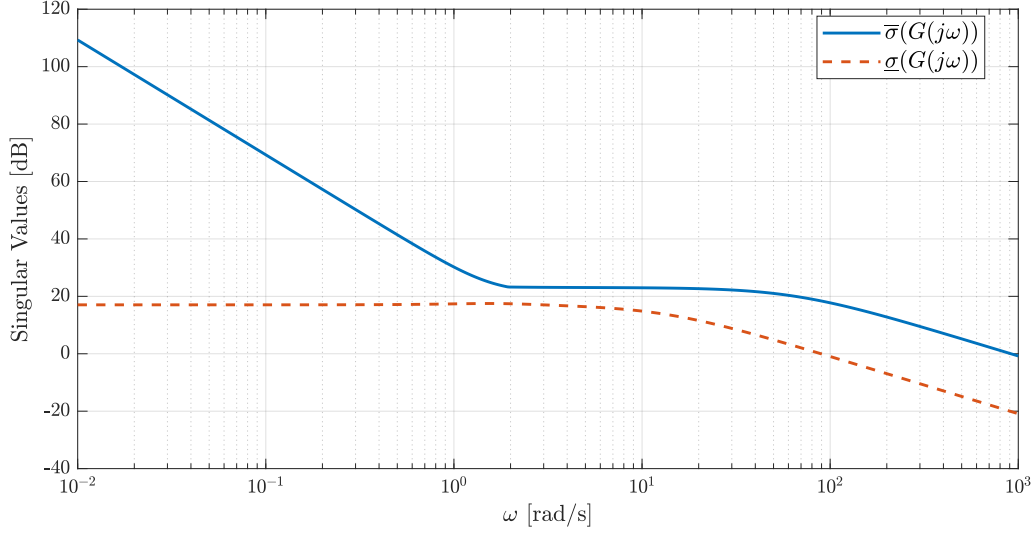


Figure 3.6: Maximum and minimum singular values of the model described by Eqs. (A.3.1) and (A.3.2).

the system's singular values is given in Fig. 3.6, and the two first components of the decomposition for $\omega = 2.5 \text{ rad} \cdot \text{s}^{-1}$ are supplied below.

$$\mathbf{G} = \underbrace{\begin{bmatrix} 0 & 0.367 - 0.058i \\ -0.006 - 0.158i & 0 \\ 0 & 0.004 - 0.144i \\ 0 & 0.169 + 0.037i \\ 0.987 - 0.039i & 0 \\ 0 & 0.901 + 0.024i \end{bmatrix}}_{\mathbf{U}} \underbrace{\begin{bmatrix} 14.413 & 0 \\ 0 & 12.935 \end{bmatrix}}_{\mathbf{\Sigma}} \underbrace{\begin{bmatrix} -0.707 & 0.317 \\ 0.707 & 0.317 \\ 0.000 & -0.867 - 0.218i \end{bmatrix}}_{\mathbf{V}^*}$$

Looking at the matrices, which are not the full decomposition but the first two columns and respective singular values, we notice that the maximum gain occurs at the *lateral* dynamics when the fore foils move at opposing rates. The second singular value is a pitch-up maneuver, as it also appears to excite heave and pitch rate dynamics. For $\omega \rightarrow 0$, $\bar{\sigma}(\mathbf{G})$ corresponds to heave excitation.

State Controllability A dynamical system is said to be state controllable if for any initial state $\mathbf{x}(0) = \mathbf{x}_0$ and any final state $\mathbf{x}_1$ there exists an input $\mathbf{u}(t)$ that transfers $\mathbf{x}_0$ to $\mathbf{x}_1$ in finite time. By defining the controllability matrix

$$\mathcal{C} = [\mathbf{B} \quad \mathbf{A}\mathbf{B} \quad \mathbf{A}^2\mathbf{B} \quad \dots \quad \mathbf{A}^{n-1}\mathbf{B}] \in \mathbb{R}^{n \times nm} \quad (3.4.11)$$

we derive a simple test for this property, specifically that $\mathcal{C}$ needs to have full row rank, or $\text{rank}(\mathcal{C}) = n$.

There's been a lot of criticism towards the concept of controllability as a *realistic* measure of any real system characteristics regarding control implementations. This is mostly due to the theoretical nature of the concept, since it makes no assumptions regarding state

trajectory, input size, constraints or even ability to hold states at x_1 as $t \rightarrow \infty$ [11]. For the studied system however, the corresponding matrix does have full row rank.

State Observability This concept is in a way dual to the previous. Roughly, it describes the possibility of estimating an unmeasured state from the output (not to be confused with *identification*, where the state matrices are unknown). Formally, a system is state observable if for any unknown initial state $\mathbf{x}(0)$, the knowledge of input u and output y time history, suffices to determine that initial state in finite time. The analogous test is the respective observability matrix having full column rank.

$$\mathcal{O} = \begin{bmatrix} \mathbf{C} \\ \mathbf{CA} \\ \mathbf{CA}^2 \\ \vdots \\ \mathbf{CA}^{n-1} \end{bmatrix} \in \mathbb{R}^{nl \times n} \quad (3.4.12)$$

It follows, that for observability to be of any concern, the measured states (output) are less than the system states ($l < n$). This will be important when designing state observers to adjust for measurement noise. As before, the studied system is observable.

Chapter 4

Control System Design

This chapter outlines the theoretical foundations relevant to the proposed control strategies, which draw upon both classical frequency-domain approaches and modern optimization-based formulations. We begin with frequency-domain analysis, examining characteristics of the system transfer functions which, along with the Singular Value Decomposition will lead to frequency domain analysis for MIMO systems that are characterized by directionality. Time-domain methods are then considered. Performance metrics will be set to benchmark the designed controllers. The notion of a performance index, formalized through quadratic cost functions and backed by optimization, leads to optimal control in Section 4.2. Both continuous and discrete variants of the Linear Quadratic Regulator are discussed, with practical considerations to aid performance in real-world settings. Constrained optimal control is introduced in Section 4.3, which naturally leads to Receding Horizon Control, or MPC in Section 4.4. Again, with a practical implementation mindset, MPC is extended for reference previewing and adaptive schemes. Finally, Section 4.5 addresses state estimation under measurement noise. Sensor noise models for simulation are developed, based on real device data. Estimator is designed as a discrete Kalman Filter, providing the foundation for output-feedback control and observer-based designs.

4.1 Problem Formulation

4.1.1 Objectives

When designing a control system, the engineer is typically concerned with specific performance requirements, known as control objectives. These include:

- **Stability:** The system ought to be stabilized by the controller at all times.
- **Regulation:** By manipulating the input u , reject disturbances d .
- **Tracking:** Keep the output close to the reference r by use of input u , also called the servo problem.
- **Robustness:** The above should hold within a margin of modeling uncertainty.

In a more modern approach, an additional objective is explicit constraint handling. Often employed in practice, is the simple clipping of the commanded input, which will be analyzed later on, along with more advanced designs.

4.1.2 Frequency Domain

Figure 4.1 provides a high-level view of the control problem in the frequency domain. A controller $\mathbf{K}$, driven by the reference $\mathbf{r}$ creates a control signal $\mathbf{u}$. The signal $\mathbf{u}$ plus unknown process disturbances $\mathbf{w}$ enter the plant $\mathbf{G}$ which has real output $\mathbf{y}$ and added measurement noise $\mathbf{v}$. The measured output at the sensors is $\mathbf{y}_m$. At this point we can define important quantities as shown in Eq. (4.1.1), derive their physical meaning and identify desired behaviors in the frequency domain.

$$\mathbf{y} = (\mathbf{I} + \mathbf{G}\mathbf{K})^{-1} \underbrace{\mathbf{G}\mathbf{K}}^{\mathbf{L}} \mathbf{r} + \underbrace{(\mathbf{I} + \mathbf{G}\mathbf{K})^{-1} \mathbf{G}}_{\mathbf{S}} \mathbf{w} - \underbrace{(\mathbf{I} + \mathbf{G}\mathbf{K})^{-1} \mathbf{G}\mathbf{K}}_{\mathbf{T}} \mathbf{v} \quad (4.1.1)$$

- $\mathbf{L}$: loop transfer function.
- $\mathbf{S}$: sensitivity function.
- $\mathbf{T}$: complementary sensitivity, or closed loop transfer function.

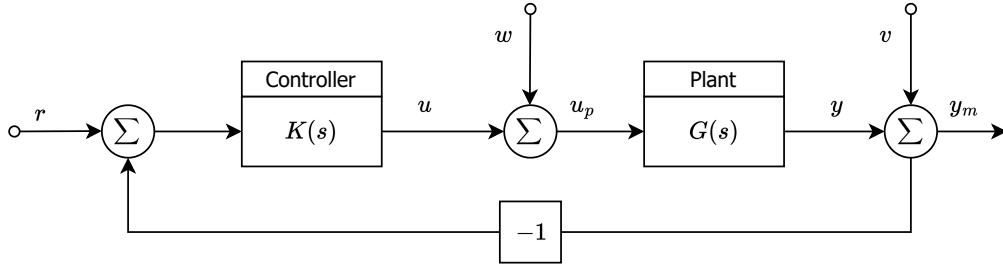


Figure 4.1: Typical controller-plant block diagram

It follows that $\mathbf{S} + \mathbf{T} = 1$ at all frequencies. With Eq. (4.1.2) giving the transfer function for the *real* output (meaning, not polluted by noise), the corresponding function for the error is given in Eq. (4.1.2).

$$\boldsymbol{\epsilon} = \mathbf{r} - \mathbf{y} = \mathbf{S} \mathbf{r} - \mathbf{S} \mathbf{G} \mathbf{w} + \mathbf{T} \mathbf{v} \quad (4.1.2)$$

By Eqs. (4.1.1) and (4.1.2), the mentioned objectives are tied to the transfer function characteristics over the frequency domain, through seemingly conflicting requirements:

- Tracking relates to large $\mathbf{T}$ and $\mathbf{L}$, and small $\mathbf{S}$.
- Disturbance attenuation relates to small $\mathbf{S}$.
- Noise rejection relates to small $\mathbf{T}$ and $\mathbf{L}$.

Note that these transfer functions depend on the definition of $\mathbf{L}$ when the system is multi-dimensional. Introducing notation such as $\mathbf{L}_o = \mathbf{G}\mathbf{K}$, $\mathbf{L}_i = \mathbf{K}\mathbf{G}$, 6 such transfer functions are defined, marked as either input or output counterparts. Following analysis made by Hespanha [16, Ch. 23] and earlier results by M. Athans, R. Kalman etc. we shall be using the input counterparts, breaking the loop at the inputs.

Bandwidth An especially important quantity, a system's bandwidth is defined as the frequency range over which the gain has decreased no more than a factor of $1/\sqrt{2} \approx 0.707$, or -3 dB [65, Ch. 6.3]. Now, for a MIMO dynamical system, this definition is lacking. Instead, we refer to bandwidth as a frequency range over which control is *effective*, and the interpretation of effectiveness regarding feedback is given as $|\epsilon/\mathbf{r}| = |\mathbf{S}|$ by Eq. (4.1.2), kept relatively small at $|\mathbf{S}| \leq 1/\sqrt{2}$. In other words, the closed loop bandwidth ω_B is defined as the frequency where $20 \log(|\mathbf{S}(j\omega)|)$ first crosses the -3 dB mark from below [11, Ch. 2.4]. Another frequency related quantity, the *crossover frequency* ω_c is defined as where $|\mathbf{L}(j\omega)|$ first crosses 1 (or 0 in dB scale) from above. Both frequencies are shown in Fig. 4.3.

Since disturbance and reference signals are usually at lower frequencies and noise at higher, the approach is unified over the frequency domain, yielding what is often described as *loop shaping*: large loop gain $|\mathbf{L}(j\omega)| > 1$ for $\omega < \omega_c$ (the crossover frequency) and the opposite for larger frequencies. Figures 4.2 and 4.3 provide qualitative insight on what a tuned controller looks like in the frequency domain. In the case of multi dimensional systems, singular values as they were introduced in Eqs. (3.4.9) and (3.4.10) give upper and lower bounds for gain in all output directions.

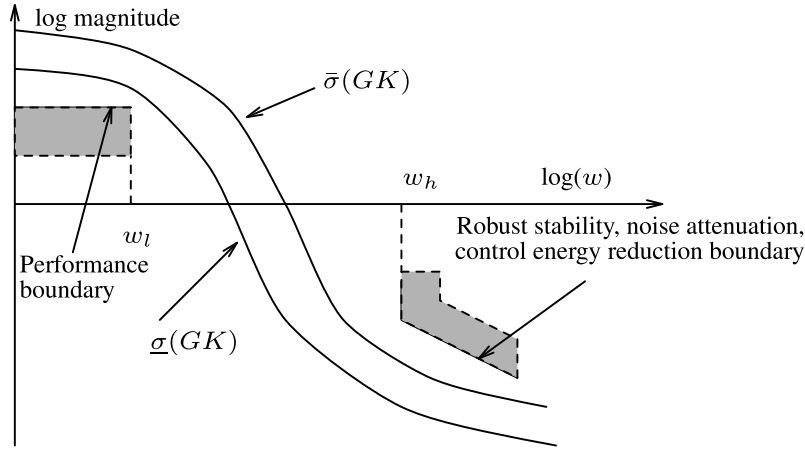


Figure 4.2: Qualitative frequency response of multivariable loop transfer function $\mathbf{L} = \mathbf{G}\mathbf{K}$ by its minimum and maximum singular values (Reproduced from [11]).

4.1.3 Time Domain

To judge the capabilities of a selected control policy, a performance measure is usually introduced. This mathematical concept spans across multiple disciplines as it is used to form an optimization problem. A simple form for a performance index is the Integral of

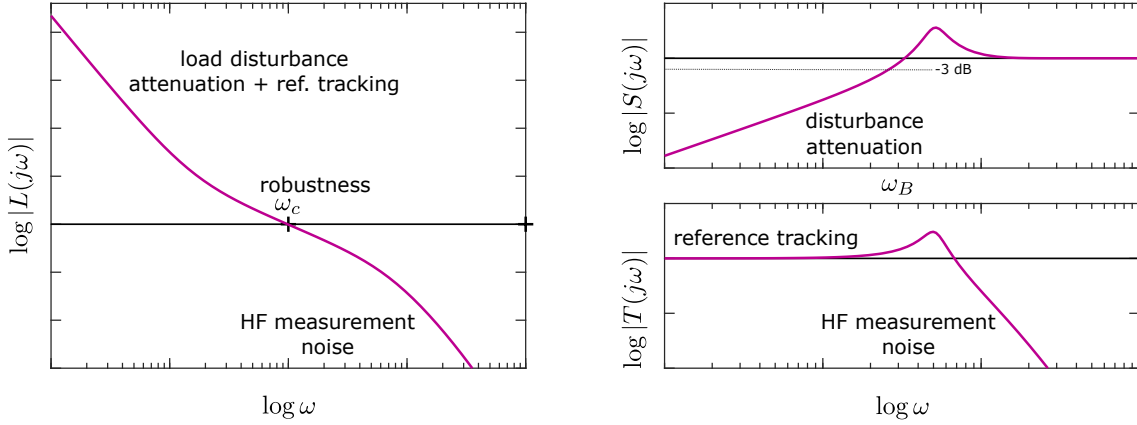


Figure 4.3: Transfer functions L , S , T for a SISO system showing results of feedback and objective regions. (Adapted from [65]).

the Square of the Error (ISE), as shown below.

$$\text{ISE} = \int_0^{\infty} (y(t) - r(t))^2 dt \quad (4.1.3)$$

There other available metrics, such as the Integral of the Absolute Error (IAE) or the Integral of the Time weighted Absolute Error (ITAE), all of which will be addressed in Chapter 5. For use in a multi-objective setting (like MIMO reference tracking), given a vector of measurements $\mathbf{y}$ and one of references $\mathbf{r}$, the error vector is $\mathbf{e}$. Also, suppose that the integral is calculated within a simulation timeframe, $[t_0, t_f]$. A diagonal weighting matrix $\mathbf{Q}$ can be used to weight errors of differing magnitudes.

$$\text{ISE} = \int_{t_0}^{t_f} [\mathbf{y} - \mathbf{r}]^T \mathbf{Q} [\mathbf{y} - \mathbf{r}] dt = \int_{t_0}^{t_f} \|\mathbf{e}\|_{\mathbf{Q}}^2 dt \quad (4.1.4)$$

The expression in Eq. (4.1.4) will be used to compare the resulting controllers in the simulation scenarios. Making a connection to the previous discussion, a high bandwidth will mean that the controller allows for fast actuator reaction (larger high0frequency content), while a lower one results in a “slower” control law that tends to be more robust. This connects the frequency domain results to expected behavior in the time domain, and vice versa.

4.2 Optimal Control

The previous reasoning regarding performance measures will now be used as an objective function to create an optimal control law for a linear time invariant system.

4.2.1 Continuous-Time LQR

By choosing a *scalar* performance index J as in Eq. (4.2.1), we are able to calculate the feedback control law to optimize against that objective function. In the continuous case,

this yields the continuous linear regulator problem [15]. The idea is to regulate the system around the origin, free of input/state constraints over an infinite horizon.

$$J = \int_0^{\infty} \mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{u}^T \mathbf{R} \mathbf{u} dt \quad (4.2.1)$$

The appropriate control sequence is calculated as the solution to the optimization problem¹

$$\begin{aligned} \mathbf{u}(t) = \arg \min_{\mathbf{u}} \quad & J(\mathbf{x}, \mathbf{u}, t) \\ \text{subj. to} \quad & \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \end{aligned} \quad (4.2.2)$$

The solution to such problems is based on Bellman's principle of optimality, and some form of a Hamilton-Jacobi-Bellman equation. Qualitatively, it is similar to introducing a lagrange multiplier $\boldsymbol{\lambda}$ which augments the cost function by $\boldsymbol{\lambda}^T (\mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} - \dot{\mathbf{x}})$ and solving an unconstrained minimization of J (see [66, Ch. 8.4]). This leads to a matrix differential equation which when solved produces a time varying feedback matrix $\mathbf{P}(t)$:

$$\dot{\mathbf{P}} + \mathbf{P}\mathbf{A} + \mathbf{A}^T \mathbf{P} - \mathbf{P}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^T \mathbf{P} + \mathbf{Q} = \mathbf{0} \quad (4.2.3)$$

In the special case of an infinite horizon however, the matrix $\mathbf{P}$ is constant, which drops the derivative and since Eq. (4.2.3) is true for all $\mathbf{x}$ it reduces to the *Algebraic Riccati Equation* (ARE).

$$\mathbf{P}\mathbf{A} + \mathbf{A}^T \mathbf{P} - \mathbf{P}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^T \mathbf{P} + \mathbf{Q} = \mathbf{0} \quad (4.2.4)$$

The resulting linear feedback law is:

$$\mathbf{u}(t) = -\underbrace{\mathbf{R}^{-1}\mathbf{B}^T \mathbf{P}}_{\mathbf{K}} \mathbf{x}(t) \quad (4.2.5)$$

The steps outlined in Eqs. (4.2.3) and (4.2.4) are also similarly described in [15, Sec. 3.12].

Frequency Domain Properties For the Linear Quadratic Regulator (LQR) scheme, stability margins have been extensively studied and proven (specifically in papers presented by Kalman or Safonov and Athans). Throughout the following sections, the loop transfer function which will be considered is $\mathbf{L}_i = \mathbf{K}\mathbf{G} = \mathbf{K}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}$ in accordance to the authors mentioned before as well as [16, Lect. 23] and [11, Ch. 9.2]. By this selection, analyzing the singular values of $\mathbf{S}$ and $\mathbf{T}$ functions provides insight on reference and disturbance to control signal relationships, effectively breaking the loop between controller and plant. It follows, that the same approach shall be followed when dealing with Model Predictive Control (MPC) frequency domain analysis (since within the linear region, MPC returns a similar control law).

Reference Input

The first addition to the previously defined feedback law, is the ability to track reference signals. For the studied system, regulation around 0 is not enough, since the vessel needs

¹The full Optimal Control Problem usually incorporates a terminal cost $\mathbf{x}(t_f)^T \mathbf{H} \mathbf{x}(t_f)$ and some cross-term $\mathbf{x}^T \mathbf{N} \mathbf{u}$, but are ignored since the final state is 0 and no input-output cross terms are penalized.

to track height signals either for obstacle avoidance (for example, waves about to hit the bow) or in order to perform pre-defined maneuvers like take-off and landing. This can be approximated by a change of origin [14, Ch. 7.5] (what follows is a MIMO generalization of the referenced works), meaning that the reference signal changes the origin around which the controller regulates. Equation (4.2.6) is the expression for the produced control signal. This is merely a *technique* to solve the servo problem, not necessarily equivalent to an optimally derived solution of a time-varying feedback law as described in [15].

$$\mathbf{u} = \mathbf{u}_{ss} - \mathbf{K}(\mathbf{x} - \mathbf{x}_{ss}) \quad (4.2.6)$$

Assuming an attainable reference input (as was shown in Fig. 1.7, this is to be decided by a different algorithm), steady state tracking requires all state derivatives being 0, and outputs coinciding with the reference. For a linear system, the steady state solution is expected to be of the form $\mathbf{x}_{ss} = \mathbf{N}_x \mathbf{r}_{ss}$ and $\mathbf{u}_{ss} = \mathbf{N}_u \mathbf{r}_{ss}$, and the system state equations are solved for $\dot{\mathbf{x}} = \mathbf{0}$, $\mathbf{y} = \mathbf{r}$ resulting in the block matrix form shown below.

$$\begin{bmatrix} \mathbf{N}_x \\ \mathbf{N}_u \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{0}_{n \times p} \\ \mathbf{I}_{p \times p} \end{bmatrix} \quad (4.2.7)$$

The inverse notation is not exactly correct, since by design the matrix to be inverted has dimensions $(n + p) \times (n + m)$ making the linear system either under or over determined. In the case of full state feedback where $p > m$, the system of equations is over determined and it may have a solution in a least squares sense. For a “tall” coefficient matrix $\mathbf{M}$ with linearly independent columns the system of equations has a solution $\mathbf{x} = \mathbf{M}^\dagger \mathbf{b}$ with $\mathbf{M}^\dagger$ denoting the *pseudo-inverse* or *Moore-Penrose inverse* of $\mathbf{M}$ [42, Ch. 11.5]. Another approach is to only use the position outputs (z, ϕ, θ) and ignore the body fixed rates, resulting into a square matrix. The second approach makes more sense in this context, since only references in positions are to be tracked while rates are regulated.

$$\mathbf{u} = \mathbf{N}_u \mathbf{r} - \mathbf{K}(\mathbf{x} - \mathbf{N}_x \mathbf{r}) = -\mathbf{K}\mathbf{x} + (\mathbf{N}_u + \mathbf{K}\mathbf{N}_x)\mathbf{r} \quad (4.2.8)$$

This can be simplified further, because tracking commands on θ are not that common, and model mismatch corrections introduced later will account for simplification errors. This makes $\mathbf{N}_u = \mathbf{0}$ and $\mathbf{N}_x = \mathbf{I}$, thus Eq. (4.2.8) reduces to Eq. (4.2.9), where $\mathbf{u}_o$ as a linearization offset is added.

$$\mathbf{u} = \mathbf{u}_o - \mathbf{K}(\mathbf{x} - \mathbf{x}_{ref}) \quad (4.2.9)$$

The concepts discussed up to this point are a form of *feed-forward* control, since the internal model is used to invert the system dynamics and calculate control actions before feedback. Tracking heave however, allows for simplifications since steady state control values remain unchanged regardless of ride height. In a maneuvering scenario, feed-forward might be beneficial.

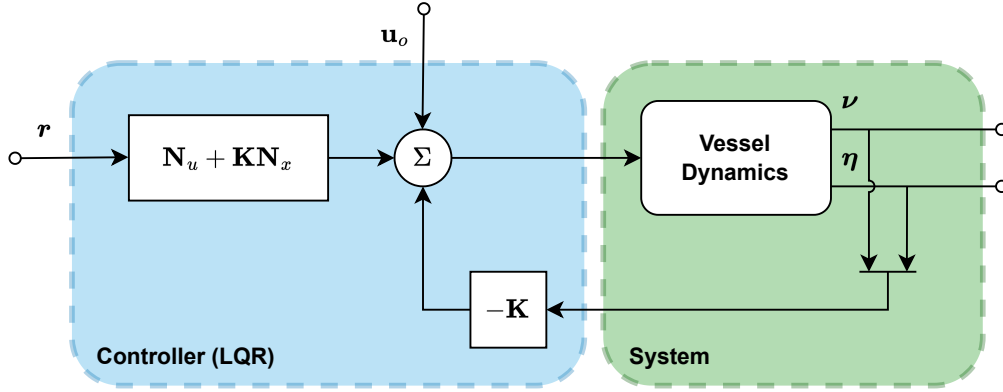


Figure 4.4: Block diagram form of tracking LQR controller.

Offset-free Tracking

While the changes made above would perfectly track a constant setpoint, they are not robust to model errors. To account for model-mismatch, the system can be augmented by integral states, providing larger margins in the lower frequency region. Once again only the position states are used for tracking (a similar approach to [27] and [11, Ch. 9.2]), and the augmented state takes the form of tracking error integral $e = \int x - r dt$. As long as $e \neq 0$, actuator output will increase. The position states are extracted using a selection matrix $\mathbf{C}_I$ of block form $[\mathbf{I}_{p \times p} \mathbf{0}_{p \times (n-p)}]$. Another approach, usually coupled with the state-space representation, is the disturbance observer which will be discussed within the MPC section.

$$\dot{\mathbf{x}}_I = \mathbf{C}_I \mathbf{x} - \mathbf{r} \Rightarrow \mathbf{x}_I = \int \mathbf{e}_r(\tau) dt. \quad (4.2.10)$$

The augmented system is defined by its state $\bar{\mathbf{A}}$ and input $\bar{\mathbf{B}}$ matrices as shown in Eq. (4.2.11). Controller gains for this system are calculated identically to the approach introduced in Section 4.2.1 and give a linear full state feedback for $n + p$ states. In literature it is often found as the Linear Quadratic Integral (LQI).

$$\begin{bmatrix} \dot{\mathbf{x}} \\ \dot{\mathbf{x}}_I \end{bmatrix} = \underbrace{\begin{bmatrix} \mathbf{A} & \mathbf{0}_{n \times p} \\ \mathbf{C}_I & \mathbf{0}_{p \times p} \end{bmatrix}}_{\bar{\mathbf{A}}} \begin{bmatrix} \mathbf{x} \\ \mathbf{x}_I \end{bmatrix} + \underbrace{\begin{bmatrix} \mathbf{B} \\ \mathbf{0}_{p \times m} \end{bmatrix}}_{\bar{\mathbf{B}}} \mathbf{u} - \begin{bmatrix} \mathbf{0}_{p \times p} \\ \mathbf{I}_{p \times p} \end{bmatrix} \mathbf{r} \quad (4.2.11)$$

Notice how the proposed scheme (also seen on the block diagram of Fig. 4.5) is structurally similar to a PID applied for multiple inputs and outputs. In fact, the gain $\mathbf{K}$ can be decomposed into separate gain matrices $\mathbf{K}_P$, $\mathbf{K}_I$, $\mathbf{K}_D$ and it was treated as such in earlier literature.

Practical Considerations

Everything described above will work as long as the actuators are not *saturated*. The proposed controller however, will produce signals without any knowledge of the system constraints.

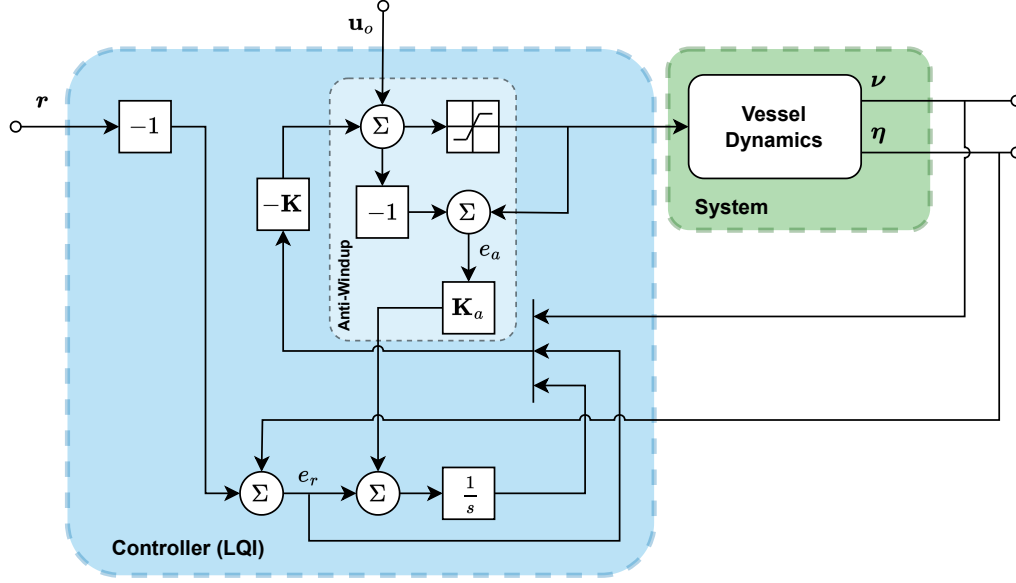


Figure 4.5: Block diagram form of LQI controller with saturation and anti-windup.

Input constraints Physical systems cannot apply arbitrary inputs, thus a common workaround is to simply “clip” the output between a lower and an upper bound causing the real system input to differ from the controller output ($\mathbf{u} \neq \mathbf{u}_c$). It is easily understood that in saturation regions, the controller is neither linear nor optimal.

Anti-windup While the constraints are imposed, the internal controller states (integrators) will continue to accumulate the error signal (windup). This will introduce a lagged response if the system goes out of the constraint region and into normal operation. To avoid this, an extra *anti-windup* loop as shown in Fig. 4.5 is designed based on guidelines presented in [14], [65]. Equation (4.2.12) shows how this is mathematically implemented ($\mathbf{u}$ is the clipped control signal). The quantity $\mathbf{u} - \mathbf{u}_c$ is also described as the actuator error $\mathbf{e}_a$. The gain $\mathbf{K}_a$ is usually selected through trial and error, and is closely related to input-output directionality of the plant. Namely, it is of $p \times m$ dimension and tunes how much each actuator error contributes to the back-calculation term (e.g, the aft flap actuator has no effect on roll correction).

$$\mathbf{u} = -\mathbf{K}_P \boldsymbol{\eta} - \mathbf{K}_D \boldsymbol{\nu} - \mathbf{K}_I \int \mathbf{e}_I + \mathbf{K}_a (\mathbf{u} - \mathbf{u}_c) dt \quad (4.2.12)$$

The largest practical consideration however remains: A digital implementation. No control law can run in a continuous manner if it is to be implemented in a digital computer (although for argument’s sake, it is possible to use analog circuits to achieve state feedback). This brings us to the following section.

4.2.2 Discrete-Time LQR

The decision to design a discrete controller for this setup, is based firstly on the eventual digital implementation and secondly because it would be counter-intuitive to compare the continuous LQR against MPC which is discrete by design and usually running at a rather slow frequency due to computational demands.

For a sampling time T_s , the continuous system is transformed to its discrete counterpart according to Eq. (4.2.13) [63].

$$\begin{aligned} \mathbf{x}[k+1] &= \mathbf{A}_d \mathbf{x}[k] + \mathbf{B}_d \mathbf{u}[k] \\ \mathbf{y}[k] &= \mathbf{C}_d \mathbf{x}[k] + \mathbf{D}_d \mathbf{u}[k] \end{aligned} \quad \text{with} \quad \begin{aligned} \mathbf{A}_d &= e^{\mathbf{A}T_s} & \mathbf{B}_d &= \left(\int_0^{T_s} e^{\mathbf{A}\tau} d\tau \right) \mathbf{B} \\ \mathbf{C}_d &= \mathbf{C} & \mathbf{D}_d &= \mathbf{D} \end{aligned} \quad (4.2.13)$$

With a relation between states and input given by Eq. (4.2.14).

$$\mathbf{x}[k] = \mathbf{A}_d^k \mathbf{x}[0] + \sum_{j=0}^{k-1} \mathbf{A}_d^j \mathbf{B}_d \mathbf{u}[k-1-j] \quad (4.2.14)$$

The optimization problem needed to be solved for calculation of the gain matrix, takes the form of Eq. (4.2.15) (brackets are replaced by subscripts, and state matrices shall refer to the *discrete* case from hereon).

$$\begin{aligned} \mathbf{u}_k &= \arg \min_{\mathbf{u}_0, \mathbf{u}_1, \dots} \sum_{k=0}^{\infty} \mathbf{x}_k^\top \mathbf{Q} \mathbf{x}_k + \mathbf{u}_k^\top \mathbf{R} \mathbf{u}_k \\ \text{subj. to } & \mathbf{x}_{k+1} = \mathbf{A} \mathbf{x}_k + \mathbf{B} \mathbf{u}_k, \quad k = 0, 1, \dots \end{aligned} \quad (4.2.15)$$

The solution is, as before, linear state feedback with the gain matrix calculated by Eq. (4.2.16) [12], where $\mathbf{P}$ is the solution to the discrete Algebraic Riccati Equation (DARE) defined in Eq. (4.2.17).

$$\mathbf{u}_k = -(\mathbf{B}^\top \mathbf{P} \mathbf{B} + \mathbf{R})^{-1} \mathbf{B}^\top \mathbf{P} \mathbf{A} \mathbf{x}_k \quad (4.2.16)$$

$$\mathbf{P} = \mathbf{A}^\top \mathbf{P} \mathbf{A} + \mathbf{Q} - \mathbf{A}^\top \mathbf{P} \mathbf{B} \left(\mathbf{B}^\top \mathbf{P} \mathbf{B} + \mathbf{R} \right)^{-1} \mathbf{B}^\top \mathbf{P} \mathbf{A} \quad (4.2.17)$$

The approach for tracking, augmentation (integral states), and constraints is identical to the continuous counterpart and follows the exact block diagram of Fig. 4.5.

Implementation

To aid in understanding and reproducibility, Algorithm 1 presents a pseudocode version implementable on a digital microcontroller.

Algorithm 1: Discrete implementation of LQI with saturation & anti-windup.

Input: Reference $\mathbf{r}$, State $\mathbf{x}$
Output: Actuator command $\mathbf{u}$
Parameters: T_s , $\mathbf{u}_{\text{trim}}$, $\mathbf{u}_{\text{min}}$, $\mathbf{u}_{\text{max}}$, $\mathbf{K}$, $\mathbf{K}_a$
Persistent : $\mathbf{x}_I$, $\mathbf{u}$, $\mathbf{u}_c$

```

1 if first call then
2   |  $\mathbf{x}_I$ ,  $\mathbf{u}$ ,  $\mathbf{u}_c \leftarrow \mathbf{0}$ 
3 end
4  $\mathbf{e}_r \leftarrow \mathbf{C}_I \mathbf{x} - \mathbf{r}$  // Tracking error
5  $\mathbf{x}_I \leftarrow \mathbf{x}_I + T_s (\mathbf{e}_r + \mathbf{K}_a (\mathbf{u} - \mathbf{u}_c))$  // Anti-windup
6  $\mathbf{x}_{\text{aug}} \leftarrow [\mathbf{x}, \mathbf{x}_I]^\top$ 
7  $\mathbf{u}_c \leftarrow \mathbf{u}_{\text{trim}} - \mathbf{K} \mathbf{x}_{\text{aug}}$ 
8  $\mathbf{u} \leftarrow \min(\max(\mathbf{u}_c, \mathbf{u}_{\text{min}}), \mathbf{u}_{\text{max}})$  // Saturation
9 return  $\mathbf{u}$  // Zero-order hold output

```

4.2.3 Designs & Tuning Parameters

The previous sections provide theory and implementation for the selected design, but sometimes it might be desired to test specific features on their own. For this reason, a naming convention is used (which also pertains to the MPC designs seen later) as shown in Table 4.1. By this convention, an LQR based controller that employs integral action, saturation limits and anti-windup will be labeled as LQI-SW.

Name	Functionality	-S	-W
LQR	Base controller (state feedback).	Input saturation to enforce actuator limits.	Anti-windup strategy for integrator during saturation events.
LQI	Added integral action. Augmentation on the tracked states.		

Table 4.1: Controller naming legend for LQR family.

The algorithm parameters give the designer a way to shape controller behavior from a high-level perspective (can be thought of as knobs that tune performance). These are specifically the gain matrices for state and anti-windup, sampling time, and actuator limits. The last two do not affect the controller under normal operation much, so we focus on the gains $\mathbf{K}$, $\mathbf{K}_a$. Since the state feedback matrix arises on the design stage, the final parameters taken into account are collected and explained in Table 4.2.

Parameter	Type	Use
$\mathbf{Q}$	Matrix, $n \times n$	State penalty matrix, typically diagonal. Bryson's rule to scale weights appropriately.
$\mathbf{R}$	Matrix, $m \times m$	Control input penalty, symmetric and positive definite.
$\mathbf{K}_a$	Matrix, $p \times m$	Anti-windup gain

Table 4.2: Tuning parameters for LQR design.

4.3 Constrained Optimal Control

It might already be evident, that the design of practical controllers often necessitates violations of optimality through heuristics or ad hoc modifications. Nonetheless, the preceding sections have highlighted two key insights that guide the subsequent analysis. First, as demonstrated in Section 4.2.1, incorporating a system model can greatly enhance performance. Second, considerable attention has been devoted in the literature to the problem of regulating systems subject to arbitrary reference trajectories and state/input constraints *within* the optimal framework. The following sections present different approaches to those problems, modeled in state space form building up to MPC. This is due to formulations adopted by Borelli et al [12], presentation slides by Bemporad [7] and the MPC Toolbox User's Guide [67]. Thus, there is a slight deviation from the transfer function standard used mostly in the process control industry (e.g DMC) found in early chapters from Maciejowski [13].

4.3.1 Constrained Finite Time Optimal Control

Posing as the backbone of MPC is the Constrained Finite Time Optimal Control (CFTOC) [12, Ch. 11], presented in Eq. (4.3.1) solved repeatedly at each sampling instant. It incorporates system dynamics, input, and state constraints while optimizing a predefined cost function over a finite prediction horizon formulated as a Quadratic Program (QP) [68, Ch. 16]. The system which we attempt to stabilize retains the discrete LTI form introduced by Eq. (4.2.13), which makes this a form of *Linear MPC*.

$$\begin{aligned}
 \mathbf{U}^* &= \arg \min_{\mathbf{U}} \quad \mathbf{x}_N^\top \mathbf{P} \mathbf{x}_N + \sum_{k=0}^{N-1} \mathbf{x}_k^\top \mathbf{Q} \mathbf{x}_k + \mathbf{u}_k^\top \mathbf{R} \mathbf{u}_k \\
 \text{s.t.} \quad &\mathbf{x}_{k+1} = \mathbf{A} \mathbf{x}_k + \mathbf{B} \mathbf{u}_k, \quad k = 0, \dots, N-1 \\
 &\mathbf{x}_k \in \mathcal{X}, \quad \mathbf{u}_k \in \mathcal{U}, \quad k = 0, \dots, N-1 \\
 &\mathbf{x}_N \in \mathcal{X}_f \\
 &\mathbf{x}_0 = \mathbf{x}(0)
 \end{aligned} \tag{4.3.1}$$

Where $\mathbf{Q} = \mathbf{Q}^\top \succeq 0$, $\mathbf{P} = \mathbf{P}^\top \succ 0$, $\mathbf{R} = \mathbf{R}^\top \succeq 0$. Constraint sets are represented by polyhedral regions, specifically $\mathcal{X}$, $\mathcal{X}_f$ for state and terminal state constraints and $\mathcal{U}$ for the set of input constraints.

4.3.2 Quadratic Program Solution

In the context of convex optimization, a quadratic program is defined such that the objective function is convex quadratic and the constraint functions are affine [42], [68].

$$\begin{aligned} \min \quad & \frac{1}{2} \mathbf{z}^\top \mathbf{H} \mathbf{z} + \mathbf{q}^\top \mathbf{z} + \mathbf{r} \\ \text{subj. to} \quad & \mathbf{G} \mathbf{z} \preceq \mathbf{w} \\ & \mathbf{A} \mathbf{z} = \mathbf{b} \end{aligned} \quad (4.3.2)$$

Where $\mathbf{P} \in \mathbb{S}_+^n$ ², $\mathbf{G} \in \mathbb{R}^{m \times n}$, $\mathbf{A} \in \mathbb{R}^{p \times n}$. Dimensions are not to be confused with the ones defined for system matrices. We can see that before attempting to solve any optimization problem, one must first pose Eq. (4.3.1) into the desired form, described by Eq. (4.3.2). Following the batch approach in [12, Ch. 8.2], we combine Eq. (4.2.14) in block matrix form to gather state constraints defined in Eq. (4.3.1) for N time steps.

$$\underbrace{\begin{bmatrix} \mathbf{x}(0) \\ \mathbf{x}_1 \\ \vdots \\ \mathbf{x}_N \end{bmatrix}}_{\mathcal{X}} = \underbrace{\begin{bmatrix} \mathbf{I} \\ \mathbf{A} \\ \vdots \\ \mathbf{A}^N \end{bmatrix}}_{\mathcal{S}^x} \mathbf{x}(0) + \underbrace{\begin{bmatrix} \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{B} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{AB} & \mathbf{B} & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{A}^{N-1}\mathbf{B} & \cdots & \mathbf{AB} & \mathbf{B} \end{bmatrix}}_{\mathcal{S}^u} \underbrace{\begin{bmatrix} \mathbf{u}_0 \\ \vdots \\ \mathbf{u}_{N-1} \end{bmatrix}}_{\mathcal{U}} \quad (4.3.3)$$

And by introducing $\bar{\mathbf{Q}} = \text{blockdiag}\{\mathbf{Q}, \dots, \mathbf{Q}, \mathbf{P}\}$, $\bar{\mathbf{R}} = \text{blockdiag}\{\mathbf{R}, \dots, \mathbf{R}\}$ the objective is now written as

$$J(\mathbf{x}(0), \mathbf{U}) = \mathcal{X}^\top \bar{\mathbf{Q}} \mathcal{X} + \mathbf{U}^\top \bar{\mathbf{R}} \mathbf{U} \quad (4.3.4)$$

By substitution of Eq. (4.3.3) into Eq. (4.3.4),

$$J = \mathbf{U}^\top \underbrace{\left(\mathcal{S}^{u\top} \mathbf{Q} \mathcal{S}^u + \mathbf{R} \right)}_{\mathbf{H}} \mathbf{U} + 2 \mathbf{x}^\top(0) \underbrace{\left(\mathcal{S}^{x\top} \mathbf{Q} \mathcal{S}^u \right)}_{\mathbf{F}} \mathbf{U} + \mathbf{x}^\top(0) \underbrace{\left(\mathcal{S}^{x\top} \mathbf{Q} \mathcal{S}^x \right)}_{\mathbf{Y}} \mathbf{x}(0) \quad (4.3.5a)$$

$$= \mathbf{U}^\top \mathbf{H} \mathbf{U} + 2 \mathbf{x}^\top(0) \mathbf{F} \mathbf{U} + \mathbf{x}^\top(0) \mathbf{Y} \mathbf{x}(0) \quad (4.3.5b)$$

Equation (4.3.5b) incorporates the equality constraints of the linear dynamical system into a quadratic program with respect to $\mathbf{U}$. But additional constraint sets $\mathcal{X}$ and $\mathcal{U}$ need to be added as inequalities, specifically as shown in Eq. (4.3.6) (for a complete derivation refer to [12, Ch. 11.2]). For affine constraints on the system inputs $\mathcal{U} = \{\mathbf{u} | \mathbf{A}_u \mathbf{u} \leq \mathbf{b}_u\}$ then the constraint set matrices become $\mathbf{G} = \text{blockdiag}(\mathbf{A}_u)$, $\mathbf{E} = \mathbf{0}$, $\mathbf{w} = [\mathbf{b}_u \dots \mathbf{b}_u]$.

$$\mathbf{G} \mathbf{U} - \mathbf{E} \mathbf{x}(0) \leq \mathbf{w} \quad (4.3.6)$$

Finally, we are able to transform Eq. (4.3.1) in its compact QP form of Eq. (4.3.7) [12, Ch. 11.3.1], solvable using standard methods available in optimization literature.

$$\begin{aligned} \mathbf{U}^* &= \arg \min_{\mathbf{U}} [\mathbf{U}^\top \mathbf{x}^\top(0)] \begin{bmatrix} \mathbf{H} & \mathbf{F}^\top \\ \mathbf{F} & \mathbf{Y} \end{bmatrix} [\mathbf{U}^\top \mathbf{x}^\top(0)]^\top \\ \text{s.t.} \quad & \mathbf{G} \mathbf{U} \leq \mathbf{w} + \mathbf{E} \mathbf{x}(0) \end{aligned} \quad (4.3.7)$$

²The positive semi-definite cone, $\mathbb{S}_+^n$ is defined as $\bigcap_{z \neq 0} \{X \in \mathbb{S}^n \mid z^\top X z \geq 0\}$ to ensure convexity of the objective function [42, Ch. 2.3.1]

Remarks Gathered below are some observations, in no particular order, regarding the previously presented mathematical rigor.

- The unconstrained QP of Eq. (4.3.5b) can be solved to obtain a different form for the dLQR solution in Eq. (4.2.16). This is a control law assuming *perfect* system knowledge, and depends only on $\mathbf{x}(0)$.
- As long as constraints remain inactive and the model is linear, the returned policy is that of LQR.
- State and output (hard) constraints were not considered for simplicity, but are also generally advised against [67] since they might lead to infeasibility of the QP that wouldn't necessarily occur on the real system.
- Further simplifying Eq. (4.3.6) to aid in understanding, the most usual constraints on inputs are $\mathbf{u}_{\min} \leq \mathbf{u} \leq \mathbf{u}_{\max}$. The constraint equations are then compactly described by Eq. (4.3.8) [7].

$$\underbrace{\begin{bmatrix} \mathbf{I} & \dots & 0 \\ 0 & \ddots & 0 \\ 0 & \dots & \mathbf{I} \\ -\mathbf{I} & \dots & 0 \\ 0 & \ddots & 0 \\ 0 & \dots & -\mathbf{I} \end{bmatrix}}_{\mathbf{G}} \mathbf{U} \leq \underbrace{\begin{bmatrix} \mathbf{u}_{\max} \\ \vdots \\ \mathbf{u}_{\max} \\ -\mathbf{u}_{\min} \\ \vdots \\ -\mathbf{u}_{\min} \end{bmatrix}}_{\mathbf{w}} \quad (4.3.8)$$

- The matrix algebra performed in Eq. (4.3.3) is not necessary, and we can include the explicit state equality constraints to solve w.r.t a solution vector $[\mathbf{x}_0 \dots \mathbf{x}_N, \mathbf{u}_0 \dots \mathbf{u}_{N-1}]$.
- All the subsequent sections revolve around extensions of the problem in Eq. (4.3.1) either as augmentations of the state space form of Eq. (4.2.14) or modifications to the objective function.

4.3.3 Receding Horizon Control

One could keep the control actions computed by the method previously described, and apply them over the prediction steps $0, 1, \dots, N$. This however, would be a naive approach as it has been shown³ that such strategies will fail to produce optimal trajectories, leading to the idea of a Receding Horizon Controller (RHC), presented in Fig. 4.6. At time t and over the horizon $t + NT_s$, an optimal input sequence $\mathbf{U}^* \in \mathbb{R}^{m \times N}$ is computed, and only the first element is applied to the system, with the procedure being repeated at time step $t + 1$. Notice the difference between predicted state and input at time t (and subsequent timesteps) and the actual values taken after the application of control inputs.

A Receding Horizon Controller that solves the CFTOC *on-line* is referred to as Model Predictive Control [12, Ch. 12.2] and it will be the main focus of Section 4.4.

³In an example by Kirk [15, Ex. 5.2-3] it was shown that piece-wise finite horizon optimal control actions will not behave appropriately. The controller “anticipates” the final time nearing, reducing control values to zero at the expense of deviations in states.

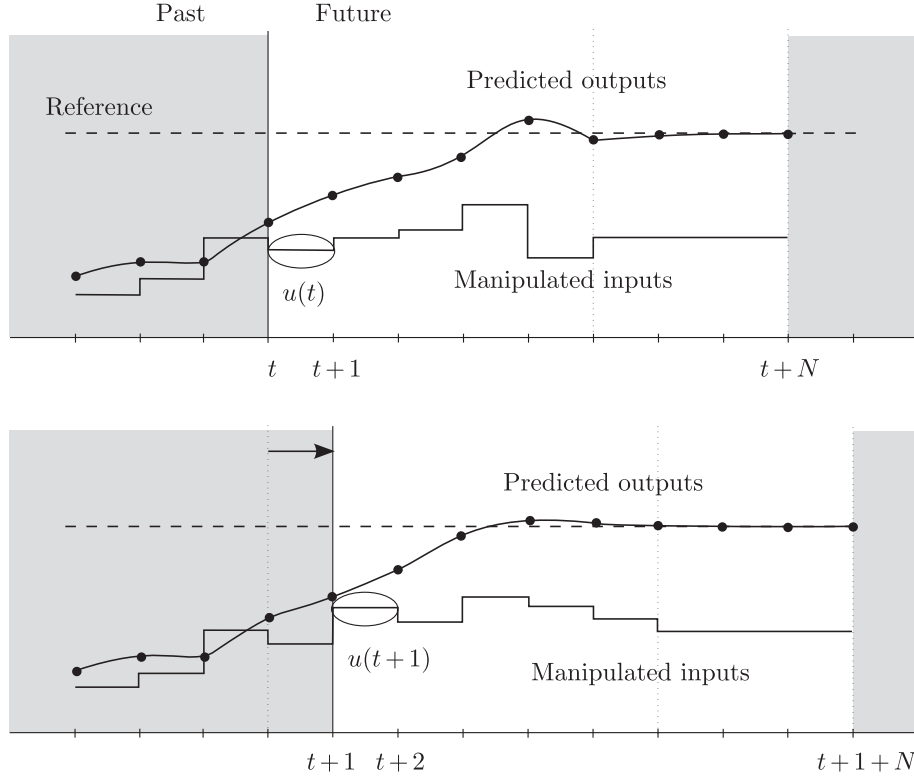


Figure 4.6: The working principle of Receding Horizon Control (Reproduced from [12]).

Offset-free Tracking

As was previously seen in Section 4.2.1 optimal control as a framework usually designs regulators, meaning the resulting controller drives the system states to the origin. Following a reference requires feed-forward action and a way to account for plant-model mismatch, both of which were introduced for the continuous LQR case using an inverse of the model dynamics for the feed-forward gain and integrators for robustness.

For an approach compliant with state space design, integral action is achievable by augmentation of the state vector with a disturbance model and use of an observer to calculate disturbance magnitudes.

$$\begin{aligned} \mathbf{x}[k+1] &= \mathbf{A}\mathbf{x}[k] + \mathbf{B}\mathbf{u}[k] + \mathbf{B}_d\mathbf{d}[k] \\ \mathbf{d}[k+1] &= \mathbf{d}[k] \\ \mathbf{y}[k] &= \mathbf{C}\mathbf{x}[k] + \mathbf{C}_d\mathbf{d}[k] \end{aligned} \quad (4.3.9)$$

For $\mathbf{d} \in \mathbb{R}^{n_d}$, the system in Eq. (4.3.9) is observable iff the pair $(\mathbf{C}, \mathbf{A})$ is observable and

$$\text{rank} \begin{bmatrix} \mathbf{A} - \mathbf{I} & \mathbf{B}_d \\ \mathbf{C} & \mathbf{C}_d \end{bmatrix} = n + n_d \quad (4.3.10)$$

This will prove to be a main limitation in accounting for model uncertainty by disturbance models. The designed observer will assume the form of Eq. (4.3.11).

$$\begin{bmatrix} \hat{\mathbf{x}}[k+1] \\ \hat{\mathbf{d}}[k+1] \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{B}_d \\ \mathbf{0} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \hat{\mathbf{x}}[k] \\ \hat{\mathbf{d}}[k] \end{bmatrix} + \begin{bmatrix} \mathbf{B} \\ \mathbf{0} \end{bmatrix} \mathbf{u}[k] + \begin{bmatrix} \mathbf{L}_x \\ \mathbf{L}_d \end{bmatrix} \left(-\mathbf{y}_m[k] + \mathbf{C}\hat{\mathbf{x}}[k] + \mathbf{C}_d\hat{\mathbf{d}}[k] \right) \quad (4.3.11)$$

The actual value of the disturbance $\mathbf{d}$ is of no concern to the controller, thus directly embedding it within the estimated states in more general formats, as shown in Eq. (4.3.12). This is the model of an input disturbance denoted by subscript id , following arbitrary linear dynamics (driven by white noise $\mathbf{w}$) and entering the plant through input matrices denoted by subscript pd .

$$\begin{aligned}\mathbf{x}_{id}[k+1] &= \mathbf{A}_{id}\mathbf{x}_{id}[k] + \mathbf{B}_{id}\mathbf{w}_{id}[k] \\ \mathbf{d}_{id}[k] &= \mathbf{C}_{id}\mathbf{x}_{id}[k] + \mathbf{D}_{id}\mathbf{w}_{id}[k]\end{aligned}\quad (4.3.12)$$

Combining Eq. (4.3.12) with plant model as in Eq. (4.3.9), we can derive a general augmented plant shown in Eq. (4.3.13) and estimate the state vector by use of a Kalman Filter observer⁴. This is one of the main disturbance models utilized in Section 4.4.

$$\begin{bmatrix} \mathbf{x}_p[k+1] \\ \mathbf{x}_{id}[k+1] \end{bmatrix} = \begin{bmatrix} \mathbf{A}_p & \mathbf{B}_{pd}\mathbf{C}_{id} \\ \mathbf{0} & \mathbf{A}_{id} \end{bmatrix} \begin{bmatrix} \mathbf{x}_p[k] \\ \mathbf{x}_{id}[k] \end{bmatrix} + \begin{bmatrix} \mathbf{B}_{pu} & \mathbf{B}_{pd}\mathbf{D}_{id} \\ \mathbf{0} & \mathbf{B}_{id} \end{bmatrix} \begin{bmatrix} \mathbf{u}[k] \\ \mathbf{w}_{id}[k] \end{bmatrix}\quad (4.3.13)$$

Interestingly this was not originally considered a trivial problem, and there's been many publications that tried to unify approaches found in industry. The specific case mentioned above is presented as a standard in [12] and also proven to be a superset of other methods by Pannocchia [69]. Generally, if the number of disturbances is equal to the number of measured outputs, then all pairs of disturbance matrices that satisfy Eq. (4.3.10) guarantee zero-offset in steady state (for non-active constraints and asymptotically stable closed loop system) [7].

Equivalence of observer design to integral action

Consider the system of Eq. (4.3.9), in continuous-time (with abuse of notation, the matrices involved are their continuous counterparts even if symbols remain identical).

$$\frac{d}{dt} \begin{bmatrix} \mathbf{x} \\ \mathbf{d} \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{B}_d \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{d} \end{bmatrix} + \begin{bmatrix} \mathbf{B} \\ \mathbf{0} \end{bmatrix} \mathbf{u}, \quad \mathbf{y} = [\mathbf{C} \ \mathbf{C}_d] \begin{bmatrix} \mathbf{x} \\ \mathbf{d} \end{bmatrix}$$

As before, the estimated disturbance is given by the observer:

$$\frac{d}{dt} \hat{\mathbf{d}} = \mathbf{L}_d(\mathbf{y} - \mathbf{C}\hat{\mathbf{x}} - \mathbf{C}_d\hat{\mathbf{d}})$$

And by integrating, the resulting control law is given by the equation below, which is an implicit integral action (based on [65, Ch. 8.5] and [7]).

$$\mathbf{u} = -\mathbf{K}\hat{\mathbf{x}} - \hat{\mathbf{d}} = -\mathbf{K}\hat{\mathbf{x}} - \mathbf{L}_d \int_0^t (\mathbf{y} - \mathbf{C}\hat{\mathbf{x}} - \mathbf{C}_d\hat{\mathbf{d}}) dt$$

**The observer method does not directly integrate tracking error as in LQR design, rather accounts for model uncertainty corrections.*

⁴To avoid repetition, the general problem of state estimation is introduced in Section 4.5, thus no effort was made to formally derive observer design before that.

4.4 Model Predictive Control

This section will cover formulations presented by the MATLAB MPC Toolbox [67], since all controllers involved in simulation scenarios, where designed within the SIMULINK environment, using the Toolbox. Each case of augmentations to the default scheme described in Sections 4.3.1 and 4.3.3, is covered by cited publications or references to the Toolbox User's Guide and its authors' presentations. Most importantly, through this section we are introduced to the MPC models simulated (and benchmarked against dLQR) in the next chapter.

4.4.1 Controller Structure

The general structure for the prediction model used in MPC Toolbox is shown in Fig. 4.7. Each state, input and output is to be made *dimensionless* by scaling factors equal to the variable's span (max – min possible values) leading to signals of order unity. These factors are denoted by s_j^x where x is the scaled variable type (usually y or u) and j refers to the j -th component. The variables involved are defined as

$$\mathbf{x}_c^\top = [\mathbf{x}_p^\top \mathbf{x}_{id}^\top \mathbf{x}_{od}^\top \mathbf{x}_n^\top] \quad \text{state vector of size } n_{xp} + n_{xid} + n_{xod} + n_{xn} \quad (4.4.1)$$

$$\mathbf{u}_o^\top = [\mathbf{u}^\top \mathbf{v}^\top \mathbf{w}_{id}^\top \mathbf{w}_{od}^\top \mathbf{w}_n^\top] \quad \text{input vector of appropriate size} \quad (4.4.2)$$

Symbol $\mathbf{u}_o$ refers to the whole input vector in the User's Guide, not to be confused with linearization offset/trim point defined in Section 3.2. The resulting state space matrices include the disturbance subsystems and are defined as in Eqs. (4.4.3) and (4.4.4).

$$\mathbf{A}_c = \begin{bmatrix} \mathbf{A}_p & \mathbf{B}_{pd}\mathbf{C}_{id} & 0 & 0 \\ 0 & \mathbf{A}_{id} & 0 & 0 \\ 0 & 0 & \mathbf{A}_{od} & 0 \\ 0 & 0 & 0 & \mathbf{A}_n \end{bmatrix}, \quad \mathbf{B}_c = \begin{bmatrix} \mathbf{B}_{pu} & \mathbf{B}_{pv} & \mathbf{B}_{pd}\mathbf{D}_{id} & 0 & 0 \\ 0 & 0 & \mathbf{B}_{id} & 0 & 0 \\ 0 & 0 & 0 & \mathbf{B}_{od} & 0 \\ 0 & 0 & 0 & 0 & \mathbf{B}_n \end{bmatrix} \quad (4.4.3)$$

$$\mathbf{C}_c = \begin{bmatrix} \mathbf{C}_p & \mathbf{D}_{pd}\mathbf{C}_{id} & \mathbf{C}_{od} & \begin{bmatrix} \mathbf{C}_n \\ 0 \end{bmatrix} \end{bmatrix}, \quad \mathbf{D}_c = \begin{bmatrix} 0 & \mathbf{D}_{pv} & \mathbf{D}_{pd}\mathbf{D}_{id} & \mathbf{D}_{od} & \begin{bmatrix} \mathbf{D}_n \\ 0 \end{bmatrix} \end{bmatrix} \quad (4.4.4)$$

Notice the similarities to the resulting Eq. (4.3.13) in previous examples of disturbance modelling. The subscript c is used to discern between the linearized model matrices and the MPC observer. Using this representation, the controller will perform state estimation, predict future trajectories, assign cost functions and finally decide on the optimal input sequence.

Observer & Estimation How observer design is incorporated within MPC has been briefly mentioned in Section 4.2.1. For a more thorough review of state estimation, Section 4.5.2 presents core concepts and the development of an estimator algorithm. The iterative process followed is illustrated in Fig. 4.8: At each time step k , the algorithm utilizes the state estimates (estimated:purple) of $k - 1$, and the measurement data of k (known:green). To aid in readability, the graphic does not use bold lettering for vectors, or subscripts to denote the specific matrices.

Within the MPC toolbox the steps are:

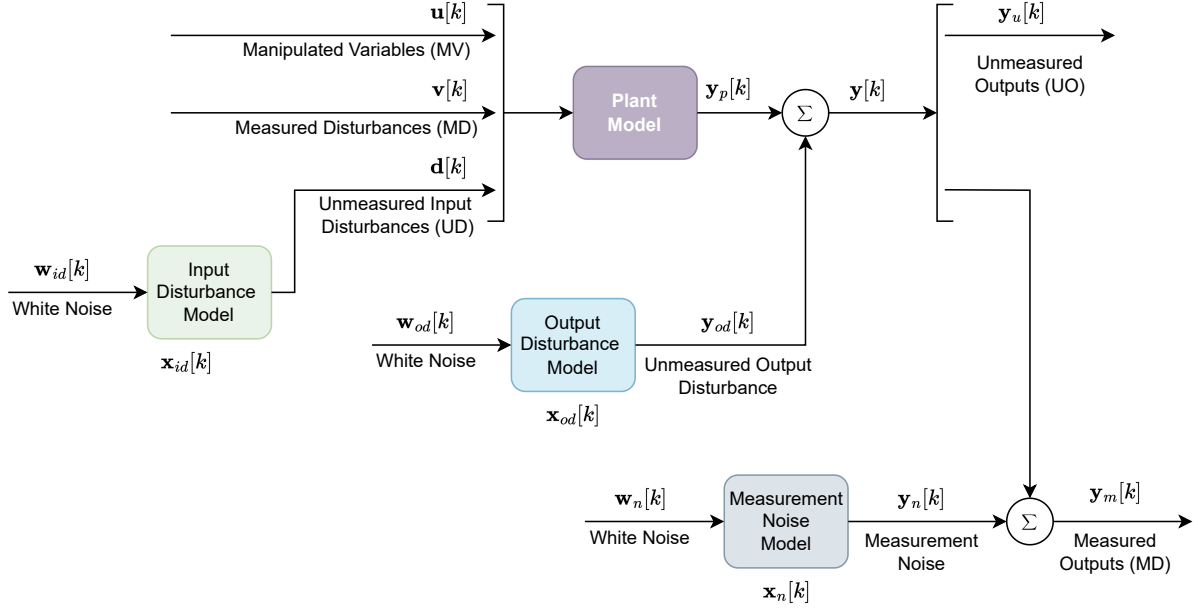


Figure 4.7: Internal model used for MPC state estimation and prediction. All sub-models are included in the state space description of Eqs. (4.4.3) and (4.4.4).

1. Obtain state estimate $\mathbf{x}_c(k|k-1)$, manipulated variable (MV) used from $k-1$ to k as $\mathbf{u}^{\text{act}}(k-1)$, measured disturbances (MD) $\mathbf{v}(k)$ and measured outputs $\mathbf{y}_m(k)$.
2. Collect columns $\mathbf{B}_u$, $\mathbf{B}_v$ of observer model $\mathbf{B}_c$ corresponding to inputs $\mathbf{u}(k)$, $\mathbf{v}(k)$ as well as rows $\mathbf{C}_m$ of $\mathbf{C}_c$ and $\mathbf{D}_{mv}$ of $\mathbf{D}_c$.
3. Compute innovation $\mathbf{e}(k) = \mathbf{y}_m(k) - (\mathbf{C}_m \mathbf{x}_c(k|k-1) + \mathbf{D}_{mv} \mathbf{v}(k))$.
4. Update the model estimates: $\mathbf{x}_c(k|k) = \mathbf{x}_c(k|k-1) + \mathbf{M} \mathbf{e}(k)$.
5. Calculate trajectory within prediction horizon $k \rightarrow k+N-1$ with $\mathbf{x}(0) = \mathbf{x}_c(k|k)$ and acquire $\mathbf{u}^{\text{opt}}(k)$.
6. Predict state values for next step: $\mathbf{x}_c(k+1|k) = \mathbf{A}_c \mathbf{x}_c(k|k-1) + \mathbf{B}_u \mathbf{u}^{\text{opt}}(k) + \mathbf{B}_v \mathbf{v}(k) + \mathbf{L} \mathbf{e}(k)$.

Prediction It may already be noticeable that there are two distinct *prediction* phases. The first, (step 5 above) refers to the CFTOC problem described in Section 4.3.1. Unknown white noise inputs are assumed to be 0, and predicted outputs are noise-free. The second, propagates the estimation model corrected by the innovation term and observer gain, and poses as the estimate used in step 1, at the next interval. Unknown inputs are assumed to be 0.

Cost Function The cost function includes terms to penalize output, input and input increment deviations as well as slack variables to handle constraint softening. Constraint

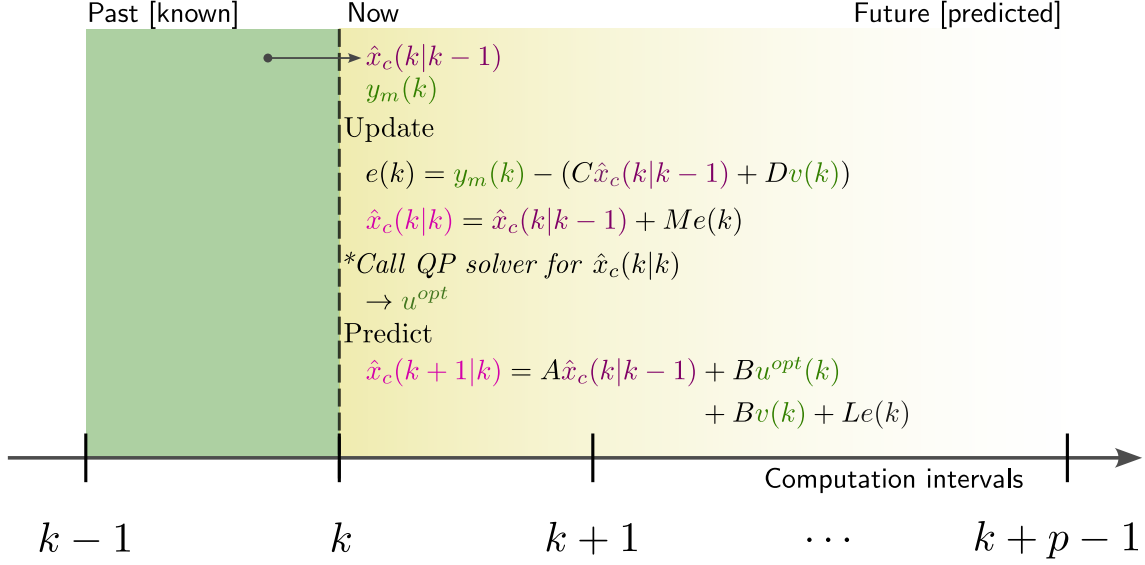


Figure 4.8: State estimation within one computation interval. Notice the distinct actions labeled *prediction*: One is used within the KF and one within the QP context.

softening refers to the tactic that seeks an optimal decision by violating constraints, usually when the QP turns out to be infeasible. The standard cost function is shown in Eq. (4.4.5) which consists of the terms outlined in Eqs. (4.4.6) to (4.4.9).

$$J(\mathbf{z}_k) = J_y(\mathbf{z}_k) + J_u(\mathbf{z}_k) + J_{\Delta u}(\mathbf{z}_k) + J_\varepsilon(\mathbf{z}_k) \quad (4.4.5)$$

As before, k is the current control interval, p is the prediction horizon z_k contains the decision variables.

- Output reference tracking. Note that $r_j(k+i|k)$ by default is constant throughout the prediction horizon.

$$J_y(z_k) = \sum_{j=1}^{n_y} \sum_{i=1}^p \left\{ \left(\frac{w_{i,j}^y}{s_j^y} \right) [r_j(k+i|k) - y_j(k+i|k)] \right\}^2 \quad (4.4.6)$$

- Manipulated variable tracking. Useful in some cases, but not within this thesis context.

$$J_u(z_k) = \sum_{j=1}^{n_u} \sum_{i=0}^{p-1} \left\{ \left(\frac{w_{i,j}^u}{s_j^u} \right) [u_j(k+i|k) - u_{j,\text{target}}(k+i|k)] \right\}^2 \quad (4.4.7)$$

- Manipulated variable move suppression. Does not allow large (aggressive) control moves.

$$J_{\Delta u}(z_k) = \sum_{j=1}^{n_u} \sum_{i=0}^{p-1} \left\{ \frac{w_{i,j}^{\Delta u}}{s_j^u} [u_j(k+i|k) - u_j(k+i-1|k)] \right\}^2 \quad (4.4.8)$$

- Constraint violation. Nonnegative, dimensionless slack variable ε_k quantifies worst-case constraint violation.

$$J_\varepsilon(z_k) = \rho_\varepsilon \varepsilon_k^2 \quad (4.4.9)$$

Notice how $\mathbf{Q}$, $\mathbf{R}$ matrices that were used up to now, are implemented within the software. For the diagonal case, we can use the weighting elements $w_{i,j}^x$ squared, denoting the weight factor of the j -th component of variable x at interval $k+i$. This can be combined with a scaling diagonal matrix employing the previously defined s_j^x factors as $\mathbf{L}_{Sx}$. Equation (4.4.10) makes the comparison with an LQR design that applies the weighting matrices to the *unscaled* states (for constant weights within the prediction horizon).

$$\mathbf{Q} = (\mathbf{L}_{Sy}^{-1} \mathbf{W}^y)^2 = (\text{diag}(s_1^y \dots s_{n_y}^y)^{-1} \times \text{diag}(w_1^u \dots w_{n_u}^u))^2, \quad \mathbf{R} = (\mathbf{L}_{Su}^{-1} \mathbf{W}^u)^2 \quad (4.4.10)$$

Optimization Variables & Constraints The selected optimization variables are chosen as $\mathbf{z}_k = [\Delta u(k|k)^\top \dots \Delta u(k+p-1|k)^\top \varepsilon_k]^\top$. This allows move blocking to occur when $H_u < H_p$, thus keeping $\Delta u = 0$ for the remainder of predictions. Constraints are usually implemented as lower/upper bounds, such that for $i = 1:p$ and $j = 1:n_u$

$$\frac{u_{j,\min}(i)}{s_j^u} - \varepsilon_k V_{j,\min}^u(i) \leq \frac{u_j(k+i-1|k)}{s_j^u} \leq \frac{u_{j,\max}(i)}{s_j^u} + \varepsilon_k V_{j,\max}^u(i) \quad (4.4.11)$$

The slack variable, combined with weighting factors $V_{j,\min}^x$, $V_{j,\max}^x$ and ρ_ε can expand the feasible region by allowing modest constraint violation, at the cost of $J_\varepsilon(z_k)$ [67, p. 2-8]. For the full QP formulation within the software, refer to [67, p. 1-7].

4.4.2 Extensions to the MPC Scheme

One of the main advantages to MPC, is its ability to handle different formulations and augmentations (either to the prediction model, or to the optimization problem) that allow for flexible control design and better descriptions of process dynamics. Within the scope of the presented work, there's been specific additions embedded into the MPC framework. Utilizing the well established observer design, actuator uncertainty and delay can be successfully modelled. Additionally, reference preview within the prediction horizon, and introduction of a time varying model to account for hydrodynamic force non-linearities are incorporated.

Disturbance Based Augmentations

The default design auto-selects a disturbance model, based on observable state descriptions. Thus, passing the linearized model to the `mpc()` command will automatically add output integrators and tune the estimator accordingly. An attempt is made to select appropriate disturbance models, which will then be evaluated in Chapter 5.

Actuator uncertainty Control surfaces are nonlinear (both with respect to angle of attack, and speed) thus making them hard to model accurately. The tools given to us by MPC design however, allow us to introduce an unmeasured disturbance acting on the input channels much like Eqs. (4.3.12) and (4.3.13). This is an especially intuitive way to

counter uncertainty: In steady state flight, the hydrodynamic coefficients will inevitably be off by some *constant*, which the controller can internally estimate.

$$\begin{aligned}\mathbf{x}_{k+1} &= \mathbf{A}_p \mathbf{x}_k + \mathbf{B}_{pu} \mathbf{u}_k + \mathbf{B}_{pd} \mathbf{d}_{id,k} \\ \mathbf{d}_{id,k} &= \mathbf{C}_{id} \mathbf{x}_{id,k} = \mathbf{C}_{id} (\mathbf{A}_{id} \mathbf{x}_{id,k-1} + \mathbf{B}_{id} \mathbf{w}_{id,k-1})\end{aligned}\quad (4.4.12)$$

The unmeasured disturbance inputs and matrices are inserted by duplicating the original plant input matrix such that $\mathbf{B}_{pu} = \mathbf{B}_{pd}$, aiding the prediction with a constant term that follows the same input dynamics.

Actuator delay The mechanical components of the vessel do not respond instantaneously to control commands, nor does lift develop at once. If the prediction model assumes the opposite, it might behave erratically in an effort to correct the real plant. A possible solution is to include the delay of the servomotors controlling the lifting surfaces, as a first order delay within the prediction model. It has already been established that a first order delay in continuous time follows Eq. (4.4.13)

$$\frac{d}{dt} u_{\text{applied}} = -\frac{1}{\tau} u_{\text{applied}} + \frac{1}{\tau} u_{\text{command}} \quad (4.4.13)$$

This is generalized for more than one inputs, and used to augment the original linearized system as in Eq. (4.4.14). Note that the matrices involved represent *continuous* dynamics, and discretization is handled by Matlab during controller creation.

$$\begin{bmatrix} \dot{\mathbf{x}} \\ \dot{\mathbf{u}}_a \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{0}_{m \times n} & -\frac{1}{\tau} \mathbf{I}_m \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{u}_a \end{bmatrix} + \begin{bmatrix} \mathbf{0}_{n \times m} \\ \frac{1}{\tau} \mathbf{I}_m \end{bmatrix} \mathbf{u}_c \quad (4.4.14)$$

The algorithm will now be able to generate input commands $\mathbf{u}_c$ and anticipate a lag in their effect, since the actual input that drives the system states is also an internal state with dynamic response. To also include the input uncertainty term from before, the input matrix can be duplicated column-wise and include $\mathbf{d}_{id}$ as UD.

Reference Preview

As was stated before, the reference term in Eq. (4.4.6) is constant within H_p . In theory, anticipative action is achieved by the equation below according to [7].

$$\min_{\mathbf{z}} \sum_{i=0}^{p-1} \left\| \mathbf{L}_{Sy}^{-1} \mathbf{W}^y (\mathbf{y}_{i+1} - \mathbf{r}(k+i)) \right\|_2^2$$

Where $r(k)$ is the current reference, and in default MPC stays constant. Unexpectedly, the way to include reference previewing within Simulink wasn't as intuitive. Addition of the *Reference Previewer* block [67, p. 4-31] allows predefined reference signals to be fed to the simulated controller, thus involving the correct terms in the cost functions throughout the horizon. The actual result of this is closer tracking of ramp references, which superficially raises the ISE score without necessarily impacting stability. Inclusion of previewing requires careful reference generation, because step signals induce aggressive commands.

Adaptive LPV-MPC

Pseudo-linear reformulations are used as a way to model complex systems without losing controller's linear structure (which might impact QP iterations or even feasibility). The approach of Linear Parameter Varying (LPV) models can be utilized as in Alcalá et al. [70] and examples from the Toolbox [67, p. 7-16]. The LPV model takes the form of Eq. (4.4.15), where state matrices are not static any more but depend on the scheduling vector $\lambda(k)$.

$$\mathbf{x}[k+1] = \mathbf{A}_p(\lambda(k))\mathbf{x}_p[k] + \mathbf{B}_{pu}(\lambda(k))\mathbf{u}[k] \quad (4.4.15)$$

In this case, $\lambda(k)$ is a scalar, specifically body-fixed forward speed u , since it is used to interpolate between the models generated as linearizations around a range of service speeds V_s .

Using Matlab's LPV block, successive linearization can be indexed by the scheduling variable, which is then used to look up the specific state-space model. Linear interpolation is used for intermediate values, while different linearization come with their respective state and input offsets. The scheduling variable can be estimated through sensor data (or directly fed from simulation, when debugging) which will be addressed in Section 4.5. During runtime $\lambda(k)$ is used to decide on prediction model, which will then remain static within H_p to formulate the QP, thus the controller remains at least piecewise linear. In general, we are not bounded by a constant scheduling vector, since in [70], λ is itself a state which evolves during prediction. Keeping it static however, greatly simplifies the design.

4.4.3 Designs & Tuning Parameters

The previous section more or less covers the different ideas put into practice for the final controller. In order to compare behavior for different cases, various designs were tested, within a developed *design*→*integrate*→*simulate* pipeline. There is a lot of room for improvement, but Table 4.3 collects the developed variants and explanations.

Name	Functionality	-I	-O	-D	-P
MPC	Base linear MPC (only linearized plant model).	Input dist. model for actuator model mismatch.	Output dist. model on measured outputs.	Actuator delay compensation model.	Anticipative action by reference previewing.
LPV-MPC	Adaptive MPC with LPV prediction model.				

Table 4.3: Controller naming legend for MPC family.

As with LQR, the MPC design holds certain “knobs” that allow us to tweak performance characteristics. It was previously shown, how the Q, R matrices are translated within the MPC framework.

Parameter	Type	Use
H_p	Scalar	Prediction horizon.
H_u	Scalar	Control horizon.
$\mathbf{W}^y$	Matrix, $n \times n$	State weights, diagonal.
$\mathbf{W}^u$	Matrix, $m \times m$	Control input weights, diagonal.
$\mathbf{W}^{\Delta u}$	Matrix, $m \times m$	Move suppression factors.
ss(ID)	Model (ss)	Input disturbance model.
ss(md1)	Model (ss)	Plant model for prediction.

Table 4.4: Tuning parameters for MPC design.

4.5 State Estimation

The central idea of full state feedback controllers, is the availability of the full system state vector through sensor measurements. When this is not possible, there are other methods available to *reconstruct* the state vector from measured outputs, such as *observers* or *state estimators*. The process of combining data from various sources to estimate unmeasured values is also referred to as *sensor fusion*. In the following section, a practical problem in control system design is considered: the selection of sensors under (theoretical) observability demands and utilization of noisy measurements. As before, an approach is given both for simulation and implementation.

4.5.1 Sensor Measurements

Shifting our focus to the studied system, the states needed to be measured have already been established, as subsets of body-fixed velocities $\boldsymbol{\nu}$ and earth-fixed positions $\boldsymbol{\eta}$. It might already be evident that:

1. The states z , w are not recoverable from angle and angular velocity measurements, as $\text{rank}(\mathcal{O})$ drops to 5 for a $\mathbf{C}$ that does not output z .
2. Direct integration of acceleration data in an effort to acquire w from an inertial measurement device will induce drift.
3. Vessel height z in the NED frame is not accurately calculated by available sensing instruments due to measurement pollution by waves.
4. An observer based on the dynamic model of Eqs. (A.3.1) and (A.3.2) introduced in Section 3.2 may prove too simple to accurately capture motion in different DOF and velocities.

These issues impose difficulties individually, which are resolved by the development of a Kalman Filter used to combine readings and filter the information stream.

Sensor Selection

The problems mentioned above require a careful sensor selection process, since inexact measurements have a direct detrimental effect on system stability (for example, the LQG

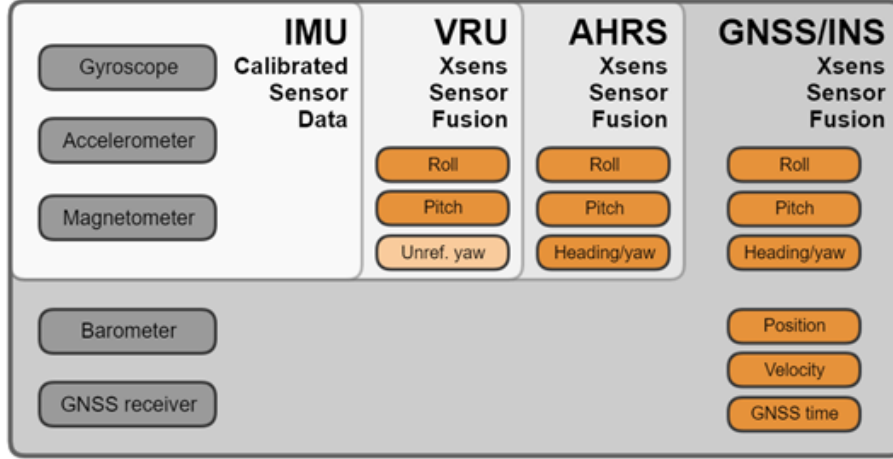


Figure 4.9: Differences between different motion tracking products and naming conventions.

controller has no stability guarantees as the LQR infinite gain margin). The proposed combination of instruments is a standard in most similar applications, supported by [1], [24], [27], [28], [30], [33]. Vessels usually are already equipped with a GPS which provides total velocity in the NED system.

$$V_{\text{GPS}} = \left\| \dot{\mathbf{p}}_{b/n}^n \right\|_2 = \sqrt{\dot{x}^2 + \dot{y}^2} \quad (4.5.1)$$

This will prove useful in providing the *scheduling parameter* previously introduced in LPV-MPC, or in general gain-scheduling setups. To acquire data regarding angular velocities (p, q, r) and orientation (ϕ, θ, ψ) , the most common choice is an IMU, or an AHRS. To estimate distance from sea level, a sonar height sensor can be utilized. Combining data from these two instruments, the w velocity component can be estimated and corrected by derivation of sonar height and integration of IMU vertical acceleration. Based on published works and results, a common selection is

- Xsens MTi-680G GNSS/INS unit combining GPS and IMU functionality [71].
- ToughSonic 14 Level & Distance sensor providing ultrasonic range readings [72].

For the purposes of this thesis, the selected sensors have internal fusion capabilities, thus achieving low noise densities. Figure 4.9 might help to clear up some confusion regarding motion tracking sensor names⁵. This comes at a higher cost when compared to a standard 9-DOF IMU (3 axis measurements from accelerometer, gyroscope, magnetometer) and external fusion within the microprocessor used in the application.

Noise Most sensor devices use electro-mechanical means to sense some physical quantity that will then be transduced into an electric signal. This signal will contain a certain level of noise, usually specified within the datasheet. A concise overview of how noise enters electrical components is given by Horowitz & Hill [73, Ch. 8]. Our usual model is that of

⁵From mtidocs.movella.com/from-imu-to-gnss/ins

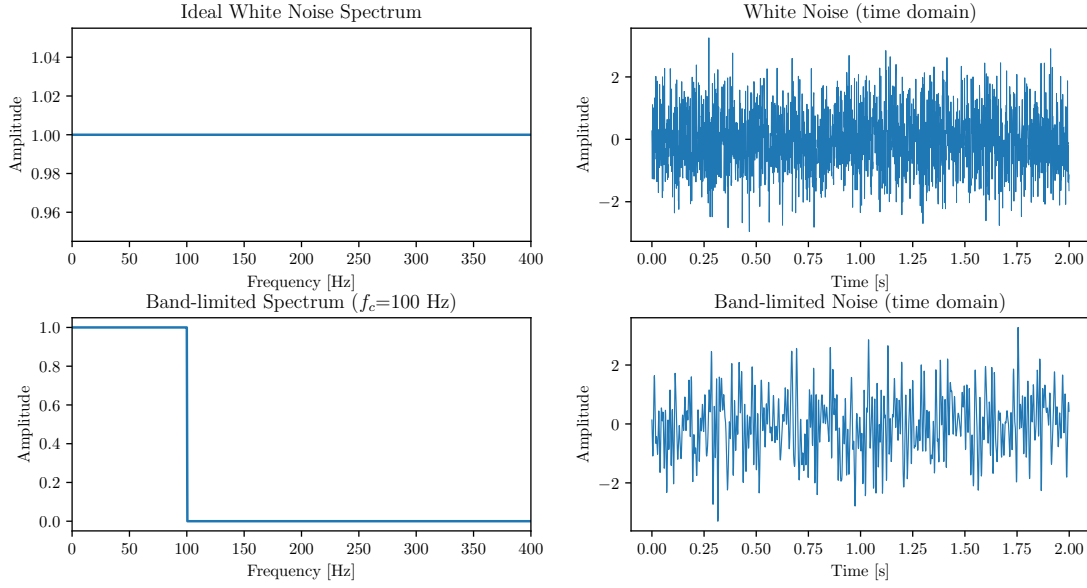


Figure 4.10: Spectral densities of white and band limited noise, and the time-domain signals they produce.

white noise, which has a spectral density that is constant at all frequencies as $S_{wn}(j\omega) = A$. It is “noise that jumps around infinitely far, infinitely fast”, or in mathematical terms, has a correlation time $\tau = 0$. As physical systems cannot produce such noise, we use *band limited white noise* which is described by a sharp cutoff at some frequency, also called a bandpass or “brick-wall” filter. Both representations are shown in Fig. 4.10.

When given noise characteristics from a manufacturer, we can calculate the expected noise variance and recreate it using software. We start with noise *density*, which is the RMS (root-mean-square) noise amplitude in an 1-Hz band with symbol e_n and units on par with the measured quantity, as $[\text{units}]/\sqrt{\text{Hz}}$. Noise *power* which is usually defined as the square of that, signifies the energy within the signal and scales with its variance, or σ^2 . Much like the discussion that was made regarding waves by Holthuijsen [46], they are connected by the exact same equation, specifically Eq. (2.4.15) (pertains to their nature as stochastic processes). Equation (4.5.2) gives a way of acquiring the variance of filtered noise from knowledge of $e_n(f)$ and a filter’s spectral response $H(f)$ [73].

$$\sigma_n^2 = \int |e_n(f)H(f)|^2 df \quad (4.5.2)$$

Which simplifies to $\text{RMS} = \sigma_n = e_n\sqrt{B}$ where B is the bandwidth, for a constant noise density and a brick-wall filter. Data has been gathered from the available datasheets, especially noise RMS, and is presented in Table 4.5 where “—” denotes values that are not given.

Simulation

The Simulink model developed in Chapter 2 has been augmented to include sensing devices with their respective noise, as given in Table 4.5. While the exact details of this

Quantity	Units	$e_n [\cdot/\sqrt{\text{Hz}}]$	B@-3 dB [Hz]	σ (RMS)
Acceleration	$\text{m} \cdot \text{s}^{-2}$	5.886×10^{-6}	500	1.316×10^{-2}
Angle	rad	–	–	3.490×10^{-3}
Ang. velocity	$\text{rad} \cdot \text{s}^{-1}$	1.222×10^{-4}	520	2.786×10^{-3}
Velocity	$\text{m} \cdot \text{s}^{-1}$	–	–	5.000×10^{-2}
Distance	m	–	≈ 20	2.000×10^{-3}

Table 4.5: Noise characteristics for different measured quantities.

implementation are given in Appendix B, note that the readings of the GNSS/INS are set to coincide with the values corresponding to the CO. In simple terms, no care was taken to include corrections for antenna placement or IMU offset from CO. For the former case, movella/xsens documentation provides corrections, while the latter is mentioned in lectures prepared by Fossen [43, Ch. 14], as point of reference transformations.

Sonar height modelling Shown in Fig. 4.11 is a very important illustration of how an ultrasonic sensor measures distance. A beam leaves the sensor, usually perpendicular to the vessel $x_b y_b$ plane, hits the sea surface and the time taken for the reflection to reach the sensor will be used to estimate the travelled distance. It has a baseline error specified in Table 4.5, an added offset due to sea-surface irregularities, and an induced error from orientation corrections, since we care about *vertical* distance in $\{n\}$ and not the actual measured h_s . In different simulation environments (like Unity3D or Gazebo) this action is described as *raycasting* and is usually fully implemented within the software. In fact, it will also be able to handle the exact surface elevation (or an average of some defined shape projection to account for beam width) where the beam meets sea level. To get the same result from Simulink, some simple derivations are needed, based on Fig. 4.11. Given the $\{b\}$ frame position as $\mathbf{p}_{b/n}^n$ along with orientation Θ_{nb} and sonar position within the vessel $\mathbf{r}_s^b$, we have for sensor position in $\{n\}$

$$\mathbf{p}_{s/n}^n = \mathbf{p}_{b/n}^n + \mathbf{R}_b^n \mathbf{r}_s^b \quad (4.5.3)$$

As the sonar sensor points towards the positive z_b axis, the appropriate unit vector in $\{b\}$ is $\mathbf{k}^b = [0, 0, 1]^\top$. This is transformed to $\{n\}$ as $\mathbf{k}^n = \mathbf{R}_b^n \mathbf{k}^b = [0, 0, \cos \theta \cos \phi]^\top$. The vector $\mathbf{p}^n$ denotes the vector sum of sensor position in $\{n\}$ and the ray $\lambda \mathbf{k}^n$ and we want its vertical component to be zero (ray touching the undisturbed sea-surface).

$$\mathbf{p}^n(\lambda) = \mathbf{p}_{s/n}^n + \lambda \mathbf{k}^n \Rightarrow (\mathbf{p}_{s/n}^n)_3 + \lambda (\mathbf{k}^n)_3 = 0 \quad (4.5.4)$$

The solution is given for $\lambda = -p_{s,z}/(\cos \theta \cos \phi)$, or:

$$\begin{aligned} h_s &= \frac{x_s \sin \theta - y_s \cos \theta \sin \phi - z_s \cos \theta \cos \phi - z_{CO}}{\cos \theta \cos \phi} \\ &= -\frac{1}{\cos \theta \cos \phi} z_{CO} + x_s \frac{\tan \theta}{\cos \phi} - y_s \tan \phi - z_s \end{aligned} \quad (4.5.5)$$

Measurements are also skewed by wave elevation, shown as h_ζ in Fig. 4.11. Also shown is the slight difference between the actual offset in height, and the one used for simulation,

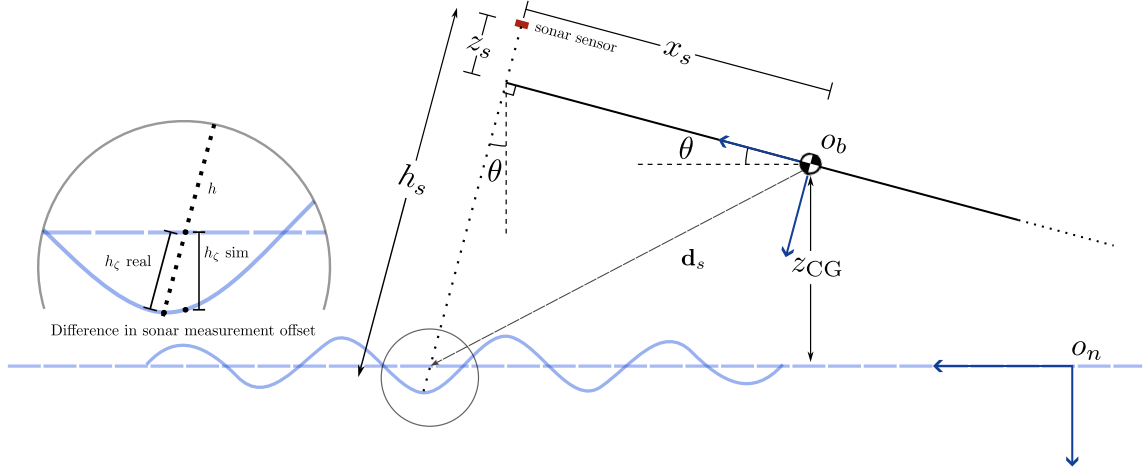


Figure 4.11: Sonar sensor placement and beam measurements. Shown magnified, is the approximation error due to the simplifications made.

as it is easier to calculate the vertical free-surface elevation given the raycast point. The sonar measurement offset h_ζ is calculated using the same tools from MSS that estimate foil submergence.

The reverse process, that of acquiring vessel height from sensor readings, is the one that interests the control engineer and the important final result is given by Eq. (4.5.8).

Sonar height measurement

A brief explanation of how to acquire z_{CG} from a sonar sensor placed on the vessel is separated between this excerpt and the *Simulation* section. Referring to Fig. 4.11, the sensor is placed at $[x_s, y_s, z_s]$ in the body-fixed system and measures a distance to sea-level h_s . Considering the 2-D longitudinal case, and flat free surface [23]:

$$z_{CO} = (h_s + z_s) \cos \theta - x_s \sin \theta \quad (4.5.6)$$

The positive sign is due to z_s being a coordinate, not absolute distance. This is generalized to include lateral deviations by introducing the vector $\mathbf{d}_s$ expressed in the $\{b\}$ frame

$$\mathbf{d}_s = [x_s, y_s, h_s + z_s] \quad (4.5.7)$$

Then, expressing the components of $\mathbf{d}_s$ in the $\{n\}$ frame and acquiring only the z component, will yield vessel height.

$$z_{CO} = (\mathbf{R}_b^n \mathbf{d}_s^b)_3 \quad (4.5.8)$$

Added Noise Sensor noise is simulated by use of two Simulink blocks. Basically, whenever data for bandwidth and noise density are given, the **band limited white noise** is selected, with a generally high sample time (usually for 2kHz), power equal to e_n^2 and a low-pass filter with cutoff at f_{-3dB} . Note that the resulting RMS will be slightly higher due to the low-pass filter shape encompassing larger frequency content. For quantities with

only σ_n known, `random number` is selected, producing numbers at the specified frequency and variance σ_n^2 .

4.5.2 Discrete Kalman Filter

As has been shown, an observer solely based on the linear system developed in Section 3.2 is inadequate for state estimation. Such an observer might be able to fuse sensor measurements within the MPC setup, provided that disturbances are slowly varying and step-like, which isn't true for height readings. For this reason, a separate observer was designed to provide noise-free state estimates to the controller. Note that the internal state estimation capabilities of MPC were not removed, since they are able to account for model mismatch in the input channels (uncertainty in lifting surface coefficients).

Kalman Filter Theory

We shall start with the mathematical description of a discrete system model whose states are to be estimated. Then, we can track how the state and its covariance propagate with time, acquiring an *expectation* for the states. Dynamically propagated *predictions* are recursively updated by measurements, giving rise to two distinct steps, what is often called the *predictor-corrector* formulation. Given the system model of Eq. (4.5.9), it is assumed that noise processes $\{\mathbf{w}_k\}$ and $\{\mathbf{v}_k\}$ are white, zero-mean and uncorrelated with known covariance matrices $\mathbf{Q}_k$ and $\mathbf{R}_k$.

$$\begin{aligned}\mathbf{x}_k &= \mathbf{F}_{k-1}\mathbf{x}_{k-1} + \mathbf{G}_{k-1}\mathbf{u}_{k-1} + \mathbf{w}_{k-1} \\ \mathbf{y}_k &= \mathbf{H}_k\mathbf{x}_k + \mathbf{v}_k\end{aligned}\tag{4.5.9}$$

Note here, that the presented equations are one of many *equivalent* formulations of Discrete Kalman Filter (DKF), with usual differences pertaining to indexing, or order of operations (update/predict). Notation will follow Simon [17, Ch. 5], but algorithm results are similar to Fossen [10, Ch. 11.3], with Åström & Murray [65, Ch. 8.4] adopting a more control-oriented approach.

To symbolically separate the two steps, we make the distinction between an *a priori* estimate $\hat{\mathbf{x}}_k^-$ based on knowledge up to but not including k and an *a posteriori* $\hat{\mathbf{x}}_k^+$ which includes measurement correction. For the state and covariance propagation by the dynamic model, Eqs. (4.5.10) and (4.5.11) yield the a priori estimate.

$$\hat{\mathbf{x}}_k^- = \mathbf{F}_{k-1}\hat{\mathbf{x}}_{k-1}^+ + \mathbf{G}_{k-1}\mathbf{u}_{k-1}\tag{4.5.10}$$

$$\mathbf{P}_k^- = \mathbf{F}_{k-1}\mathbf{P}_{k-1}^+\mathbf{F}_{k-1}^\top + \mathbf{Q}_{k-1}\tag{4.5.11}$$

An initialization step will usually set $\hat{\mathbf{x}}_0^+ = \mathbb{E}[\mathbf{x}_0]$ (the expected value of the initial state) and $\mathbf{P}_0^+ = \mathbb{E}[(\mathbf{x}_0 - \hat{\mathbf{x}}_0^+)(\mathbf{x}_0 - \hat{\mathbf{x}}_0^+)^\top]$ (uncertainty in the initial estimate of $\mathbf{x}_0$). After a measurement has been acquired, Eqs. (4.5.12) to (4.5.14) provide the a posteriori state and covariance estimates, with the $(\mathbf{y} - \mathbf{H}\hat{\mathbf{x}})$ term often named as the *innovation*.

$$\mathbf{K}_k = \mathbf{P}_k^- \mathbf{H}_k^\top \left(\mathbf{H}_k \mathbf{P}_k^- \mathbf{H}_k^\top + \mathbf{R}_k \right)^{-1} \quad (4.5.12)$$

$$\hat{\mathbf{x}}_k^+ = \hat{\mathbf{x}}_k^- + \mathbf{K}_k (\mathbf{y}_k - \mathbf{H}_k \hat{\mathbf{x}}_k^-) \quad (4.5.13)$$

$$\mathbf{P}_k^+ = (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k) \mathbf{P}_k^- (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k)^\top + \mathbf{K}_k \mathbf{R}_k \mathbf{K}_k^\top \quad (4.5.14)$$

Implementation

Implementation of the above theoretical concepts within the scope of this work firstly requires the acknowledgment that the KF in its simplest form is *not* an optimal predictor for this case. Overall the dynamics present with non-linearities breaking the assumption of a linear model, driven by correlated, colored noise which breaks the assumption of white zero-mean and un-correlated. Since the filter gains are not steady-state constant however, the implementation overall achieves a certain degree of smoothing over which will be quantified in the next chapter.

A separate, practical issue is that measurements arrive at differing intervals, thus we can't perform complete update steps on all measured states. In fact, due to the low refresh rate of the ultrasonic sensor, prediction steps will have to run many times before a measurement correction is made [10, Ch. 11.3]. In this case, the DKF is normally developed, but the measurement step is fed the appropriate matrices corresponding to the available measurements and called only when these measurements arrive.

Model Prediction The two states of interest, z and w will be used along with u , the longitudinal speed component in $\{b\}$, as it is the scheduling parameter. The dynamic model is described in Eq. (4.5.15), where Eqs. (4.5.15b) and (4.5.15c) are discrete integrators of body-fixed acceleration measurements treated as plant inputs. As for Eq. (4.5.15a), it is the linear, discrete counter part of Eq. (3.1.3a).

$$z_k = z_{k-1} - T_s \theta u_{k-1} + T_s w_{k-1} \quad (4.5.15a)$$

$$u_k = u_{k-1} + T_s \dot{u}_{k-1} \quad (4.5.15b)$$

$$w_k = w_{k-1} + T_s \dot{w}_{k-1} \quad (4.5.15c)$$

Or, in matrix form, showing the state and input matrices:

$$\begin{bmatrix} z \\ u \\ w \end{bmatrix}_k = \begin{bmatrix} 1 & -T_s \theta & T_s \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} z \\ u \\ w \end{bmatrix}_{k-1} + \begin{bmatrix} 0 & 0 \\ T_s & 0 \\ 0 & T_s \end{bmatrix} \begin{bmatrix} \dot{u} \\ \dot{w} \end{bmatrix}_{k-1} \quad (4.5.16)$$

In order to build the noise covariance matrix $\mathbf{Q}$ we essentially need the variance of process noise during the prediction step. Notice that process noise acting on z is a linear combination of u and w noise. Consider the following derivations, based on probability theory presented by Simon [17, Ch. 2.4.2].

The process of Eq. (4.5.16) experiences white, zero-mean, uncorrelated in time noise ξ_u and ξ_w entering through the input channels by the system matrix $\mathbf{G}$. It follows that,

$$\boldsymbol{\xi} = [\xi_u, \xi_w] \sim \mathcal{N}(0, \boldsymbol{\xi}), \text{ where } \boldsymbol{\xi} = \text{diag}(\sigma_u^2, \sigma_w^2)$$

The total noise entering the system is a linear combination of the above (θ is treated as a constant).

$$\mathbf{w} = T_s \begin{bmatrix} -\theta & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \boldsymbol{\xi} \quad (4.5.17)$$

Finally, the covariance matrix $\mathbf{Q}$ given an affine transformation $\mathbf{A}$ is calculated as $\mathbf{A} \boldsymbol{\xi} \mathbf{A}^\top$.

$$\mathbf{Q} = \mathbf{w} \mathbf{w}^\top = T_s^2 \begin{bmatrix} \theta^2 \sigma_u^2 + \sigma_w^2 & -\theta \sigma_u^2 & \sigma_w^2 \\ -\theta \sigma_u^2 & \sigma_u^2 & 0 \\ \sigma_w^2 & 0 & \sigma_w^2 \end{bmatrix} \quad (4.5.18)$$

Before moving to measurements, its deemed best to clear up any confusion. In general, measurements can be treated as inputs and used on the prediction step when they do not directly hold information on the selected states. Measurement noise still needs to be propagated as it was just shown using the accelerations.

Measurement Update For this implementation, only two directly measured quantities affect the filter: Height estimate from sonar and the longitudinal component of speed in $\{n\}$ measured by the Global Positioning System (GPS). The latter is fairly well approximated⁶ while the former may need additional caution. By analyzing the kinematic relationships of Section 2.2 and specifically Eq. (2.2.7) for the components in x_n , the horizontal speed component for $\psi = 0$ and negligible lateral deviations will give Eq. (4.5.19b).

$$z_{CO,k} = z_k \quad (4.5.19a)$$

$$V_{x,k} = u_k + \theta w_k \quad (4.5.19b)$$

Or, in matrix form:

$$\begin{bmatrix} z_{CO} \\ V_x \end{bmatrix}_k = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & \theta \end{bmatrix} \begin{bmatrix} z \\ u \\ w \end{bmatrix}_k \quad (4.5.20)$$

The measurement covariance matrix $\mathbf{R}$ shall also be constant, diagonal with elements taken from Table 4.5 for distance and speed variance.

4.5.3 Tuning Parameters

The available tuning parameters are the $\mathbf{Q}$, $\mathbf{R}$ matrices as they have already been defined, the sampling time T_s and less importantly the initial guesses. Algorithm 2 is directly implementable on microcontroller software, and makes use of the exact DKF equations described previously. The different sensor updates are handled by isolating the appropriate $\mathbf{H}$, $\mathbf{R}$ components whenever a new measurement arrives (by a timestamp check for example).

⁶The sensor's internal fusion does provide a fair estimate for vertical speed as well, so the exact combination of components and measurements should be field tested

Algorithm 2: Corrector-predictor KF for different measurement frequencies

Input: Model Input $\mathbf{u}$, Measurement $\mathbf{y}$
Output: State Estimate $\hat{\mathbf{x}}$
Parameters: T_s , $\mathbf{Q}$, $\mathbf{R}$, $\mathbf{P}_0$, $\hat{\mathbf{x}}_0$
Persistent : $\mathbf{P}$, $\hat{\mathbf{x}}$, $\mathbf{y}_{\text{last}}$

```

1 if first call then
2   |  $\hat{\mathbf{x}} \leftarrow \hat{\mathbf{x}}_0$ 
3   |  $\mathbf{P} \leftarrow \mathbf{P}_0$ 
4 end
5 Set current  $\mathbf{F}$ ,  $\mathbf{G}$ 
6  $[\hat{\mathbf{x}}, \mathbf{P}] \leftarrow \text{kfPredict}()$  // Prediction step
7 if  $\text{getTime}(\mathbf{y}) \neq \text{getTime}(\mathbf{y}_{\text{last}})$  then
8   | Set appropriate  $\mathbf{H}$  and  $\mathbf{R}$ 
9   |  $[\hat{\mathbf{x}}, \mathbf{P}] \leftarrow \text{kfUpdate}()$  // Update step
10  |  $\mathbf{y}_{\text{last}} \leftarrow \mathbf{y}$ 
11 end
12 return  $\hat{\mathbf{x}}$ 
13 Function  $\text{kfUpdate}(\mathbf{y}, \hat{\mathbf{x}}, \mathbf{H}, \mathbf{P})$  is
14 | Eqs. (4.5.12) to (4.5.14)
15 end
16 Function  $\text{kfPredict}(\hat{\mathbf{x}}, \mathbf{P}, \mathbf{F}, \mathbf{G})$  is
17 | Eqs. (4.5.10) and (4.5.11)
18 end

```

Chapter 5

Simulation Results

Although the previous analysis introduced various controller designs and parameters, no final numerical values were presented. This is deliberate: selecting a control architecture without first evaluating time-domain responses or frequency-domain characteristics would be meaningless. The following sections present a series of scenarios and performance visualizations providing benchmarks that guide the controller selection process. Section 5.2 addresses the problem of tracking slowly varying heave references in the absence of disturbances or model mismatch (aside from inherent nonlinearities). Section 5.3 examines disturbance rejection, modeled using long-crested regular or random waves, and discusses the case of general irregular seas. State estimation is also seen functioning in short-crested seas. Section 5.4 introduces perturbed models and evaluates controller response over multiple iterations as a statistical measure. Finally, Section 5.5 presents a combined analysis and discussion of the results.

5.1 Setup & Objectives

The simulated controllers will adhere to the naming conventions of Table 4.1 and Table 4.3. As the LQR design is unusable outside of linear simulations, only the LQI variants are considered, especially LQI-SW. As for MPC, preliminary results use the base linear MPC scheme to explore augmentation effects, with LPV-MPC inheriting the optimal base design. It will be shown throughout this chapter, that selection of an architecture that guarantees stability at the presence of uncertainty, is a multi objective optimization problem that rarely has an “one-size-fits-all” solution. As is the case with robust control approaches such as $\mathcal{H}_\infty$, the optimal controller may provide stability against worst case margins of uncertainty, but will perform “worse” for the nominal model. By this, the tone of the remaining chapter is set: It is merely an attempt at providing various designs, their strengths and possible weaknesses, quantitatively measured with specified metrics, as a baseline for a practical implementation.

Controller Parameters The controller parameters, as given in Tables 4.2 and 4.4 were decided mostly through trial and error. A good starting point is usually given by Bryson’s rule [16], [74] utilized by Furrer [30] and Belime [1]. The topic of parameter optimization

lies outside the scope of the current work for reasons that will be discussed in Section 5.5. Tables 5.1 and 5.2 collect the base parameters that all controller designs share. The anti-windup gain $\mathbf{K}_a$ is calculated based on the resulting LQI gains, inverting the controllers integral part. All results are given for a sampling time of 0.1s. The $\mathbf{L}$ matrices for MPC are the scale factors for non-dimensionalizing the internal signals. Lastly, the input disturbance and internal plant models are given based on previous discussion.

Parameter	Value
$\mathbf{Q}$	$\text{diag}(4.00, 0.74, 2.05, 0.16, 0.16, 0.16, 4.00, 1.31, 5.25)$
$\mathbf{R}$	$\text{diag}(1.31, 1.31, 1.31)$
$\mathbf{K}_a$	$-2\mathbf{K}_I^\dagger$

Table 5.1: Final controller parameters for LQI.

Parameter	Value
H_p	10
H_u	2
$\mathbf{W}^y$	[4, 3, 4, 0.1, 0.1, 0.1]
$\mathbf{W}^u$	[0.2, 0.2, 0.2]
$\mathbf{W}^{\Delta u}$	[0.15, 0.15, 0.15]
$\mathbf{L}^y$	[0.2, 0.1745, 0.1745, 0.5, 0.5, 0.5]
$\mathbf{L}^u$	[0.1745, 0.1745, 0.1745]
ss(ID)	Input integrators, $\mathbf{B} = \text{diag}(0.04, 0.04, 0.04)$
ss(md1)	Eqs. (A.3.1) and (A.3.2) without u state

Table 5.2: Final controller parameters for MPC.

Scaling It is important to scale measured and commanded signals so every channel is unitless and of order one before applying control metrics. This happens as a normalization process to apply singular value decomposition, and it does not affect the controller creation process, as the parameters were defined previously. Normalization is performed by defining diagonal scaling matrices $\mathbf{D}_e$ and $\mathbf{D}_u$ (positive, typically with entries equal to the maximum expected or allowed deviation for each output/error and input respectively) and scaling signals as

$$\mathbf{y} = \mathbf{D}_e^{-1}\hat{\mathbf{y}}, \quad \mathbf{e} = \mathbf{D}_e^{-1}\hat{\mathbf{e}}, \quad \mathbf{u} = \mathbf{D}_u^{-1}\hat{\mathbf{u}}.$$

For a plant $\hat{\mathbf{G}}$ (mapping $\hat{\mathbf{u}} \mapsto \hat{\mathbf{y}}$), the scaled plant is

$$\mathbf{G}_s = \mathbf{D}_e^{-1} \hat{\mathbf{G}} \mathbf{D}_u. \quad (5.1.1)$$

The designed controller $\hat{\mathbf{K}}$ is also transformed into unitless $\mathbf{K}$ so that $\mathbf{u} = \mathbf{K}\mathbf{x}$ for full state feedback.

$$\mathbf{K} = \mathbf{D}_u^{-1} \hat{\mathbf{K}} \mathbf{D}_e \quad (5.1.2)$$

Same ofcourse applies to all important transfer function quantities, such as the loop transfer function $\mathbf{L} = \mathbf{D}_e^{-1} \hat{\mathbf{G}} \hat{\mathbf{K}} \mathbf{D}_e$. Thus, computing performance and robustness metrics (e.g.,

singular values, condition numbers, pole locations interpreted against allowable deviations) should be performed on the scaled system; otherwise, norms mix inconsistent physical units and yield misleading results [11].

5.1.1 Performance Metrics

In order to comparatively assess multiple controllers, and evaluate performance of a single design at different tasks, there is a need to include some sort of metric. Examples of control metrics are compactly presented in a preprint by Sarhadi [75], which also outlines the hardships of evaluation in practical applications. Equations (5.1.3) to (5.1.6) describe the exact mathematical nature of the used metrics, as well as their usual names in literature. Since we are interested in error signals of more dimensions, the control signal is a column vector, scaled as mentioned previously, giving $\mathbf{D}_e^{-1}\mathbf{e}(t) = [e_1(t), \dots, e_n(t)]^\top$. For control effort, either the commanded setpoint or actuator position may be used, but it is important to scale and remove offsets such that $\mathbf{u}(t) = \mathbf{D}_u^{-1}(\mathbf{u}_a(t) - \mathbf{u}_o(t))$ for example. Removing trim offsets makes sense, as they are a baseline value that the controller has to produce in order to maintain stability. By these assumptions, the produced metrics are in a sense, signal norms and can be treated as such [11][65].

$$\text{ISE: } \int_{t_0}^{t_f} \sum_i |e_i(t)|^2 dt = \|\mathbf{e}(t)\|_2^2 \quad (5.1.3)$$

$$\text{IAE: } \int_{t_0}^{t_f} \sum_i |e_i(t)| dt = \|\mathbf{e}(t)\|_1 \quad (5.1.4)$$

$$\text{ITAE: } \int_{t_0}^{t_f} t \sum_i |e_i(t)| dt \quad (5.1.5)$$

$$\text{Actuator Effort: } \int_{t_0}^{t_f} t \sum_i |u_i(t)|^2 dt = \|\mathbf{u}(t)\|_2^2 \quad (5.1.6)$$

$$(5.1.7)$$

In the frequency domain, we will briefly examine the singular values of the closed loop transfer functions, especially $\bar{\sigma}(\mathbf{S}_i(j\omega))$ and $\underline{\sigma}(\mathbf{S}_i(j\omega))$ to evaluate performance. Other important values are the maximum overshoot, as well as rise and settling time, but these are generally visually discernible using just the plots while the specific setting does not set critical requirements on such metrics (e.g, no hard limit on overshoot or settling time).

5.2 Reference Tracking

At the absence of exogenous disturbances and state measurement corruption by noise, the results of this section bear the most importance in choosing the optimal design. For this reason, we will inspect augmentations that will prove ineffective, and eventually decide on

three characteristic designs, one from the LQ class, one from linear MPC, and an adaptive LPV-MPC.

5.2.1 Step Response

The following experiments provide insight on tracking a step signal in heave. Starting with the simple LQI in Fig. 5.1, the three designs are compared, with -SW being the superior choice, as expected. Not using saturation creates unbounded signals, while not using anti-windup ends up with delayed removal of saturated input. Both problems are visible, with the latter appearing on the command signals, where the -S design delayed the fore starboard actuator reduction. Exact metrics for the experiment are presented in Table 5.3

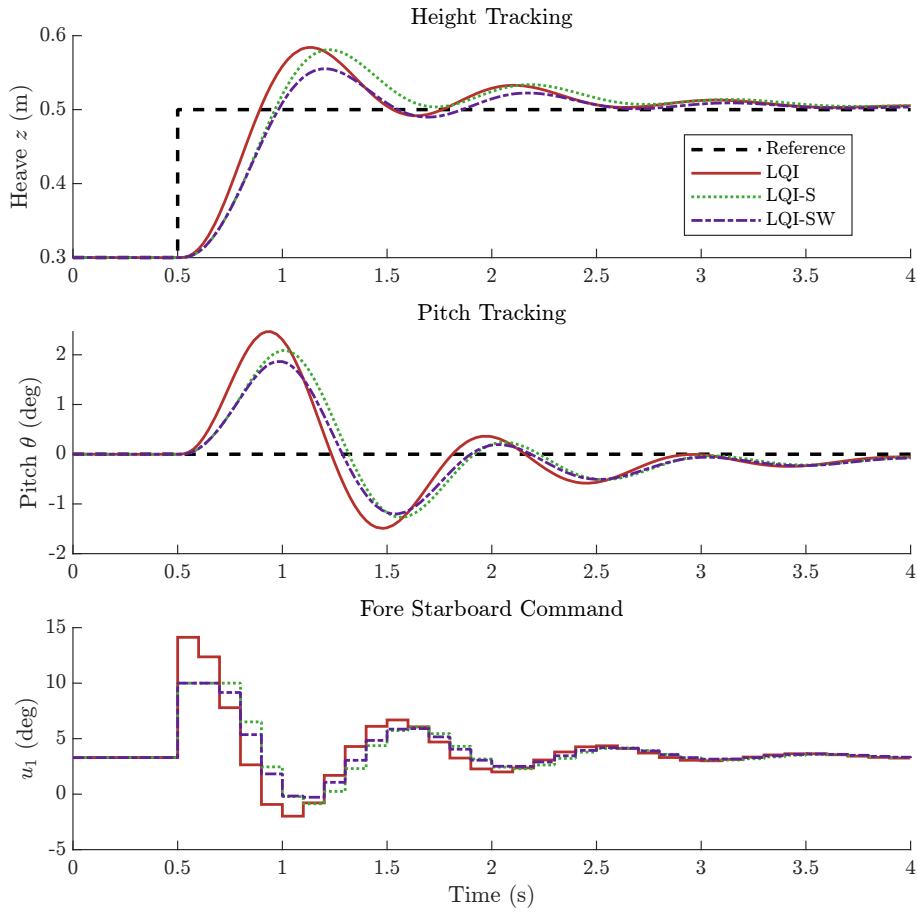


Figure 5.1: Response to step change of +0.2m in heave tracking for LQI controllers. Visible saturation and integrator windup effects.

Comparisons for the MPC class are not as straightforward. Taking a look at Fig. 5.2, input and output disturbance designs seem to fall behind compared to an augmented model accounting for actuator delay. At the same time, -O model seems to perform better than the rest of the prediction augmentations. But we should not be quick to judge. Firstly, the diagram simply illustrates that “perfect” model knowledge yields better performance,

Controller	ISE	IAE	ITAE	Act. Effort
LQI	1.3105	1.5771	1.9944	1.9372
LQI-S	1.4553	1.6928	2.2177	1.4160
LQI-SW	1.2841	1.4097	1.6821	1.2387

Table 5.3: Step response performance metrics for LQI controllers.

but it provides no insight into robustness. Secondly, a step response is far from a practical control system requirement. What *is comparable*, is that MPC has the potential to outperform LQI, especially in actuator effort, visible in Table 5.4 and in overshooting requirements. This brings us to a previously introduced MPC benefit: anticipative action. All final controller objects have reference previewing enabled, producing the responses in Fig. 5.3. Once again, the -IDP variant proves superior, completely eliminating overshoot while conserving energy spent on actuation Table 5.5. In a way, this design anticipates both the reference trajectory, as well as the actuator dynamics.

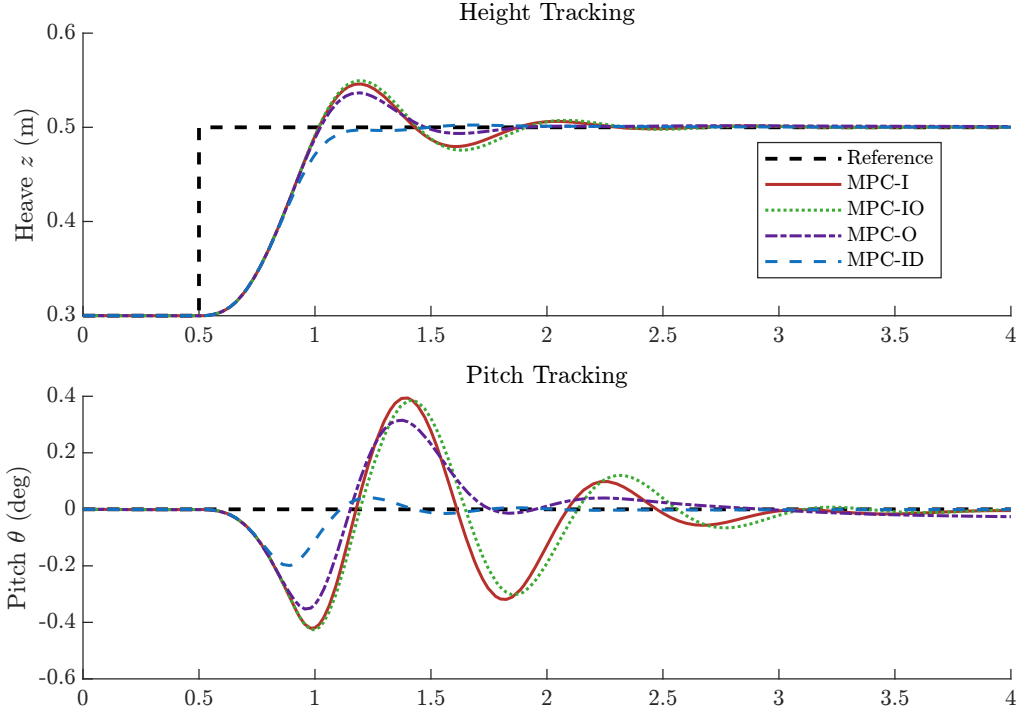


Figure 5.2: Response to step change of +0.2m in heave tracking for MPC controllers with no reference previewing. Generated input signals are omitted as these are not final designs.

To close this section, and as an introduction to trajectory effects, consider Fig. 5.4, where the inspection of Fig. 5.4a and Fig. 5.4c was what gave rise to considering actuator response, and including the delay model in predictions. A controller that has no way to account for command signal delay, expects the plant to act instantaneously, inducing aggressive over correction when this is not the case, visible in Fig. 5.4a. Even if a smoother reference signal is used as in Fig. 5.4c, this behavior is only damped but not corrected. Figures 5.4b and 5.4d present an optimal design that accounts for delays, but it is im-

Controller	ISE	IAE	ITAE	Act. Effort
MPC-I	1.3167	1.0583	0.9320	1.5396
MPC-IO	1.3326	1.0911	0.9844	1.5830
MPC-O	1.2927	0.9725	0.8117	1.2935
MPC-ID	1.2728	0.8512	0.6116	0.9284

Table 5.4: Step response performance metrics for MPC controllers without reference previewing.

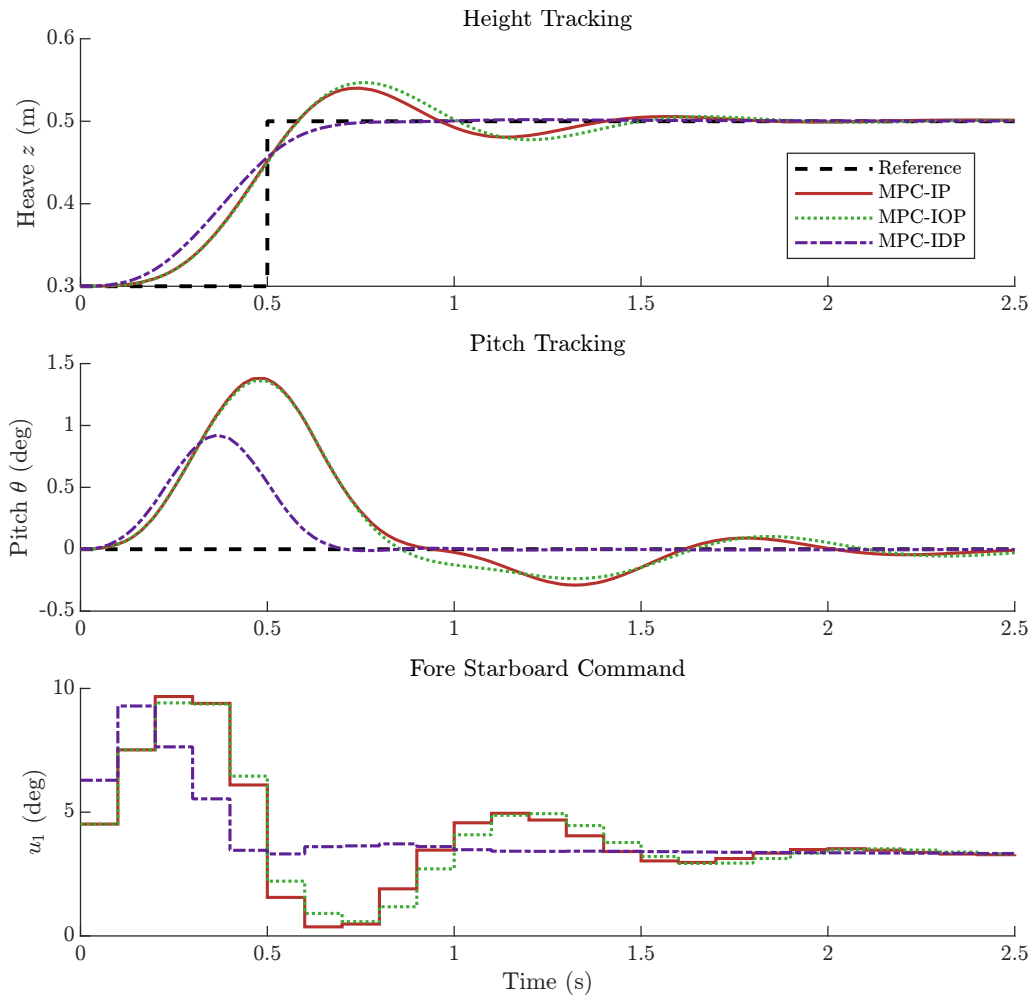


Figure 5.3: Response to step change of +0.2m in heave tracking for MPC controllers with reference previewing active. Significantly smaller settling time.

Controller	ISE	IAE	ITAE	Act. Effort
MPC-IP	0.1304	0.4467	0.3185	1.0725
MPC-IOP	0.1494	0.4922	0.3683	1.0730
MPC-IDP	0.1395	0.2923	0.1284	0.6470

Table 5.5: Step response performance metrics for MPC controllers with reference previewing active.

portant to note that if the underlying plant changes delays, the problem will persist. All plots include the internally calculated optimal control sequence for $H_c = 2$ and predicted trajectory for $H_p = 10$.

Generated Trajectory Effects The reader is once again pointed to Fig. 1.7, as to how a reference is generated separately from the controller. Providing an optimal setpoint reference is a wholly different problem, pertaining to path planning and motion control. However, since many results will require some setpoint to be given, a brief consideration is appropriate. Figure 5.5 shows the resulting behavior when the reference changes from a step to a ramp or a “smooth” ramp. Also compared are the settling times of MPC vs LQI again, as well as the preview capabilities of the former.

The smooth ramp signal is internally calculated as an interpolant between a final value y_f occurring at t_f and an initial (y_0, t_0) . A cubic Hermite spline was chosen having an analytical expression of Eq. (5.2.1) in the unit interval $[0, 1]$.

$$\mathbf{p}(t) = (2t^3 - 3t^2 + 1) \mathbf{p}_0 + (t^3 - 2t^2 + t) \mathbf{m}_0 + (-2t^3 + 3t^2) \mathbf{p}_1 + (t^3 - t^2) \mathbf{m}_1 \quad (5.2.1)$$

Without loss of generality, the initial point can be picked as 0, while assuming 0 derivatives at starting end ending points, and a change of variable such that $t_n = (t - t_0)/(t_f - t_0)$, then Eq. (5.2.2) yields the smooth ramp interpolant for reference tracking.

$$y(t_n) = y_0 + (3t_n^2 - 2t_n^3)y_f \quad (5.2.2)$$

Performance metrics are omitted because they do not provide a meaningful comparison basis for this case. Upon visual inspection of the graphs, the ramp response appears to require less energy for tighter tracking, but this may come at a cost of actuator wear and ride comfort. This of course introduces yet another metric, regarding the generated motion velocities and accelerations and how they affect ride quality, but this topic lies well beyond the current scope.

5.2.2 Multiple Output Reference

The previous tests implicitly required that all states should follow their given reference, however the generated reference for all but heave, was zero. Unfortunately it is once again that we find ourselves facing the issue of setpoint generation described in Fig. 1.7. The designed controllers are capable of driving multiple outputs to a given reference, as shown in Fig. 5.6 for the LQI-SW variant, with apparent beneficial effects for overall performance. This tactic also falls under the category of feed-forward control, as it is required to have

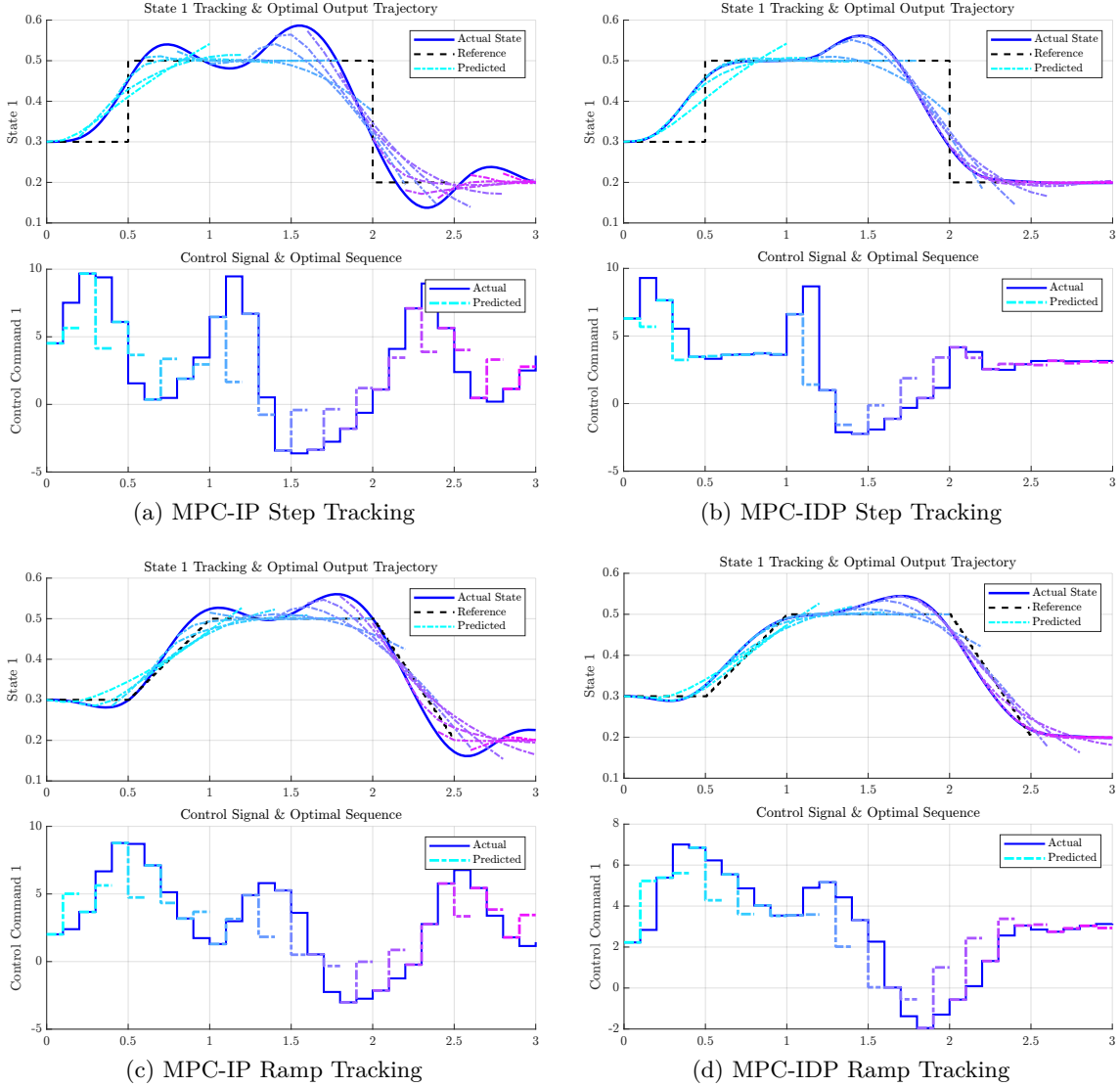


Figure 5.4: Effect of actuator delay modelling in MPC response. Controller output and optimal input predictions are shown at $f = 5$ Hz.

some means of estimating appropriate trajectories for all outputs, given a desired steady-state. In the previously mentioned figure for example, introducing a pitch reference results to better transient behavior while at the same time reducing unneeded integrator increase which in turn produces oscillations. In a more quantitative view, while the single reference tracking task achieved an $ISE=1.46$ and control effort of 1.19 , including the pitch reference lowered these metrics to 0.8 and 0.77 respectively. Given the previous discussion on results, the final selected controllers for subsequent analysis shall be LQI-SW from the LQ family, and MPC-IDP/-IP, MPC-LPV-IDP/-IP from the MPC family.

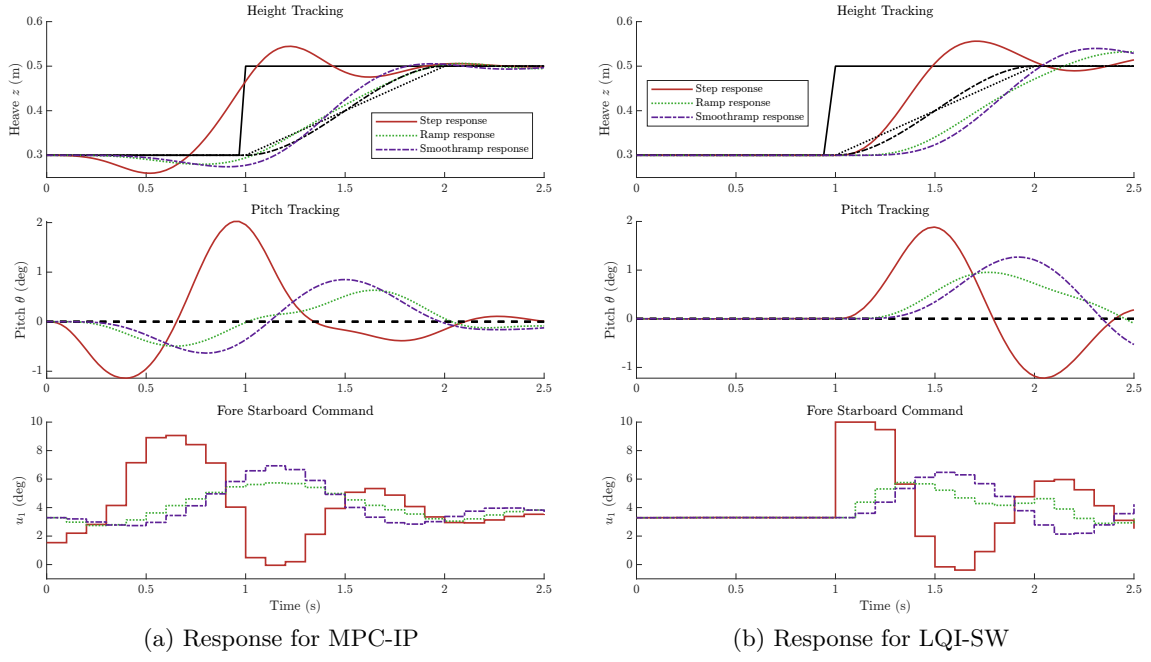


Figure 5.5: Effect of different reference trajectories.

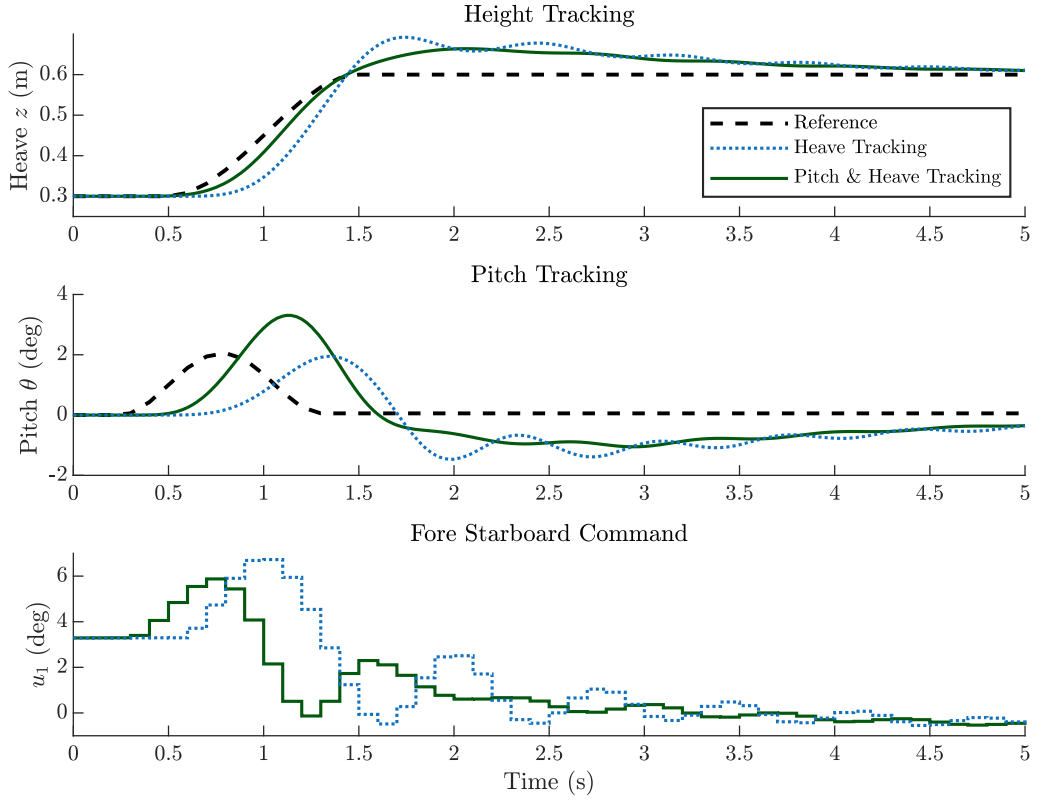


Figure 5.6: Response for multiple output reference tracking, LQI-SW. The dotted lines are trajectories generated for zero pitch reference.

5.3 Disturbance Rejection

This section presents the controlled system behavior subject to various type of disturbances. Performance is evaluated using frequency domain tools, cumulative data diagrams and statistical methods. Disturbance causes that will be investigated include

- Sea state, modelled by regular and irregular waves.
- Speed dependent uncertainty of the underlying dynamics.
- Force/Moment disturbances, such as lateral gusts.
- Lack of “perfect” state knowledge and injection of sensor noise.

General robustness tests and model uncertainty considerations, are presented in a separate section, but may also fall under the disturbance rejection category.

5.3.1 Frequency Response

Disturbance rejection is often studied in the frequency domain, which is why the developed MPC variants were cast into linear state feedback in state space form. By appropriate scaling, results are comparable with the simple LQI designs by looking at the singular value plots in Fig. 5.7. Unfortunately, such an analysis *does not* include saturation effects, as these are non-linear in nature. There is a noticeable similarity with the “desired” behavior dictated by Figs. 4.2 and 4.3, with the distinct difference that frequencies of interest are defined as a range, bounded by the minimum and maximum singular values.

The plots concern the input transfer functions as discussed in Section 4.2.1, while the Nyquist frequency ω_N for the specified sampling rate is also plotted as a vertical line. In general, signals that include components at frequencies larger than ω_N will experience aliasing, and fold into distorted signals of less perceived frequency. This might cause the controller to respond to noise as if it was a disturbance, which is why anything above ω_N should be filtered before entering the closed loop. Regarding important frequency properties, the MPC-ID design for example, presents with a cutoff $\omega_c = 8 \text{ rad} \cdot \text{s}^{-1}$ to $11 \text{ rad} \cdot \text{s}^{-1}$ and a bandwidth defined as in Section 4.1 falling between $\omega_B = 13 \text{ rad} \cdot \text{s}^{-1}$ to $17 \text{ rad} \cdot \text{s}^{-1}$, both ranges lying at higher bands than the other designs. Given the nominal model, we expect superior reference tracking and disturbance rejection from MPC-ID, while there is a visible sensitivity peak for the LQI-SW design close to $\omega = 10 \text{ rad} \cdot \text{s}^{-1}$, further hindering performance.

The LPV variants are missing from the analysis because they are once again non-linear in nature, and perform exactly the same as their basic counterparts, for the nominal model at nominal speed.

5.3.2 Waves

Waves and their sinusoidal nature are tightly related to the frequency response behavior dictated by the singular value analysis. Using regular waves at a frequency range $\omega_w = 1.8 \text{ rad} \cdot \text{s}^{-1}$ to $6.5 \text{ rad} \cdot \text{s}^{-1}$ and amplitude $\zeta_a = 0.1 \text{ m}$, the heave and pitch response amplitudes were calculated for the three different designs at head-on and following seas. Details on the selection of wave parameters are given in Appendix A.2, following theoretical results of Section 2.4.2. Results are plotted in Fig. 5.8 against frequency of encounter,

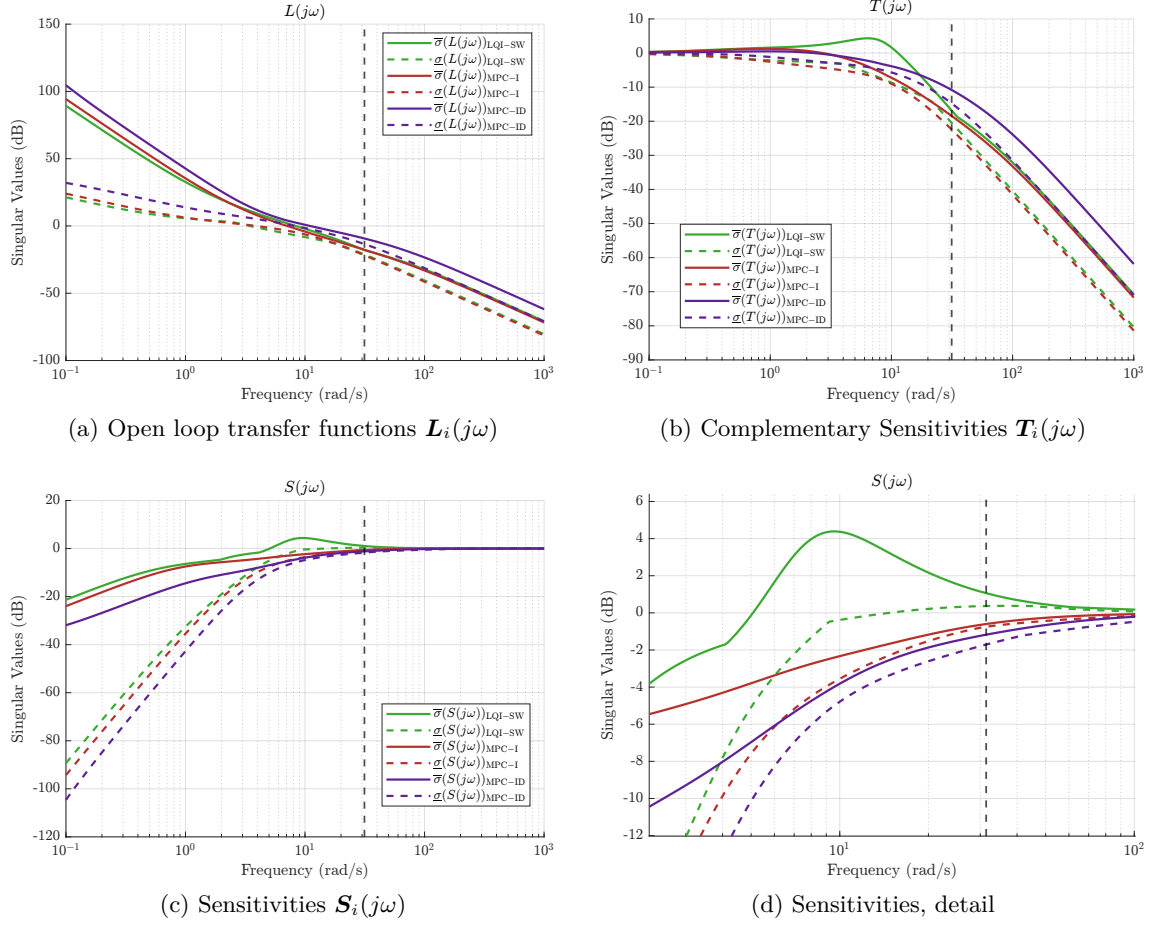
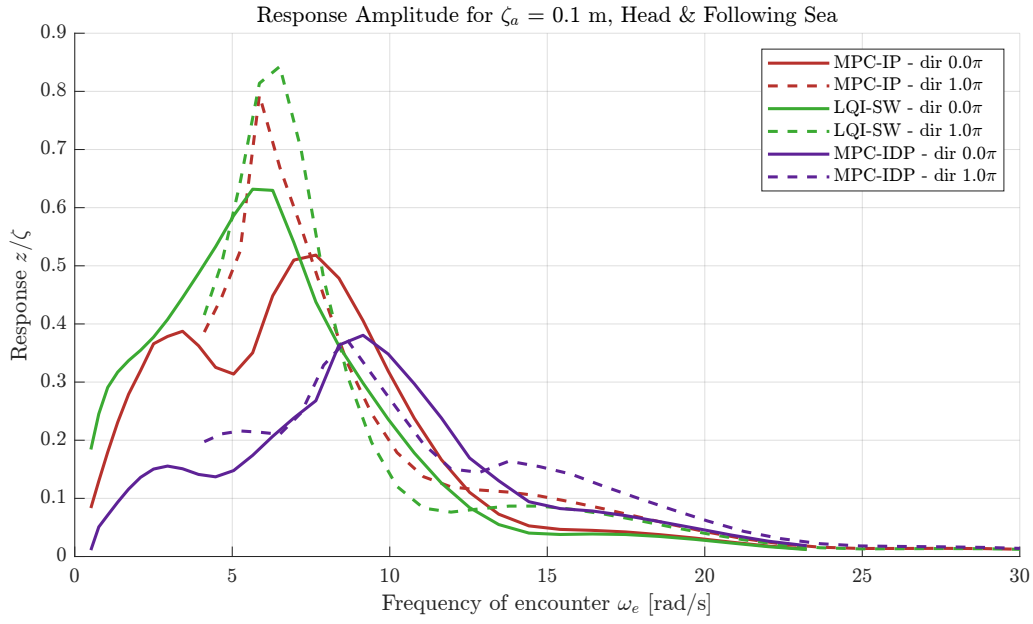
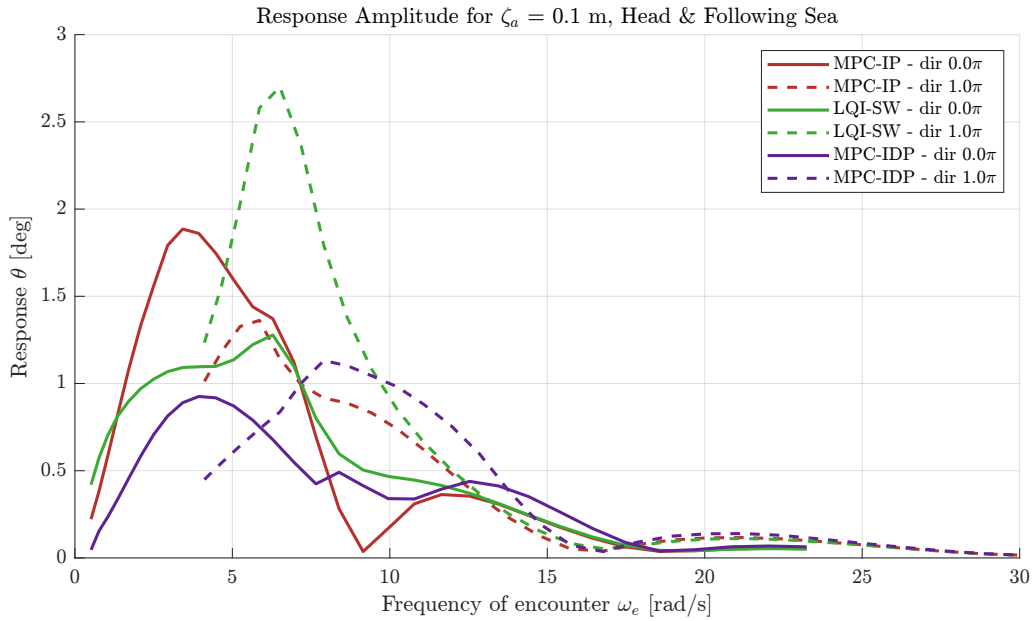


Figure 5.7: Comparison of singular values for different designs. Vertical line signifies the Nyquist frequency $\omega_N \approx 31.4 \text{ rad} \cdot \text{s}^{-1}$ for $f_s = 10 \text{ Hz}$.

according to Eq. (2.4.13). The same results are also plotted in Fig. 5.9 against the encountered wavelength, using Eq. (2.4.7) at the encounter frequency. As expected, the MPC-ID achieves overall less amplitudes for the exact same wave excitation. As a realistic sea will be represented by a multi-chromatic assortment of wave components, both the MPC variant seem capable of achieving better overall ride quality.

Effects of resonance appear in both motions for Figs. 5.8 and 5.9, related to the controller's response characteristics. At the absence of a hydrostatic restoring force, which is usually dictating the resonant peaks, the controller sets the restoring coefficients. Behavior such as that was also outlined by Hatzakis [26]. At the same time, since there is no selection between platforming and contouring modes, lower frequencies (thus higher wavelengths) cause larger response peaks in following seas which are also discussed by Kim et. al [23], [24].

(a) Heave amplitude normalized to ζ_a .

(b) Pitch amplitude.

Figure 5.8: Comparison for regular wave disturbance on a frequency sweep, head and following sea, x -axis referring to encounter frequency for $V_s = 7 \text{ m} \cdot \text{s}^{-1}$.

5.3.3 Additional Scenarios

Some additional disturbance scenarios are included to further assess the reliability of the proposed designs.

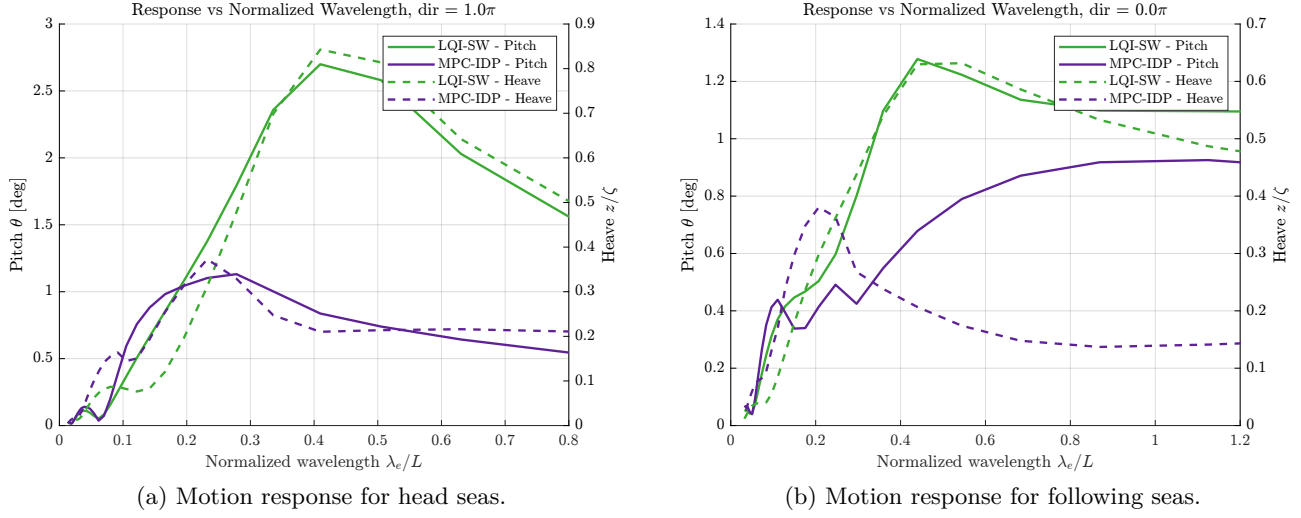


Figure 5.9: Response amplitudes for various wave frequencies with respect to encountered wavelength.

Take-off & Touchdown An important maneuver is of course that of taking off and “landing” again, accompanied by a sudden increase and decrease of velocity. The model does not include hull dynamics, which means that the results are only indicative of the moments just after take off and just before touchdown. This is depicted in Fig. 5.10, where body-fixed forward velocity fluctuates at the same with a heave reference. All designs are able to complete the maneuver, but LQI shows a delay in adapting to the changing system dynamics whereas MPC based designs closely track the given reference, rejecting both wave and speed disturbances.

Roll Control While the lateral dynamics of the simulated vessel are probably not adequately modelled, for most simulations the non-linear model was free in the lateral DOF as well. Figure 5.11 presents such a case, where a $400 \text{ N} \cdot \text{m}$ disturbance acts on the x_b axis moment, inducing a large deviation in roll. The vessel is also traveling at a less than nominal speed ($5 \text{ m} \cdot \text{s}^{-1}$ to $6 \text{ m} \cdot \text{s}^{-1}$), fighting with actuator constraints. The controller reacts by lowering the port foil to negative values, returning to normal operation when the disturbance is removed. The same test, when run with a locked lateral deviation (v velocity component set to 0), expectedly showed lesser heave deviations induced by roll motion.

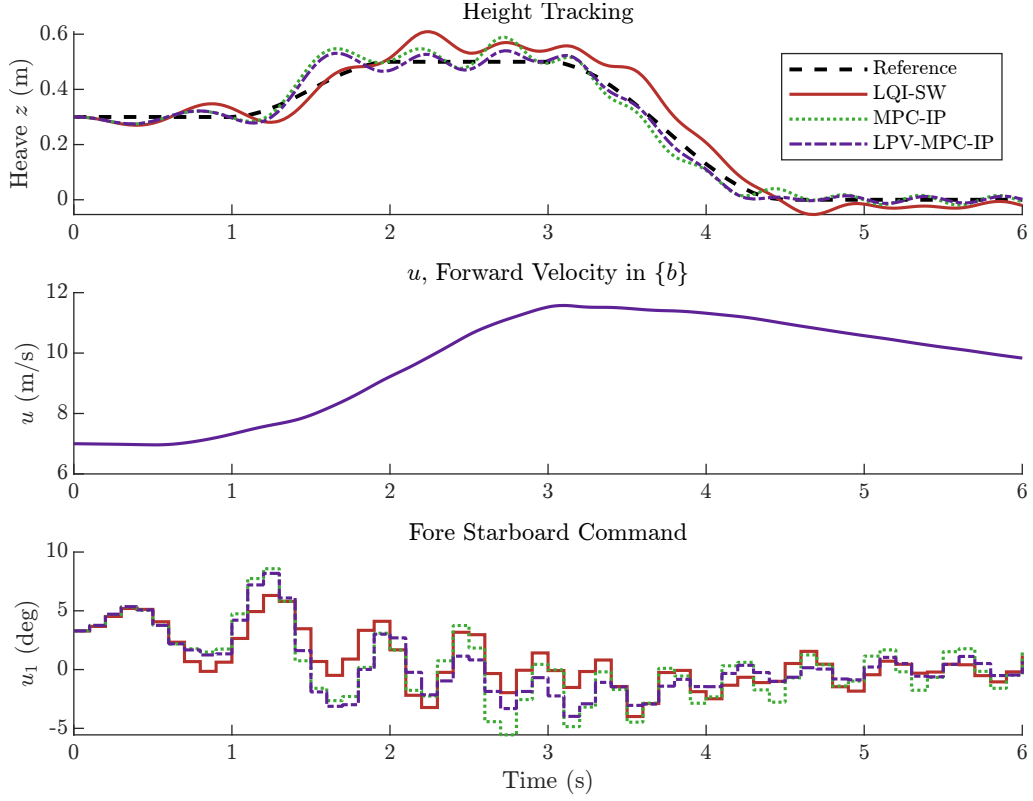


Figure 5.10: Controller performance at a take-off/touchdown maneuver with regular wave disturbance of $\omega_e = 4 \text{ rad} \cdot \text{s}^{-1}$ and wave direction 0° .

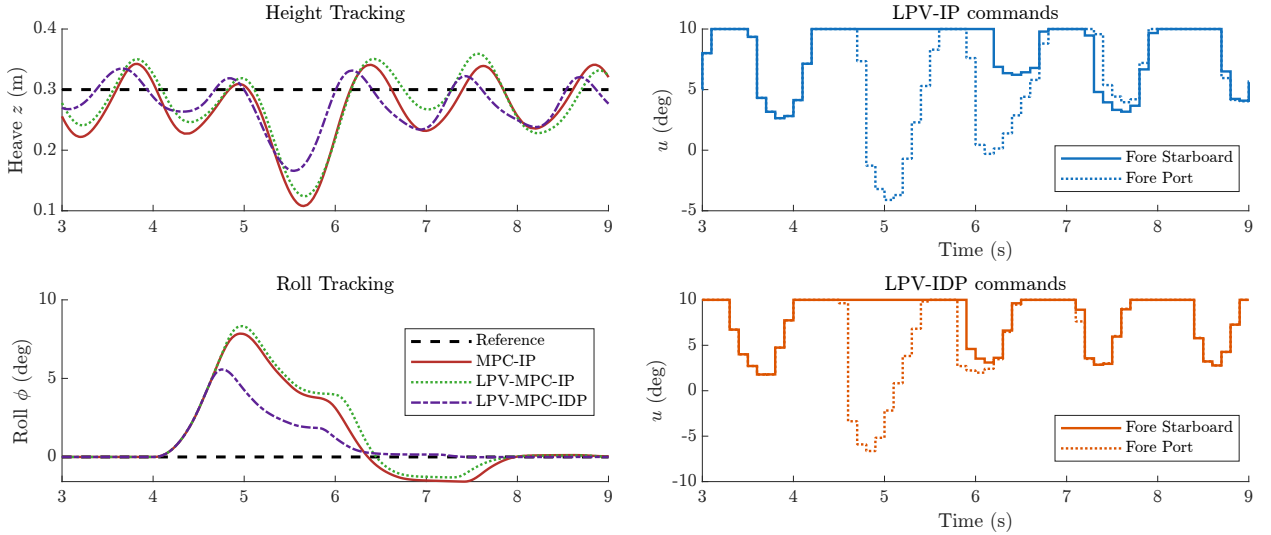


Figure 5.11: Controller performance at a disturbance in K moment, applied for $t \in [4, 5.5] \text{ s}$ and regular wave disturbance of $\omega_e = 4 \text{ rad} \cdot \text{s}^{-1}$ for wave direction 0° .

5.3.4 Sensor Feedback

Up to this point, full state feedback had perfect knowledge of all states. In a realistic scenario, sensors such as the ones introduced in Section 4.5 inject noise in the signals used for feedback. At the same time, some states are not known at all, and the use of state estimators introduces further distortion acting as a combination of noise sources. Simulations were run for short-crested irregular seas based on the ITTC spectrum defined in the MSS toolbox. Selected parameters were $H_s = 0.4$ m, cutoff frequency $\omega = 6$ rad·s⁻¹, mean wave direction of π , 40 frequencies in grid and a direction cutoff of $\pi/8$. Before the measurement signal reaches the DKF, there is also a second order butterworth filter implemented, with a cutoff frequency at 2 rad·s⁻¹. The results are given in Figs. 5.12 and 5.13. Notice that both figures present measurements on a *separate* y -axis, to avoid shrinking the estimates. Figure 5.12 compares all estimated quantities (z, u, w) with the actual state values and the available measurements. As the DKF runs at 100 Hz, it is able to respond to changes in smaller intervals.

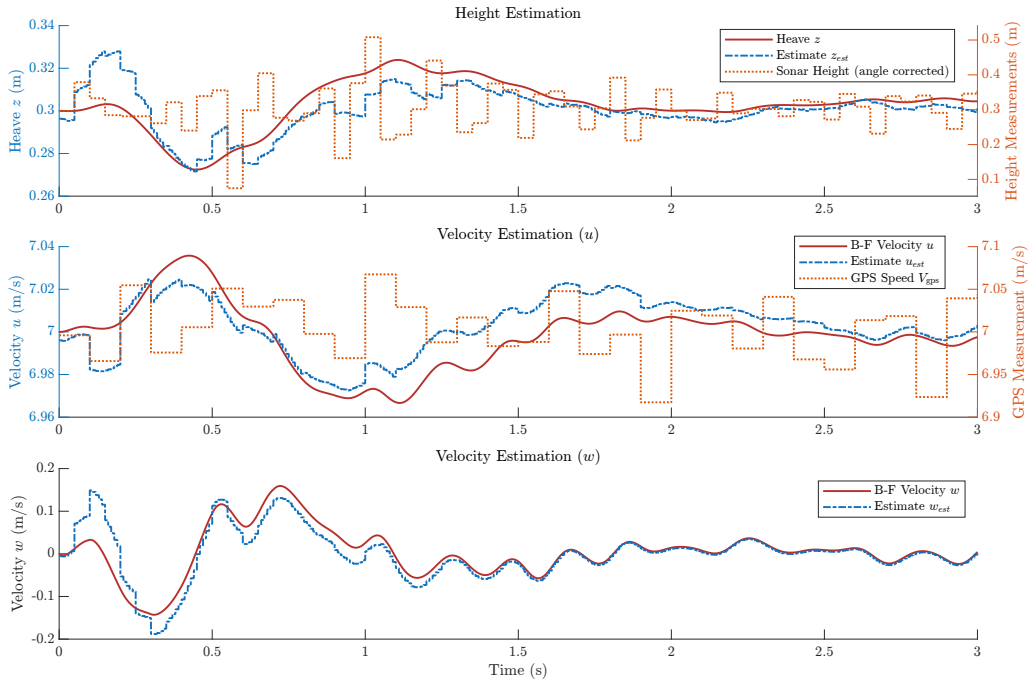
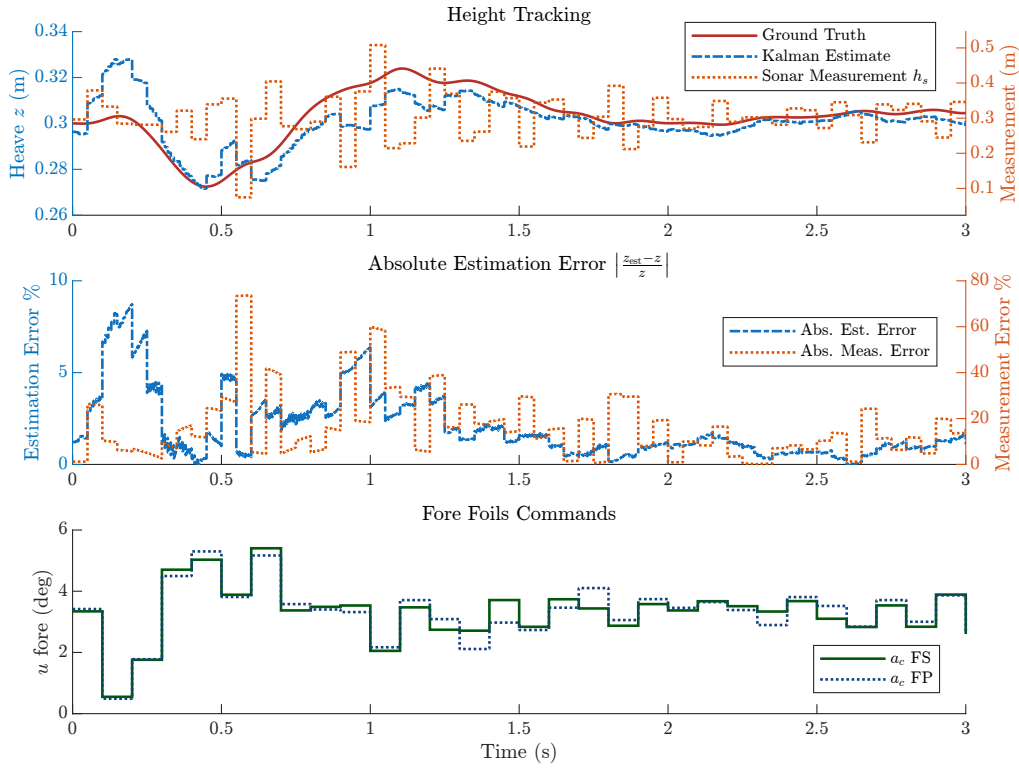
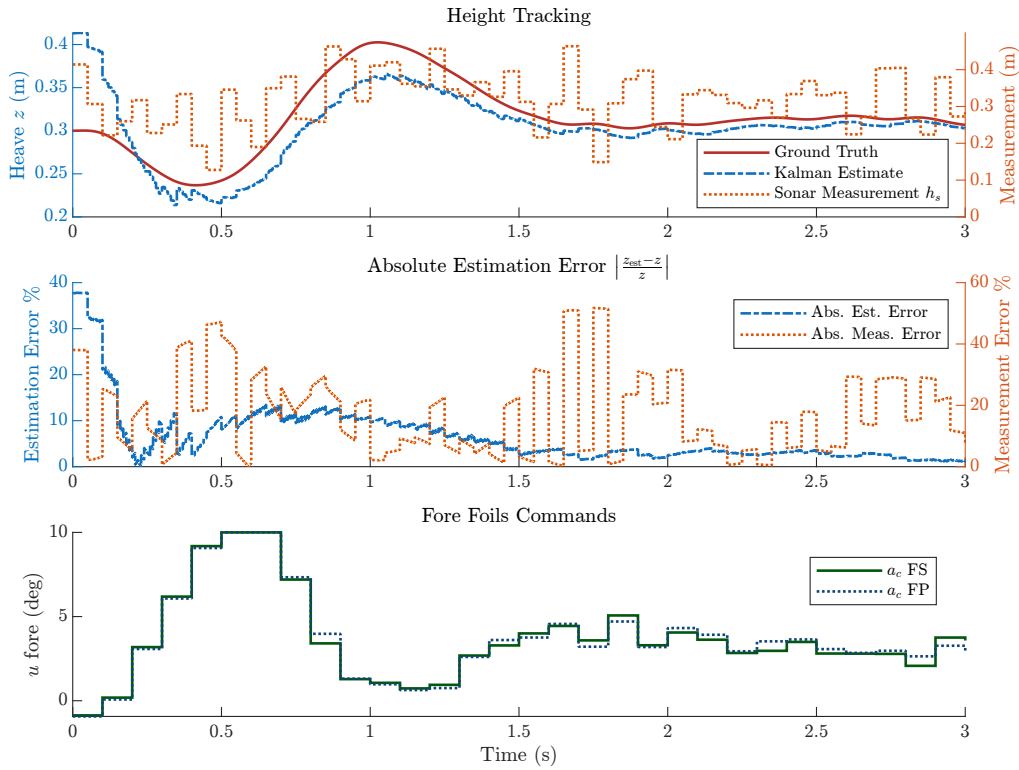


Figure 5.12: Comparison of ground truth, estimated and measured quantities, occurring at different sample rates.

The estimator's convergence rate is also dependent on the initial guess, which is set equal to the first measurement. Shown in Fig. 5.13, a “bad” initial guess takes time to be corrected, inhibiting the filter's error reduction. While a good initial guess keeps absolute estimation error well below 10% from the beginning, the same levels may take more time to be reached from a bad initial guess. In both cases however, the filter is able to reduce error significantly, providing adequate correction to the measured data.



(a) Initial estimate close to real value.



(b) Initial estimate lying far from the real value.

Figure 5.13: Effect of initial guess on eventual DKF convergence.

5.4 Model Uncertainty

What is the case in most real-world applications, is unfortunately either not touched upon in many control textbooks, or hidden under difficult to navigate mathematical rigor. The model on which the controller is designed, will *never* coincide with the real plant on which it will be deployed. What we can hope to do, is devise ways to either obtain better estimates via system identification, or “close” that modelling gap, by adaptive strategies. This was discussed in Chapter 4, and adaptive behavior against changing dynamics has already been established in the previous section. Another method to assess robustness against parametric uncertainty is the introduction of probabilistic tools. This has been the topic of many publications by R. Tempo, which led to valuable insights, merging uncertain linear system theory with probability [76]. The following also took inspiration from Liu et al. [77], where the controller was designed on perturbed systems and then used to control the nominal. Subsequent discussion draws upon chapters 6 and 7 of [76].

Given a nominal system $f(\mathbf{x}(t), \mathbf{u}(t))$ and its parametrically perturbed counterparts $f(\mathbf{x}(t), \mathbf{u}(t), \mathbf{p})$ for time invariant parameters, we wish to assess stability of a controller $\mathbf{K}$ designed on the nominal plant against all possible perturbations. This set-up deviates from the usual $M - \Delta$ format, but is directly implementable on the time-domain Simulink model. It is a way of calculating stable regions based on Monte Carlo sampling.

For N randomized trials, we perform the following

- Randomly select parameter vector from a uniform distribution such that $\mathbf{p} \sim U(\mathbf{p}_l, \mathbf{p}_u)$.
- Initialize the perturbed model $f(\mathbf{x}(t), \mathbf{u}(t), \mathbf{p})$, and simulate response with the pre-defined controller $\mathbf{K}$.
- Assess steady state convergence, $\mathbb{E}[\mathbf{e}(t)] < \mu_{\text{tol}}$ and $\text{Var}(\mathbf{e}(t)) < \sigma_{\text{tol}}^2$ for $t > 0.9t_{\text{sim}}$.
- Collect metrics for converging runs.

The results which will be analyzed were obtained for $N = 400$ runs, on LQI-SW, MPC-IP, LPV-MPC-IP and LPV-MPC-IDP controllers. Selected parameters were mass m , CG offset $\mathbf{r}_g^b$ and inertia tensor $\mathbf{I}_g$, with the exact parameters and bounds shown in Table 5.6. These bounds were further restricted to 70% of their original values, but the level of restriction is configurable. No waves are generated, and there is an increase of speed by $3 \text{ m} \cdot \text{s}^{-1}$ occuring at $t = 1 \text{ s}$.

Parameter	Lower	Upper	Units
m	165.0000	265.0000	kg
x_g	-0.5500	0.5500	m
y_g	-0.3100	0.3100	m
z_g	-0.1500	0.1500	m
I_{xx}	377.2500	628.7500	$\text{kg} \cdot \text{m}^2$
I_{yy}	23.2500	38.7500	$\text{kg} \cdot \text{m}^2$
I_{xy}	0.0000	8.0000	$\text{kg} \cdot \text{m}^2$

Table 5.6: Parameter bounds for probabilistic stability assessment.

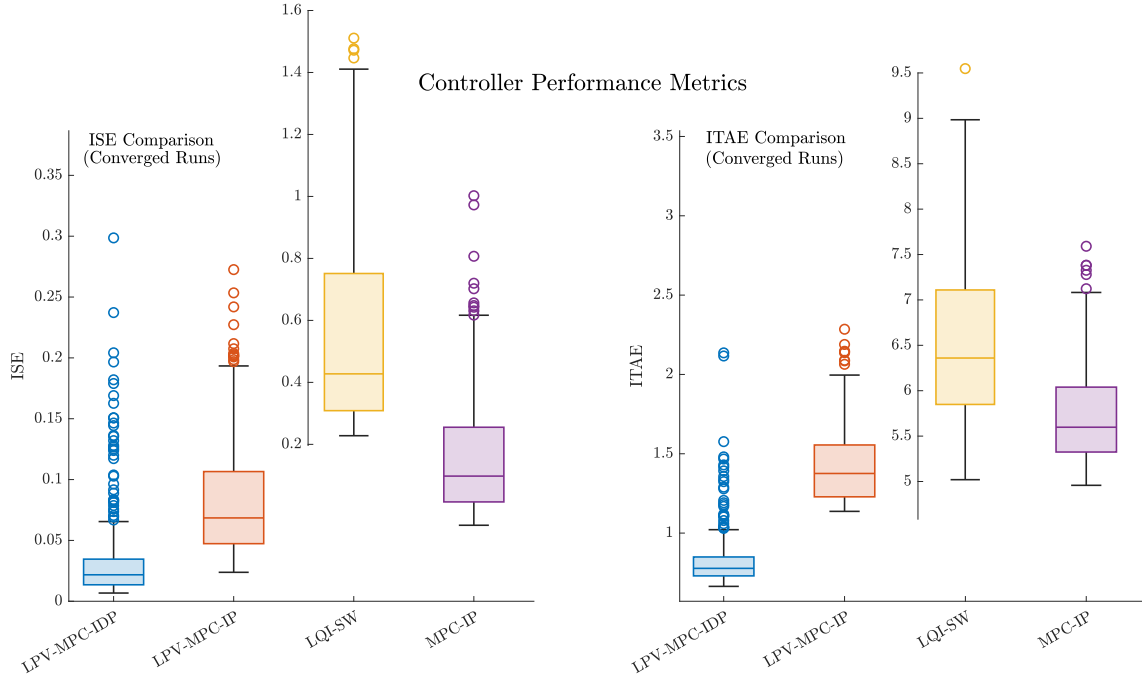


Figure 5.14: Boxplot analysis of different control designs at $N = 400$ trials. Visible outliers and vertical scale offset for LQ-SW.

To avoid presenting too many similar plots, only the responses of LPV-MPC-IP and LQI-SW were selected, but overall results will relate to all four designs. In Fig. 5.14, box plots illustrate the results based on the ISE and ITAE metrics. The latter is important, as the controller will deviate in the beginning for large parameter offsets, adjusting for the new system. Thus, sustained errors should be weighed heavily. The LQ design is almost not visible at the plot's scale as it acquired significantly higher error scores, which is why it is presented on its own y -axis.

Figures 5.15 and 5.16 summarize some of the most important findings in this work. The adaptive MPC with internal disturbance observer successfully tracks the given reference for about 98% of the sampled plants. Given its simplicity, LQI performs fairly well, failing to stabilize at a very similar region. The rate of non-convergence (defined as N_{unstable}/N) for LPV-MPC-IDP was 1.5%, for the same adaptive design without delay prediction it reached 2.75% which was about the same for the basic MPC-IP (2%). The same metric for LQI stood at 15%, as is visible in Fig. 5.16b.

Lastly, Figs. 5.17 and 5.18 characterize the controller's behavior subject to deviations in mass and horizontal placement of the center of gravity.

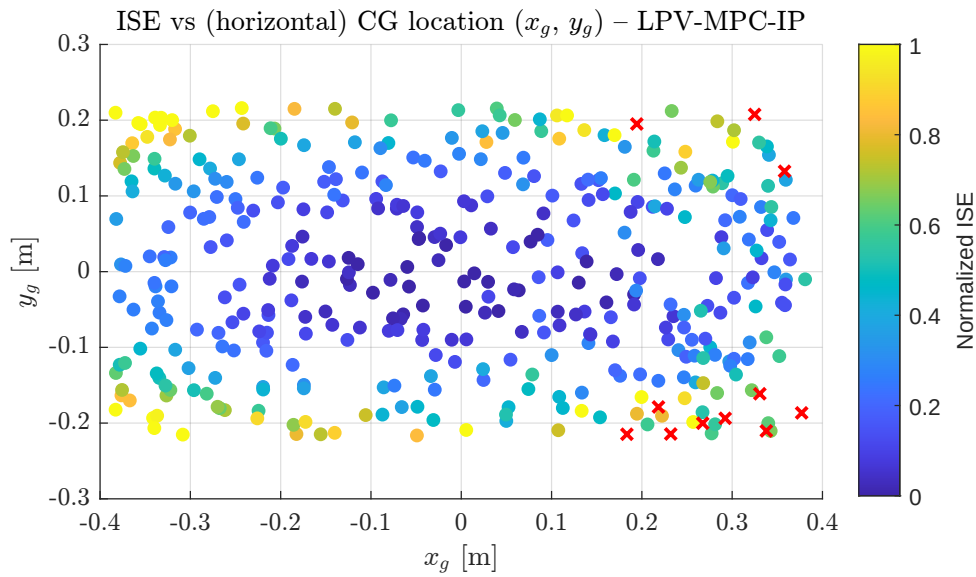
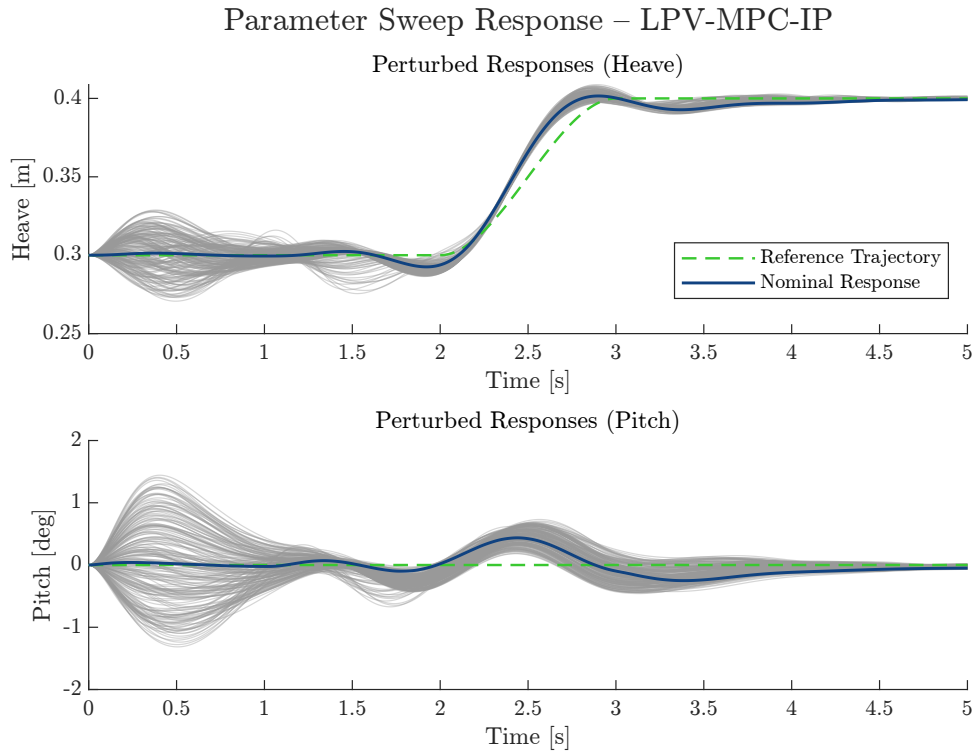
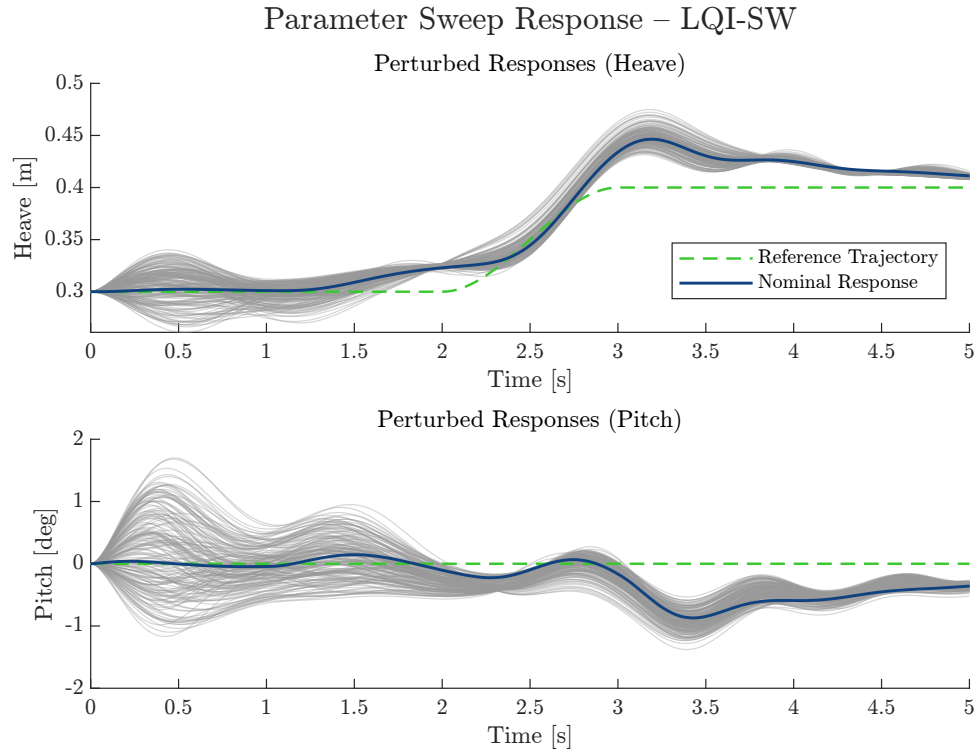
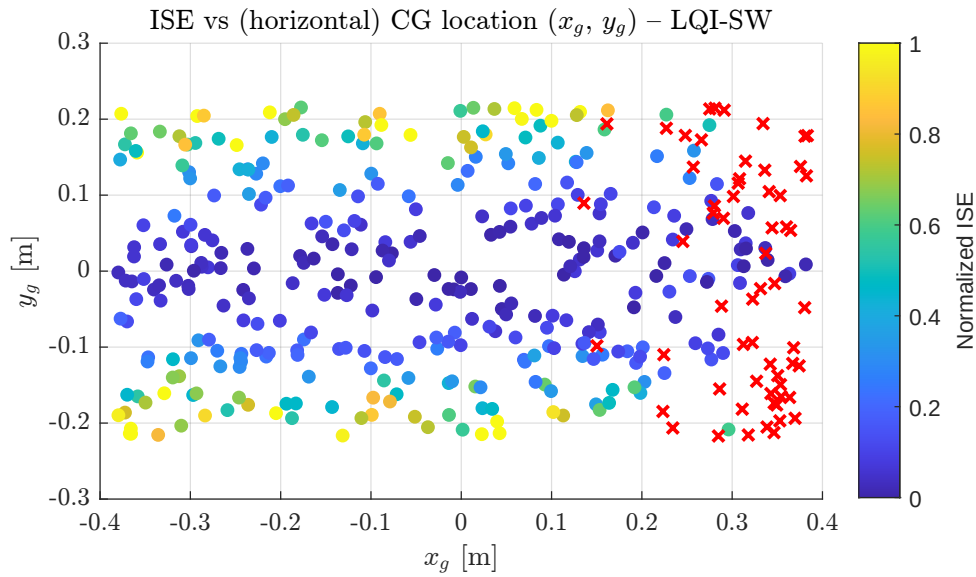


Figure 5.15: Response for reference tracking and disturbance rejection under model uncertainty, LPV-MPC-IP.



(a) Heave and pitch response of 200 trials.



(b) Performance heatmap for 400 trials. Unconverged are marked 'x'

Figure 5.16: Response for reference tracking and disturbance rejection under model uncertainty, LQI-SW.

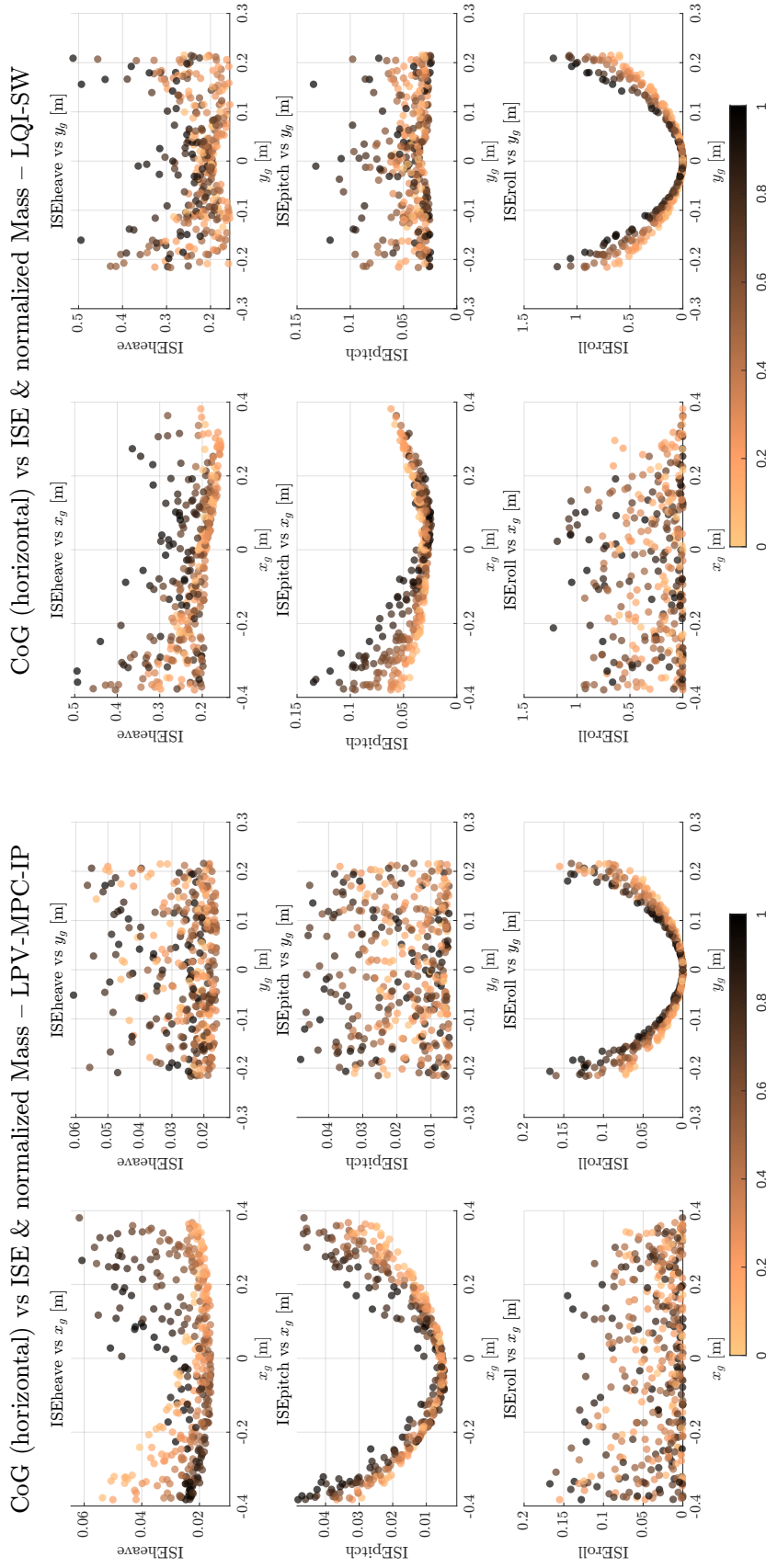


Figure 5.17: Correlation of mass and horizontal CG position to performance: LPV-MPC-IP.

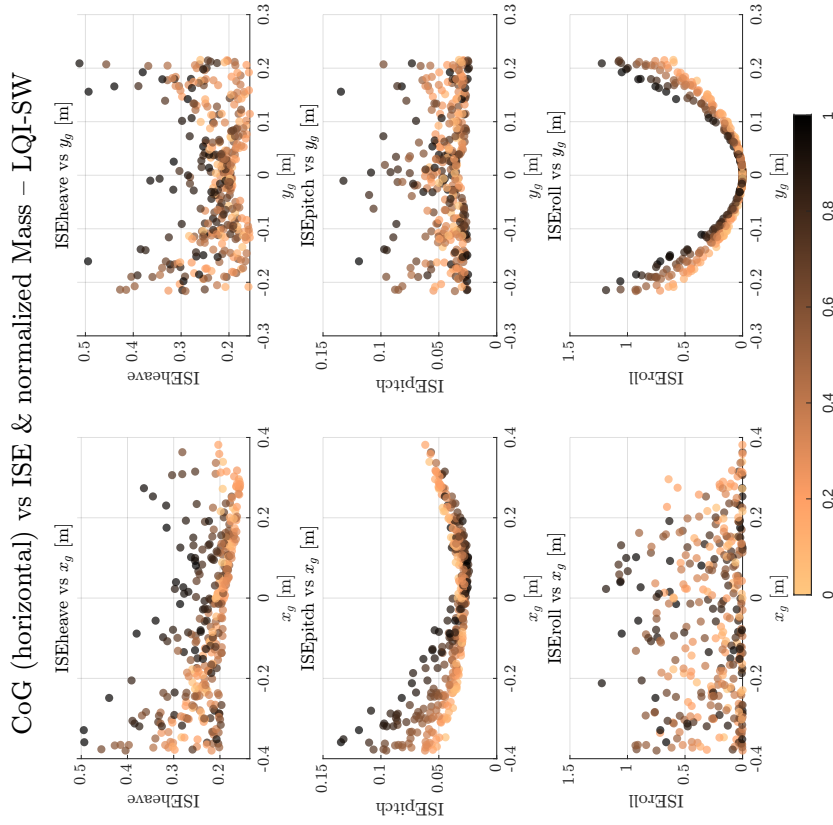


Figure 5.18: Correlation of mass and horizontal CG position to performance: LQI-SW.

5.5 Discussion

Before presenting the cumulative conclusions of this thesis, it is imperative to discuss the findings of the previous sections. This will be a brief section consisting mostly of remarks, as most results have already been analyzed in their respective section.

Controller Parameter Selection The final controller parameters were admittedly decided arbitrarily, but in the case of low order models little can be gained from optimizing performance by parameter selection. After all, even the decision of the objective function is ill-defined, as no single metric holds all important information [75], and applying a multi-objective optimization problem would require computational effort for minimal improvement in performance. What remains important, is the comparison between various architectures of the same parameters, guiding us into appropriate methods, further tuned on the real system.

Performance Metrics There are many ways to quantitatively verify an algorithm's performance, and the ones picked are only a small subset. The selected metrics depend on the project's specifications, available computational resources, modelling error margins and so on. It goes without saying that differing metrics, yield different optimal designs.

Wave Response The wave responses shown in Fig. 5.8 that were re-plotted against λ/L in Fig. 5.9 are comparable to the ones in Fig. 5.19. There is a need for caution however, as the diagrams by Faltinsen refer to uncontrolled flight, and there are apparent deviations between them and the graphs presented earlier. This comparison however, including the collection of Response Amplitude Operators from real operational or experimental data, can prove beneficial in validating the Simulink model.

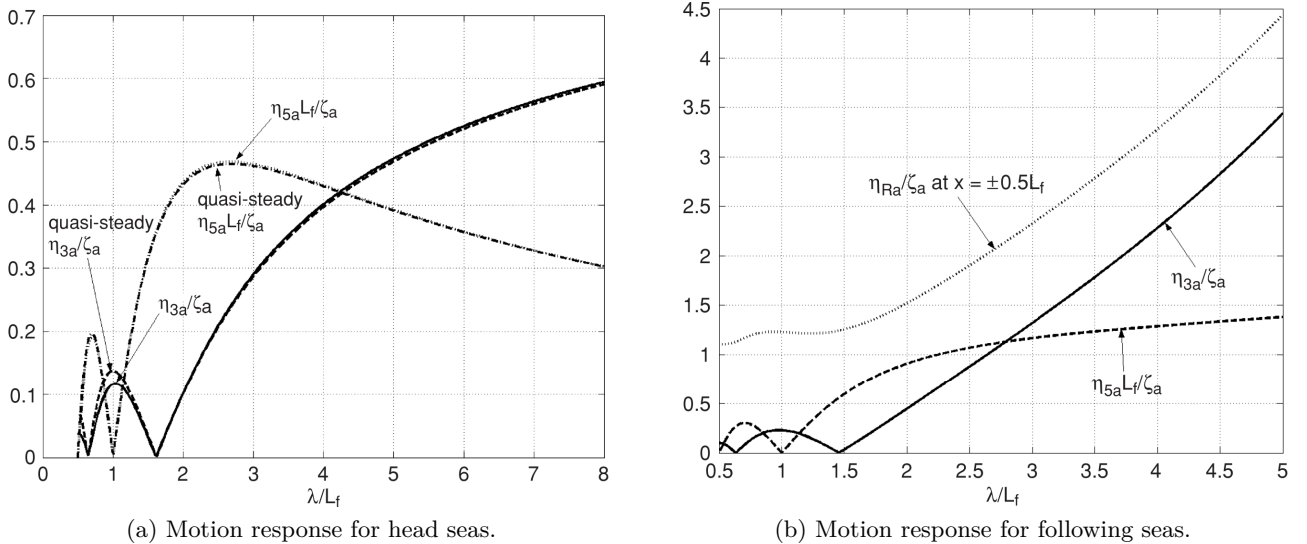


Figure 5.19: Heave and pitch amplitudes at increasing encountered wavelength. Uncontrolled case (Reproduced from [9]).

Sensors & Noise Effects The designed estimator seems to converge adequately fast, meaning that it can recover quickly even if there's signal loss. However, most trials were in the linear region, including only small deviations in pitch or roll. As the signal given to the Kalman Filter is corrected for euler angles before entering, it might be interesting to test performance in the non-linear region.

Care should be taken when using the butterworth filter to reject high frequency sinusoids from the sonar sensor readings. Any low-pass filter introduces delays, and especially if its cutoff frequency falls within the controllers bandwidth ω_B [10, Ch. 11], it might result in removing signal frequencies on which the controller should act.

Robustness This term should be used sparingly, as the proposed architectures are admittedly *not* robust, either by design or mathematically proven. While Section 5.4 lightly touched upon this subject, Robust-MPC is a subject on it's own, while the previous results only inform us that the controller has stabilizing authority over some design region which is not fully explored.

The heatmaps of Fig. 5.16b and Fig. 5.15b present with interesting, albeit expected findings. Moving the CG closer to the fore foils reduces their effective moment. The controller is able to correct for constant disturbances in the plant inputs, but at some point it will reach saturation and lead the system to instability. Figures 5.17 and 5.18 show another view of the same results in a point cloud form. Some results are expected, but what stood out was how there is an evident correlation between longitudinal weight distribution and tracking performance. Any changes in mass should be made towards the stern, such that an increase moves the CG towards the aft, while a decrease towards the bow.

Chapter 6

Conclusions & Future Work

This section is in a way a continuation of Section 1.3, as the defined objectives have now been more or less satisfied.

6.1 Dynamic Modelling & Simulation Environment

The proposed model of Fig. 2.25 provided a non-linear representation of the vessel dynamics, capable of producing perturbed models and performing linearizations. While the longitudinal dynamics are adequately modelled, the lateral degrees of freedom and in extension the coupling between them require further work. As an addition, the inclusion of Simulink blocks the MSS toolbox allowed for realistic short crested seas, which can provide important model assessment against real-world data.

Vessel Hydrodynamics By setting the lifting surface coefficients as lookup tables and not using their linear counterparts, larger operating regions can be described. At the same time, simply switching from 2D lookup tables to 3D, one can incorporate experimental data including *Re* sweeps.

Including the thrust forces in the model, allowing speed to vary according to the longitudinal force equilibrium added a degree of complexity that is often omitted in previous studies. While usually not detrimental to the overall results, the trimming and linearization will surely provide skewed results if thrust is not included.

6.2 Proposed Control Algorithms

The designed controllers overall seem to perform well, especially when used on the nominal model. Figure 5.14 along with the previously calculated failure rates provides the most comprehensive view on the benefits in using MPC. It is important to note however, that given the computational complexity both for the embedded microprocessor running the algorithm and the control engineer devising the code, LQI might prove to be the obvious choice. Given a fairly good approximation of model dynamics and coupled with the integral action and saturation, Algorithm 1 is exceptionally simple compared to what the MPC

Toolbox will provide when asked for C code generation. Once again, different criteria will select a different design. The developed estimator performs adequately as well, irregardless of the various assumptions that the system is breaking (colored noise, non-linearities, etc).

An important finding is how output estimation behaves. The hydrofoil craft can achieve dynamic stability at a range of foil depth. Including an output disturbance model essentially informs the controller that sensor readings are *not* trustworthy, and subject to constant bias of random walk character. This may prove as a dangerous approach to offset-free tracking, because tracking errors will be attributed to sensor bias, and be “accounted for” by the disturbance term. In short, the algorithm instead of correcting the output deviation by actuating the system, updates its state estimate to match the current equilibrium state. Thus, only input disturbances are estimated.

A more subtle caveat, which might have gone unnoticed in real world tests, is how the objective function decides on policy in real-time. While the error becomes smaller for the tracking states (heave in this case), further actuation might involve larger overall cost since it requires changes in $\mathbf{u}$ and increase of velocity states w, q . This in turn drives the controller to hold the stable state with marginal error as the most optimal with respect to the criteria set. This effect can be minimized by reduction of weights, but in a realistic, dynamic environment, it will probably not have time to manifest (no real asymptotic steady state outside of simulations).

Generally, development of higher fidelity models for the MPC internal plant is not preferable when the only benchmark used is a simulated vessel. This problem arises in machine learning as *overfitting*, where an algorithm adapts too well to specific training data, but fails to generalize in realistic settings. In this case, including too much of the simulation model to the internal plant, will yield increased performance but only “on paper”. This is why there is an overarching hesitation to declare the LPV-MPC-IDP design as the best suited for this task. The first order delay *is not* a realistic actuator response, which leaves us with little to assume about robustness when coupled with an unknown actuator.

6.3 Next Steps

Proposals for future work can focus specifically on one area of the two presented in this work. Unfortunately not much can be said about the simulation environment without comparison with experimental, or otherwise verified data. Having gone through the process of constantly altering it and improving it however, I believe that relying on pure Simulink up to the hydrodynamic level was not the optimal choice. An interesting alternative, closely related to the work of Godo [57] is the development of a non-linear dynamic lifting line model to augment the parametric modelling capabilities without relying on pre-computed coefficients.

For control & estimation, there are also numerous improvements that can be conducted. As the base prediction model used throughout this work has been unstable, experimentation with a pre-stabilized model [7], coupling MPC with the base LQ may improve stability margins.

Equally important, is the fact that generating trajectories to control more than one outputs, will have a beneficial effect on tracking and control effort. Going a step further, it may be essential to define appropriate set-points for the rates and not only the positions

to minimize control effort, taking advantage of the system's dynamics. The current design aims to eliminate rates (at a lower penalty of course) while also tracking heave movement, which is counter intuitive.

Development of robust MPC variants, such as Tube-MPC, to optimize performance against uncertain parameters [12] is also an alternative that will provide more guarantees in behavior. It may also be coupled with the design of a Moving Horizon Estimator (MHE) comparing performance to DKF.

Lastly, given a platform to experiment, such as a small scale vessel many more applications can be tested, such as:

- System identification and comparison to linearization output.
- Implementation of algorithms on an embedded microprocessor and *hardware-in-the-loop* verification before deployment.
- Real time data collection for comparison to simulated runs.

Appendices

Appendix A

Models, Parameters, Concepts

A.1 Test Vessel Parameters

To conduct simulations, a test vessel was picked based on data availability and since it provided a baseline design for MEBC solar class boats. Data has been collected from public documents such as [1] and official repositories [61]. The physical constants used in the simulation are already defined in the Nomenclature. Appropriate deviations from the nominal values are used to define perturbed models, as a way to study robustness without explicitly introducing parametric uncertainty as with $\mathcal{H}_\infty$ or μ -synthesis [29]. After all, the approach taken in Chapter 4 did not provide a *robust* MPC controller by definition.

Table A.1 collects all parameters used to initialize the model, as they were defined in Fig. 2.3 and Fig. 4.11. Positions are given as $d\{\text{axis}\}_{\text{object}}$, meaning the absolute distance from the vessel's CO *at nominal setup*. Moving the mass center as a perturbation automatically includes correction for positions of objects, since the origin of $\{b\}$ remains unaltered. The actual vectors defining position should take into account the axis definitions of Fig. 2.3.

Symbol	Description	Value	Units
m	Vessel mass	215	kg
I_y	Inertia moment about y_b axis	503	$\text{kg} \cdot \text{m}^2$
I_x	Inertia moment about x_b axis	31	$\text{kg} \cdot \text{m}^2$
dx_s	Distance to CG along x_b , sonar sensor	1.6	m
dy_s	Distance to CG along y_b , sonar sensor	0	m
dz_s	Distance to CG along z_b , sonar sensor	0.1	m
Hydrofoils			
dx_f	Distance to CG along x_b , fore foils	1.1	m
dy_f	Distance to CG along y_b , fore foils	0.62	m
dz_f	Distance to CG along z_b , fore foils	0.75	m
dx_a	Distance to CG along x_b , aft foil	2.45	m
dy_a	Distance to CG along y_b , aft foil	0	m
dz_a	Distance to CG along z_b , aft foil	0.96	m
S_F	Foil projected surface, fore	0.048	m^2
S_A	Foil projected surface, aft	0.046	m^2
c	Foil chord	0.08	m
$c_{s,f}$	Strut chord, fore	0.08	m
$c_{s,a}$	Strut chord, aft	0.1	m
b	Foil span	0.76	m
AR	Aspect ratio	≈ 12	
Propeller			
w	Wake number	0.05	
dx_p	Placement, along x_b axis	2.45	m
dz_p	Placement, along z_b axis	0.96	m
D	Diameter	0.3	m
z	Number of blades	3	
P/D	Pitch ratio	1.1	
A_E/A_O	Expanded area ratio	0.35	

Table A.1: Parameter definitions for nominal simulation model.

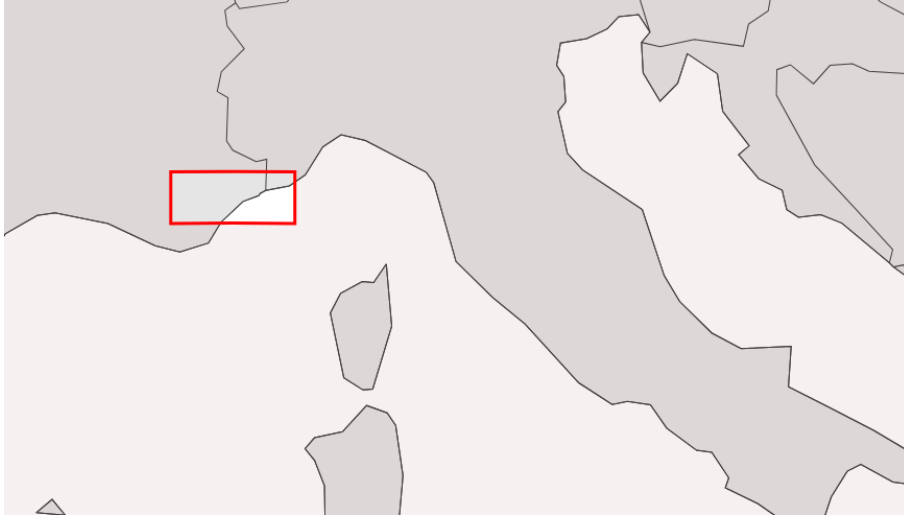


Figure A.1: Geographical window used for attaining sea state parameters.

A.2 Selection of Wave Parameters

It is probably evident by now, that sea state generation will play an important role in assessing control system performance. To get a feel of realistic sea state conforming to the operational profile of the studied vessel, one can make use of oceanographic data. One such dataset from the Copernicus Marine Service was used, based on research conducted by Korres et al. [78].

Figure A.2 presents aggregated results of this study, for the summer period of 2024 over the geographical window depicted in Fig. A.1. While it may seem narrow and fairly close to land, it marks roughly the position where MEBC takes place, as such vessels are not fit for traveling large distances in the open sea. The results of Fig. A.2 are mean values over the geographical window and were used to define both the regular wave components and the irregular short crested seas used in Section 5.3.

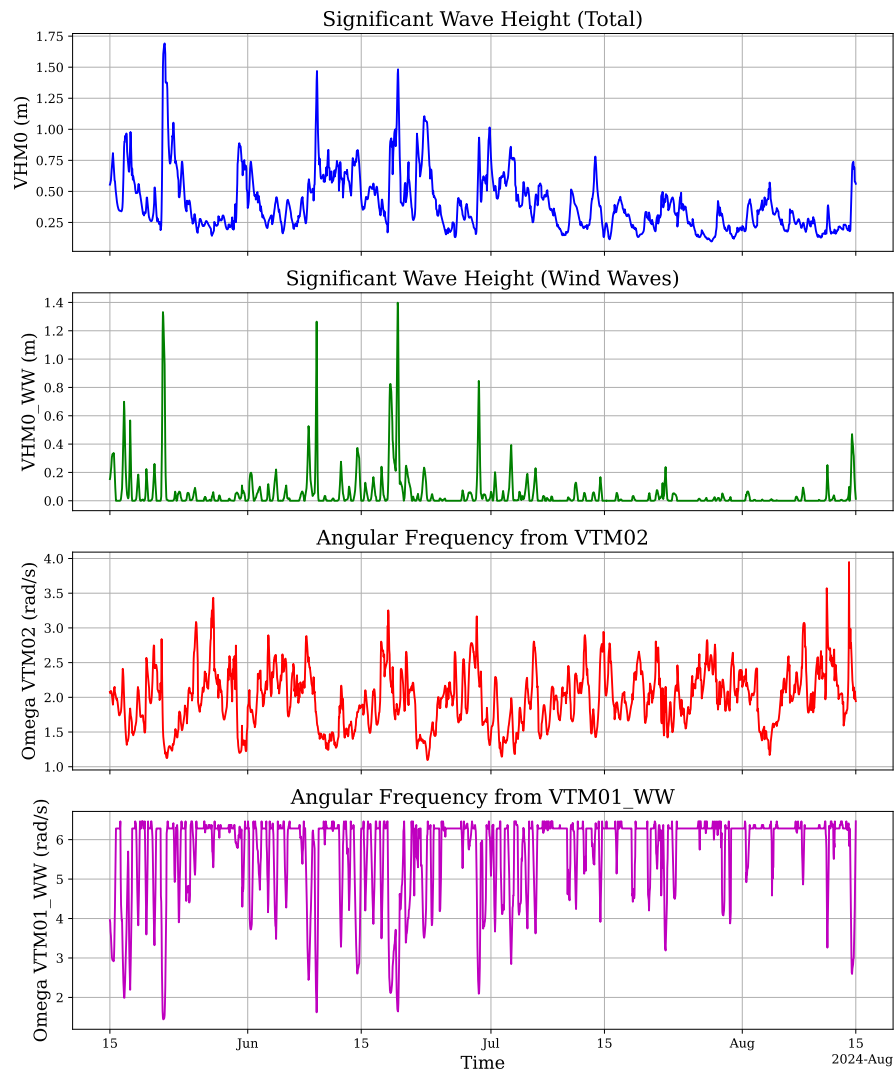


Figure A.2: Sea state over the summer of 2024 averaged over a specified geographical domain in the Mediterranean.

A.3 Nominal Model Matrices

$$\mathbf{Ax} = \begin{bmatrix} 0 & 0 & -7.0 & 0 & 1.0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1.0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1.0 \\ 0 & 0 & -9.81 & -1.179 & 0.8501 & 0 & -0.02685 \\ 0 & 0 & 0 & -2.803 & -12.86 & 0 & 3.894 \\ 0 & 0 & 0 & 0 & 0 & -64.36 & 0 \\ 0 & 0 & 0 & -0.4837 & 0.02095 & 0 & -15.44 \end{bmatrix} \begin{bmatrix} z \\ \phi \\ \theta \\ u \\ w \\ p \\ q \end{bmatrix} \quad (\text{A.3.1})$$

$$\mathbf{Bu} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1.365 & -1.365 & -1.129 & 0.007056 \\ -30.15 & -30.15 & -29.41 & 0 \\ -129.6 & 129.6 & 0 & 0 \\ 13.91 & 13.91 & -31.1 & 0.002895 \end{bmatrix} \begin{bmatrix} \delta_{fs} \\ \delta_{fa} \\ \delta_e \\ \delta_p \end{bmatrix} \quad (\text{A.3.2})$$

$$\mathbf{G}(s) = \begin{bmatrix} \frac{-30(s+14)(s+3)(s+1.1)}{s(s+15)(s+13)(s+1.6)(s-0.2)} & \frac{-30(s+14)(s+3)(s+1.1)}{s(s+15)(s+13)(s+1.6)(s-0.2)} \\ \frac{-130}{s(s+64)} & \frac{130}{s(s+64)} \\ \frac{14(s+13)(s+1.5)}{(s+15)(s+13)(s+1.6)(s-0.2)} & \frac{14(s+13)(s+1.5)}{(s+15)(s+13)(s+1.6)(s-0.2)} \\ \frac{-1.4(s+30)(s+14)(s+3)}{(s+15)(s+13)(s+1.6)(s-0.2)} & \frac{-1.4(s+30)(s+14)(s+3)}{(s+15)(s+13)(s+1.6)(s-0.2)} \\ \frac{-30(s+14)(s+1.8)(s-0.7)}{(s+15)(s+13)(s+1.6)(s-0.2)} & \frac{-30(s+14)(s+1.8)(s-0.7)}{(s+15)(s+13)(s+1.6)(s-0.2)} \\ \frac{-130}{(s+64)} & \frac{130}{(s+64)} \\ \frac{14(s+13)(s+1.5)s}{(s+15)(s+13)(s+1.6)(s-0.2)} & \frac{14(s+13)(s+1.5)s}{(s+15)(s+13)(s+1.6)(s-0.2)} \\ \frac{-29(s+18)(s-5.4)(s+1.1)}{s(s+15)(s+13)(s+1.6)(s-0.2)} & \frac{-0.029(s+20)(s-0.06)}{s(s+15)(s+13)(s+1.6)(s-0.2)} \\ 0 & 0 \\ \frac{-31(s+13)(s+1.3)}{(s+15)(s+13)(s+1.6)(s-0.2)} & \frac{0.0029(s+13)(s+0.18)}{(s+15)(s+13)(s+1.6)(s-0.2)} \\ \frac{-1.1(s+38)(s+17)(s-5.4)}{(s+15)(s+13)(s+1.6)(s-0.2)} & \frac{0.0071(s+15)(s+13)(s-0.26)}{(s+15)(s+13)(s+1.6)(s-0.2)} \\ \frac{-29(s+20)(s^2+s+1.2)}{(s+15)(s+13)(s+1.6)(s-0.2)} & \frac{-0.0085(s+36)(s-0.26)}{(s+15)(s+13)(s+1.6)(s-0.2)} \\ 0 & 0 \\ \frac{-31(s+13)(s+1.3)s}{(s+15)(s+13)(s+1.6)(s-0.2)} & \frac{0.0029(s+13)(s+0.18)s}{(s+15)(s+13)(s+1.6)(s-0.2)} \end{bmatrix} \quad (\text{A.3.3})$$

A.4 Mathematical Concepts

Rigid Body Kinetics

$$\begin{aligned} \mathbf{M}_{RB} &= \begin{bmatrix} m \mathbf{I}_{3 \times 3} & -m \mathbf{S}(\mathbf{r}_g^b) \\ m \mathbf{S}(\mathbf{r}_g^b) & \mathbf{I}_b \end{bmatrix} \\ &= \begin{bmatrix} m & 0 & 0 & 0 & mz_g & -my_g \\ 0 & m & 0 & -mz_g & 0 & mx_g \\ 0 & 0 & m & my_g & -mx_g & 0 \\ 0 & -mz_g & my_g & I_x & -I_{xy} & -I_{xz} \\ mz_g & 0 & -mx_g & -I_{yx} & I_y & -I_{yz} \\ -my_g & mx_g & 0 & -I_{zx} & -I_{zy} & I_z \end{bmatrix} \end{aligned} \quad (\text{A.4.1})$$

$$\mathbf{C}_{RB}(\mathbf{v}) = \begin{bmatrix} \mathbf{0}_{3 \times 3} & -m \mathbf{S}(\mathbf{v}_1) - m \mathbf{S}(\mathbf{v}_2) \mathbf{S}(\mathbf{r}_g^b) \\ -m \mathbf{S}(\mathbf{v}_1) + m \mathbf{S}(\mathbf{r}_g^b) \mathbf{S}(\mathbf{v}_2) & -\mathbf{S}(\mathbf{I}_b \mathbf{v}_2) \end{bmatrix} \quad (\text{A.4.2})$$

$$\mathbf{C}_{RB}(\mathbf{v}) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ -m(y_g q + z_g r) & m y_g p + w & m z_g p - v \\ m(x_g q - w) & -m(z_g r + x_g p) & m z_g q + u \\ m(x_g r + v) & m y_g r - u & -m(x_g p + y_g q) \\ m(y_g q + z_g r) & -m(x_g q - w) & -m(x_g r + v) \\ -m(y_g p + w) & m(z_g r + x_g p) & -m(y_g r - u) \\ -m(z_g p - v) & -m(z_g q - u) & m(x_g p + y_g q) \\ 0 & -I_{zy} q - I_{zx} p + I_z r & -I_y q + I_{yz} r + I_{zx} p \\ I_{yz} q - I_z r + I_{zx} p & 0 & I_x p - I_{xy} p - I_{xz} q \\ -I_{xy} p + I_y q - I_{yz} r & -I_x p + I_{xy} q + I_{xz} r & 0 \end{bmatrix} \quad (\text{A.4.3})$$

A.5 Vessel Design & Dynamic Stability

Discussed in the main body, a problem that arose during system modeling and simulation, was the counter-intuitive behavior of the longitudinal modes. To quantify the effects of weight placement (essentially a design variable, which can't be altered through a control law), an investigation of the model's physical characteristics was done based on works by B. Etkin [3] and J. Roskam [2].

For a vessel to be statically stable in pitch and height, we solved Newton's equations of motion at rest. Looking back at Fig. 3.1, dynamic stability demands what is called in literature "positive pitch stiffness", describing a *restoring moment* created as a response to pitch disturbances. This is depicted in Fig. A.3, as the balanced and positive stiffness curve, having negative slope for α , or $\frac{\partial C_m}{\partial \alpha} < 0$. We shall now derive an algebraic approximation for C_{m_α} in the context of a hydrofoil craft.

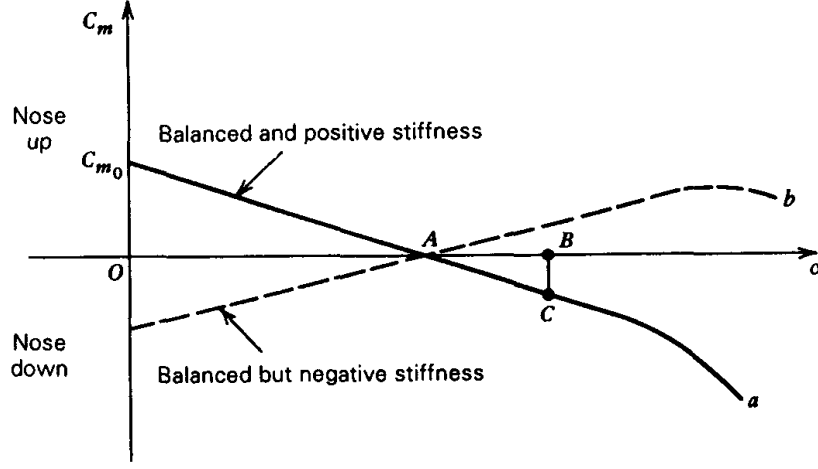


Figure A.3: Pitch stiffness (Reproduced from [3]).

Always referring to Fig. 3.1 and Eqs. (3.2.9) and (3.2.10), expanding lift and drag terms to include dynamic viscosity and surface and dividing by the term $\frac{1}{2}\rho V^2 S_w c_w$, leads to the non dimensional forms of Eqs. (A.5.1) and (A.5.2).

$$C_{m_w} = C_{m_{ac,w}} + \frac{l_w}{c_w} (C_{L_w} + C_{D_w} \alpha_\infty) + \frac{z_w}{c_w} (C_{L_w} \alpha_\infty - C_{D_w}) \quad (\text{A.5.1})$$

$$C_{m_t} = \frac{S_t}{S_w} C_{m_{ac,t}} - \frac{l_t S_t}{c_w S_w} (C_{L_t} + C_{D_t} \alpha_\infty) - \frac{z_t S_t}{c_w S_w} (C_{D_t} - C_{L_t} \alpha_\infty) \quad (\text{A.5.2})$$

For the whole vessel, we also include pitching moment due to propulsion elements (e.g, the propeller), leading to

$$C_m = C_{m_w} + C_{m_t} + C_{m_p} \quad (\text{A.5.3})$$

Taking the derivative with respect to α (here, the α_∞ angle, considering fixed controls), and by removing the terms containing α_{infly} for being negligible, as well as C_{m_p} for having constant moment, we acquire the following form for the whole vessel pitching moment.

$$\frac{\partial C_m}{\partial \alpha} = \frac{\partial C_{m_w}}{\partial \alpha} + \frac{\partial C_{m_t}}{\partial \alpha} + \frac{\partial C_{m_p}}{\partial \alpha} \quad (\text{A.5.4})$$

$$= \frac{\partial}{\partial \alpha} \left[C_{m_{ac,w}} + \frac{S_t}{S_w} C_{m_{ac,t}} + \frac{l_w}{c_w} C_{L_w} - \frac{z_w}{c_w} C_{D_w} - \frac{l_t S_t}{c_w S_w} C_{L_t} - \frac{z_t S_t}{c_w S_w} C_{D_t} \right] \quad (\text{A.5.5})$$

Each hydrodynamic term in Eq. (A.5.5) is already calculated (as a linear regression of foil aero data), since it concerns wing characteristics. To quantify the effect of longitudinal CG position on the dynamic stability, Eq. (A.5.5) is solved for the characteristics of the simulated system, while moving the center of gravity between the fore and aft foils. Figure A.4 is a graph of that function, marking three different design choices that were subsequently simulated with results shown in Fig. A.5, verifying the assumption stated in the main body: the selected design appears to be marginally stable (not in the theoretical control sense), thus not having the expected behavior.

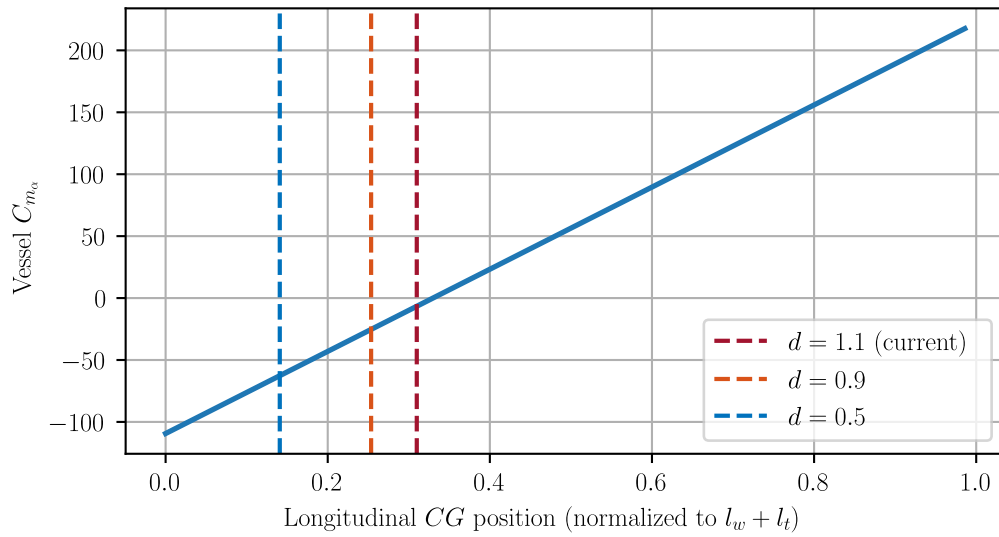


Figure A.4: Vessel $C_{m\alpha}$ for varying CG position, vertical lines corresponding to design choices simulated in Fig. A.5.

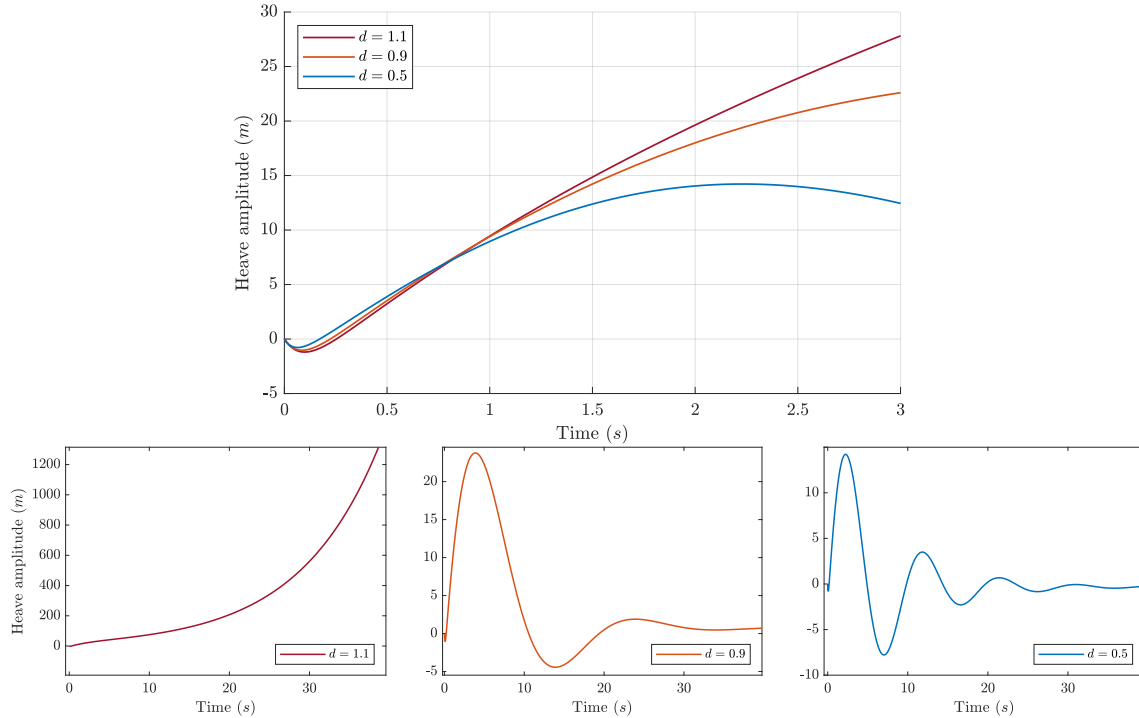


Figure A.5: Heave response for elevator impulse input at different CG placements (distance d from fore foils).

Appendix B

Developed Simulation Environment

This is a brief overview of the code that was developed to support the works within this thesis. It is divided between modelling and control, as is the theoretical part, as well as the MATLAB implementation. The complete Simulink model that forms the base for simulations and experiments is shown in Fig. B.1, with added explanatory remarks on each subsystem.

The codebase is divided between `implementation_control` and `implementation_dynmodel`, to separate functionalities regarding the non-linear model and linearization, from control design. A mix of source code and function files handle the model parameter definitions, general setup, controller creation and of course the simulations and benchmarks of Chapter 5.

B.1 Parametric Model

The parametric model, shown at the center of Fig. B.1, is none other than the one described by the diagram of Fig. 2.25. Actuator delay is handled by the *control delay* block before entering the vessel dynamics. Sea state is generated as a superposition of sinusoidal signals, defined by the *Waves* subsystem. State trajectories are recorded for simulation and data processing. Propeller rpm can be automatically set by including a PI controller.

B.2 Control & Estimation

Changing between different control architectures and real or estimated state feedback is done by Matlab's *Variant Subsystems*. Given a *workspace* variable, the Simulink model will switch between different controllers and turn state estimation on or off. There were 13 designs in total, of which only specific ones were considered for Chapter 5. All designs run on the same sampling time (10 Hz), with LQI-SW implementing the block diagram of Fig. 4.5 as shown in Algorithm 1. For state estimation, Algorithm 2 is used with additive noise being driven by band limited noise blocks connected in series with low-pass filters at the bandwidth frequency, and power spectral densities derived by Table 4.5.

B.3 Main Functionalities

Below is an overview of the important functions and scripts defined to aid in experimentation for the purposes of this thesis. They are also presented in roughly the same logical sequence that the program should be operated, in order to reach a proposed control design.

model_parameters Initializes the model with all parameters from Table A.1, including lookup tables for wing lift, drag and propeller open water characteristics. Also holds scaling matrices and sensor noise characteristics according to Table 4.5.

dimensionalize_model() Given m , $\mathbf{r}_g^b$, $\mathbf{I}_g$ create the appropriate $\mathbf{M}_{RB}$ matrices for EOM at the CO.

make_linmodel_array This is how automatic linearization is handled. Using **find_operpoint()** and **find_linear_model()**, a simpler simulink model of reduced DOF, in the form of Eq. (3.1.1) is linearized at various operating points. The result of this is illustrated in Fig. 3.2.

calculate_mpc, calculate_lqr Controllers are designed based on the linearized models, using pre-defined weighting matrices. For MPC, **generate_mpc()** serves as the link between scripts and the MPC Toolbox functionalities, to avoid repetitions.

sim_setup() Simulation specifics is set with predefined or given arguments. Includes model, simulation time, reference, sea state.

reference_generator() Generates a trajectory for the controller to follow over the simulation window. Allows for easy definition of steps, ramps, or cubic hermite smoothed ramps at a given point with specific duration and final value.

benchmark() Runs a simulation directly from the script level and returns logged signals for further analysis. Calculates *metrics*, including ISE, ITAE, IAE, controlEffort etc.

measureAmplitude() Used for assessing steady state oscillatory behavior under sinusoidal disturbances such as waves. Returns the motion amplitude.

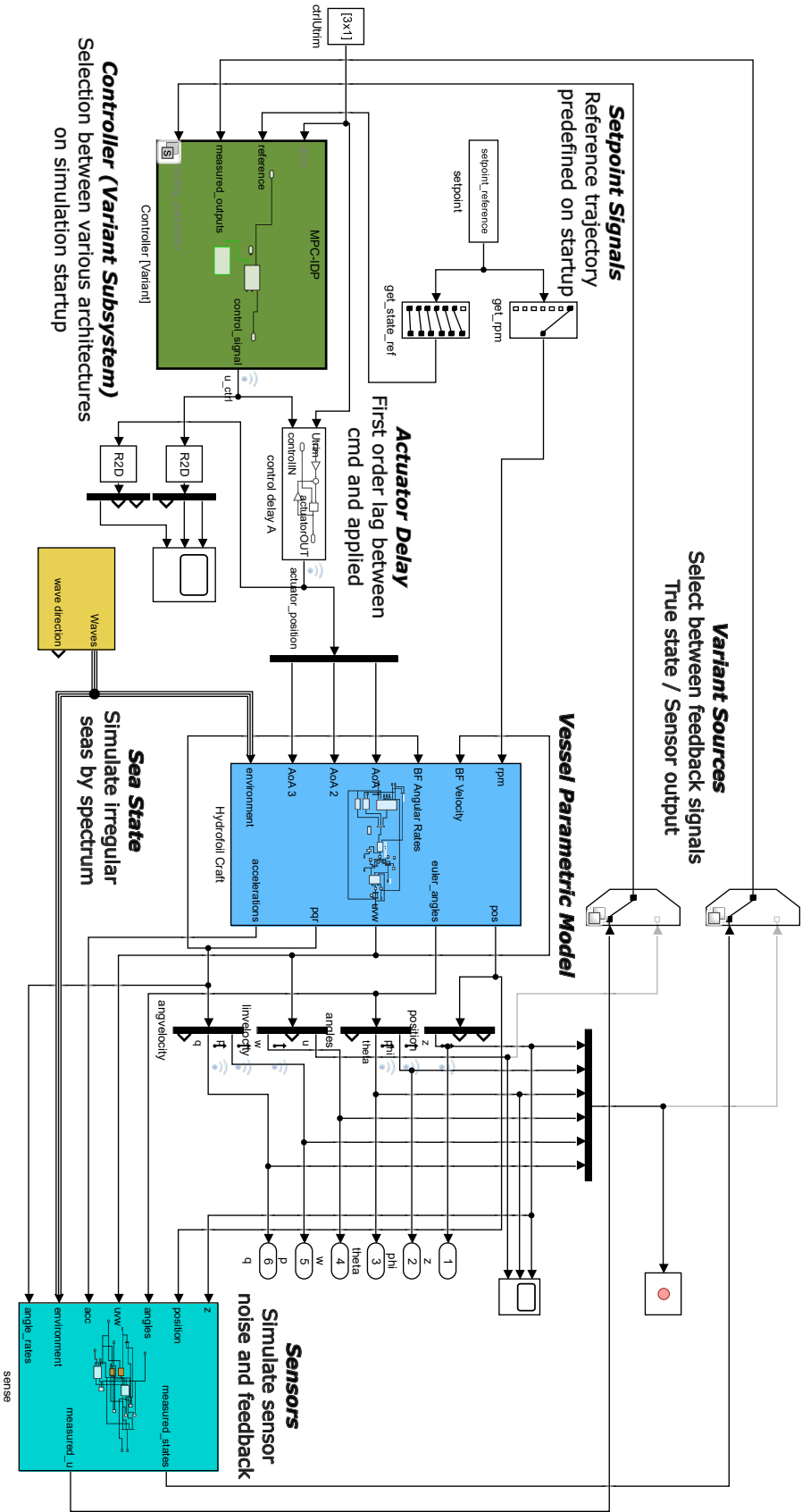


Figure B.1: Simulink model

Bibliography

- [1] B. J. Belime, “An hydrofoil control system for the SR02 vessel in the scope of the técnico solar boat project”, Master’s thesis, Instituto Superior Técnico, Lisbon, Portugal, Oct. 2020.
- [2] J. Roskam, *Airplane Flight Dynamics and Automatic Flight Controls*. 2001, vol. Part 1, Publication Title: Airplane Flight Dynamics and Automatic Flight Controls.
- [3] B. Etkin, *Dynamics of Atmospheric Flight*. John Wiley and Sons, Inc, 1972, Issue: 746 Publication Title: The Aeronautical Journal.
- [4] R. W. Beard and T. W. McLain, *Small unmanned aircraft: theory and practice*. Princeton, NJ: Princeton Univ. Press, 2012, 300 pp.
- [5] D. R. Stark and I. A. Hirsch, “Control system for hydrofoil”, pat. 3,886,884, Jun. 3, 1975.
- [6] “CFI brief: Flight controls of a typical commercial airliner”. [Online]. Available: <https://learntoflyblog.com/cfi-brief-flight-controls-of-a-typical-commercial-airliner/>.
- [7] A. Bemporad. “Model Predictive Control”. [Online]. Available: http://cse.lab.imtlucca.it/~bemporad/mpc_course.html.
- [8] J. D. Anderson, *Fundamentals of Aerodynamics* (McGraw-Hill series in aeronautical and aerospace engineering), Sixth edition. New York, NY: McGraw Hill Education, 2017, 1130 pp.
- [9] O. M. Faltinsen, *Hydrodynamics of high-speed marine vehicles*, Reprinted. Cambridge: Cambridge Univ. Press, 2007, 454 pp.
- [10] T. I. Fossen, *Handbook of Marine Craft Hydrodynamics and Motion Control*, 1st ed. Wiley, Jun. 2021. DOI: [10.1002/9781119575016](https://doi.org/10.1002/9781119575016).
- [11] S. Skogestad and I. Postlethwaite, *Multivariable Feedback Control: Analysis and Design*, 2nd ed. Hoboken, NJ: John Wiley, 2005, 574 pp.
- [12] F. Borrelli, *Predictive Control for Linear and Hybrid Systems*, 1st ed, in collab. with A. Bemporad and M. Morari. Cambridge: Cambridge University Press, 2017.
- [13] J. M. Maciejowski, *Predictive Control: With Constraints*, 1st ed. Harlow: Prentice Hall, 2001, 331 pp.

- [14] G. F. Franklin, J. D. Powell, and A. Emami-Naeini, *Feedback Control of Dynamic Systems*, Eighth edition. Ny, NY: Pearson, 2019, 902 pp.
- [15] D. E. Kirk, *Optimal Control Theory: An Introduction* (Dover Books on Electrical Engineering Series). Mineola, N.Y: Dover Publications, 2004.
- [16] J. P. Hespanha, *Linear Systems Theory*, Second edition. Princeton: Princeton University Press, 2018, 1 p.
- [17] D. Simon, *Optimal State Estimation: Kalman, H [infinity], and Nonlinear Approaches*. Hoboken, N.J: Wiley-Interscience, 2006, 526 pp.
- [18] M. Martin, “Equations of motion for hydrofoil craft”, Hydronautics Incorporated, Laurel, Maryland, Technical Report, 1962.
- [19] M. Martin, “Hydrofoil craft dynamics in a realistic sea including automatic control”, Hydronautics Incorporated, Laurel, Maryland, Technical Report, 1967.
- [20] L. J. Keuning, “A calculation method for the heave and pitch motions of a hydrofoil boat in waves”, Delft University of Technology, Ship Hydromechanics Laboratory, 483-P, Jul. 1979.
- [21] I. E. Garrick, *On some reciprocal relations in the theory of nonstationary flows*, NTRS Report/Patent Number: NACA-TR-629, Jan. 1, 1938.
- [22] H. Yamato, T. Koyama, A. Fushimi, and S.-h. Kim, “A study on unified automatic control system for longitudinal motion of jetfoil based on optimal servo theory”, *Journal of the Society of Naval Architects of Japan*, vol. 1997, no. 181, pp. 241–247, 1997. DOI: [10.2534/jjasnaoe1968.1997.241](https://doi.org/10.2534/jjasnaoe1968.1997.241).
- [23] S.-H. Kim and H. Yamato, “An experimental study of the longitudinal motion control of a fully submerged hydrofoil model in following seas”, *Ocean Engineering*, vol. 31, no. 5, pp. 523–537, Apr. 1, 2004. DOI: [10.1016/j.oceaneng.2003.10.003](https://doi.org/10.1016/j.oceaneng.2003.10.003).
- [24] S.-H. Kim and H. Yamato, “The estimation of wave elevation and wave disturbance caused by the wave orbital motion of a fully submerged hydrofoil craft”, *Journal of Marine Science and Technology*, vol. 10, no. 1, pp. 22–31, Mar. 1, 2005. DOI: [10.1007/s00773-004-0183-9](https://doi.org/10.1007/s00773-004-0183-9).
- [25] G. D. Bousquet, M. S. Triantafyllou, and J.-J. E. Slotine, “Control of a flexible, surface-piercing hydrofoil for high-speed, small-scale applications”, presented at the 2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), ISSN: 2153-0866, Sep. 2017, pp. 4203–4208. DOI: [10.1109/IROS.2017.8206282](https://doi.org/10.1109/IROS.2017.8206282).
- [26] I. Chatzakis, “Motion control of high-speed hydrofoil vessels using state-space methods”, Master’s thesis, Massachusetts Institute of Technology, Cambridge, Massachusetts, 2004.
- [27] R. Bencatel, S. Keerthivarman, I. Kolmanovsky, and A. R. Girard, “Full state feedback foiling control for america’s cup catamarans”, *IEEE Transactions on Control Systems Technology*, vol. 29, no. 1, pp. 1–17, Jan. 2021. DOI: [10.1109/TCST.2019.2955059](https://doi.org/10.1109/TCST.2019.2955059).
- [28] G. P. van Marrewijk and J. K. Schonebaum, “An experimentally validated dynamical model of a single-track hydrofoil boat”, *Naval Engineers Journal*, vol. 130, pp. 113–134, Mar. 2018.

- [29] T. Kapnisis, “Modelling and control of experimental scale hydrofoil craft”, Master’s thesis, Delft University of Technology, 2022, 128 pp.
- [30] F. Furrer and P. J. Gohl, “Control of an autonomous sailing catamaran on the concept of hydrofoils”, Semester Thesis, Swiss Federal Institute of Technology, Zurich, 2010.
- [31] M. Calmon et al., “L’hydroptere: How multidisciplinary scientific research may help break the sailing speed record”, *Proceedings of the 2nd INTERNATIONAL CONFERENCE ON INNOVATION IN HIGH PERFORMANCE SAILING YACHTS*, Jan. 1, 2010.
- [32] H. Stromqvist, “Robust motion control strategies for hydrofoil-equipped naval vessels considering scaling effects”, Master’s thesis, KTH Royal Institute of Technology, Stockholm, Sweden, Jun. 24, 2019.
- [33] A. Thalin, “Extended kalman filter as observer for a hydrofoiling watercraft”, Master’s thesis, KTH Royal Institute of Technology, Stockholm, Sweden, 2022.
- [34] E. Anderberg, “Stabilizing a single strut hydrofoil using linear-quadratic control and gain scheduling”, Master’s thesis, KTH Royal Institute of Technology, Stockholm, Sweden, Mar. 7, 2024.
- [35] Y. Shang and J. Horrillo, “Advanced heave control of a hydrofoil-based autonomous surface vehicle: CFD modeling with integrated variable propeller revolution control”, *Ocean Engineering*, vol. 315, p. 119 800, Jan. 1, 2025. DOI: [10.1016/j.oceaneng.2024.119800](https://doi.org/10.1016/j.oceaneng.2024.119800).
- [36] Y. Shang, J. Horrillo, and A. Jose, “Utilizing moving-mass actuated stabilizers for pitch and roll control in hydrofoil-based autonomous surface vehicles: A direct CFD analysis integrated with auto-control algorithm”, *Ocean Engineering*, vol. 335, p. 121 692, Aug. 15, 2025. DOI: [10.1016/j.oceaneng.2025.121692](https://doi.org/10.1016/j.oceaneng.2025.121692).
- [37] J. M. K. Godø, “Zero-emission hydrofoil fast ferries: Design, modelling, and performance evaluation”, Doctoral thesis, Norwegian University of Science and Technology, Trondheim, Dec. 2024.
- [38] 26th ITTC Specialist Committee on Uncertainty Analysis, “Fresh water and seawater properties”, International Towing Tank Conference, 7.5-02-0-03, Sep. 2011.
- [39] “An overview of reference frames and coordinate systems (in the SPICE context)”. [Online]. Available: https://naif.jpl.nasa.gov/pub/naif/toolkit_docs/Tutorials/pdf/individual_docs/17_frames_and_coordinate_systems.pdf.
- [40] T. I. Fossen and T. Perez, *Marine systems simulator (MSS)*, 2004. [Online]. Available: <https://github.com/cybergalactic/MSS>.
- [41] T. I. Fossen, “A nonlinear unified state-space model for ship maneuvering and control in a seaway”, *International Journal of Bifurcation and Chaos*, vol. 15, no. 9, pp. 2717–2746, Sep. 2005. DOI: [10.1142/S0218127405013691](https://doi.org/10.1142/S0218127405013691).
- [42] S. P. Boyd and L. Vandenberghe, *Introduction to Applied Linear Algebra – Vectors, Matrices, and Least Squares*. Cambridge, UK ; New York, NY: Cambridge University Press, 2018.
- [43] T. I. Fossen. “Lecture notes TTK 4190 guidance, navigation and control of vehicles”. [Online]. Available: <https://github.com/cybergalactic/FossenHandbook>.

- [44] M. Touw, “Prediction of the longitudinal stability and motions of a hydrofoil ship with a suspension system between the wings and the hull using a state-space model”, Master’s thesis, Delft University of Technology, 2020.
- [45] E. B. Piene, “Disturbance rejection of a high speed hydrofoil craft using a frequency weighted h2-optimal controller”, Master’s thesis, Norwegian University of Science and Technology, 2018.
- [46] L. H. Holthuijsen, *Waves in Oceanic and Coastal Waters*, 1st ed. Cambridge University Press, Jan. 25, 2007. DOI: [10.1017/CBO9780511618536](https://doi.org/10.1017/CBO9780511618536).
- [47] J. N. Newman, *Marine Hydrodynamics*, 40th anniversary edition. Cambridge: MIT press, 2017.
- [48] M. Drela, “XFOIL: An analysis and design system for low reynolds number airfoils”, in *Low Reynolds Number Aerodynamics*, T. J. Mueller, Ed., Springer Berlin Heidelberg, 1989, pp. 1–12. DOI: [10.1007/978-3-642-84010-4_1](https://doi.org/10.1007/978-3-642-84010-4_1).
- [49] G. Leishman, *Introduction to Aerospace Flight Vehicles*. Embry-Riddle Aeronautical University, 2022. DOI: [10.15394/eaglepup.2022.1066](https://doi.org/10.15394/eaglepup.2022.1066).
- [50] E. S. Filippas, G. P. Papadakis, and K. A. Belibassakis, “Free-surface effects on the performance of flapping-foil thruster for augmenting ship propulsion in waves”, *Journal of Marine Science and Engineering*, vol. 8, no. 5, p. 357, May 2020, Number: 5 Publisher: Multidisciplinary Digital Publishing Institute. DOI: [10.3390/jmse8050357](https://doi.org/10.3390/jmse8050357).
- [51] H. E. Taha and A. S. Rezaei, “On the dynamics of unsteady lift and circulation and the circulatory-non-circulatory classification”, in *AIAA Scitech 2019 Forum*, San Diego, California: American Institute of Aeronautics and Astronautics, Jan. 7, 2019. DOI: [10.2514/6.2019-1853](https://doi.org/10.2514/6.2019-1853).
- [52] K. A. Belibassakis and G. K. Politis, “Hydrodynamic performance of flapping wings for augmenting ship propulsion in waves”, *Ocean Engineering*, vol. 72, pp. 227–240, Nov. 1, 2013. DOI: [10.1016/j.oceaneng.2013.06.028](https://doi.org/10.1016/j.oceaneng.2013.06.028).
- [53] J. D. DeLaurier, “An aerodynamic model for flapping-wing flight”, *The Aeronautical Journal*, vol. 97, no. 964, pp. 125–130, Apr. 1993. DOI: [10.1017/S0001924000026002](https://doi.org/10.1017/S0001924000026002).
- [54] S. L. Brunton and C. W. Rowley, “Empirical state-space representations for theodorsen’s lift model”, *Journal of Fluids and Structures*, vol. 38, pp. 174–186, Apr. 1, 2013. DOI: [10.1016/j.jfluidstructs.2012.10.005](https://doi.org/10.1016/j.jfluidstructs.2012.10.005).
- [55] A. V. Oppenheim, A. S. Willsky, and S. H. Nawab, *Signals and systems* (Prentice-Hall signal processing series), 2. ed. Upper Saddle River, NJ: Prentice Hall, 1997, 957 pp.
- [56] J. O. Scherer, “Experimental and theoretical investigation of large amplitude oscillating foil propulsion”, Hydronautics Incorporated, Laurel, Maryland, May 1968.
- [57] J. M. K. Godø, S. Steen, and O. M. Faltinsen, “A resistance model for hydrofoil fast ferries with fully submerged foil systems”, *Ocean Engineering*, vol. 301, p. 117 503, Jun. 1, 2024. DOI: [10.1016/j.oceaneng.2024.117503](https://doi.org/10.1016/j.oceaneng.2024.117503).
- [58] M. S. Triantafyllou and F. S. Hover, *Maneuvering and Control of Marine Vehicles*. Cambridge, Massachusetts USA: Department of Ocean Engineering. Massachusetts Institute of Technology, Nov. 5, 2003.

- [59] “Solar Boat Twente (website): Boat 2020-21”. [Online]. Available: <https://www.solarboattwente.nl/en/boot-20-21/>.
- [60] M. M. Bernitsas, D. Ray, and P. Kinley, “KT, KQ and efficiency curves for the wageningen b-series propellers”, Department of Naval Architecture and Marine Engineering, University of Michigan, Ann Arbor, Michigan, 237, May 1981.
- [61] “Técnico Solar Boat Open Source Content”. [Online]. Available: <https://gitlab.com/tecnico.solar.boat>.
- [62] “B-series propeller generator”. [Online]. Available: <https://www.wageningen-b-series-propeller.com/>.
- [63] C.-T. Chen, *Linear System Theory and Design* (The Oxford series in electrical and computer engineering), 3rd ed. New York: Oxford University Press, 1999, 334 pp.
- [64] M. Dahleh, M. Dahleh, and G. Verghese, “Chapter 27: Poles and zeros of MIMO systems”, in *6.241J Dynamic Systems and Control*, ser. MIT OpenCourseWare, Creative Commons BY-NC-SA., Massachusetts Institute of Technology, 2011.
- [65] K. J. Åström and R. M. Murray, *Feedback Systems: An Introduction for Scientists and Engineers*. Princeton: Princeton University Press, 2008.
- [66] S. L. Brunton and J. N. Kutz, *Data-Driven Science and Engineering: Machine Learning, Dynamical Systems, and Control*, 2nd edition. Cambridge, 2022.
- [67] A. Bemporad, N. L. Ricker, and M. Morari, *Model Predictive Control Toolbox™ User’s Guide*. 1 Apple Hill Drive Natick, MA: The MathWorks, Inc.
- [68] J. Nocedal and S. J. Wright, *Numerical Optimization* (Springer series in operations research and financial engineering), Second edition. New York, NY: Springer, 2006, 1 p.
- [69] G. Pannocchia, “Offset-free tracking MPC: A tutorial review and comparison of different formulations”, in *2015 European Control Conference (ECC)*, Jul. 2015, pp. 527–532. DOI: [10.1109/ECC.2015.7330597](https://doi.org/10.1109/ECC.2015.7330597).
- [70] E. Alcalá, V. Puig, and J. Quevedo, “LPV-MPC control for autonomous vehicles”, *IFAC-PapersOnLine*, 3rd IFAC Workshop on Linear Parameter Varying Systems LPVS 2019, vol. 52, no. 28, pp. 106–113, Jan. 1, 2019. DOI: [10.1016/j.ifacol.2019.12.356](https://doi.org/10.1016/j.ifacol.2019.12.356).
- [71] *MTi-680g*, Movella, May 2023. [Online]. Available: <https://www.xsens.com/hubfs/Downloads/Leaflets/MTi-680G.pdf>.
- [72] *ToughSonic 14 level & distance sensor*, Senix, Jul. 1, 2019. [Online]. Available: <https://senix.com/content/DataSheets/Data%20Sheet%20Senix%20ToughSonic%2014%20-%2030mm.pdf>.
- [73] P. Horowitz and W. Hill, *The Art of Electronics*, Third edition. Cambridge, New York: Cambridge University Press, 2024, 1230 pp.
- [74] A. E. Bryson and Y.-C. Ho, *Applied Optimal Control: Optimization, Estimation, and Control*, 1st ed. Routledge, May 4, 2018. DOI: [10.1201/9781315137667](https://doi.org/10.1201/9781315137667).
- [75] P. Sarhadi, *On the standard performance criteria for applied control design: PID, MPC or machine learning controller?*, version: 3, Oct. 1, 2025. DOI: [10.48550/arXiv.2503.14379](https://doi.org/10.48550/arXiv.2503.14379).

- [76] R. Tempo, G. Calafiore, and F. Dabbene, *Randomized Algorithms for Analysis and Control of Uncertain Systems: With Applications* (Communications and Control Engineering). London: Springer, 2013. DOI: [10.1007/978-1-4471-4610-0](https://doi.org/10.1007/978-1-4471-4610-0).
- [77] C. Liu, S. Shi, and B. De Schutter, “Stability and performance analysis of model predictive control of uncertain linear systems”, 2024, Publisher: arXiv Version Number: 3. DOI: [10.48550/ARXIV.2405.15552](https://doi.org/10.48550/ARXIV.2405.15552).
- [78] G. Korres, C. Oikonomou, D. Denaxa, and M. Sotiropoulou, *Mediterranean sea waves analysis and forecast (copernicus marine service MED-waves, MEDWAM4 system, version 1*, 2023. DOI: [10.25423/CMCC/MEDSEA_ANALYSISFORECAST_WAV_006_017_MEDWAM4](https://doi.org/10.25423/CMCC/MEDSEA_ANALYSISFORECAST_WAV_006_017_MEDWAM4).